SCF

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5G FAPI: PHY API specification

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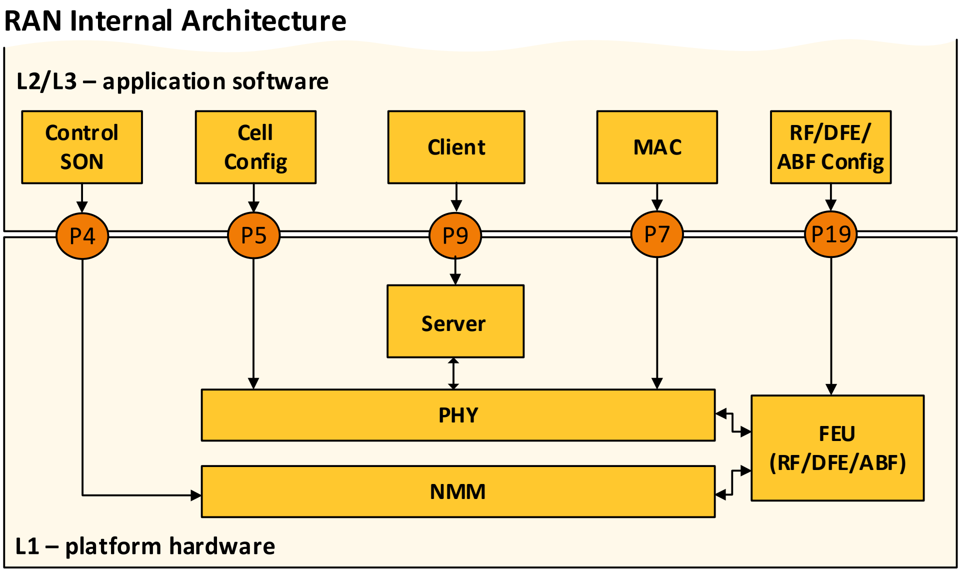
Scope

The functional application platform interface (FAPI) is an initiative within the small cell industry released by Small Cell Forum (SCF), which establishes interoperability and innovation among suppliers of platform hardware, platform software and application software. It does so by providing a common API around which suppliers can create a competitive ecosystem. In doing this, we support a long and distinguished engineering tradition of providing an 'interchangeability of parts' to ensure that the systems vendors can take advantage of the latest innovations in silicon and software with minimum barriers to entry, and the least amount of custom re-engineering. Hence the specification helps support an innovative and competitive ecosystem for vendors of 5G small cell hardware, software and equipment.

For 5G, the FAPI suite comprises five specification documents covering the following APIs:

- '5G FAPI: PHY API' – main data path (P7) and PHY mode control (P5) interface [SCF222]
- '5G FAPI: RF and Digital Front End Control API' – (P19) for Frontend Unit control [SCF223]
- 'Network Monitor Mode API' – (P4) for 2G/3G/4G/5G [SCF224]
- '5G nFAPI Specification' [SCF225]
- '5G OAM Specification' [SCF229]



SON (Self Organizing Networks), MAC (Medium Access Control), NMM (Network Monitor Mode), FEU (Front End Unit), including DFE (Digital Front End) and ABF (Analog Beam Forming)

The 5G FAPI defines internal interfaces and is agnostic to RAN architecture.

Historical placeholder APIs P1, P2, P3, P6 and M1 are outside the scope of this document.

The table below summarizes all of SCF's FAPI API specifications, covering 5G, 4G, 3G and 2G radio access technologies (RATs).

			SCF FAPI Support					
Brand name		**3GPP RAT Type** [3GPP TS 29.274]	**PHY API**	**Network Monitor Mode**	**RF / Digital Front End**	**network FAPI PHY/MAC split**	**Small cell (PNF/RU) management model**	**OAM API**
2G	GSM	GERAN		[SCF224]				
3G	UMTS	UTRAN	[SCF048]	[SCF224]				
3G	HSPA	HSPA Evolution	[SCF048]	[SCF224]				
4G	LTE	EUTRAN (WB-E-UTRAN)	[SCF082]	[SCF224]	[SCF082]	[SCF082]	[SCF167]	
4G	LTE-NB-IoT	EUTRAN-NB-IoT	[SCF082]	[SCF224]	[SCF082]	[SCF082]	[SCF167]	
4G	LTE-M	LTE-M	[SCF082]	[SCF224]	[SCF082]	[SCF082]	[SCF167]	
5G	5G NR	NR	[SCF222]	[SCF224]	[SCF223]	[SCF225]	[SCF227] [1]	[SCF229]

SCF FAPI support for different radio access technologies – the green boxes indicate new for 5G

[1] Currently in scoping

5G PHY API executive summary

This document provides the 5G PHY API specification for an application platform interface (API) between the MAC and PHY protocol layers within the small cell ecosystem. This functional application platform interface (FAPI) is an internal interface within an integrated or disaggregated small cell.

This version is for release 15 of 3GPP specification, as well select release 16 features. Among others, it supports:

- DL and UL SU-MIMO and MU-MIMO (including Massive-MIMO), based on SRS
- eMIMO[23], including mTRP, low-PAPR RS, and MU-CSI.
  - MU-CSI support: UCI payloads can be transparent to FAPI.
- 2-step RACH [24][25].
- NR positioning [27]

It also supports optimized processing:

- Control Plane/User Plane in-message separation
- Word-aligned padding

Features relevant to multiple split architectures are also supported:

- Delay management with or without timestamps
- L2 awareness of the L1 interface between PHY and FEU
  - Number and scope of spatial streams
  - Connectivity and synchronization events

Upper layer virtualization and PHY resource partitioning are supported via Virtual Functions and PHY Groups. PHY implementations may allow pooling of a PHY Group's resources.

Bitstream-level backward compatibility is also supporting via extensions, back to FAPIv7 and – without extensions – FAPIv6.

The 5G PHY API specification defines a control interface (P5) and a user plane or data path interface (P7). The document is structured as follows:

- **Introduction**: This section provides the high-level overview and governing principles of the API.
- **PHY API Procedures**: This section details procedures to configure and control a PHY entity.
- **PHY API Messages**: This section defines the FAPI specific messages and structures to implement these procedures

Contents

Tables

Figures

1. Introduction

1.1 5G new radio (NR)

5G NR is standardized by 3GPP (http://www.3gpp.org) and designed as an evolution to the current 4G LTE wireless network. Requirements for 5G include higher bandwidth, lower latency, improved reliability and increased density of users/devices – although not necessarily all at the same time. Instead, three service types are defined: enhanced mobile broadband (eMBB), ultra-reliable low latency (URLLC) and massive internet of things (mIoT). This wide range of use cases results in flexibility, which is a key feature of 5G.

Figure 1-1 shows the architecture of a 5G network. It consists of only two elements: the 5G Core (consisting of the access and mobility function (AMF), user plane function (UPF), etc.) and the 5G Node B (gNB). The 5G FAPI resides within the gNB element. The standardized interfaces in 5G network are called NG, Xn and F1. The L1 is not involved in these interfaces.

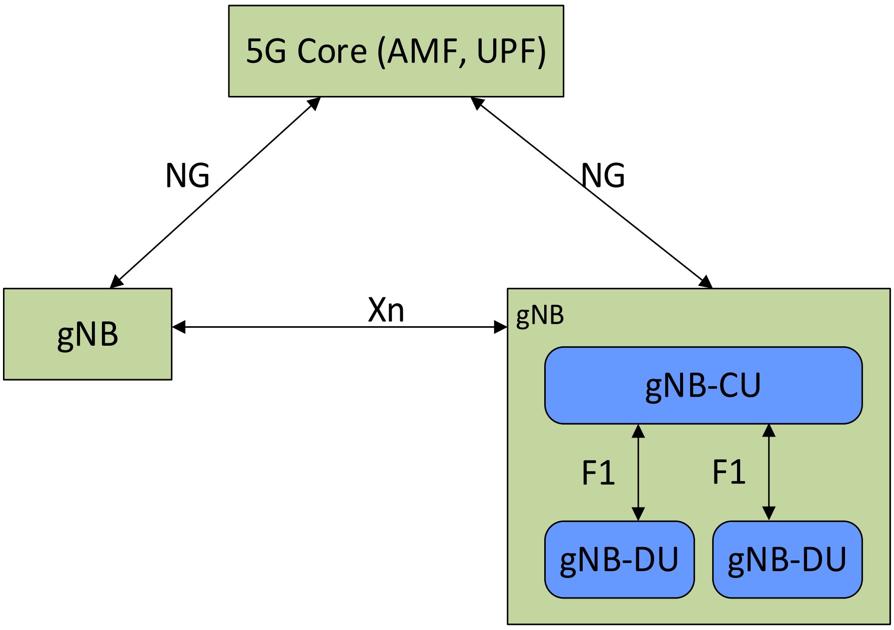


Figure 1-1 5G architecture

For 5G FAPI the MAC and PHY reside is a single element, resulting in FAPI existing as an internal interface within the gNB, as shown in Figure 1-2.

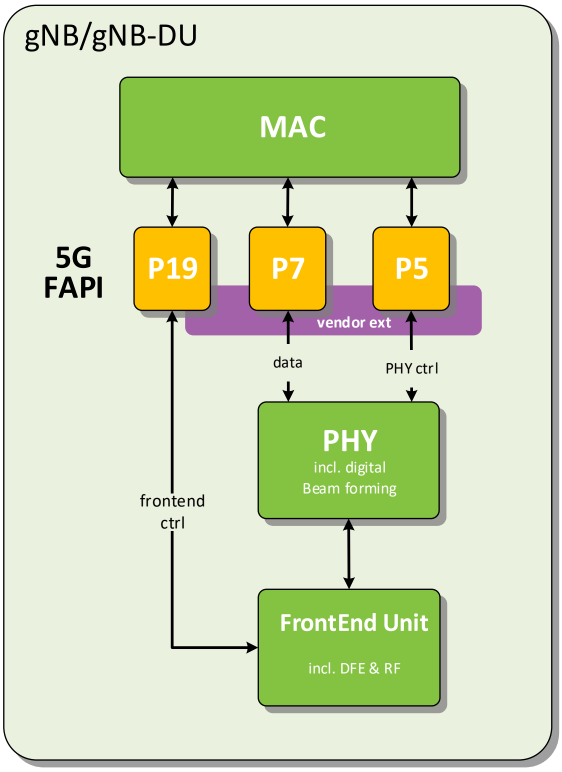


Figure 1-2 5G FAPI architecture

The introduction of 5G nFAPI for split 6 – with MAC and PHY residing in different physical locations, a distributed unit (S-DU) and radio unit (S-RU) – is outside the scope of this document.

1.2 PHY API

The 5G FAPI PHY API, defined in this document, resides within the gNB/gNB-DU component and Figure 1-3 shows the protocol model for the gNB defined in the 5G-RAN architectural standard 3GPP TS 38.401 [1]. It highlights the separation of control- and data-plane information, which is maintained throughout the 5G NR network. Both control- and data-plane information is passed through the PHY API. However, each API message contains either control- or data-plane information, but never both.

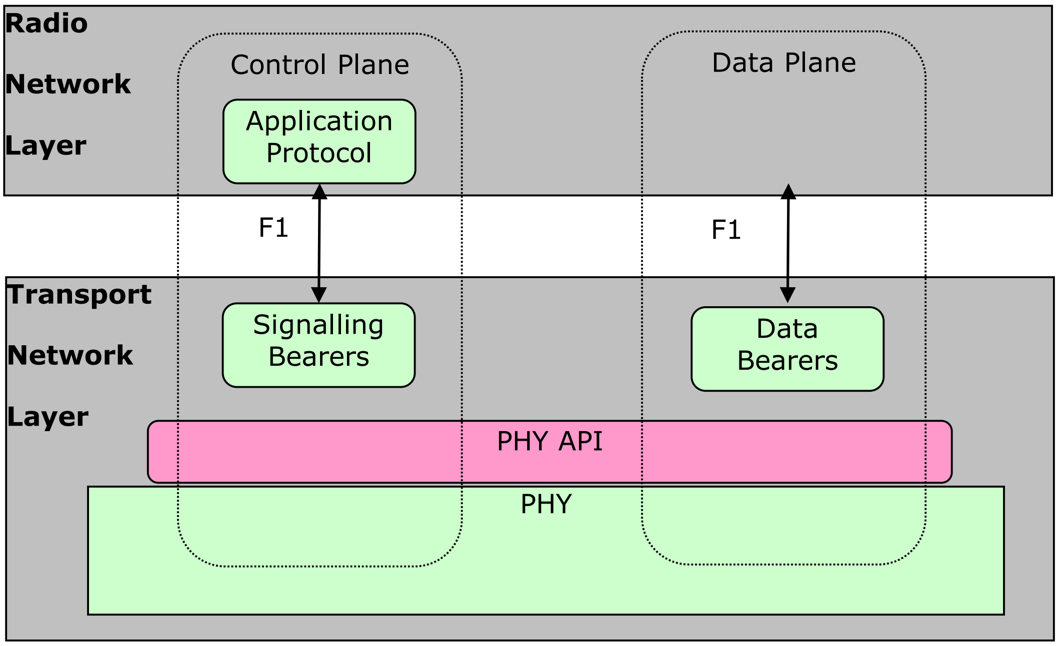


Figure 1-3 5G RAN protocol model [from 3GPP TS 38.401]

Figure 1-4 provides an example of how the different L2/L3 protocol layers will interact with the PHY API. In this example, a PHY control entity is responsible for configuration procedures (P5). The MAC layer is responsible for the exchange of data-plane messages with the PHY (P7). The PHY configuration sent over the P5 interface may be determined using SON techniques, information model parameters sent from an OAM system, or a combination of both methods. If carrier aggregation is supported, then one instance of the PHY API exists for each component carrier, as defined in 3GPP TS 38.104 [7].

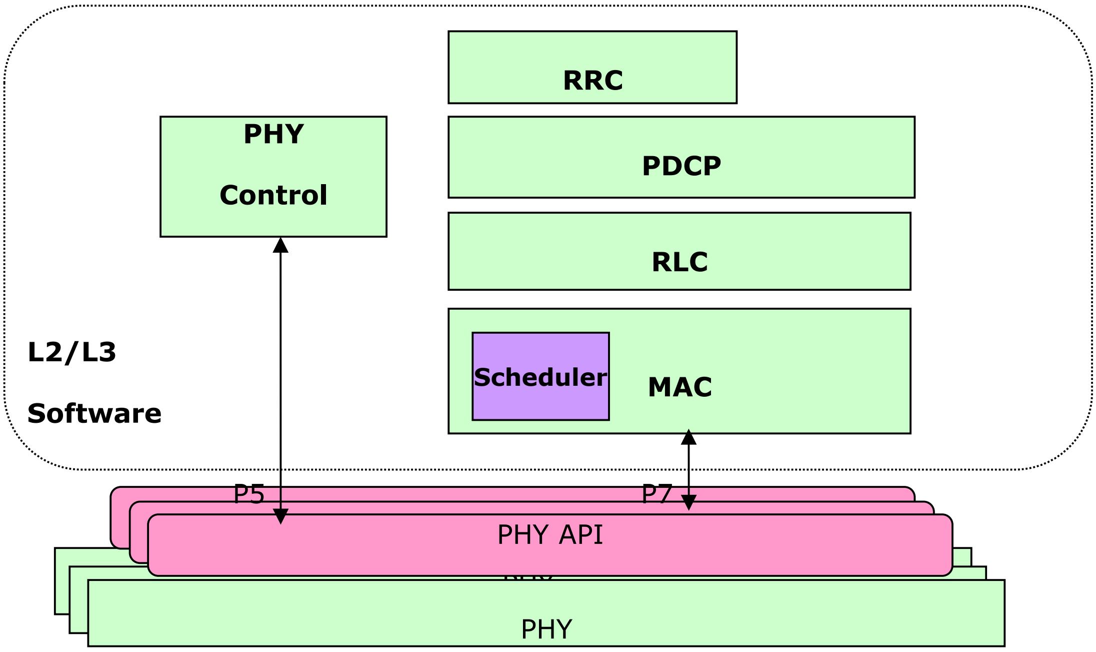


Figure 1-4 PHY API interactions

2. PHY API Procedures

This section gives an overview of the procedures which use the PHY API. These procedures are split into two groups: configuration procedures and slot procedures. Configuration procedures handle the management of the PHY layer and are expected to occur infrequently. Slot procedures determine the structure of each slot and operate with a periodicity based on the subcarrier spacing numerology, namely 125us, 250us, 500us or 1ms periodicity. Unless otherwise noted, these procedures apply per PHY.

The set of PHY supported by the PHY layer is partitioned into PHY Groups.

When defined, PHY ID #255 corresponds to one of:

- a supervisory entity (e.g. an SoC-wide concept, hosting a non-PHY specific, common context) that cannot be in PHY RUNNING state and cannot terminate P7. In nFAPI, as specified in SCF-225 [10], PNF also terminates messages with PHY ID#255. Common context is specific to each virtual function, when virtual functions are defined.
- a PHY Group context, that cannot be in PHY RUNNING state and cannot terminate P7.

Digital beamforming (for uplink and downlink channels) and digital precoding (only for PDSCH, in this version of the specification) relies on both configuration and slot procedures:

- Configuration procedure: L2/L3 configures a table of digital beam weights (DBT: digital beam table) applicable to baseband ports, as well as a table of precoder weights (PMT: precoder matrix table) applicable to logical ports.
- Slot procedures: L2/L3 signals per-slot, per-channel beam indices. For PDSCH, L2/L3 also signals precoder matrix indices.

RF beamforming is supported in P19 API, per SCF-223 [9].

Virtual Functions allow the PHY layer to support multiple – possibly virtualized – L2/L3 functions.

2.1 Configuration Procedures

The configuration procedures supported by PHY API (except for API handled at common or PHY Group context) are:

- Initialization
- Termination
- Restart
- Reset
- Query
- Error notification

PHY APIs applicable to the common or PHY Group context are described in 2.1.10

These procedures will move the PHY layer through the IDLE, CONFIGURED and RUNNING states, as shown in Figure 2-1. A list of the PHY API configuration messages which are valid in each state is given in Table 2-1 .

L2/L3 shall determine from L1 Capability (e.g. TLV 0x0176) whether a PHY's State machines can change independently of other PHYs or not.

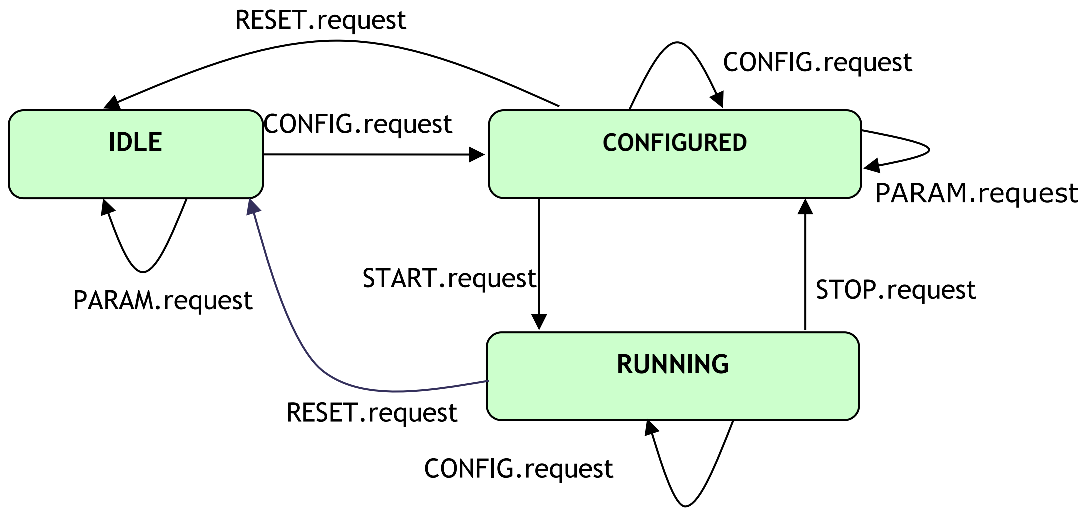


Figure 2-1 PHY layer state transactions on PHY API configuration messages

Idle state	Configured state	Running state
`PARAM.request`	`PARAM.request`	`CONFIG.request`
`CONFIG.request`	`CONFIG.request`	`STOP.request`
	`START.request`	`RESET.request`
	`RESET.request`	

Table 2-1 PHY API configuration request messages valid in each PHY state

When timeouts for P5 request procedures are supported, P5 procedures (from request to response/indication that signals the end of the procedure) are expected to last no more than the time value declared in TLV 0x015F. If a procedure lasts more than that amount, L2/L3 software shall trigger a `RESET.request` for those P5 procedures that change PHY state.

2.1.1 Initialization

The initialization procedure moves the PHY from the IDLE state to the RUNNING state, via the CONFIGURED state. An overview of this procedure is given in Figure 2-2, the different stages are:

- The PARAM message exchange procedure
- The CONFIG message exchange procedure
- The START message exchange procedure

For SFN/SL Synchronization, the initialization procedure is completed when the PHY sends the L2/L3 software a `SLOT.indication` message.

For Delay Management with Timestamps, the initialization procedure is completed when the PHY sends the L2/L3 software a `START.response` message.

For Delay Management without Timestamps, the initialization procedure is completed when the PHY sends the L2/L3 software a `TIMING.indication` message.

The remainder of this section describes the PARAM, CONFIG and START message exchange procedures.

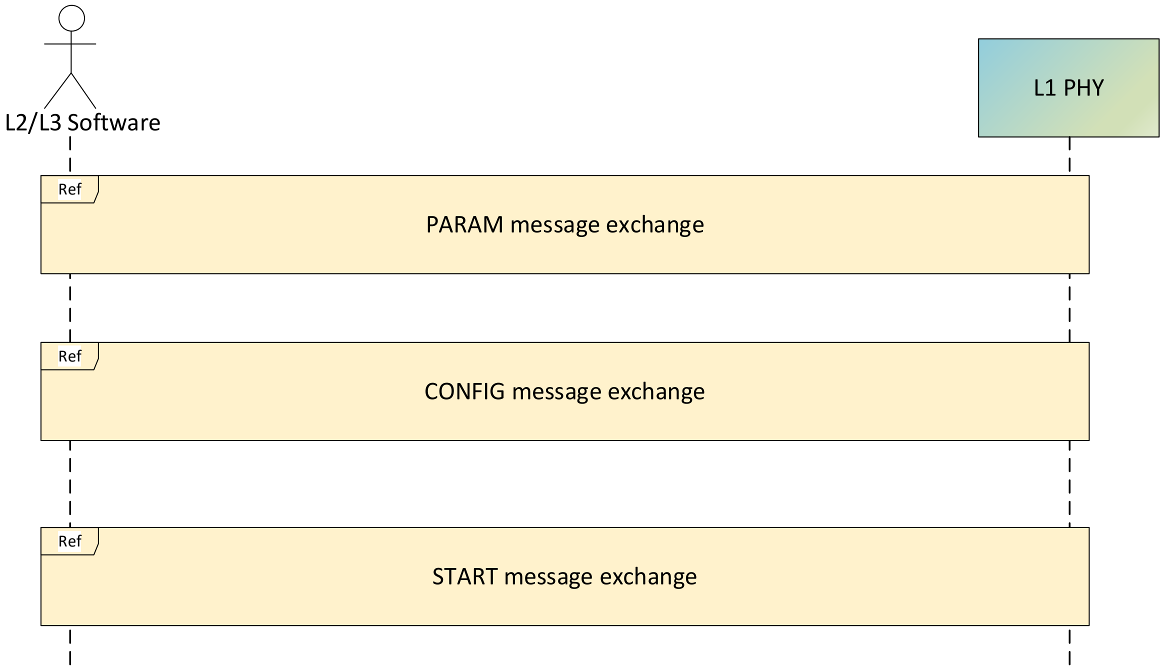


Figure 2-2 Initialization procedure

2.1.1.1 Param message exchange procedure

The PARAM message exchange procedure is shown in

Figure 2-3. Its purpose is to allow the L2/L3 software to collect information about the PHY configuration and current state. The information returned by the PHY depends on its state and is described in Table 2-2. The PARAM message exchange is optional.

PHY state	Information returned by PHY
IDLE	The PHY indicates which capabilities it supports
CONFIGURED	The PHY returns its current configuration
RUNNING	The PHY returns invalid state

Table 2-2 Information returned by the PHY during a PARAM message exchange

From

Figure 2-3, it can be seen that the PARAM message exchange procedure is initiated by the L2/L3 software sending a `PARAM.request` message to the PHY. It is recommended that the L2/L3 software starts a guard timer to wait for the response from the PHY. If the PHY is operating correctly it will return a `PARAM.response` message. In the IDLE and CONFIGURED states, this message will include the current PHY state and a list of configuration information, as described in Table 2-2. In the RUNNING state, this message will indicate an INVALID_STATE error. To determine the PHY capabilities it must be moved to the CONFIGURED state using the termination procedure. If the guard timer expires before the PHY responds, this indicates the PHY is not operating correctly. This must be rectified before further PHY API commands are used; the rectification method is outside the scope of this document.

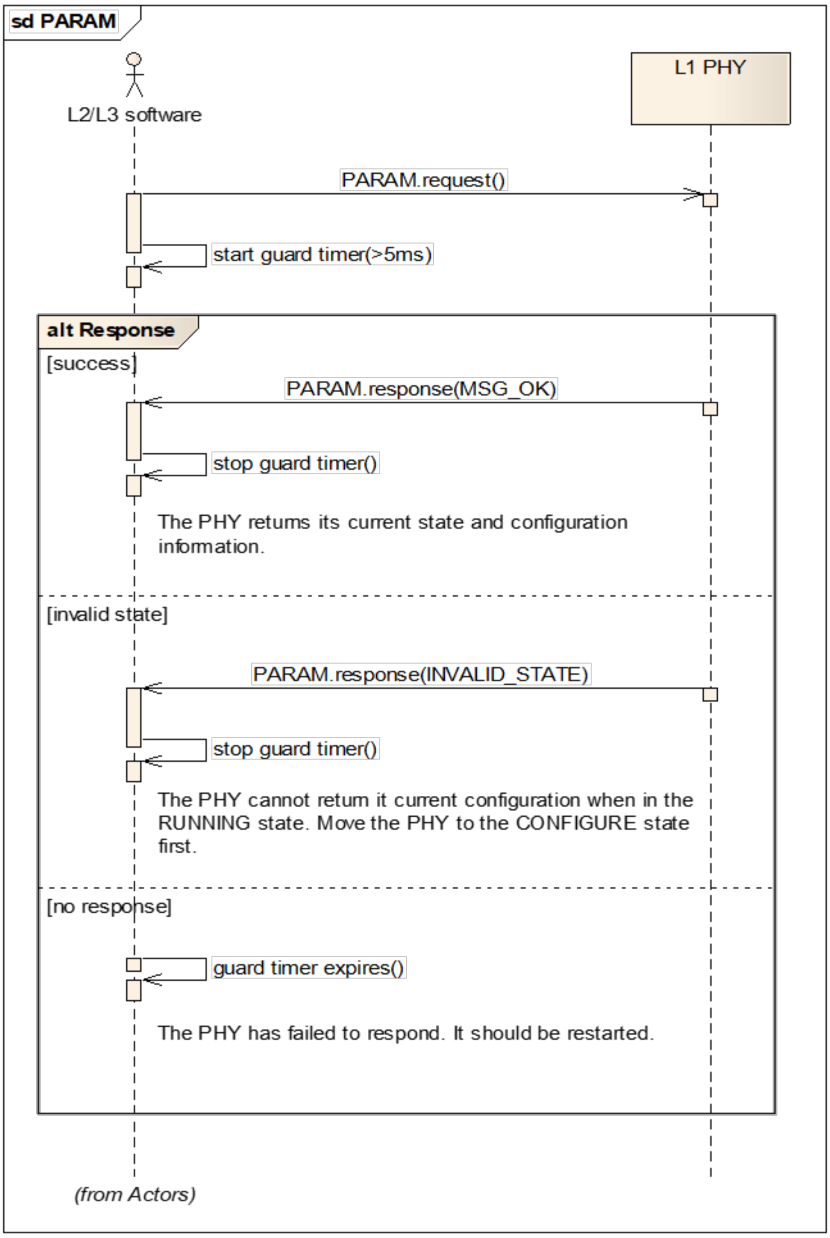


Figure 2-3 PARAM message exchange

2.1.1.2 Config message exchange procedure

The CONFIG message exchange procedure is shown in Figure 2-4. Its purpose is to allow the L2/L3 software to configure the PHY. It can be used when the PHY is in any state. The procedure has slight differences depending on the PHY state. For clarity, each case is described separately.

If the PHY is in the IDLE state, the `CONFIG.request` message, sent by the L2/L3 software, must include all mandatory TLVs. The mandatory TLVs are indicated by the PHY in the `PARAM.response` message. If all mandatory TLVs are included, and set to values supported by the PHY, L1 will return a `CONFIG.response` message, indicating it is successfully configured and has moved to the CONFIGURED state. If the `CONFIG.request` message has missing mandatory TLVs, invalid TLVs, or unsupported TLVs, the PHY will return a `CONFIG.response` message, indicating an incorrect configuration. In this case, it will remain in the IDLE state and all received TLVs will be ignored.

If the PHY is in the CONFIGURED state, the `CONFIG.request` message, sent by the L2/L3 software, may include only the TLVs that are required to change the PHY to a new configuration. If the PHY supports these new values, it will return a `CONFIG.response` message, indicating it has been successfully configured. However, if the `CONFIG.request` message includes invalid TLVs, or unsupported TLVs, the PHY will return a `CONFIG.response` message, indicating an incorrect configuration. In this case, all received TLVs will be ignored and the PHY will continue with its previous configuration. In both cases, if the PHY receives a `CONFIG.request` while in the CONFIGURED state it will remain in the CONFIGURED state.

If the PHY is in the RUNNING state, then a limited subset of CONFIG TLVs may be sent in a `CONFIG.request` message. The permitted TLVs are indicated by the PHY in `PARAM.response`. If the `CONFIG.request` message has invalid TLVs, or TLVs which must not be reconfigured in the RUNNING state, the PHY will return a `CONFIG.response` message, indicating an incorrect configuration. In this case, it will remain in the RUNNING state and all received TLVs will be ignored.

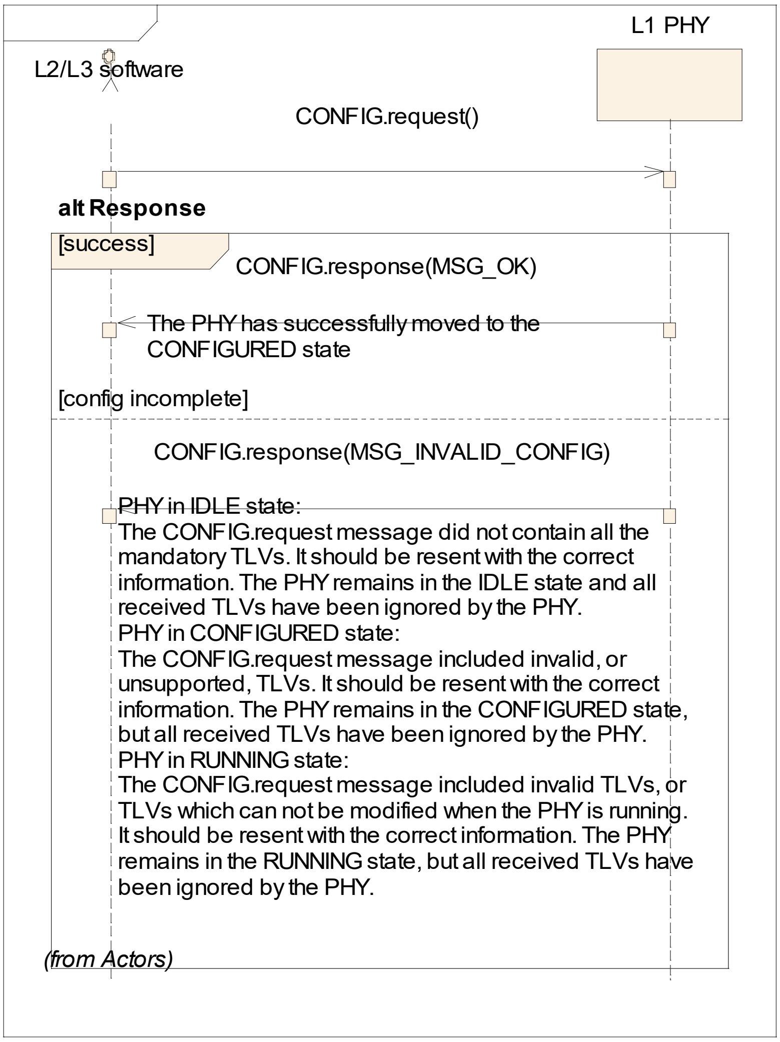


Figure 2-4 CONFIG message exchange

2.1.1.3 Start message exchange procedure

The START message exchange procedure is shown in Figure 2-5. Its purpose is to instruct a configured PHY to start transmitting as a gNB. The L2/L3 software initiates this procedure by sending a `START.request` message to the PHY.

If the PHY is in the CONFIGURED state, and it supports SFN/SL-based synchronization, it will issue a SLOT indication. After the PHY has sent its first `SLOT.indication` message, it enters the RUNNING state.

If the PHY is in the CONFIGURED state, and it supports Delay Management with Timestamps (see section 2.2.1.1), it will issue a `START.response`, after which it enters the RUNNING state.

If the PHY is in the CONFIGURED state, and it supports Delay Management without Timestamps (see section 2.2.1.2), it will issue a `TIMING.indication`, after which it enters the RUNNING state.

If the PHY receives a `START.request` in either the IDLE or RUNNING state, it will return an `ERROR.indication` including an INVALID_STATE error.

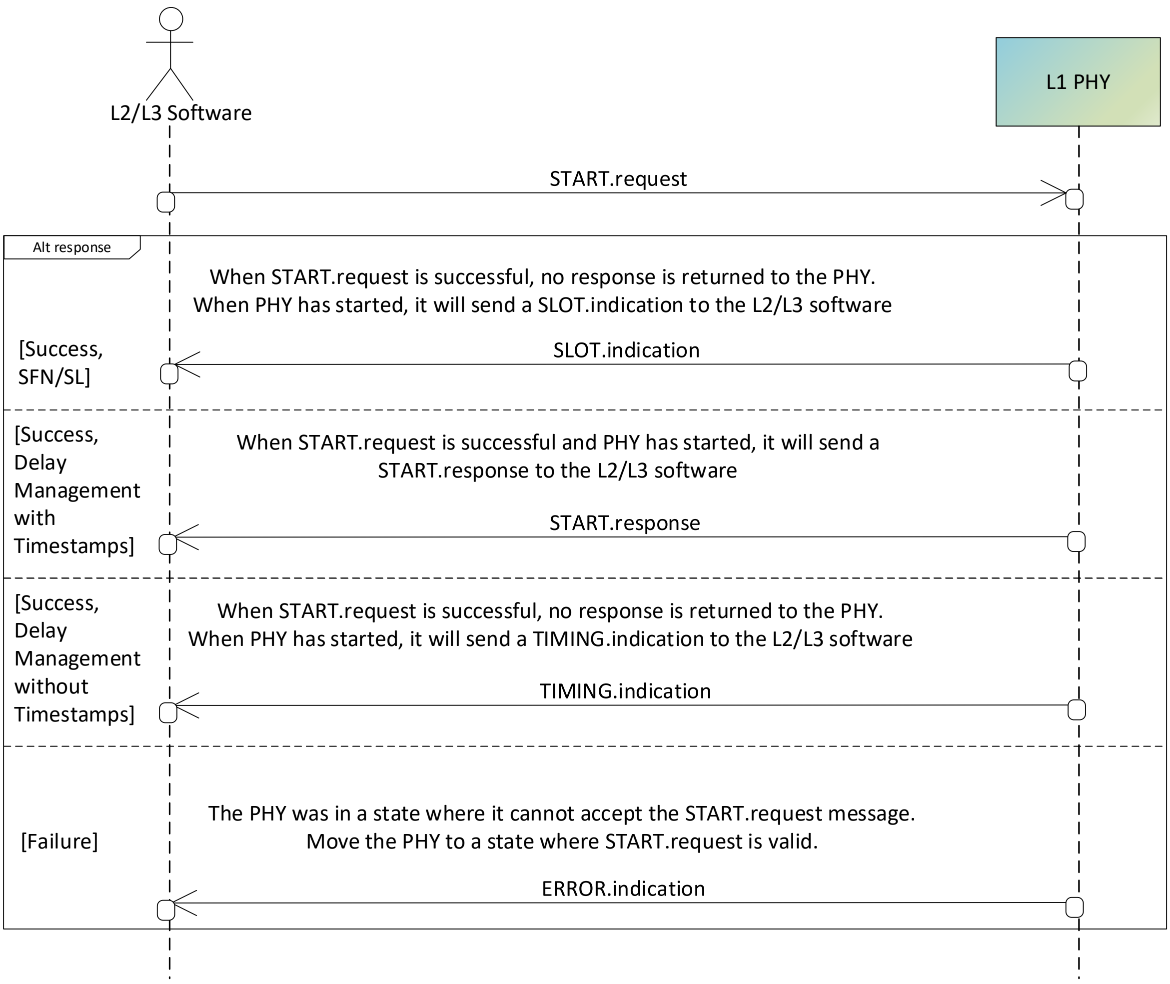


Figure 2-5 START message exchange

2.1.2 Termination

The termination procedure is used to move the PHY from the RUNNING state to the CONFIGURED state. This stops the PHY transmitting as a gNB. The termination procedure is shown in Figure 2-6 and initiated by the L2/L3 software sending a `STOP.request` message.

If the `STOP.request` message is received by the PHY while operating in the RUNNING state, it will stop all TX and RX operations and return to the CONFIGURED state. When the PHY has completed its stop procedure, a `STOP.indication` message is sent to the L2/L3 software.

If the `STOP.request` message was received by the PHY while in the IDLE or CONFIGURED state, it will return an `ERROR.indication` message including an INVALID_STATE error. However, in this case the PHY was already stopped.

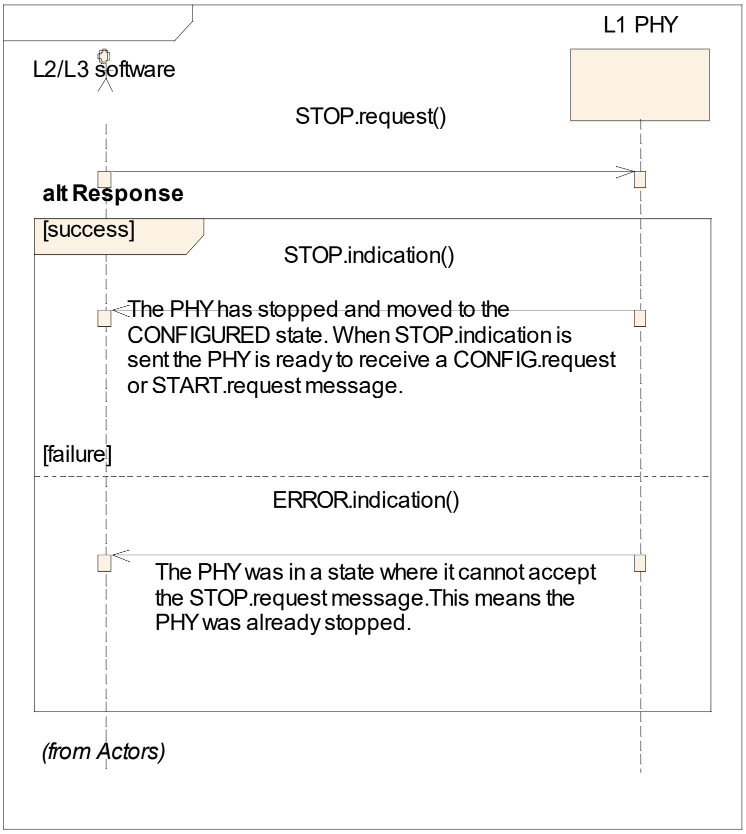


Figure 2-6 Termination procedure

2.1.3 Restart

The restart procedure is shown in Figure 2-7. It can be used by the L2/L3 software when it needs to stop transmitting, but later wants to restart transmission using the same configuration. To complete this procedure, the L2/L3 software can follow the STOP message exchange shown in Figure 2-6. This moves the PHY to the CONFIGURED state. To restart transmission, it should follow the START message exchange, shown in Figure 2-5, moving the PHY back to the RUNNING state.

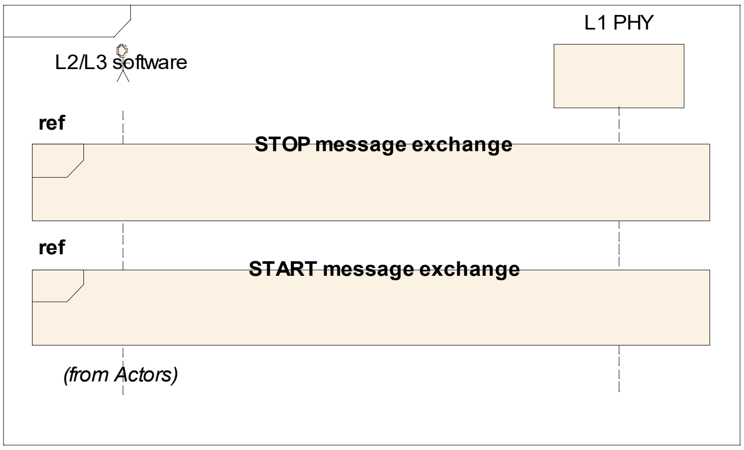


Figure 2-7 Restart procedure

2.1.4 Reset

The reset procedure is shown in Figure 2-8. This procedure is used when the L2/L3 software wants to return the PHY to the IDLE state. Under normal conditions, this can only be achieved by terminating the PHY (as shown in Figure 2-6) and then resetting the PHY.

When L2 detects P5 timeout, it instead invokes `RESET.request` directly.

For a dedicated PHY or PHY Group contexts, the RESET procedure is complete when `RESET.indication` is received. For common context, the RESET procedure is complete when `CCONTEXT_READY.indication` is received.

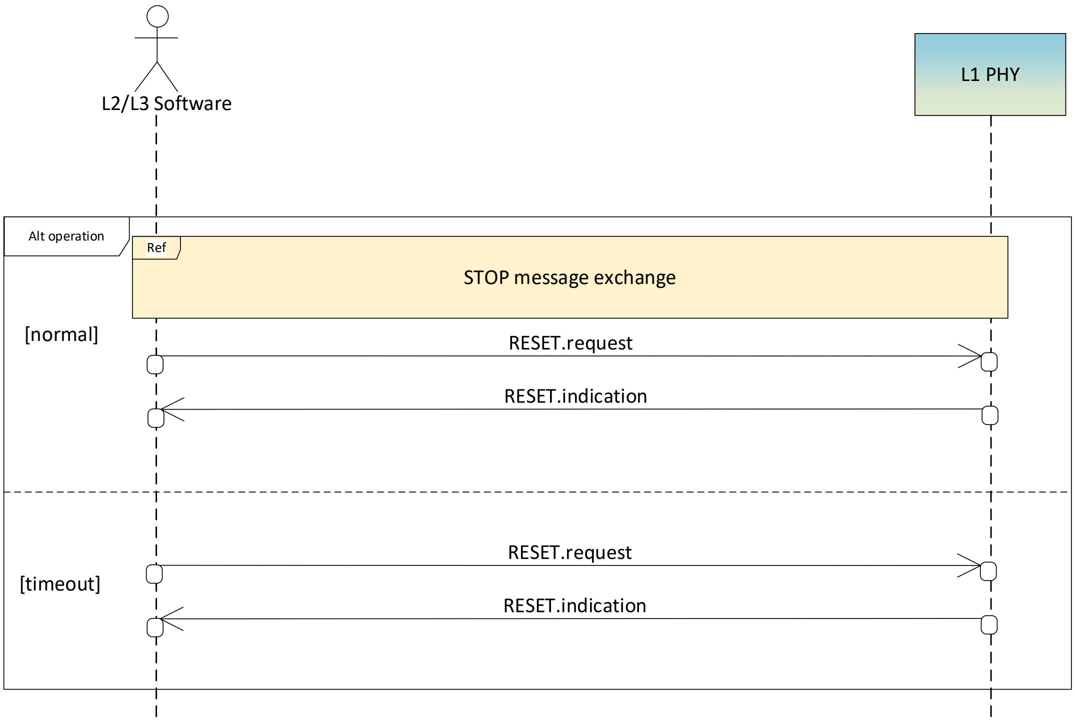


Figure 2-8 Reset procedure

2.1.5 Reconfigure

Two methods of reconfiguration are supported by the PHY:

- a major reconfiguration where the PHY is stopped; and
- a minor reconfiguration where the PHY continues running.

The major reconfigure procedure is shown in Figure 2-9. It is used when the L2/L3 software wants to make significant changes to the configuration of the PHY. The STOP message exchange, shown in Figure 2-6, is followed to halt the PHY and move it to the CONFIGURED state. The CONFIG message exchange, shown in Figure 2-4, is used to reconfigure the PHY. Finally, the START message exchange, shown in Figure 2-5, is followed to start the PHY and return it to the RUNNING state.

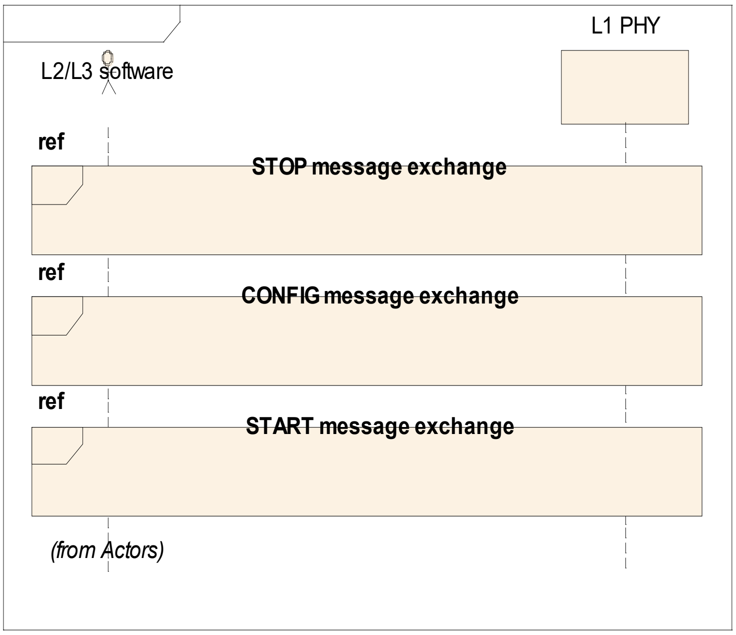


Figure 2-9 Major reconfiguration procedure

The minor reconfiguration procedure is shown in Figure 2-10. It is typically used in conjunction with a RRC system information update.

In the slot where the L2/L3 software requires the configuration change it sends the `CONFIG.request` message to the PHY. Only a limited subset of CONFIG TLVs may be sent, these are indicated by the PHY in `PARAM.response`. TLVs included in the `CONFIG.request` message for slot N will be applied at the SFN/SL given in the `CONFIG.request` message. Reconfiguring the PHY while in the RUNNING state has a further restriction, the `CONFIG.request` message must be sent before the `DL_TTI.request` and `UL_TTI.request` message.

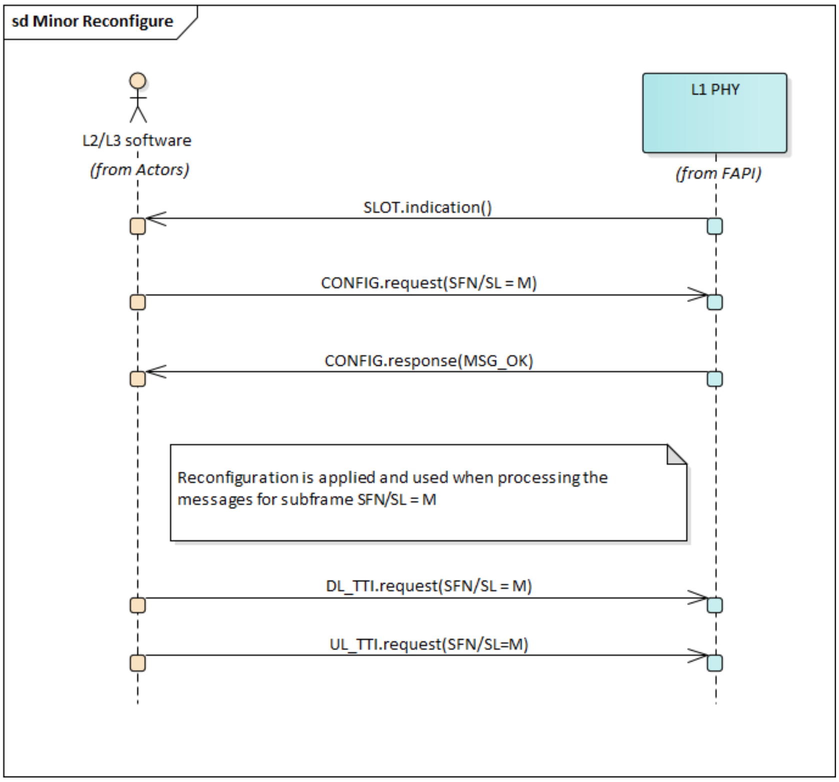


Figure 2-10 Minor reconfigure procedure

2.1.6 Query

The query procedure is shown in Figure 2-11. It is used by the L2/L3 software to determine the configuration and operational status of the PHY. The PARAM message exchange, shown in Figure 2-3, is used. This signalling sequence can be followed when the PHY is stopped, in the IDLE state and, optionally, the CONFIGURED state.

When terminated at Common Context or in PHY Group Context (see section 2.1.10) , the PARAM exchange procedure can return capabilities that are not dependent on particular PHY IDs in the context or on PHY states, in particular the following capabilities:

- Any parameter in the Capability Validity table (Table 3-32)
- Any parameter in the PHY Support table (Table 3-33)
- Any parameter in the PHY/DFE Validity Map table (Table 3-34)
- Any other parameter that is valid for all PHY IDs in the context, regardless of state. When the PHY ID is 255, the parameter applies to all the PHYs.

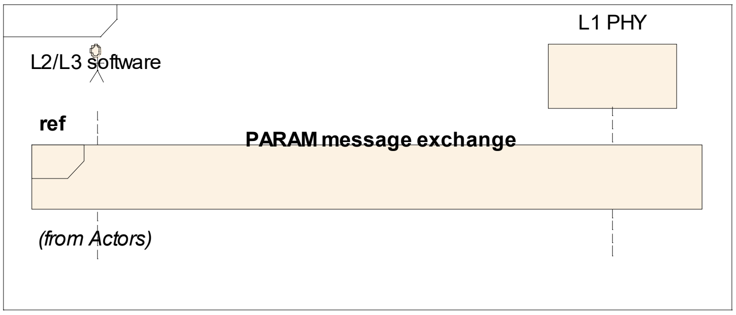


Figure 2-11 Query procedure

2.1.7 Notification

The notification procedure is shown in Figure 2-12. The PHY sends a notification message when it has an event of interest for the L2/L3 software. Currently, there is one notification message called `ERROR.indication`.

The `ERROR.indication` message has already been mentioned in multiple procedures. It is used by the PHY to indicate that the L2/L3 software has sent invalid information to the PHY.

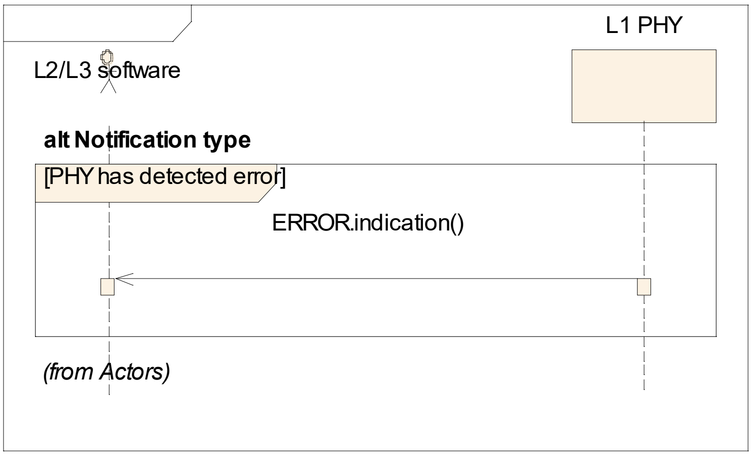


Figure 2-12 Notification procedures

2.1.8 Protocol Negotiation

The protocol negotiation procedure is shown in Figure 2-13. The procedure allows L2/L3 and PHY to ensure that the same FAPI version is used. In particular:

- Protocol negotiation is triggered by L2/L3 sending a `PARAM.request` message (step 1), with a *protocolVersion* indicating a FAPI protocol version supported by L2/L3.
    - Note: typically, this would be the highest FAPI version that L2/L3 supports.
- PHY replies to MAC with a `PARAM.response` message (step 2) that contains:
    - *phyFapiProtocolVersion*: the highest FAPI protocol version that PHY supports

- *phyFapiNegotiatedProtocolVersion*: the FAPI version that PHY assumes in subsequent P5/P7 exchanges with L2/L3.

After step 2, both L2/L3 and PHY are assumed to use the same protocol version, as indicated by *phyFapiNegotiatedProtocolVersion*.

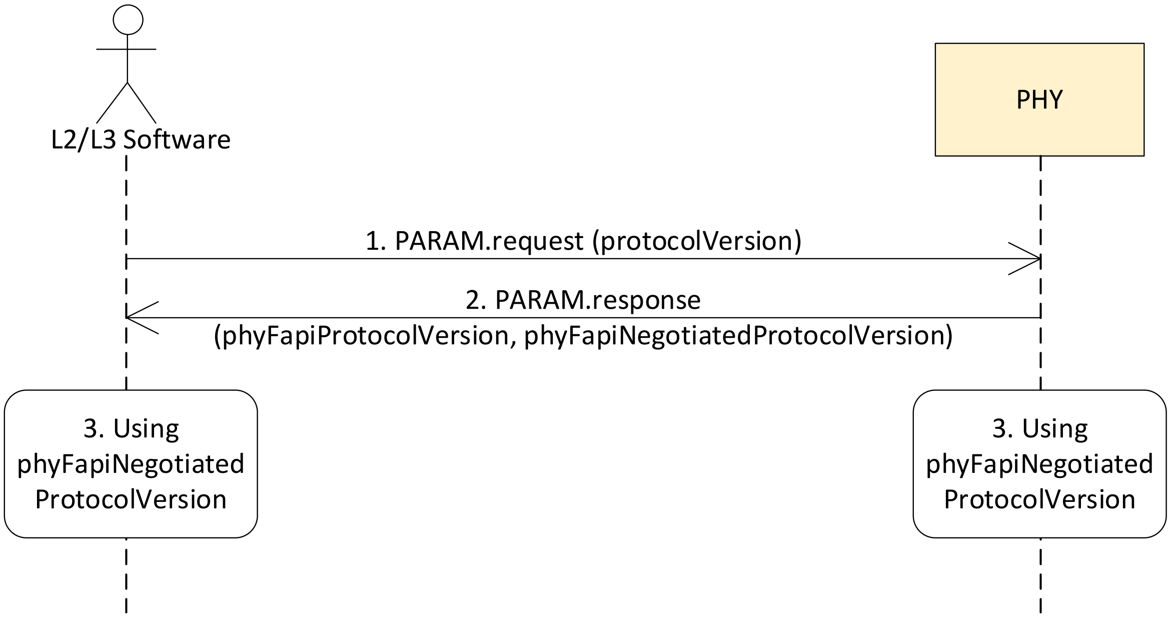


Figure 2-13 Protocol negotiation

Regardless of *phyFapiNegotiatedProtocolVersion*, PHY whose *phyFapiProtocolVersion* is higher than or equal to this FAPI release can always follow the above procedure to negotiate the protocol version.

Protocol negotiation can only happen when all defined PHYs are in IDLE state. Regardless of whether any PHYs are defined, FAPI protocol negotiation can occur with respect to the common context. When supervisory PHY (i.e. PHY ID 255, or PNF in the nFAPI context) is defined, the common context is hosted by supervisory PHY, but common context does not require a supervisory PHY.

The *phyFapiProtocolVersion* is assumed for all exchanges with any PHY IDs sharing the common context.

The PHY handling of any past or future P5 exchanges is only defined if the P5 exchanges are performed according to *phyFapiNegotiatedProtocolVersion*.

2.1.8a PHY Groups

The set of all PHYs supported by the PHY Layer is partitioned into PHY Groups. The orchestration of PHY Groups is outside the context of this release.

PHY Groups may terminate P5 APIs, as described in section 2.1.10. P5 APIs terminating in one PHY Group cannot impact or characterize other PHY Groups or PHYs belonging to other PHY Groups.

2.1.9 PHY Instantiation

This process describes the PHY Profile discovery and selection process, which leads to defining PHYs (and corresponding PHY IDs). It is illustrated in Figure 2-14.

The PHY Instantiation procedure has the following steps:

PHY Profile Discovery: L2/L3 uses the PHY Query Procedure (steps 1 and 2 in the figure) towards Common or PHY Group Context (see section 2.1.10), to discover the supported PHY Profiles and a map indicating which PHY and DFE profiles are pairable.

- Note: Some PHY implementations may pre-define PHY IDs (e.g. < 255), in which case this procedure is not needed.

DFE Profile Discovery: L2/L3 uses the DFE Query Procedure in SCF-223 [9] (steps 3 and 4 in the figure) towards DFE, to discover the supported DFE Profiles

- Note: The order in which DFE and PHY Profile Query procedures are invoked is not subject to this specification.

PHY Profile Selection: L2/L3 selects a PHY profile, via the `CONFIG.request` message (step 5) towards Common or PHY Group Context.

- Note: Common or PHY Group Context does not support PHY (IDLE, CONFIGURED, RUNNING) states, so the state diagram from Figure 2-1 does not apply to PHY ID#255.

PHY Definition: PHY IDs < 255 are defined (but cannot be configured), based on the selected profile (step 6).

- `CONFIG.response` (step 7) announces successful configuration, if PHY Definition succeeds. If unsuccessful, an error code will indicate the failure cause and the PHY Instantiation fails.

DFE Profile Selection: L2/L3 selects a DFE profile, via the DFE `CONFIG.request` message (step 8).

- Note: The order in which DFE and PHY Profile Selections are invoked is not subject to this specification.
- DFE `CONFIG.response` (step 9) announces successful configuration. If unsuccessful, an error code will indicate the failure cause the PHY Instantiation fails.

FEU Configuration and Initialization: L2/L3 executes the Configuration and Initialization procedures towards DFE and RF (step 8).

- The Configuration and Initialization procedures towards FEU components is documented in SCF-225. [10]

PHY Instantiation: When the DFE and RF have both entered RUNNING state, PHY IDs defined through Profile selection become configurable (step 11), after successful DFE and PHY Profile selection.

- If newly defined, these PHY IDs may be considered IDLE.

If PHY IDs existed in any state (IDLE/CONFIGURED/RUNNING) prior to any of the PHY Profile Selection and DFE Profile Selection steps, they remain in the same states, as long as none of their capabilities or configurations are not impacted by the PHY or DFE Profile Selections, otherwise `ERROR.indication` will be triggered by these PHY IDs.

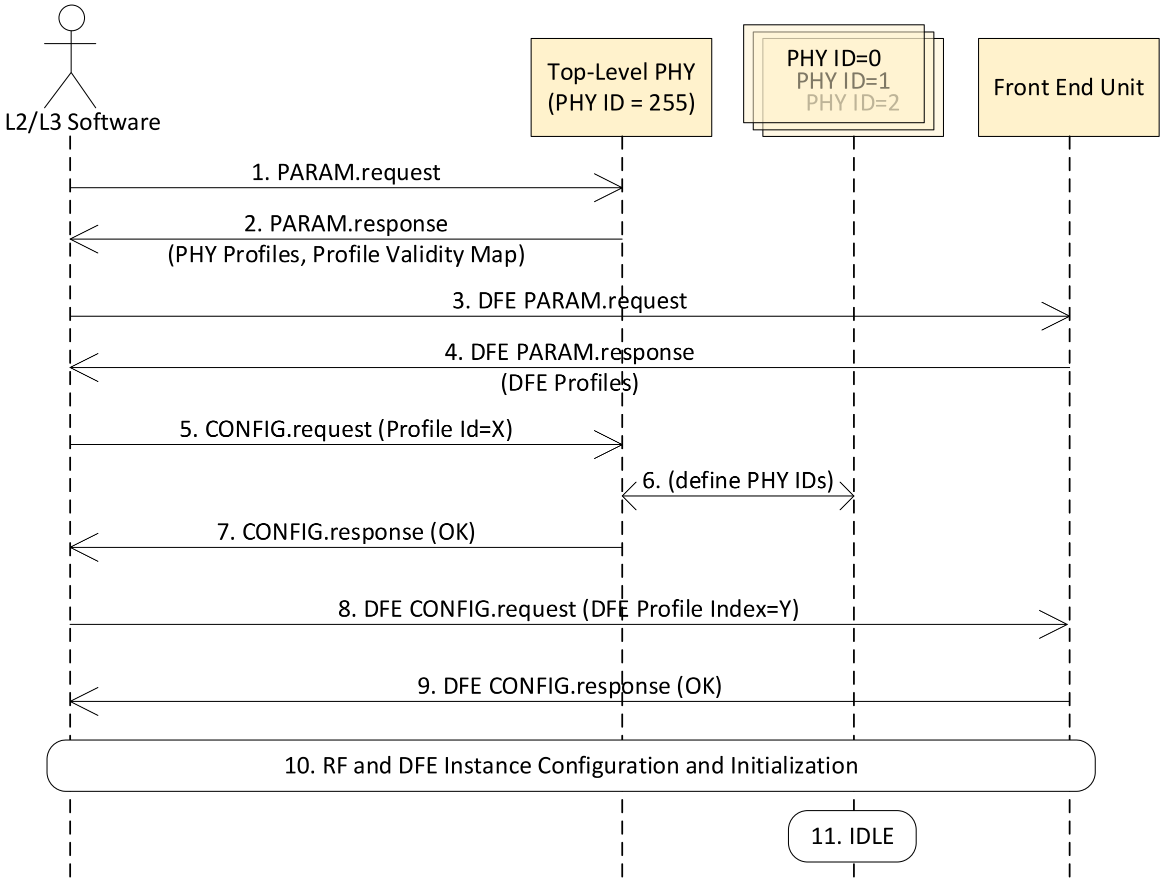


Figure 2-14 PHY instantiation procedure

In general, PHY instantiation is a procedure that controls the definition and for each PHY ID and instantiation of each PHY, as illustrated in Figure 2-15. In particular, a particular PHY ID *x*:

- PHY ID *x* is defined when L2/L3 selects a PHY profile that includes *x*
- A defined PHY ID *x* is associated with a PHY state machine for the PHY associated with PHY ID *x.*
    - For a newly defined PHY ID, the corresponding PHY state machine is in IDLE state.
- A PHY (with a defined PHY ID) can be assumed CONFIGURABLE when it is associated FEU and when all the FEUs (DFE and RF instances) that it associated with are initialized.
- A PHY ID stops being defined when L2/L3 selects a PHY profile that excludes *x*.
    - this can only happen if PHY Id *x* is in CONFIGURED or IDLE state, otherwise the profile selection is rejected.

In addition, L1 rejects any PHY profile if such a change results in any capability, configuration or FEU mapping for a PHY ID x that is in PHY RUNNING state.

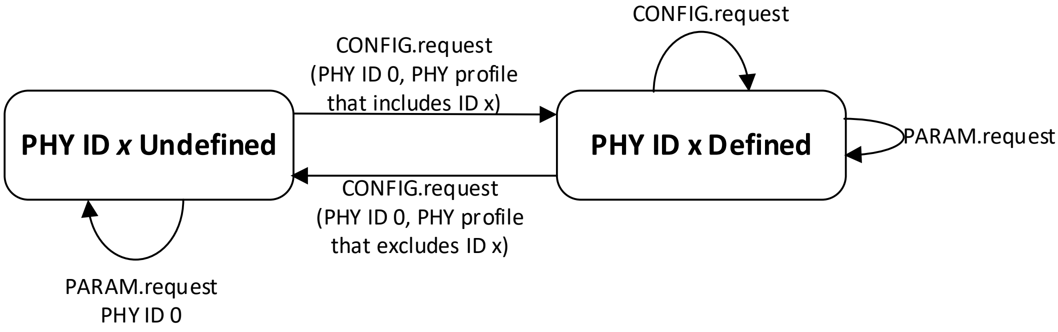


Figure 2-15 PHY ID state machine

2.1.10 Common and PHY Group Contexts

As discussed in 2.1, P5 APIs can terminate in:

- the Common Context
- a PHY Group Context
- a specific PHY.

When a supervisory PHY concept (e.g. PHY ID 255 or PNF in SCF-225 [10]) is supported, the Common Context is supported in this supervisory PHY.

Common Context and PHY Group Context:

- only terminate P5, but not P7
- cannot be in PHY RUNNING state
    - i.e., cannot terminate the P5 START or STOP procedures
- can be queried (see section 2.1.6)
- can be configured
- can be reset

For configuration, Common Context and PHY Group Context can terminate or initiate the following procedures:

- CONFIG exchange, for the PHY Config TLVs (see Table 3-46)
- If PNF is supported, any PNF_xxx procedures from SCF-225 [10]

The scope of a P5 API is determined as follows:

- A specific PHY: PHY ID is dedicated (e.g. PHY ID < 255). No PHY Group may be associated with the API.
- Common Context: PHY Group Identifier is absent and PHY ID is common (e.g. PHY ID = 255)
- PHY Group Context: PHY Group Identifier is present; in this case PHY ID must be common (e.g. PHY ID = 255).

2.1.11 PHY/FEU Interface

In some cases (e.g. split 7.2x, as documented in section 4.5.3.2 of [26]), the interface between PHY and FEU may be subject to disconnection or synchronization events. These events may be observable at L2/L3 software, via PHY (P5 interface), or via FEU (P19-C interface in SCF-223 [9]).

An L2 that can observe such events, shall take the following actions:

a. When a PHY's associated FEU is determined to have become disconnected:
   - L2/L3 software shall transition the corresponding PHY and FEU to Idle, via the mechanisms in section 2.1.
   - L2/L3 software shall re-query the FEU's capability (as well as any associated PHY's capability), after the FEU is determined to have been reconnected.

b. When a PHY or its associated FEU is determined to have become desynchronized:
   - L2/L3 software shall stop any RUNNING PHY, via the mechanisms in section 2.1
   - When the PHY and its associated FEU are both determined to be synchronized, L2 may transition the PHY to RUNNING state, via the mechanisms in section 2.1

When PHY/FEU interface disconnection or synchronization events are observable by PHY, respectively FEU, the observations shall be made known to L2/L3 software via the P5 CONNECTIVITY.Indication (section 3.3.6.3) and P19-C Connectivity Indication (in SCF-223 [9]) APIs, as illustrated in Figure 2-16.

Figure 2-16 illustrates a disconnection/reconnection scenario and a synchronization loss/recovery scenario. In the case of reconnection, the reconnection observation can be coupled with a synchronization loss, or with a synchronization gain.

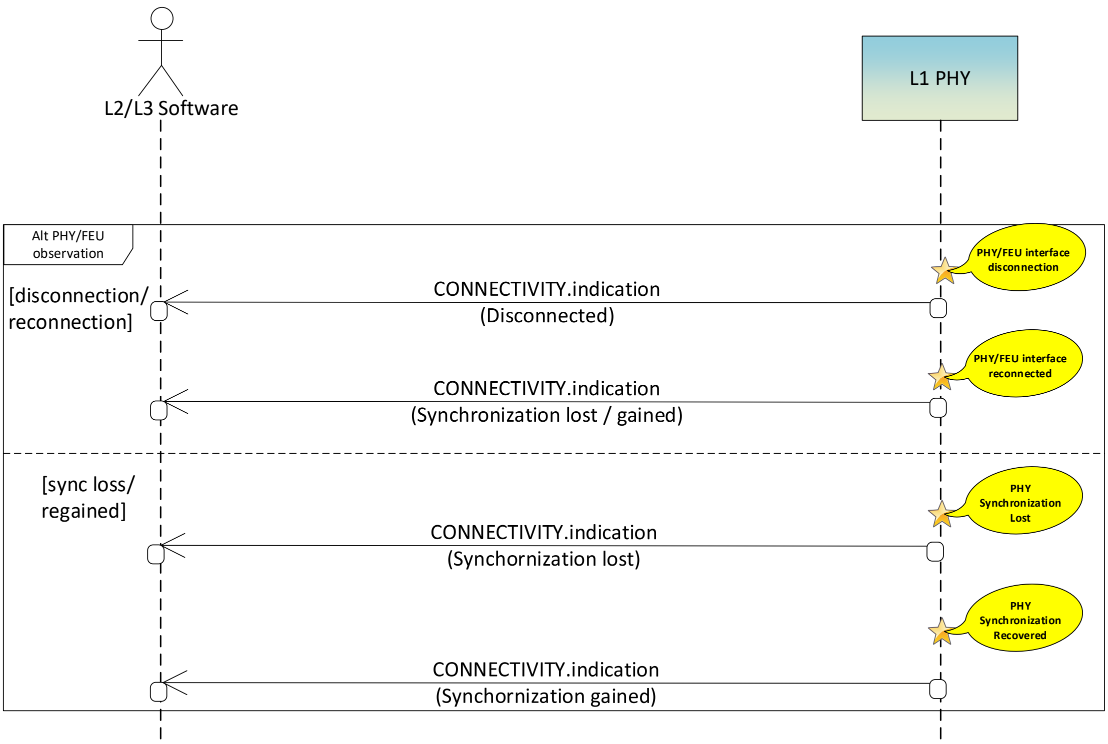


Figure 2-16 CONNECTIVITY.Indication call flow

2.2 Slot Procedures

The slot procedures have two purposes. Firstly, they are used to control the DL and UL frame structures. Secondly, they are used to transfer the slot data between the L2/L3 software and PHY. The slot procedures supported by the PHY API are:

- Transmission of a 62.5us, 125us, 250us, 500us or 1ms SLOT message
- Synchronization of SFN/Slot between the L2/L3 software and PHY
- Transmission of the BCH transport channel
- Transmission of the PCH transport channel
- Transmission of the DL-SCH transport channel
- Transmission of the downlink control information (DCI)
- Transmission of the CSI reference signal
- Reception of the RACH transport channel
- Reception of the UL-SCH transport channel
- Reception of the uplink control information (UCI)
- Reception of the sounding reference signal

1.1.0 Delay Management

This section describes two delay management mechanisms, based on receive window maintained at L1, per message type.

Note that receive windows are anchored to slots. The slot being anchored to is specific to each of the PDU(s) carried in the message.

- e.g. if a `DL_TTI.request` carries a 15 kHz PDSCH PDU and a 30 kHz SSB PDU, the 15 PDSCH PDU reception shall be handled according to the Rx window for the 15 kHz slot and the SSB PDU reception shall be handled according to the Rx window for the 30 kHz slot, relevant to the `DL_TTI.request` message.

Special slot anchoring is assumed, as follows:

- PRACH PDUs for Long preambles shall be anchored to the 15 kHz PRACH numerology.
- If supported (see capability TLV *DL_TTI RxWindow for SSB*) and configured, SSB PDUs shall target the `DL_TTI.request` Rx Window for the highest numerology for PDSCH, if lower (in kHz) than the SSB numerology.
- If supported (see capability TLV *DL_TTI RxWindow for PRACH*) and configured, PRACH PDUs shall target the `UL_TTI.request` Rx Window for highest numerology for PUSCH, if lower (in kHz) than the PRACH numerology.

2.2.1.1 Delay Management with Timestamps

For PHYs supporting both Delay Management and the nFAPI message format, the supported slot procedures are as documented in section 2.1.3 of SCF-225 [10].

2.2.1.2 Delay Management without Timestamps

Where L2 and L1 cannot rely on time stamps, it is still possible to support Delay Management, as described in this section, to ensure that P7 slot procedures occur in a timely fashion.

Similarly to Delay Management procedure documented in 2.2.1.1:

a. a PHY instance expects P7 slot-based messages from VNF to reach the instance in a Receive Time Window interval, that PHY maintains to buffer and process time-critical P7 messages (`DL_TTI.request`, `UL_TTI.request`, `UL_DCI.request`, `TX_DATA.request`), to apply at their targeted slots.
b. a Receive Timing Window is characterized by the following parameters, illustrated in Figure 2-11 of SCF-225 [10]:
    A. a size (*Timing Window*) and
    B. offset prior to the targeted slot (<msg> *Timing offset*, where <msg> is one of the above-listed time-critical messages)
c. relevant P7 messages for Slot S received by PHY Receive Window mechanism:
    A. inside the Timing Window interval: are processed for transmission, reception or configuration at Slot S;
    B. outside the Timing Window interval: are marked by PHY as "too early" or "too late", as appropriate, and used to construct a *timing report* to L2.

Unlike the Delay Management procedure documented in 2.2.1.1:

d. the headers of P7 messages are not assumed to signal any time stamps.
e. the L1 → L2 *timing report* from c.B above is `TIMING.indication`, as described in section 3.4.1A, which requires no time stamps in P7 messages.
    A. `TIMING.indication` may be event-driven (by too-early, too-late events), or periodic, or both, per configuration.
f. an initial `TIMING.indication` message is triggered by transition of PHY into RUNNING state.
    A. DL/UL Node Sync messages are not needed for the timestamp-free Delay Management mechanism in this section.
    B. slot-level timing awareness between L2 and L1 may make use of `TIMING.indication`
    C. L2 can adjust its transmission window based on `TIMING.indication`

Delay Management without timestamps is illustrated in Figure 2-17, for `DL_TTI.request` messages.

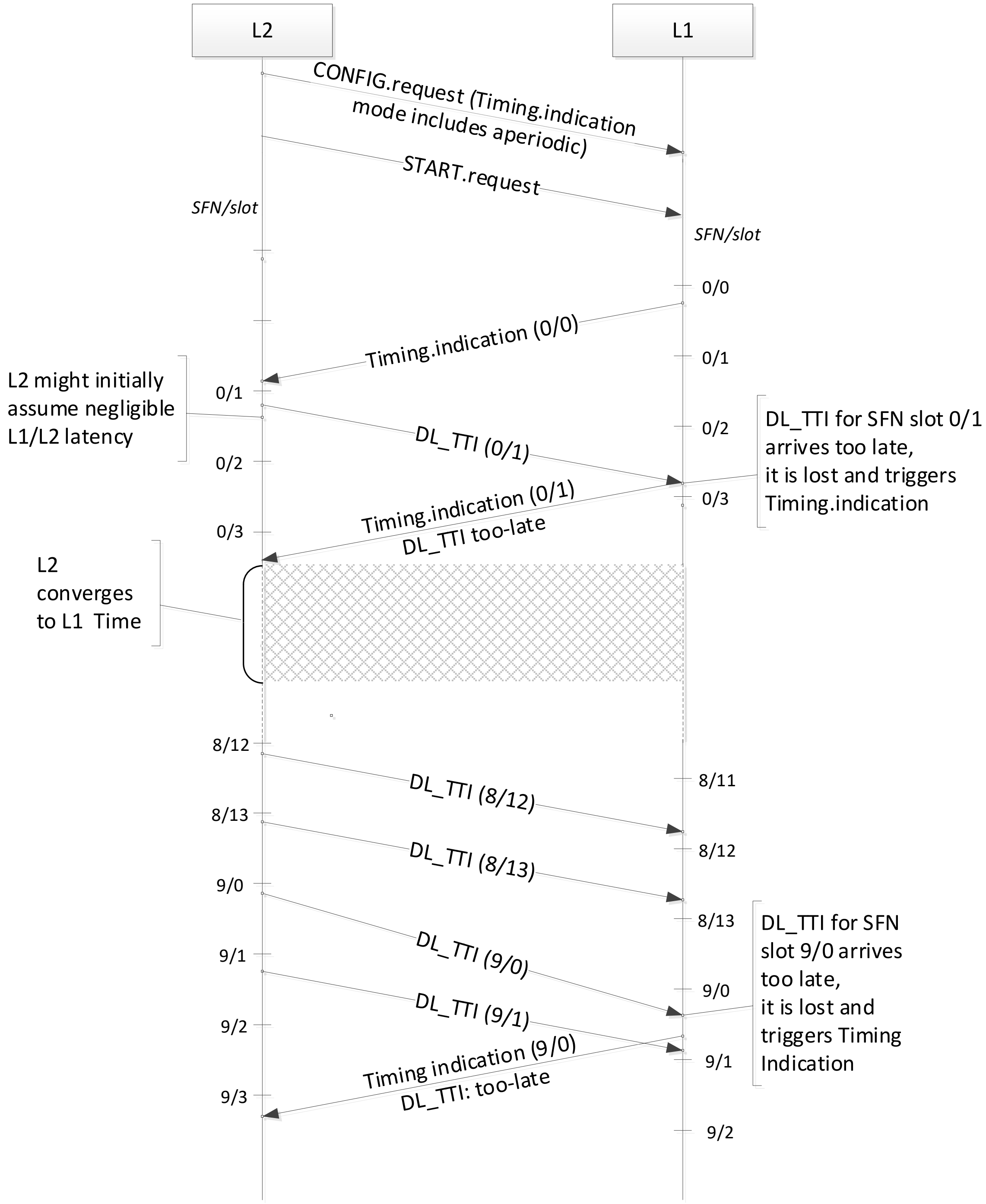


Figure 2-17 Delay management timing sequence without timestamps, with DL_TTI example

2.2.2 SLOT signal

This section only applies to PHYs supporting SFN/SL synchronization, as documented in section 2.2.3.

A `SLOT.indication` message is sent from the PHY, to the L2/L3 software, indicating the start of a slot.

The periodicity of the SLOT.indication message is dependent on the numerology and illustrated in Figure 2-18, where the number of SLOT.indication messages per subframe is highlighted. The case where μ_{max} = 4 is not illustrated on that figure.

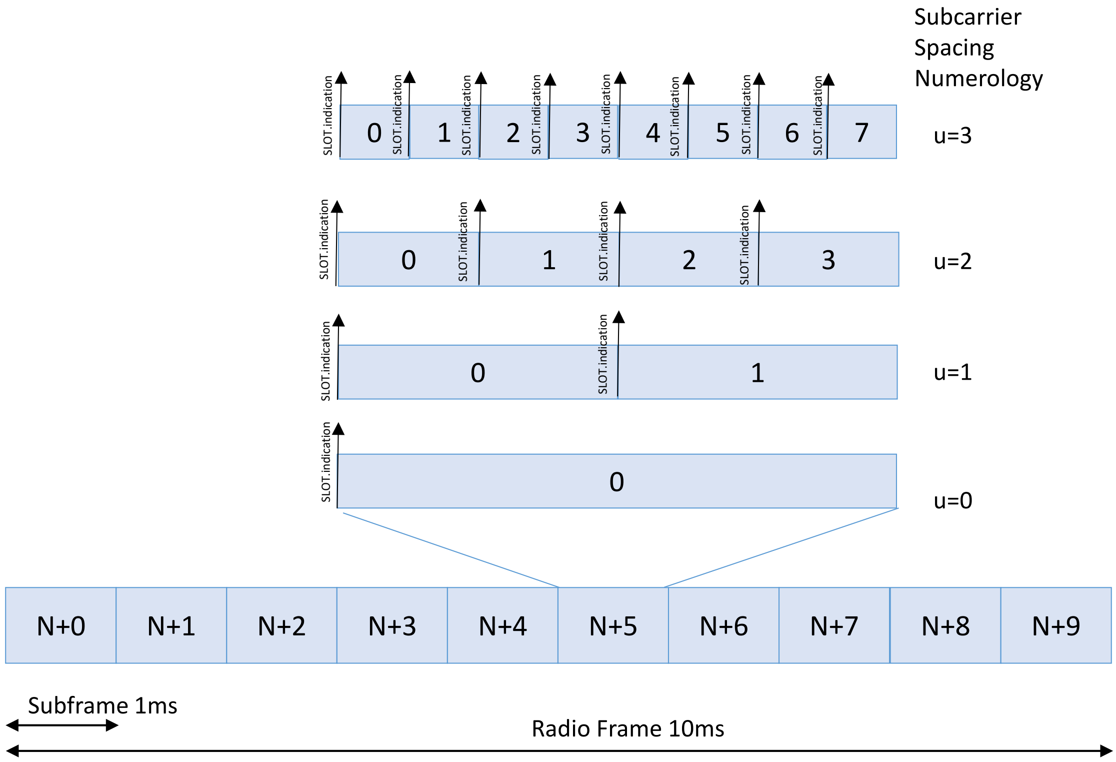


Figure 2-18 Periodicity of SLOT.indication messages, illustrated for μ = 0, 1, 2, 3

2.2.3 SFN/SL Synchronization

For PHYs supporting Delay Management, see section 1.1.0.

The SFN/SL synchronization procedure is used to maintain a consistent SFN/SL value between the L2/L3 software and PHY. Maintaining this synchronization is important since different subframes and slots have different structures.

Two options are provided by the PHY API; the first option configures the PHY to use the SFN/SL value provided by the L2/L3 software. The second option configures the PHY to initialize the SFN/SL and ensure the L2/L3 software remains synchronous. The synchronization option is selected at compile time. For each option, two procedures are described: the initial start-up synchronization and the maintenance of the synchronization.

2.2.3.1 L2/L3 software is master

The SFN/SL synchronization start-up procedure, where the L2/L3 software is master, is given in Figure 2-19. The start-up procedure followed is:

- After successful configuration the L2/L3 software sends a START.request message to move the PHY to the RUNNING state
- When the L2/L3 software is configured as master the initial PHY SFN/SL = M, where M could be any value. In the SLOT.indication message, SFN/SL = M

- The L2/L3 software sends a `DL_TTI.request` message to the PHY containing the correct SFN/SL = N
- The PHY uses the SFN/SL received from the L2/L3 software. It changes its internal SFN/SL to match the value provided by the L2/L3 software

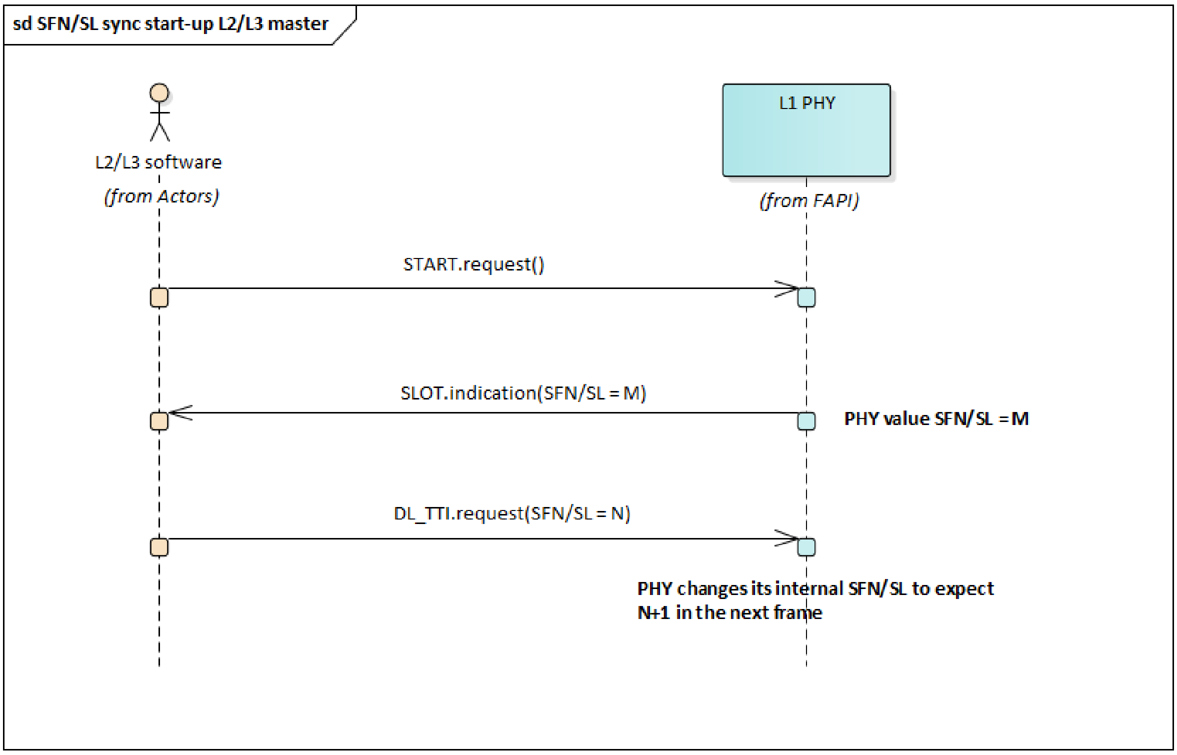


Figure 2-19 SFN/SL synchronization start-up with L2/L3 master

The SFN/SL synchronization maintenance procedure is shown in Figure 2-20. In this example, the L1 PHY is expecting the next `DL_TTI.request` to contain information regarding frame M. The procedure followed is:

- The PHY sends the `SLOT.indication` message with SFN/SL = M.
- The L2/L3 software sends a `DL_TTI.request` or `UL_TTI.request` message to the PHY containing SFN/SL = N
- If SFN/SL M = N
    - The PHY received the SFN/SL it was expecting. No SFN/SL synchronization is required
- If SFN/SL M ≠ N
    - The PHY received a different SFN/SL from the expected value. SFN/SL synchronization is required
    - The PHY uses the SFN/SL received from the L2/L3 software. It changes its internal SFN/SL to match the value provided by the L2/L3 software
    - The PHY returns an `ERROR.indication` message indicating the mismatch

This SFN/SL synchronization procedure assumes the L2/L3 software is always correct. However, it's possible the SFN/SL synchronization was unintended, and due to a L2/L3 software issue. The generation of an `ERROR.indication` message, with expected and received SFN/SL values, should allow the L2/L3 software to perform a correction with a further SFN/SL synchronization.

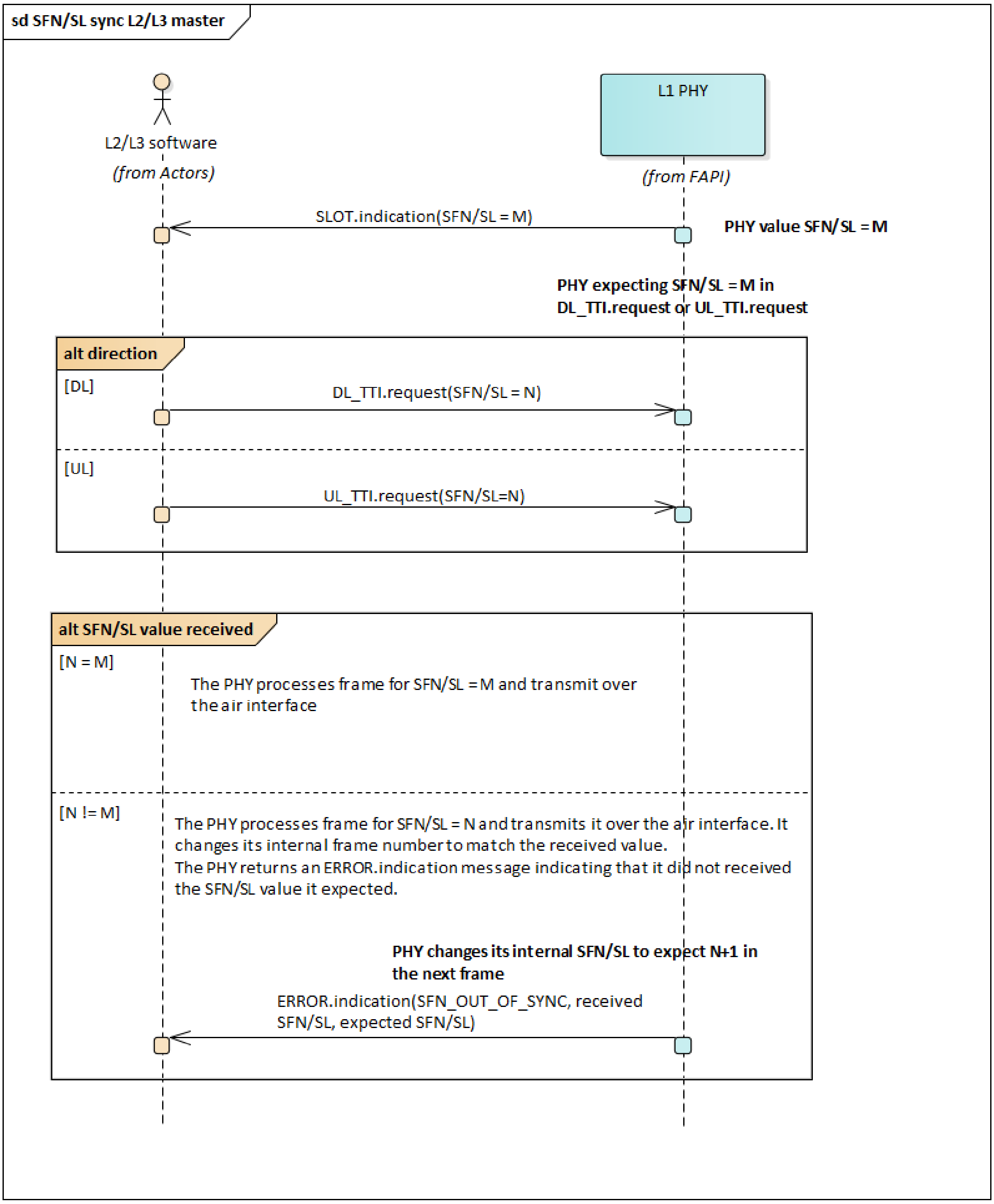


Figure 2-20 SFN/SL synchronization with L2/L3 master

2.2.3.2 L1 PHY is master

The SFN/SL synchronization start-up procedure, where the L1 software is master, is given in Figure 2-21. The start-up procedure followed is:

- After successful configuration the L2/L3 software sends a `START.request` message to move the PHY to the RUNNING state
- If the L1 software is configured as master, the initial PHY SFN/SL = M. The value of M is not deterministic, and could have been set by an external mechanism, such as GPS. The PHY sends a `SLOT.indication` message to the L2/L3 software, with SFN/SL = M. The L2/L3 software uses the SFN/SL received from the PHY. It changes its internal SFN/SL to match the value provided by the PHY
- The L2/L3 software sends a `DL_TTI.request` or `UL_TTI.request` message to the PHY containing SFN/SL = M

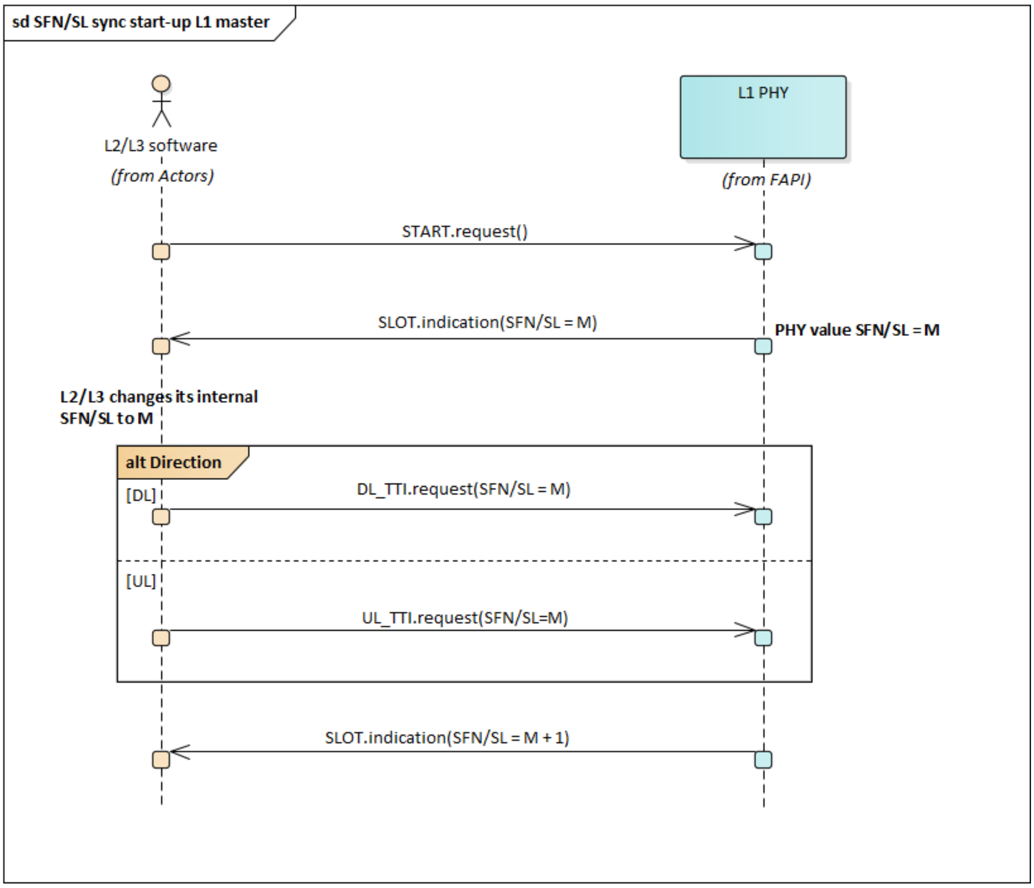


Figure 2-21 SFN/SL synchronization start-up with L1 master

The SFN/SL synchronization maintenance procedure is shown in Figure 2-22. In this example, the L1 PHY is expecting the next `DL_TTI.request` or `UL_TTI.request` to contain information regarding frame M. The procedure followed is:

- The PHY sends a `SLOT.indication` message to the L2/L3 software, with SFN/SL = M
- The L2/L3 software sends a `DL_TTI.request` or `UL_TTI.request` message to the PHY containing SFN/SL = N
- If SFN/SL M = N
    - The PHY received the SFN/SL it was expecting. No SFN/SL synchronization is required
- If SFN/SL M ≠ N

- The PHY received a different SFN/SL from the expected value. SFN/SL synchronization is required
- The PHY discards the received `DL_TTI.request` or `UL_TTI.request` message
- The PHY returns an `ERROR.indication` message indicating the mismatch

This SFN/SL synchronization procedure will continue to discard `DL_TTI.request request` or `UL_TTI.request` messages and emit `ERROR.indication` messages until the L2/L3 software corrects its SFN/SL value.

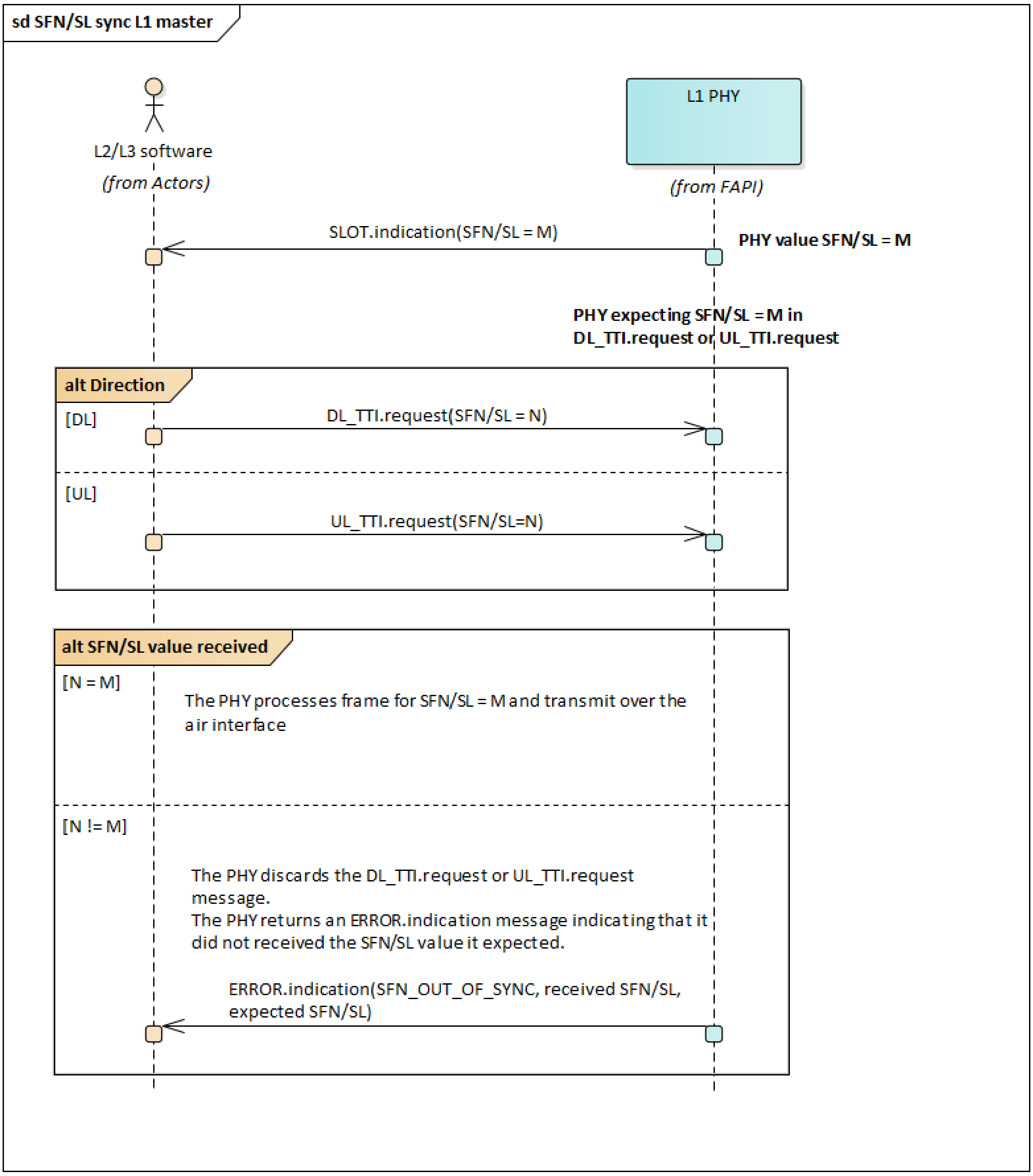


Figure 2-22 SFN/SL synchronization with L1 master

2.2.3a Slot Numerology

L2-initiated Slot-carrying P7 API shall set that slot value according to the highest numerology configured, among all the numerologies, across all uplink and downlink traffic directions and channels.

L2-initiated Channel-specific PDUs shall only be carried in those P7 API whose slot starts (according to the P7 slot numerology) coincide with the start of the channel slot (according to the Channel PDU numerology).

- E.g. If the highest P7 numerology is 60 kHz, 15 kHz PDUs (e.g. PUSCH PDU) are only expected in APIs (UL_TTI.request) where values of 60 kHz slots are divisible by 4

2.2.4 API message order

PHYs supporting Delay Management shall ignore any constraints relating to `SLOT.indication` or the relative order in which messages arrive for a particular slot. Instead, messages are expected in to arrive according to their Receive Windows, as documented in section 2.1.3.5 of SCF-225 [10].

The PHY API has constraints of when certain messages can be sent, or will be received, by the L2/L3 software.

The downlink API message constraints are shown in Figure 2-23:

- The SFN/SL included in the `SLOT.indication` message is expected in the corresponding `DL_TTI.request`
- If the PHY is being reconfigured using the `CONFIG.request` message, this must be the first message for the slot.
- If the *Skip_blank_DL_CONFIG* option is indicated during the PARAM procedure, `DL_TTI.request` can be optionally skipped when there is nothing to schedule. Otherwise, `DL_TTI.request` must be sent for every downlink slot and must be the next message.
- If the *Skip_blank_UL_CONFIG* option is indicated during the PARAM procedure, `UL_TTI.request` can be optionally skipped when there is nothing to schedule. Otherwise, `UL_TTI.request` must be sent for every uplink slot and must be the next message.
- The `TX_DATA.request` and `UL_DCI.request` messages are optional. It is not a requirement that they are sent in every downlink slot.
- L1 may indicate whether it can expect more than one instance of `DL_TTI.request`, `UL_TTI.request`, `UL_DCI.request` or `TX_DATA` for a slot. If supported by L1, L2 may configure L1 to expect only 1 `DL_TTI.request`, 1 `UL_TTI.request` and 1 `UL_DCI.request` for a slot.
- There can be more than one instance of the `TX_DATA.request` if mini-slot is used. This allows the PHY to minimize latency.

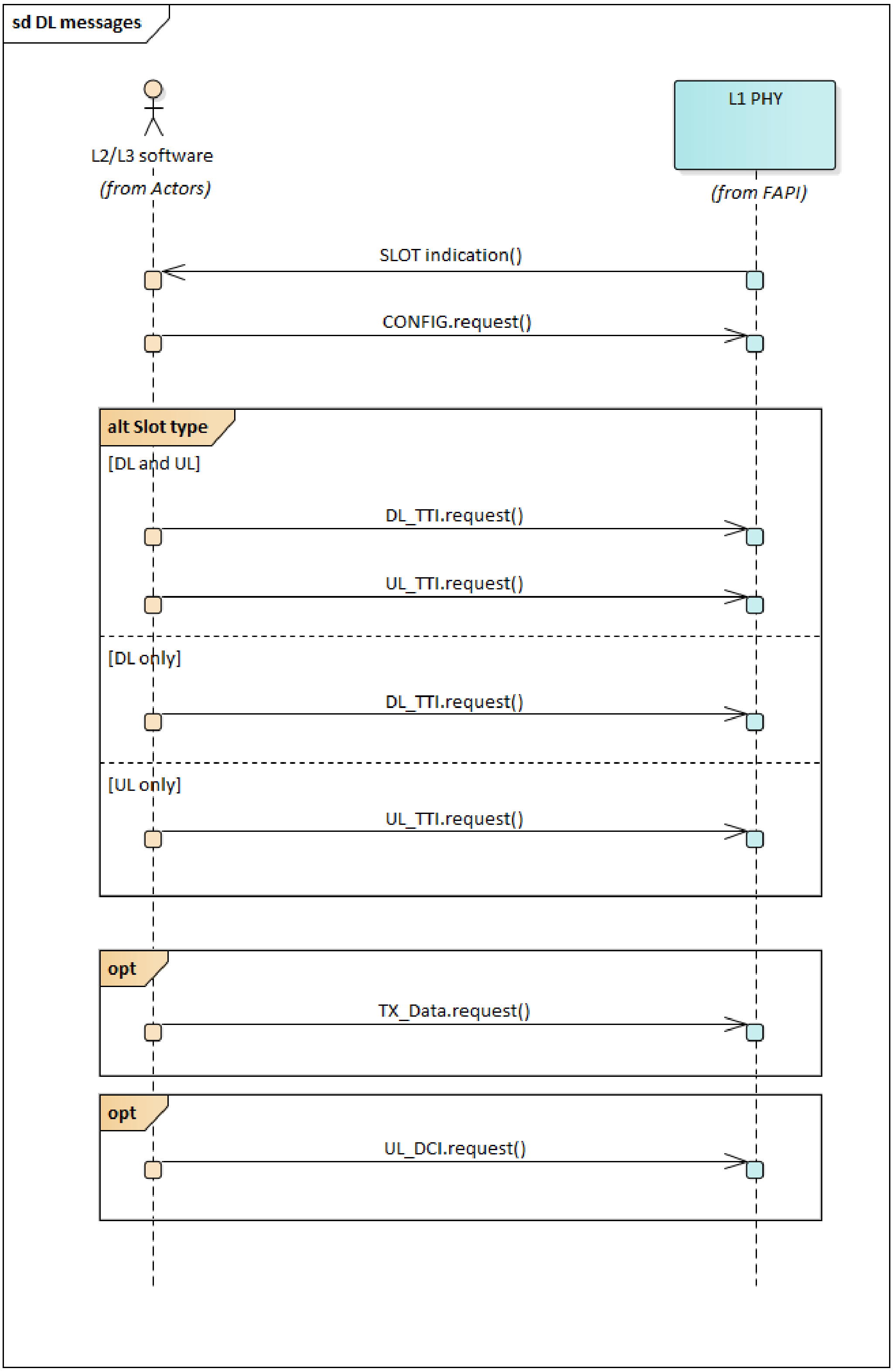


Figure 2-23 DL message order

The uplink API message constraints are shown in Figure 2-24:

- The UL API messages are optional. It is not a requirement that they are sent in every slot.

- If present, the messages can be in any order
    - The CRC.indication message is included if uplink data PDUs were expected in the slot.
    - The RX_DATA.indication message is included if uplink data PDUs were expected in the slot.
    - The UCI.indication message is included if UCI PDUs were expected in the slot.
    - The RACH.indication message is included if any RACH preambles were detected in the slot
    - The SRS.indication message is included if any sounding reference symbol information is expected in the slot.
- L1 may indicate whether it can generate more than one instance of CRC.indication, RX_DATA.indication, UCI.indication, RACH.indication, SRS.indication or DL_TTI.response per slot and subcarrier spacing. If supported by L1, L2 may configure L1 to limit itself to only one instance of the CRC.indication, RX_DATA.indication, UCI.indication, RACH.indication, and SRS.indication messages per slot and subcarrier-spacing.
- There can be more than one instance of the CRC.indication, RX_DATA.indication, UCI.indication, RACH.indication, and SRS.indication messages if more than one subcarrier spacing finishes in the same slot. As an example, from Figure 2-18, slot 7 for u=3 completes at the same time as slot 1 for u=1.
- There can be more than one instance of the CRC.indication and RX_DATA.indication if mini-slot is used. This allows the PHY to minimize latency.

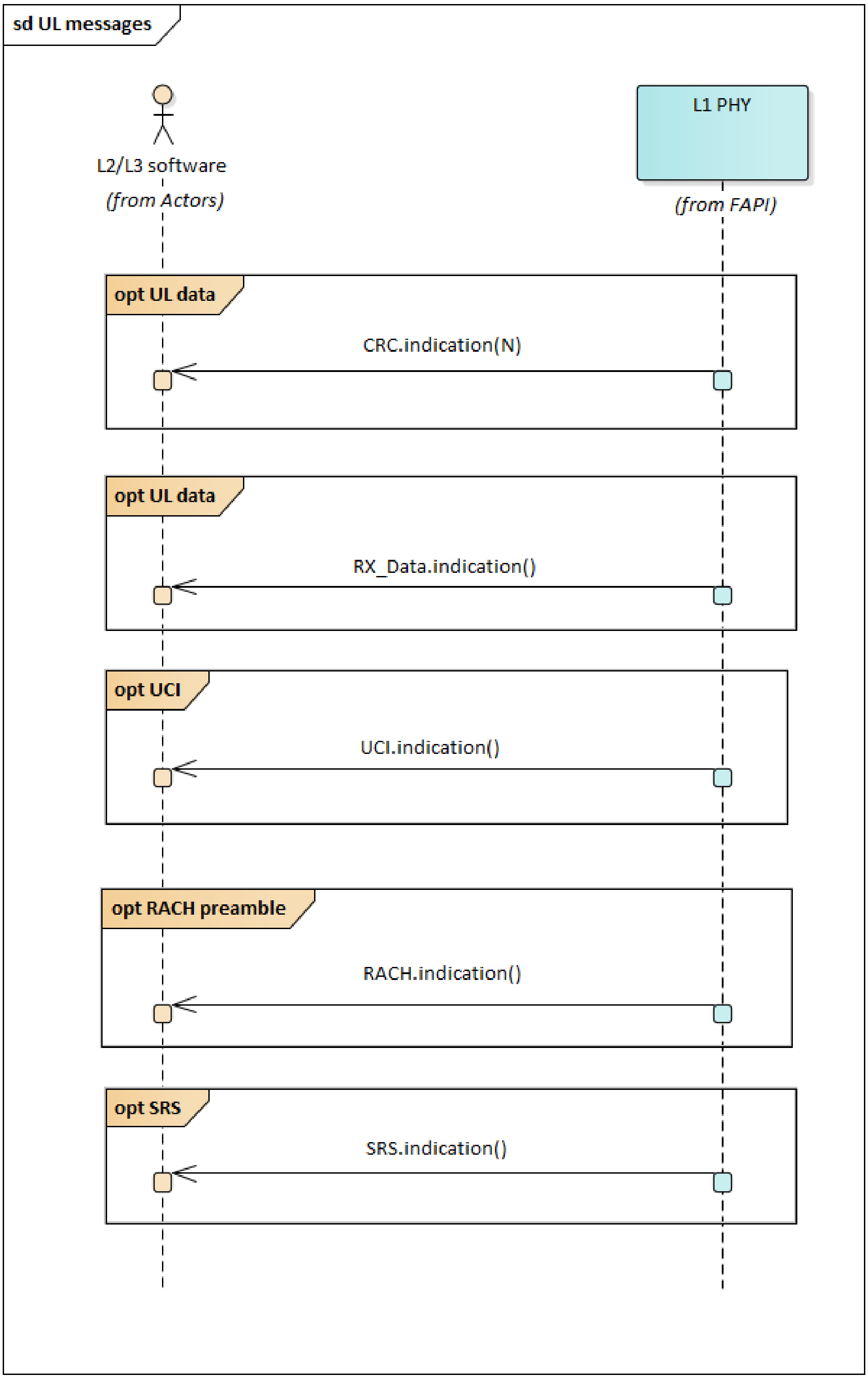


Figure 2-24 UL message order

2.2.5 Downlink

The procedures relating to downlink transmission are described in this Section.

2.2.5.1 BCH

The BCH transport channel is used to transmit the Master Information Block (MIB) information to the UE, per 3GPP [3GPP TS 38.331 [6] . It has a periodicity of 80ms. BCH payload is carried using the PBCH physical channel. Together with the PSS and the SSS synchronization signals, PBCH forms the Synchronization Signal Block (SSB), as shown in Figure 2-25 in 3GPP TS 38.300 [8].

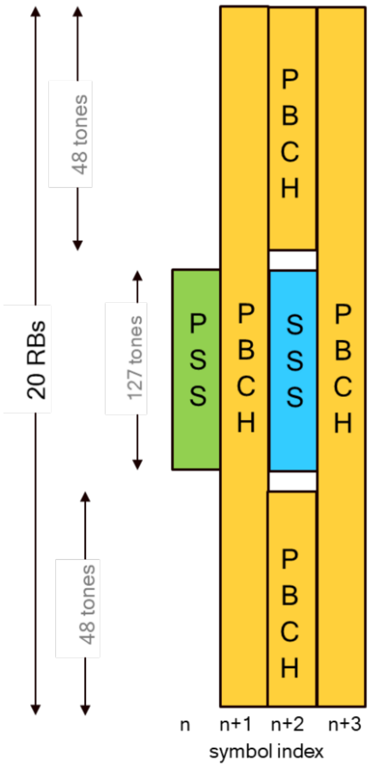


Figure 2-25 Time and frequency structure of the synchronization signal block (SSB)

SS blocks can be transmitted in different beams in a time-multiplexed fashion. The set of SS blocks within a beam-sweep is called an *SS burst.*

The periodicity of the SS burst set varies from 5 to 160ms. All SS blocks in an SS burst set needs to be transmitted within 5ms. The SS blocks can be transmitted only in predetermined time slots. Depending on the band of deployment the location of the SSB allowed slots vary as shown in Figure 2-26 and Figure 2-27.

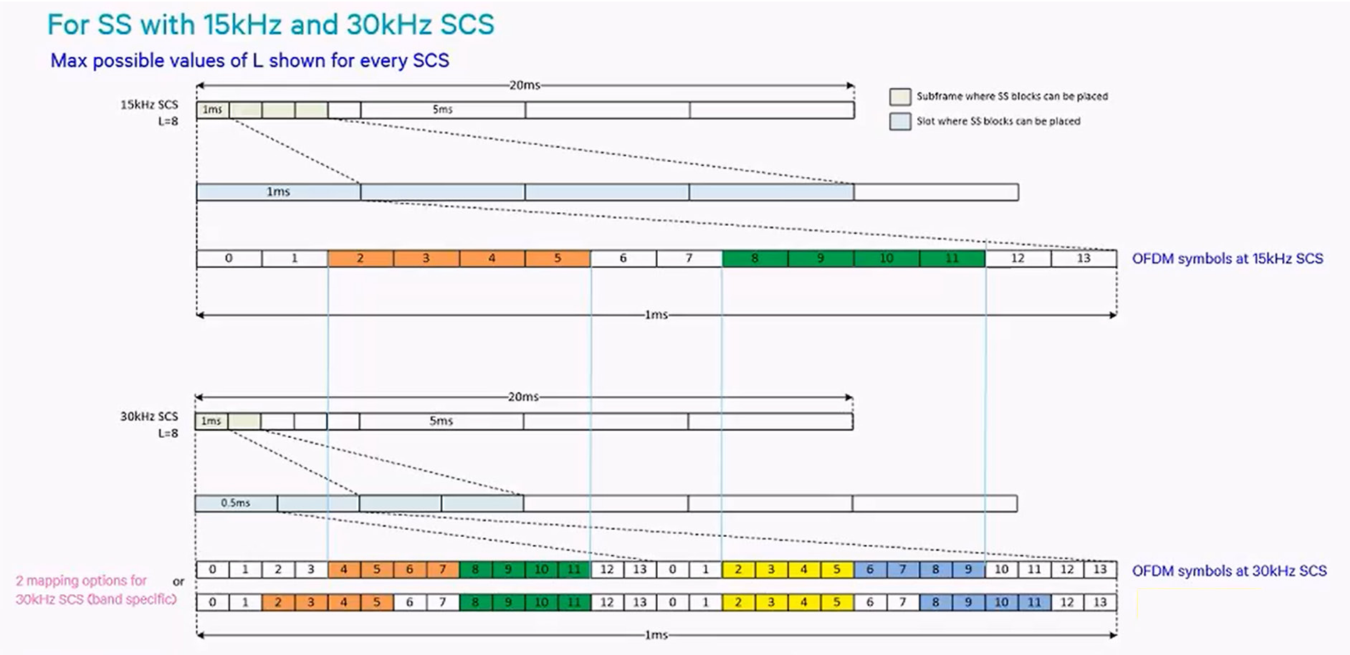


Figure 2-26 Possible SS block locations for FR1 (sub6 GHz) deployments

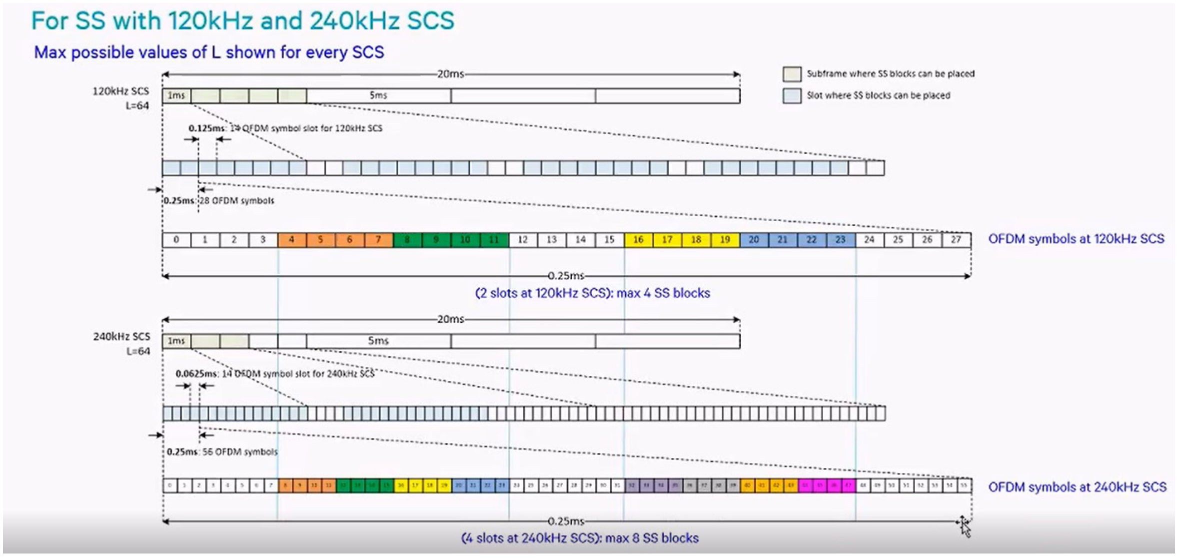


Figure 2-27 Possible SS block locations for FR2 deployments

2.2.5.2 PCH

The PCH transport channel is used to transmit paging messages to the UE. The UE has specific paging occasions where it listens for paging information Figure 2-28. The L2/L3 software is responsible for calculating the correct paging occasion for a UE. The PHY is only responsible for transmitting PCH PDUs when instructed by the `DL_TTI.request` message.

The PCH procedure is shown in Figure 2-28. To transmit a PCH PDU the L2/L3 software must provide the following information:

- In `DL_TTI.request` a PDSCH PDU and PDCCH PDU are included.

- In `TX_DATA.request` a MAC PDU containing the paging message is included.

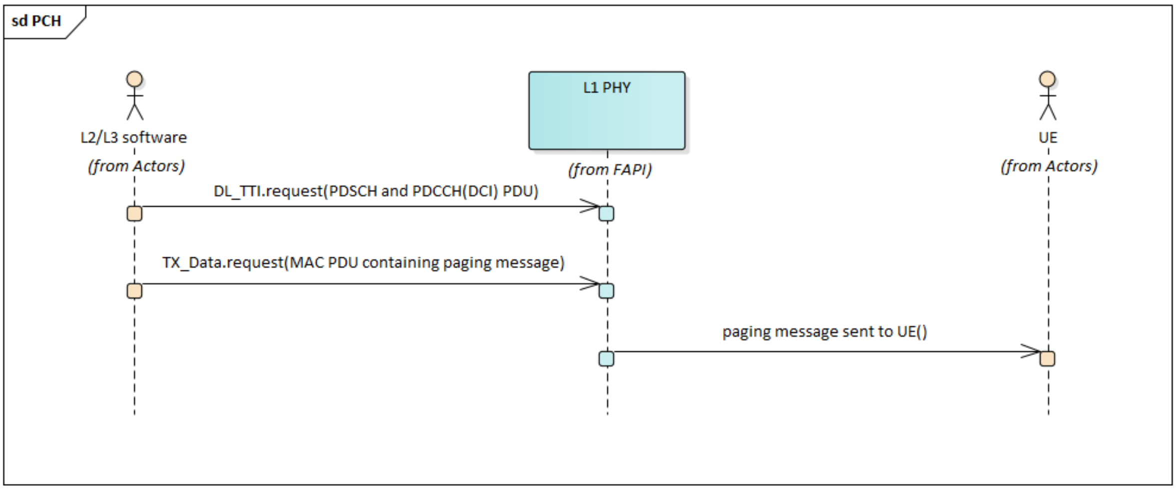


Figure 2-28 PCH procedure

2.2.5.3 DL-SCH

The DL-SCH transport channel is used to send data from the gNB to a single UE. HARQ is always applied on the DL-SCH transport channel at least for unicast PDSCH. Therefore, when scheduling a downlink PDSCH transmission which will require HARQ-ACK feedback from UE, the L2/L3 software has to schedule uplink transmission on PUCCH or PUSCH for the UE to feed back an ACK/NACK response.

The procedure for the DL-SCH transport channel is shown in Figure 2-29.To transmit a DL-SCH PDU, the L2/L3 software must provide the following information:

- In `DL_TTI.request` a PDSCH PDU and PDCCH PDU are included. The PDCCH PDU contains control regarding the DL frame transmission
- In `TX_DATA.request` a MAC PDU containing the data is included
- At the expected slot UCI HARQ information is included in a later `UL_TTI.request`, where the timing of this message is variable. The HARQ can be sent on the PUSCH or PUCCH, therefore, the information of the HARQ response on the uplink is sent in either:

    - PUSCH PDU – is used if the UE is scheduled to transmit data and the ACK/NACK response
    - PUCCH PDU – is used if the UE is just scheduled to transmit the ACK/NACK response

- The PHY will return the ACK/NACK response information in the `UCI.indication` message

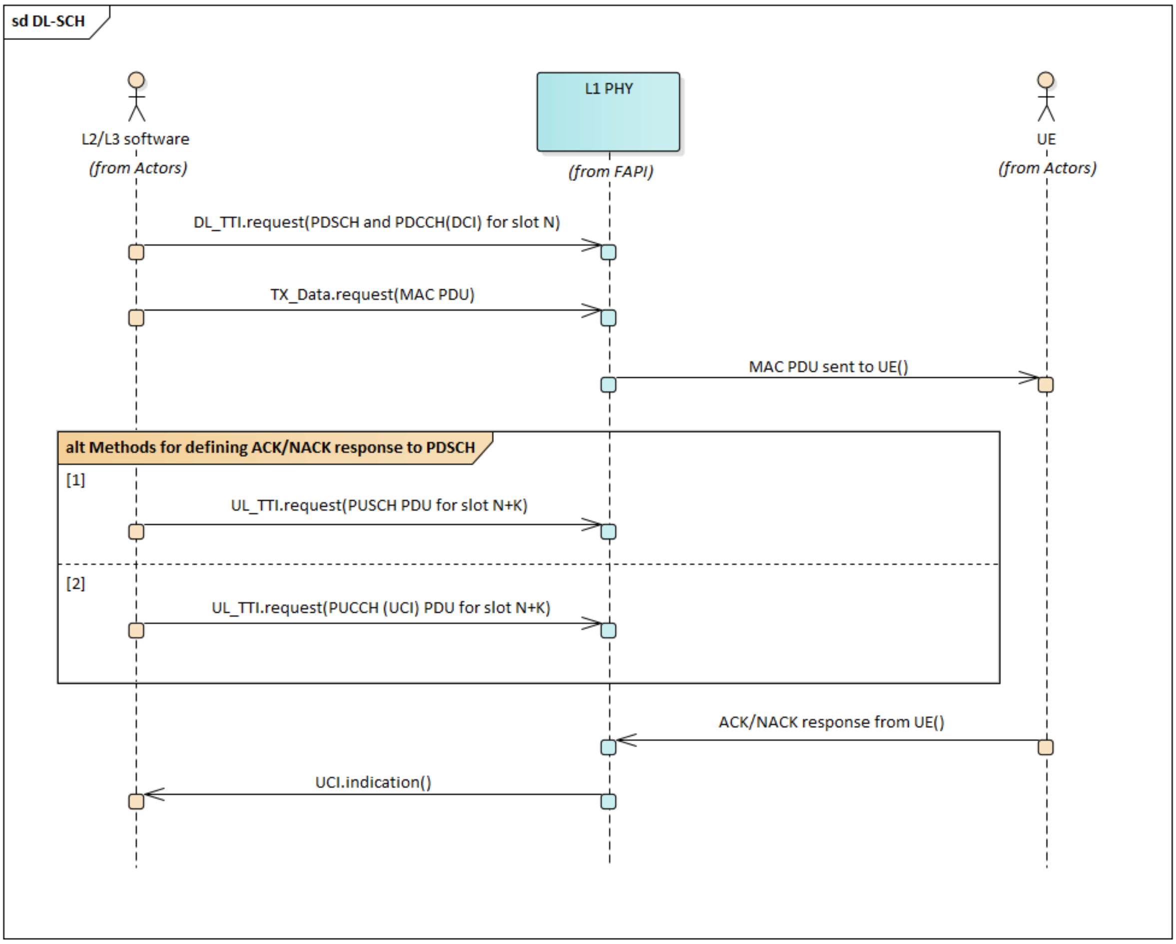


Figure 2-29 DLSCH procedure

With DCI Format 1-1, the DL SCH channel can send two data transport blocks to a UE. This requires a single PDCCH (DCI) PDU, a single PDSCH PDU and two MAC PDUs. The procedure is shown in Figure 2-30. To initiate a transmission of two data transport blocks, the L2/L3 software must provide the following information:

- In the first transmission slot in the `DL_TTI.request` a PDSCH and PDCCH (DCI) PDU is included. The PDCCH PDU contains control regarding the DL frame transmission. A PDSCH PDU includes two codewords, one for each transport block specified in the DCI PDU.
- In `TX_DATA.request` two MAC PDUs containing the data are included.

(The remaining behaviour is identical to single-layer transmission.)

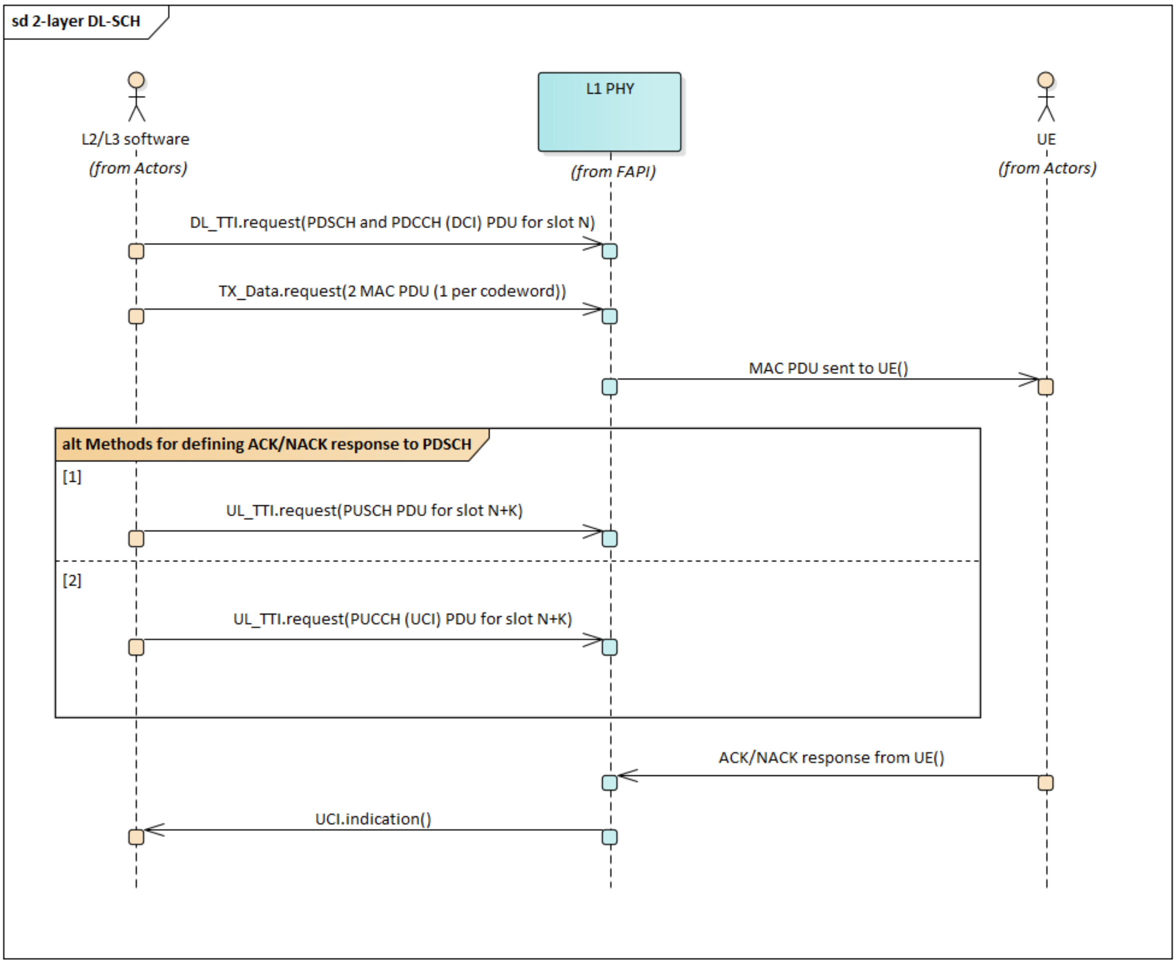


Figure 2-30 2-layer DLSCH procedure

Multi-slot transmission is also an option for the DL-SCH, where the same MAC PDU is transmitted for N slots. The procedure is shown in Figure 2-31, and the L2/L3 must provide the following information:

- In `DL_TTI.request` a PDSCH PDU and PDCCH (DCI) PDU is included. The PDCCH PDU contains control regarding the DL frame transmission and the UE will have previously been configured to be aware that multi-slot transmission is used.
- In `TX_DATA.request` a MAC PDUs containing the data is included.
- The multi-slot transmission can be over 2,4 or 8 slots and for the remaining slots the L2/L3 software must provide the following information:
  - In `DL_TTI.request` a PDSCH PDU is included. The PDSCH PDU contains the information for this slot, for example, in multi-slot transmission the incremental redundancy version changes per slot.
  - In `TX_DATA.request` MAC PDU(s) containing the data is/are included

(The remaining behaviour is identical to single-layer transmission.)

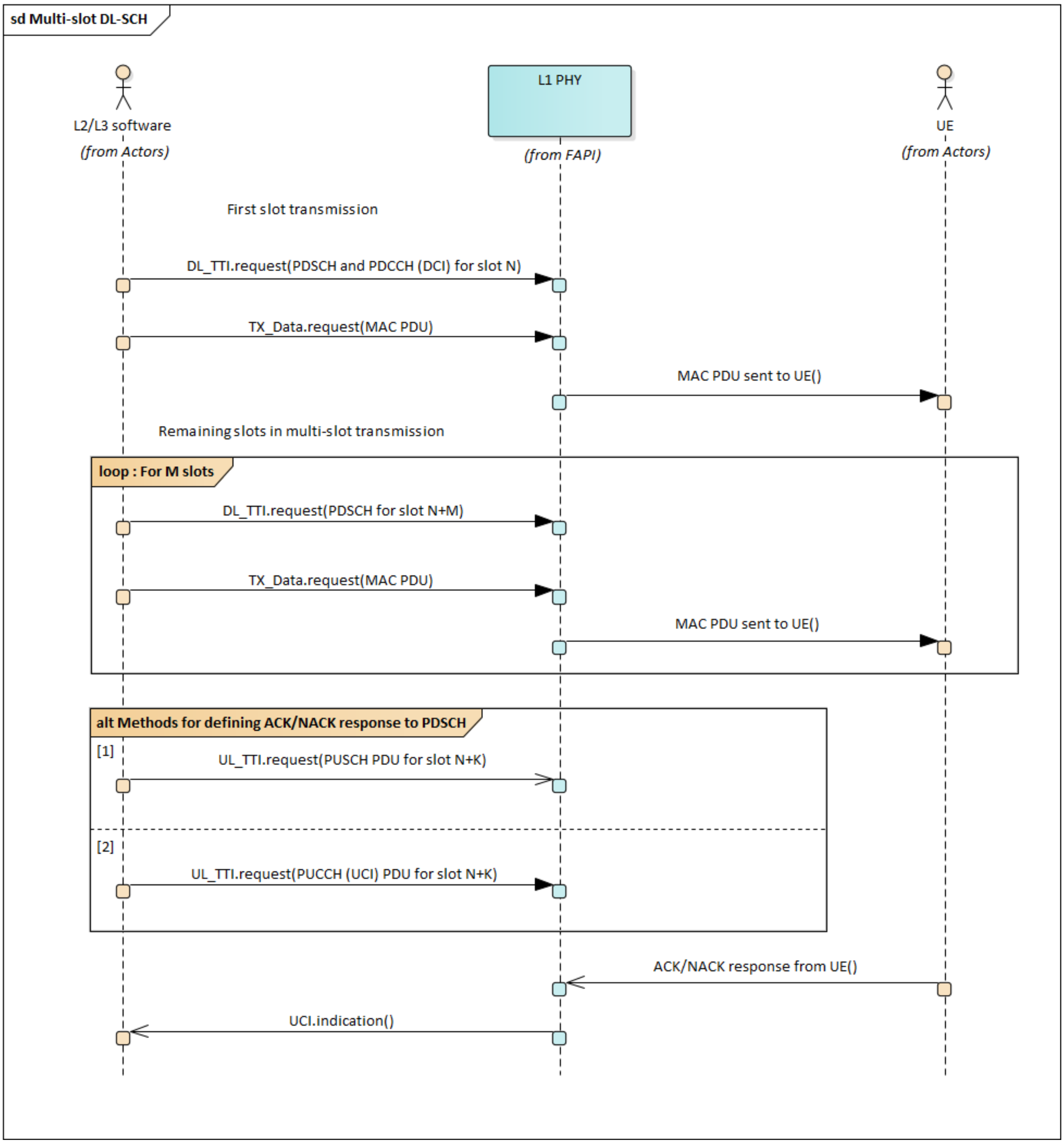


Figure 2-31 Multi-slot DL-SCH procedure

2.2.5.4 Downlink Reference Signals

The downlink includes several reference signals:

- DMRS for PDSCH and PDCCH
- PTRS for PDSCH
- CSI-RS
- PRS

The reference signals transmitted for either PDSCH or PDCCH are included in the `DL_TTI.request` PDSCH or PDCCH PDUs, respectively. However, to transmit a CSI-RS, respectively PRS the `DL_TTI.request` includes a CSI-RS PDU, respectively a PRS PDU. This behaviour is shown below in Figure 2-32, where:

- For a DL data transmission DMRS is included for both PDSCH and PDCCH
    - In `DL_TTI.request` the PDSCH and PDCCH PDUs are included, these PDUs include DMRS information.
    - A TX_DATA.request is sent to the PHY including the MAC PDU
    - The MAC PDU is sent to the UE with DMRS in both the PDSCH and PDCCH transmission.
- For a DL data transmission optionally PTRS can be enabled
    - In `DL_TTI.request` the PDSCH and PDCCH PDUs are included. The PDSCH PDUs will indicate that optional PTRS is included and both PDUs include DMRS information.
    - A `TX_DATA.request` is sent to the PHY including the MAC PDU
    - The MAC PDU is sent to the UE with PTRS in the PDSCH transmission and DMRS in both the PDSCH and PDCCH transmission.
- CSI-RS and PRS transmissions are not dependent on PDSCH transmission, or on each other. Instead, CSI-RS or PRS can be sent to a UE in the same slot as a PDSCH transmission, or in a slot with no PDSCH transmission for the UE. There is also no dependency between CSI-RS and PRS transmissions.
    - For CSI-RS, in `DL_TTI.request` the CSI-RS PDU is included. This results in a CSI-RS transmission to the UE.
    - For PRS, in `DL_TTI.request` the PRS PDU is included. This results in a PRS transmission to the UE.

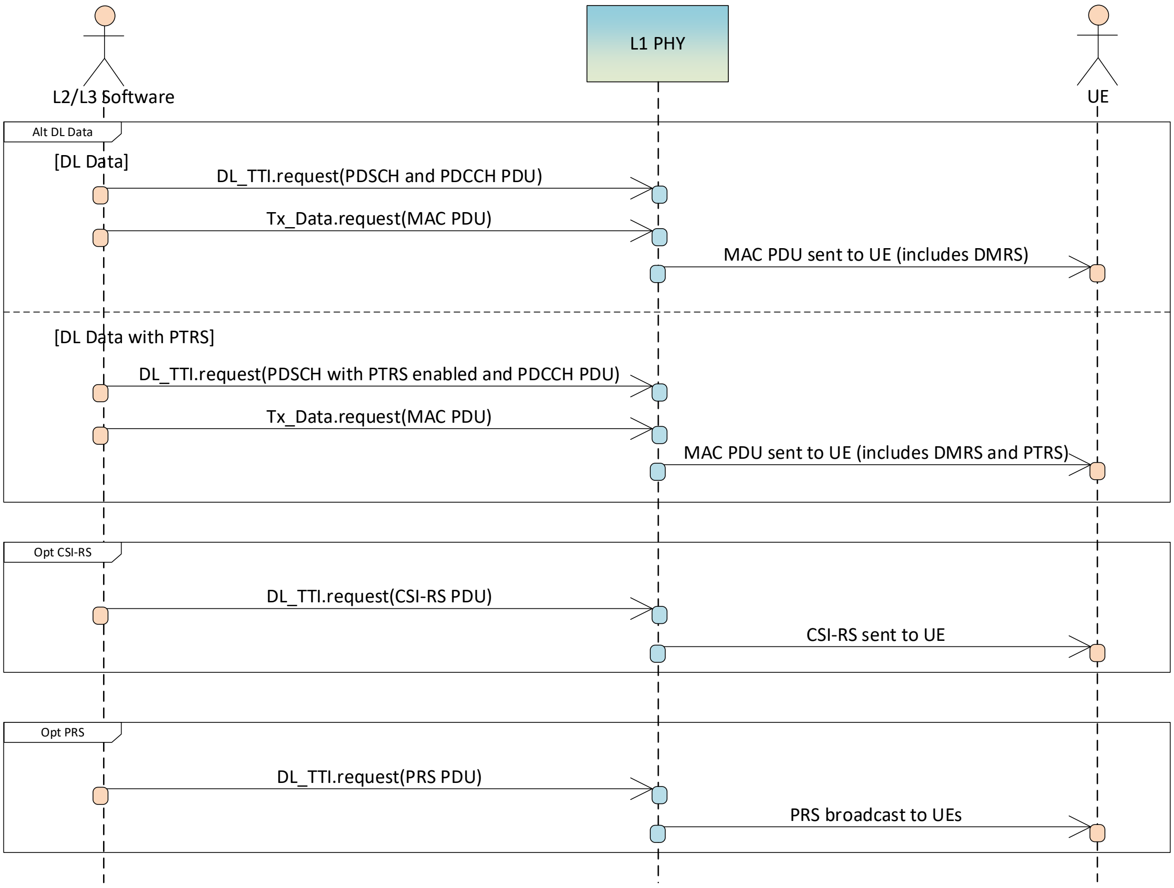


Figure 2-32 Downlink reference signals

2.2.5.5 Transmit Power Levels

In this specification version, two power profiles are supported to signal the transmit power levels in the `DL_TTI.request` message (PDSCH, PDCCH, CSI-RS and SSB) PDUs:

- ProfileNR: relative power levels are signalled based on 3GPP NR specification relationships.
- ProfileSSS: DMRS, CSI-RS and PSS relative power levels are signalled with respect SSS. Data and PT-RS power levels may also be signalled with respect to SSS, or based on 3GPP NR specification relationships. EPRE metrics used in this profile are averaged across all mapped channel/signal REs, where each RE's energy is summed over all ports or layers mapped to the RE.
    - Note: the EPRE definitions for ProfileNR and ProfileSSS are aligned.

In all cases:

- L1 may offer the capability to configure baseband scaling of SS/PBCH with respect to the nominal SSS signal generation in 3GPP TS 38.211 [2], section 7.4.2.3.1, in dB or dBFS units. In case of dBFS units, the scaling applies per RE to the fullscale wideband PSD allocation.

ProfileNR

The power offset parameters used for ProfileNR are shown in Figure 2-33. The PDUs and respective power offset values included are:

- PDCCH PDU
    - powerControlOffsetSS[ProfileNR]
- PDSCH PDU
    - powerControlOffsetSS[ProfileNR]
    - powerControlOffset[ProfileNR]
    - β_{DMRS} (numDmrsCdmGrpsNoData and dmrsConfigType)
- CSI-RS PDU
    - powerControlOffsetSS[ProfileNR]
    - powerControlOffset[ProfileNR]
- PRS PDU
    - prsPowerOffset
- SSB PDU
    - β_{PSS} – (betaPSS[ProfileNR])

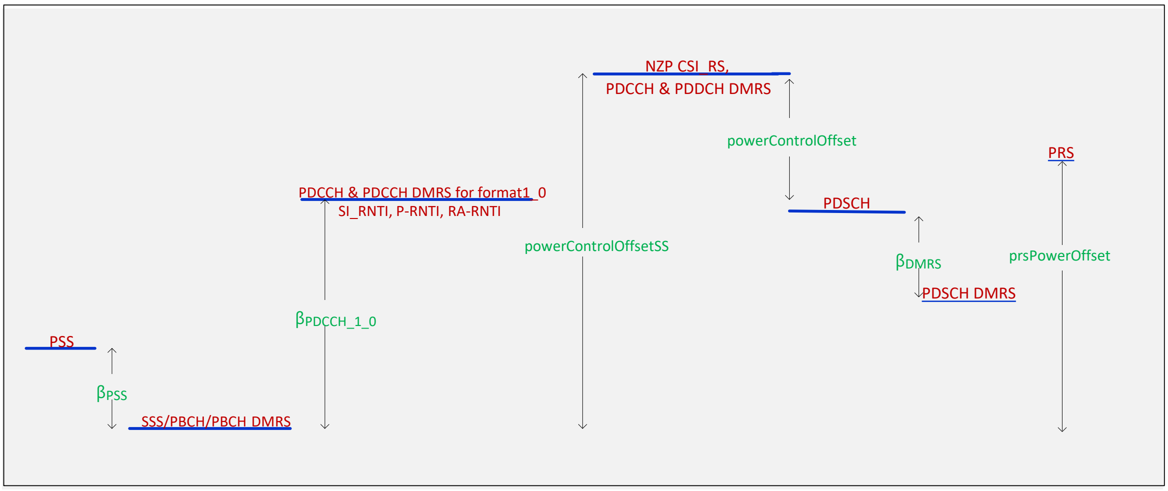


Figure 2-33 ProfileNR: Transmit power levels

ProfileSSS

The power offset parameters used for ProfileSSS are shown in Figure 2-34. The PDUs and respective power offset values included are:

- PDCCH PDU
    - pdcchDmrsPowerOffset[ProfileSSS]
    - pdcchDataPowerOffset[ProfileSSS]
- PDSCH PDU

- pdschDmrsPowerOffset[ProfileSSS]
- pdschDataPowerOffset[ProfileSSS]
- pdschPtrsPowerOffset[ProfileSSS]

- CSI-RS PDU

  - csiRsPowerOffset[ProfileSSS]

- PRS PDU

  - prsPowerOffset

- SSB PDU

  - β_{PSS} – (betaPSS[ProfileSSS])

In the cases of some signals, the L1 maybe be signalled – if supported, and configured – to make use of 3GPP-based power offsets, in which case ProfileNR parameters are used for those signals.

1. pdcchDataPowerOffset[ProfileSSS]
2. pdschDataPowerOffset[ProfileSSS]
3. pdschPtrsPowerOffset[ProfileSSS]
4. betaPSS[ProfileSSS]

In Figure 2-34, these parameters are marked in *[bracketed, italicized captions]*.

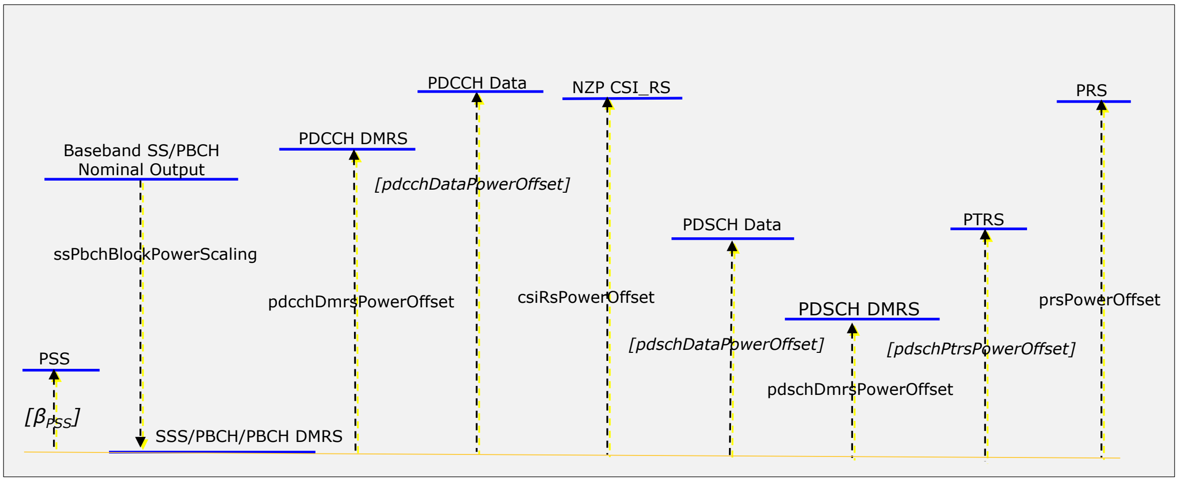


Figure 2-34 ProfileSSS: Transmit power levels

2.2.6 Uplink

The procedures relating to uplink reception are described in this Section.

2.2.6.1 RACH

The RACH transport channel is used by the UE to send data to the gNB when it has no scheduled resources. Also, the L2/L3 software can indicate to the UE that it should initiate a RACH procedure.

In the scope of the PHY API, the RACH procedure begins when the PHY receives an `UL_TTI.request` message indicating the presence of a PRACH slot.

The RACH procedure is shown in Figure 2-35. To configure a RACH procedure the L2/L3 software must provide the following information:

- The `UL_TTI.request` is sent to the PHY including a RACH PDU.
- If a UE decides to RACH, and a preamble is detected by the PHY:
    - The PHY will include 1 RACH PDU for each FD and TD occasion in the `RACH.indication` message. This RACH PDU includes all detected preambles
- If no RACH preamble is detected by the PHY, then no `RACH.indication` message is sent

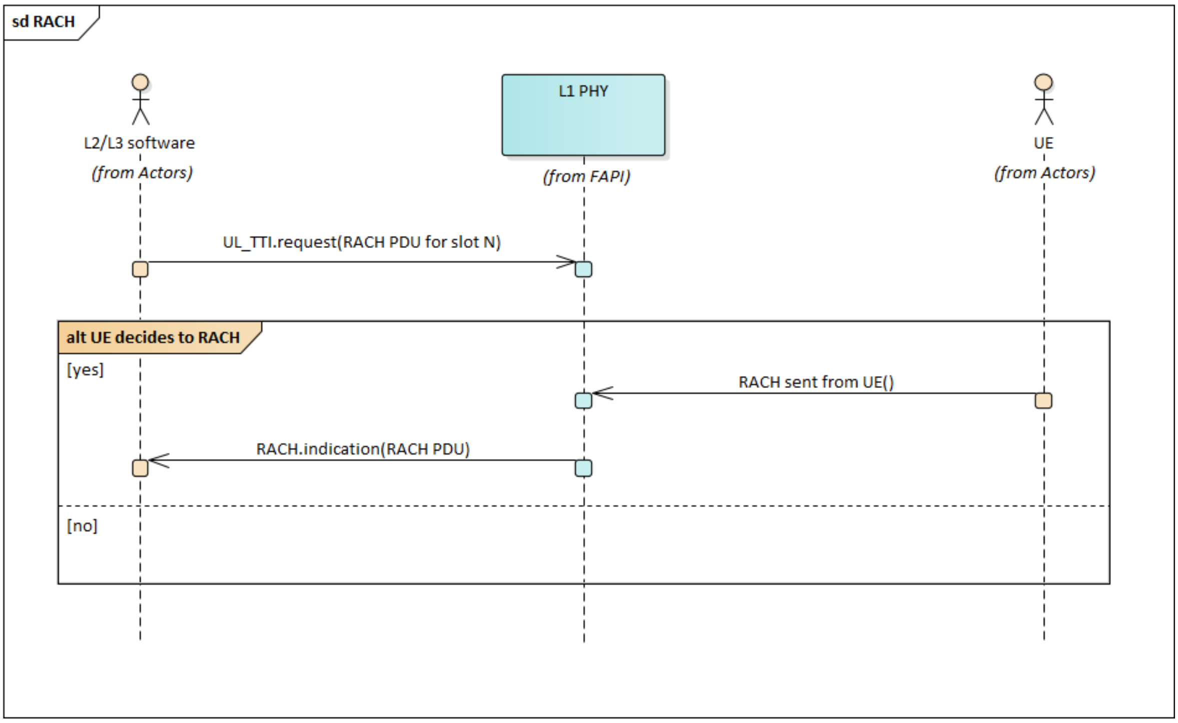


Figure 2-35 RACH procedure

In addition to the Release-15 RACH procedure, this FAPI release supports a 2-step RACH procedure, per 3GPP Rel-16 specifications [24][25], based on the a map linking MsgA-PRACH preambles and MsgA-PUSCH & associated DMRS occasions. The map may be:

- configured semistatically (in P5 MsgA-PUSCH configuration)
- signalled at slot resolution (in PRACH PDU in `UL_TTI.request`)

Conceptually, the 2-step RACH Procedure combines the RACH procedure above, with MsgA-PUSCH PDU signalling similar to UL-SCH section 2.2.6.2, where:

- the `UL_DCI.request` / control info is replaced by the above map configuration/signalling.

- The `UL_TTI.request` → PUSCH PDU is replaced by `UL_TTI.request` → MsgA-PUSCH PDU.

In particular, as illustrated in Figure 2-36

- For PHYs supporting semistatic configuration of a map linking MsgA-PRACH preambles and MsgA-PUSCH & associated DMRS occasions, MAC may choose to configures the map in P5 `CONFIG.request`, via the *Multi MsgA-PUSCH Config TLV*.
- An `UL_TTI.request` is sent to the PHY including a RACH PDU.
  - If and only if a semistatic map MsgA-PRACH preambles and MsgA-PUSCH & associated DMRS occasions was not configured via `P5 CONFIG.request`, a map shall be configured dynamically in the PRACH PDU.
- An `UL_TTI.request` is set to PHY including a MsgA-PUSCH PDU corresponding to the map signalled or configured earlier.
- Upon observing MsgA PRACH Preamble, PHY issues a `RACH.indication`
- Upon observing MsgA PUSCH corresponding to the preamble, PHY issues a CRC.indication for the carried MsgA PUSCH payload.
  - PHY determines the correspondence between the MsgA PRACH Preamble and the MsgA PUSCH observation from the earlier configured or signalled map.
  - If the CRC is successful, PHY also issues an RX_DATA.indication to signal the MsgA PUSCH payload.

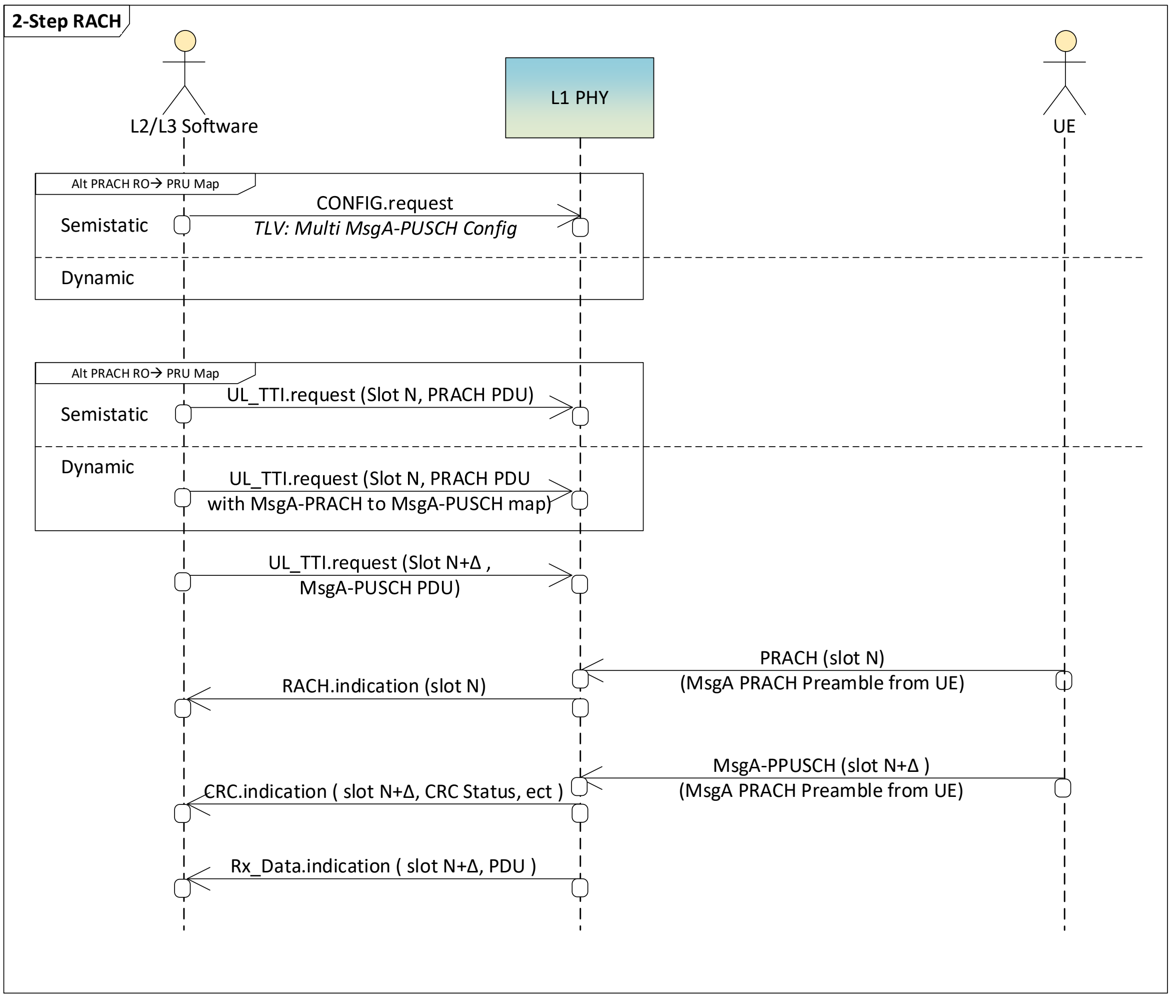


Figure 2-36 2-Step RACH Procedure

Linking MsgA-PRACH preambles and MsgA-PUSCH & associated DMRS occasions

For P5-based MsgA-RACH → MsgA-PUSCH Maps:

- Each instance of the MsgA-PUSCH resource table (Table 3-52) in TLV 0x1032 applies to a single group (A or B)
- A separate PRACH configuration (pointed to by msgA-prachResConfigIndex) is needed to map to each Msg-A PUSCH resource in Table 3-52, via the map in Table 3-53.

For P7-based MsgA-RACH → MsgA-PUSCH Maps:

- A PRACH PDU may contain maps (Table 3-122) for multiple groups (A, B or dedicated), where multiple group-specific MsgA-PUSCH may be mapped in in a single invocation of mapping Table 3-52.

A very simplified mapping illustration is based on the configuration in Figure 2-37:

- Msg-A PRACH:
  - consists of two time-domain occasions (rTD0 and rTD1) and two frequency-domain occasions (rFR_0 and rFD_1).

- Accommodates two groups (A and B).
- The height of the resources represents the number of preambles in each resources (e.g. here group B contains twice as many preambles as group A) – see section 8.1A of 3GPP TS 38.213 [22] for the detailed order of PRACH resources.

- Msg-A PUSCH:

  - Group-A consists of two time-domain occasions (uTD0 and uTD1) and two DMRS resources (uDR_a, uDR_b)
  - Group-B consists of two time-domain occasions (uTD2 and uTD3) and one DMRS resource (uDR_a)
  - The height of the resources represents the number of frequency-domain occasions in each resources.

    - Here, Group-A and Group-B are assigned the same number of frequency-domain occasions.
    - See section 8.1A of 3GPP TS 38.213 [22] for the detailed order of MsgA-PUSCH resources. The ordered indices of *MsgA-A PUSCH Occasions and associated DMRS resources* [22] constitute the space of PRU Preamble Indices, in this document.

  - Here, it can be seen that there are:

    - half as many Msg-A PUSCH frequency domain occasions than there are preambles in Msg-A PRACH Group-A
    - => 2 Msg-A PRACH preambles → 1 MsgA PUSCH resource for Group A
    - a quarter as many Msg-A PUSCH frequency domain occasions than there are preambles in Msg-A PRACH Group-B

      - => 4 Msg-A PRACH preambles → 1 MsgA PUSCH resource for Group B

For this extremely simplified illustration, the Msg-A PRACH → Msg-A PUSCH mapping is shown in Figure 2-38:

- P5-based mapping requires two invocations of mapping Table 3-53 (one for each Group A or B)
- P7-based mapping requires one invocation of mapping Table 3-53.

Notes:

- this example was specifically chosen to allow highly repetitive mapping for intuitive illustration. In general, P7-based mapping can require multiple invocations of mapping Table 3-53.
- the ordered list of Msg-A PUSCH resources in each Group constitutes *Pru Preamble Space* for the Group.

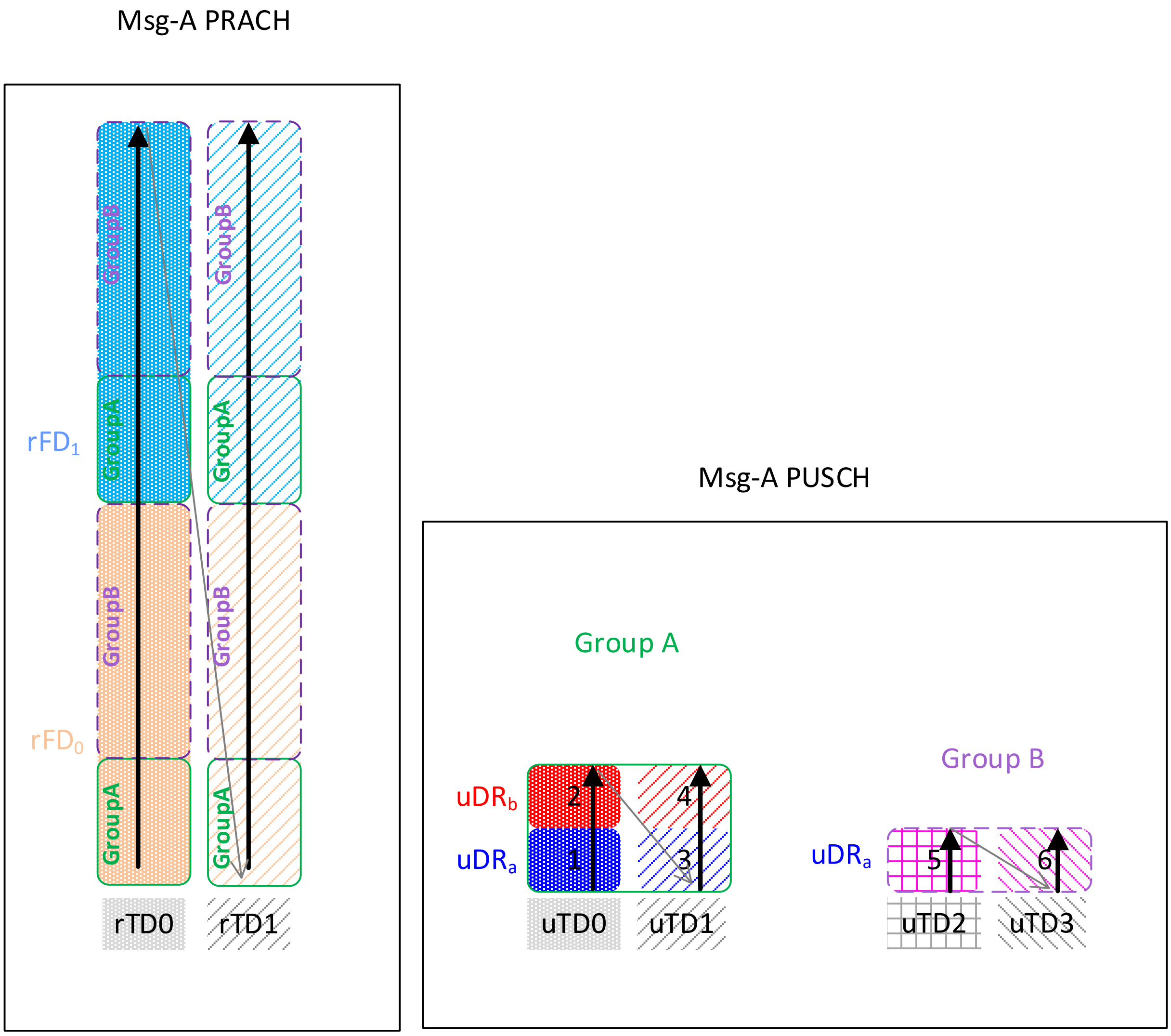


Figure 2-37 2-Step RACH Mga-A PRACH and MsgA-PUSCH illustration

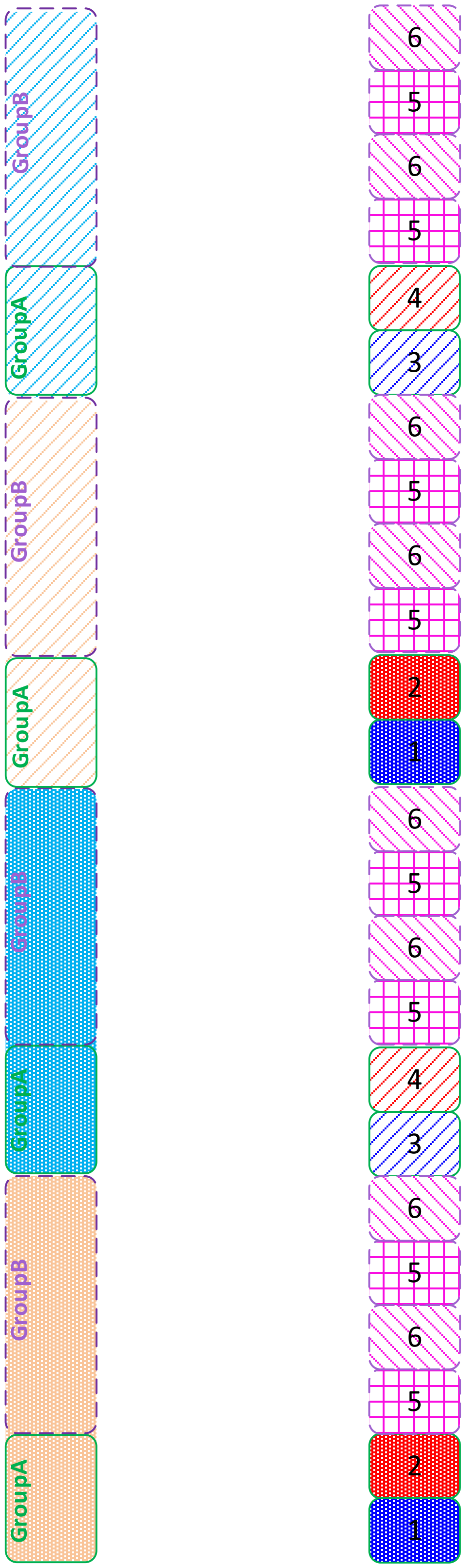


Figure 2-38 MsgA-PRACH → MsgA-PUSCH mapping illustration

2.2.6.2 UL-SCH

The UL-SCH transport channel is used to send data from the UE to the gNB. HARQ is always applied on the UL-SCH transport channel and the ACK/NACK value is indicated when the next transmission for the HARQ process is scheduled via DCI. Therefore, included in the next iteration of this procedure.

The procedure for the UL-SCH transport channel is shown in Figure 2-39. To transmit an UL-SCH PDU, the L2/L3 software must provide the following information:

- Within the `UL_DCI.request` for slot N a DCI PDU is included. The DCI PDU contains control information regarding the UL frame transmission being scheduled and indicates new data or retransmission for a specified HARQ process.
- In `UL_TTI.request` for slot N+K1 an ULSCH PDU is included, where the value of K1 is variable.
- The PHY will return CRC information for the received data in the `CRC.indication` message
- The PHY will return the received uplink data in the `RX_DATA.indication` message.
- If UCI information was expected in the uplink slot, the PHY will return the `UCI.indication` message.

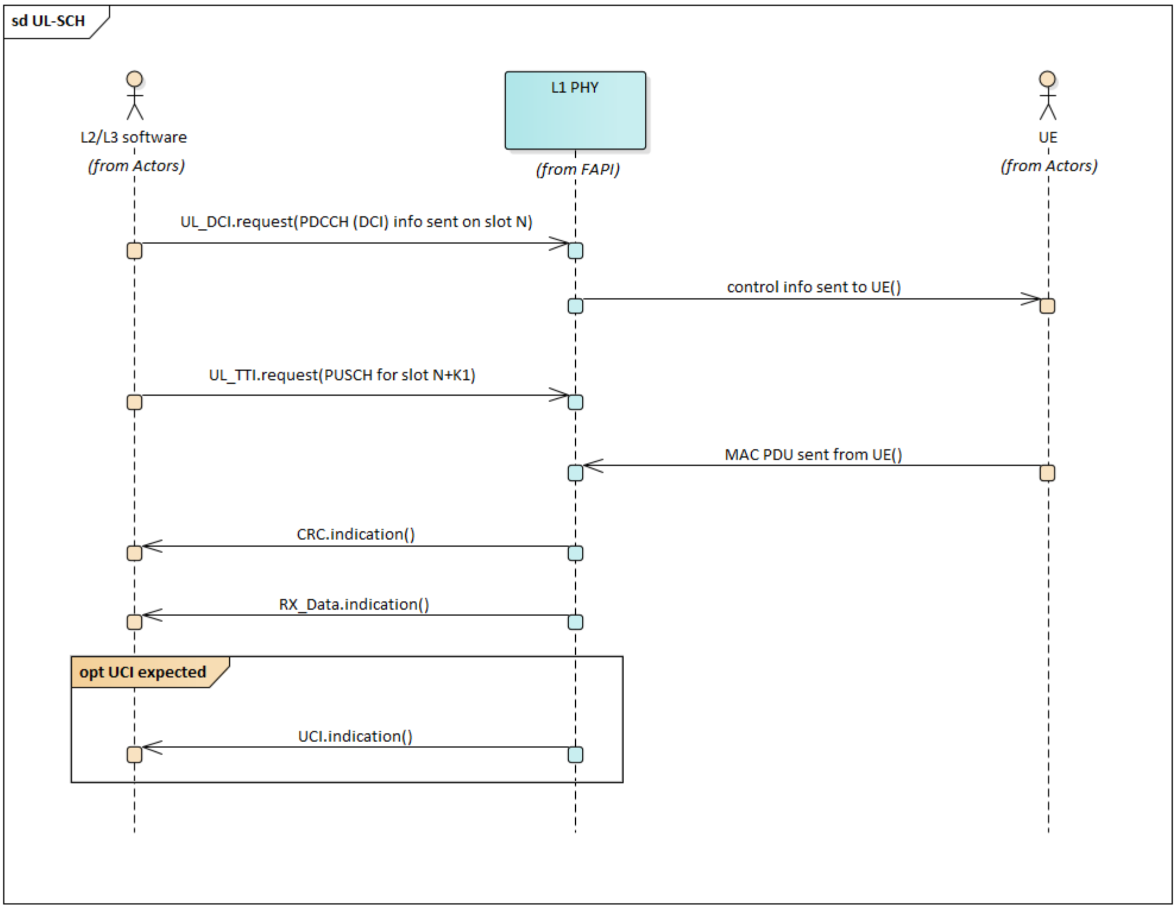


Figure 2-39 ULSCH procedure

Multi-slot transmission is also an option for the UL-SCH with configured grant, where the same MAC PDU is transmitted for N slots. The procedure is shown in Figure 2-40, and the L2/L3 must provide the following information:

- For each slot in the PUSCH transmission
    - In `UL_TTI.request` for slot N+M a PUSCH PDU is included, the UE will have previously been configured to be aware that multi-slot transmission is used.
    - The PHY will return CRC information for the received data in a the `CRC.indication` message
    - The PHY will return the received uplink data in the `RX_DATA.indication` message.

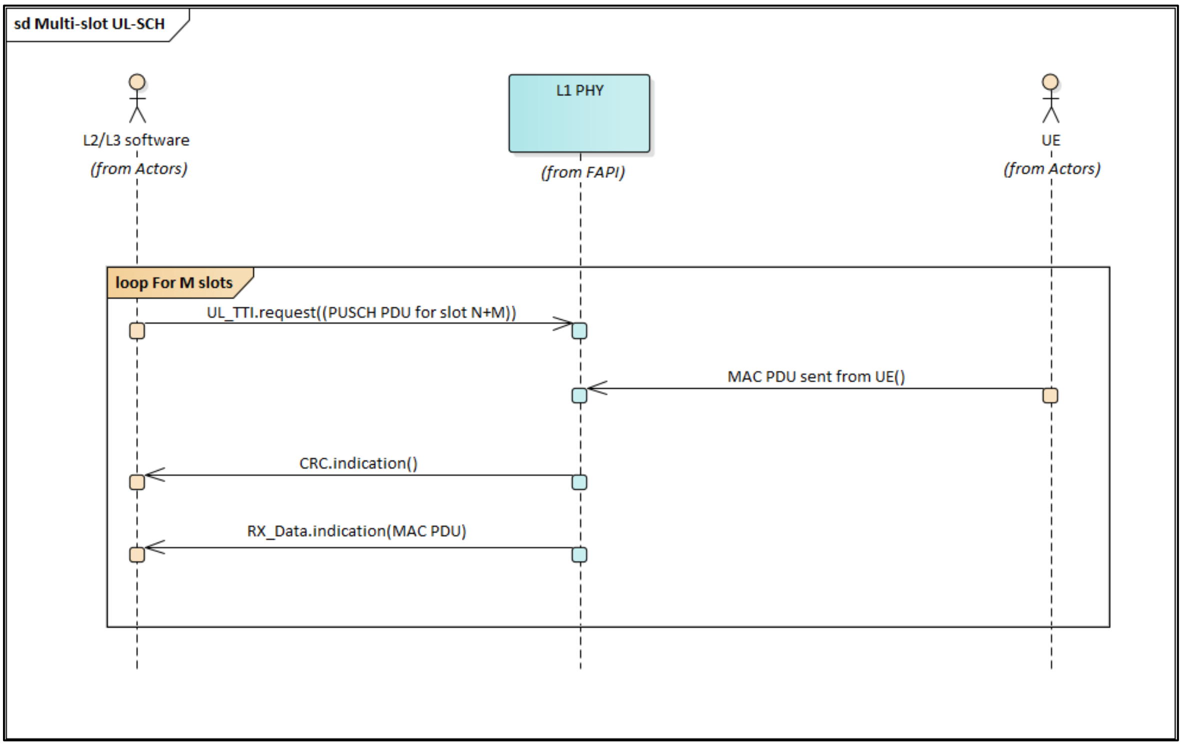


Figure 2-40 Multi-slot UL-SCH procedure

2.2.6.3 SRS

The sounding reference signal (SRS) is used by L2/L3 software to determine the quality of the uplink channel. The SRS procedure is shown in Figure 2-41. To schedule a SRS the L2/L3 software must provide the following information:

- In `UL_TTI.request` a SRS PDU is included.
- The PHY will return the SRS response to the L2/L3 software in the `SRS.indication` message

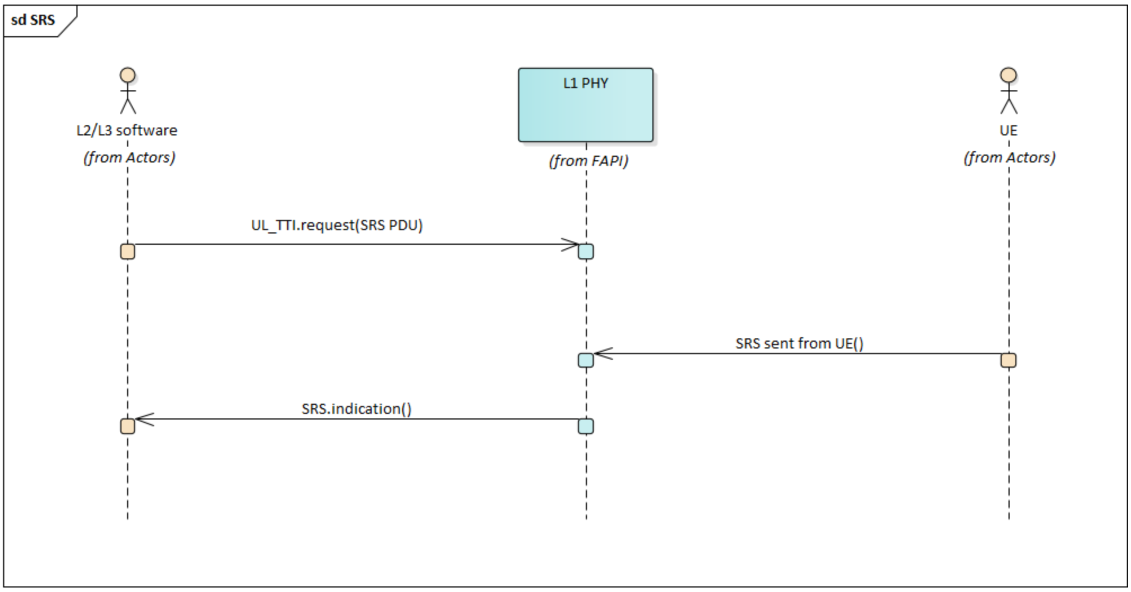


Figure 2-41 SRS procedure

2.2.6.4 CSI

The CSI reporting mechanism is used by the L2/L3 software to determine the quality of the downlink channel. The CSI reporting procedure is shown in Figure 2-42. To schedule a CSI report the L2/L3 software must provide the following information:

- For an aperiodic report the PDCCH PDU is included in the `UL_DCI.request`. This instructs the UE to send a CSI report. For periodic CSI report no explicit DCI information is sent.
- In the `UL_TTI.request`, where the L2/L3 software is expecting a CSI report:
  - If a PUSCH is scheduled for UCI then a PUSCH PDU is included with the CSI configuration information.
  - Otherwise, a PUCCH is scheduled with the CSI configuration information.
- The PHY will return the CSI report to the L2/L3 software in the `UCI.indication` message

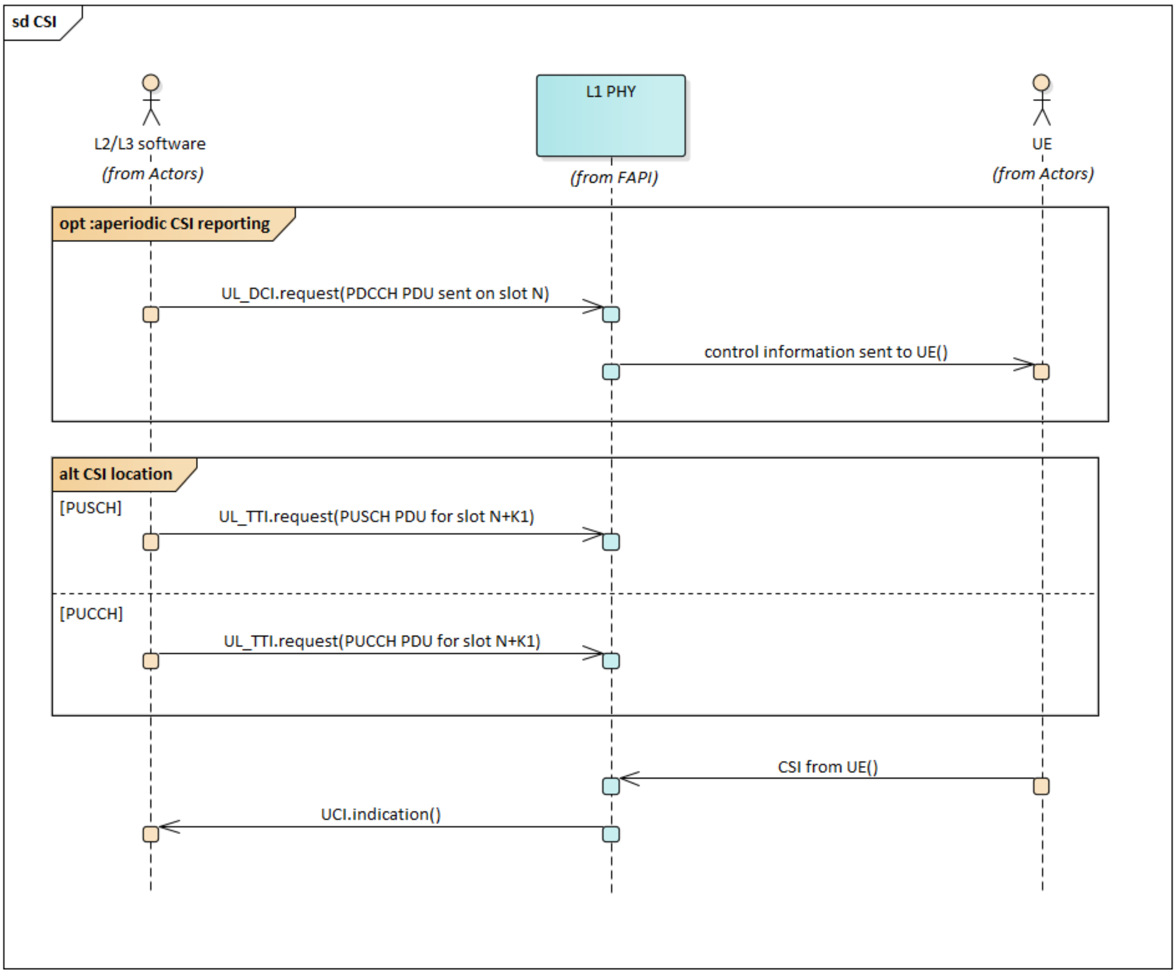


Figure 2-42 CSI procedure

2.2.6.5 SR

The scheduling request (SR) procedure is used by the UE to request additional uplink bandwidth. The L2/L3 software configures the SR procedure during the RRC connection procedure. The SR procedure is shown in Figure 2-43. To schedule a SR the L2/L3 software must provide the following information:

- In `UL_TTI.request` a PUCCH PDU is included.
- The PHY will return the SR to the L2/L3 software in the `UCI.indication` message.

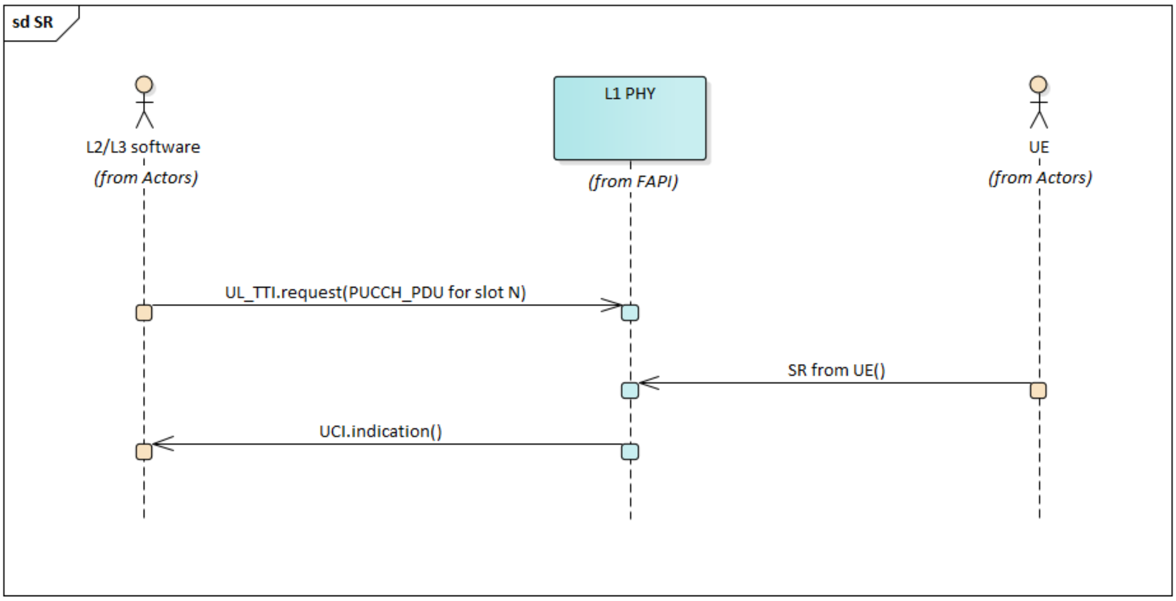


Figure 2-43 SR procedure

2.2.6.6 Uplink Reference Signals

The downlink includes several reference signals:

- DMRS for PUSCH and PUCCH
- PTRS for PUSCH
- SRS

The reference signals transmitted for either PUSCH or PUCCH are included in the `UL_TTI.request` PUSCH or PUCCH PDUs, respectively. However, to transmit a SRS the `UL_TTI.request` includes a SRS PDU. The SRS PDU is described in Section 2.2.6.3, while the other uplink reference signals are described below in Figure 2-44, where:

- For a UL data transmission DMRS is included for PUSCH
  - In `UL_TTI.request` the PUSCH PDU is included, this PDU includes DMRS information.
  - The MAC PDU is sent from the UE with DMRS.
- For a UL data transmission optionally PTRS can be enabled
  - In `UL_TTI.request` the PUSCH PDUs will indicate that optional PTRS is included and DMRS is included.
  - The MAC PDU is sent from the UE with PTRS and DMRS.
- For a UL UCI transmission DMRS is included for PUCCH
  - In `UL_TTI.request` the PUCCH PDU is included, this PDU includes DMRS information.
  - The UCI is sent from the UE with DMRS.

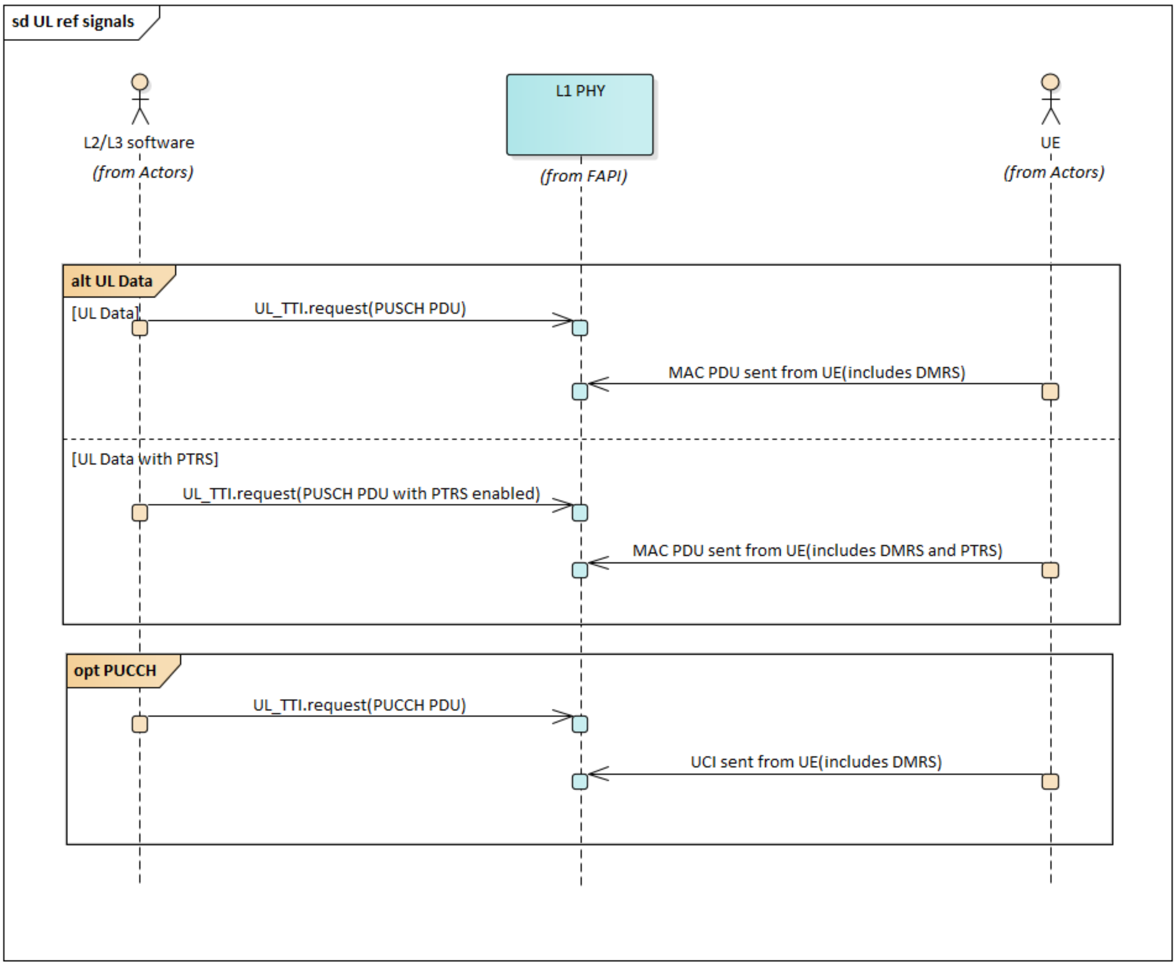


Figure 2-44 Uplink reference signals

2.2.7 Spatial Multiplexing

Spatial multiplexing of channels and layers can be achieved either via precoding and beamforming tables configurable to PHY, or via dynamic precoder computation by PHY.

The spatial multiplexing parameters are defined to be flexible and support varying levels of sophistication.

2.2.7.1 Precoding and Beamforming Based on Semistatic Tables

For semi-static spatial multiplexing, the precoding and beamforming is defined in tables loaded into the PHY at configuration time. In slot messages a precoding index and/or beamforming index is included in each PDU.

Precoding and digital beamforming can be viewed as a cascade of two matrix multiplications PM and DB as depicted in Figure 2-45. For precoding operation an index to a pre-stored precoding matrix PM(idx) is specified in the message. Digital beamforming is represented by a matrix DB. For efficient messaging columns of DB are picked from a pre-stored digital beamforming table (DBT) as illustrated in Figure 2-46. It suffices to specify the index of the beam to be applied for each input port in the message. Baseband ports shown in Figure 2-45 are logical entities that may be further processed by a digital frontend (DFE) unit. Within the context of 5GNR a baseband port typically corresponds to component carrier that is handled by a specific RF path (i.e. TXRU).

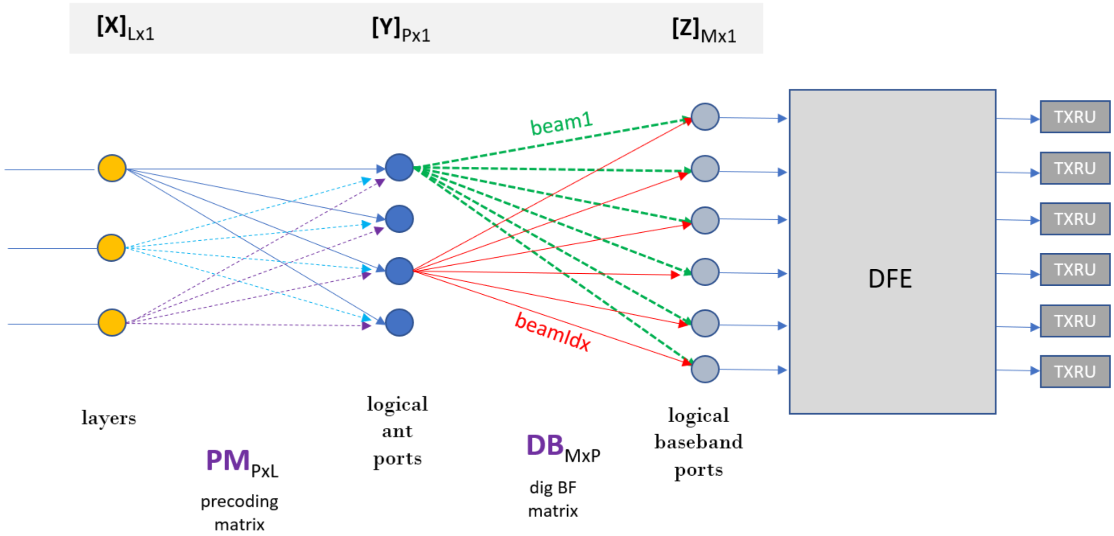


Figure 2-45 Precoding and digital beamforming flow

X: precoder input vector of size (L×1)
Y: precoder output vector of size (P×1)
Z: digital beamformer output vector of size (M×1)
PM: Precoding matrix of size (P×L)
DB: digital beam forming matrix of size (M×P)

$$Y = PM \times X$$
$$Z = DB \times Y$$

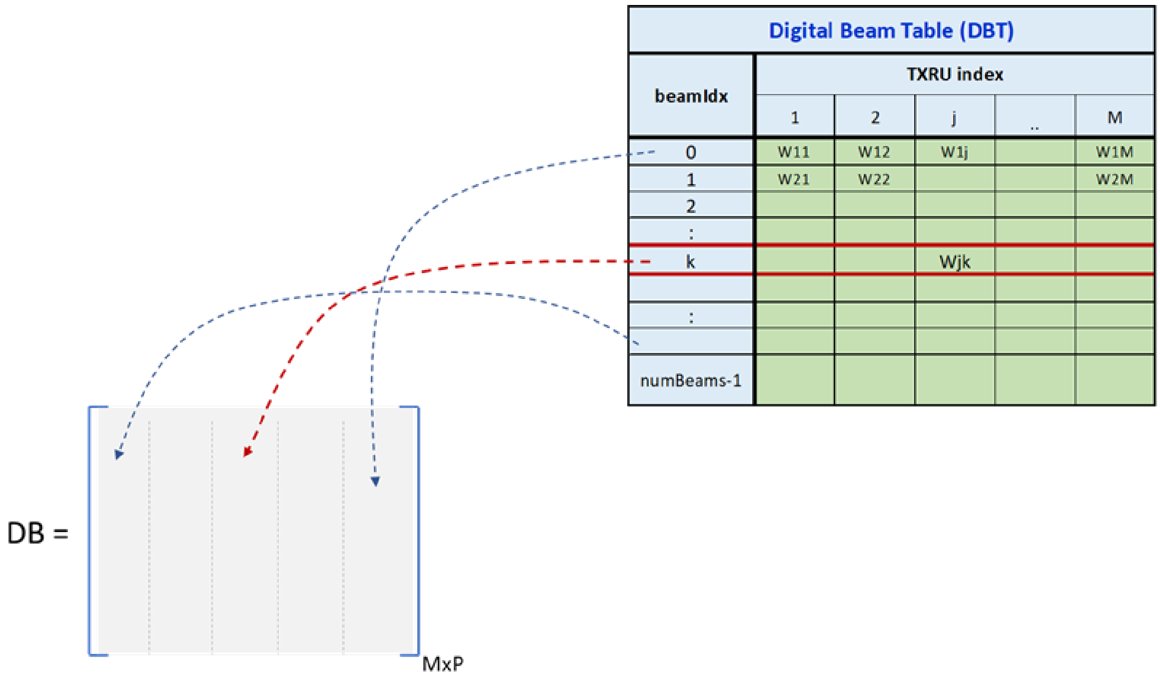


Figure 2-46 Overview of digital beam table (DBT)

2.2.7.2 Precoding based on PHY-determined weights

Alternatively, downlink precoders/uplink combiners can be determined in PHY. To help PHY in such spatial multiplexing, PHY collects SRS samples which can be used to form uplink codebook-based combiners. Under channel reciprocity conditions, SRS samples can also be used to form downlink precoders and non-codebook uplink combiners.

To assist L2 select UEs and layers for spatial multiplexing in a slot, L1 reports a frequency-subsampled set of channel estimates to L2, in the form of e.g. the channel matrix H for the UE, or an SVD decomposition thereof, as described in Table 2-3.

MU-MIMO direction	5G NR SRS type	Type of channel abstraction (in `SRS.indication`)	Layer signalling for PHY precoder/combiner determination
DL	Antenna-switch	SVD representation: H = U S V (under channel reciprocity assumptions) U: Rx_{UE} X Rx_{UE} left-singular vector matrix S: Rx_{UE} X Rx_{UE} diagonal matrix of singular values V: Rx_{UE} X Tx_{gNB} matrix of right-singular vectors for Rx_{UE} singular values.	Selection, in `DL_TTI.request` of UEs' Downlink (e.g. PDSCH) ,PDU to schedule, and of singular values in S diagonal matrix, for each scheduled UE
UL	Codebook-based	H	Selection of UEs to schedule, and of UL-TPMI, for each scheduled UE.
UL	Non-codebook-based	H (under weak channel reciprocity assumptions)	Selection of UEs to schedule, and of Set of SRS TX ports, for each scheduled UE.

Table 2-3 SRS-based reporting & layer selection, per direction & usage type

Digital precoding (for downlink) is depicted in Figure 2-47, and digital combining (for uplink) is depicted in Figure 2-48. Downlink precoder determination and uplink combiner determination are based on a signalling of UEs and layers (see Table 2-3) for joint spatial multiplexing over the same set of RBs and symbols, grouped into MU-MIMO groups, as described in section 2.2.6.2a.

Baseband ports shown in Figure 2-47 and Figure 2-48 are logical entities that may be additionally processed by a digital frontend (DFE) unit. Within the context of 5GNR a baseband port typically corresponds to component carrier that is handled by a specific RF path (i.e. TXRU).

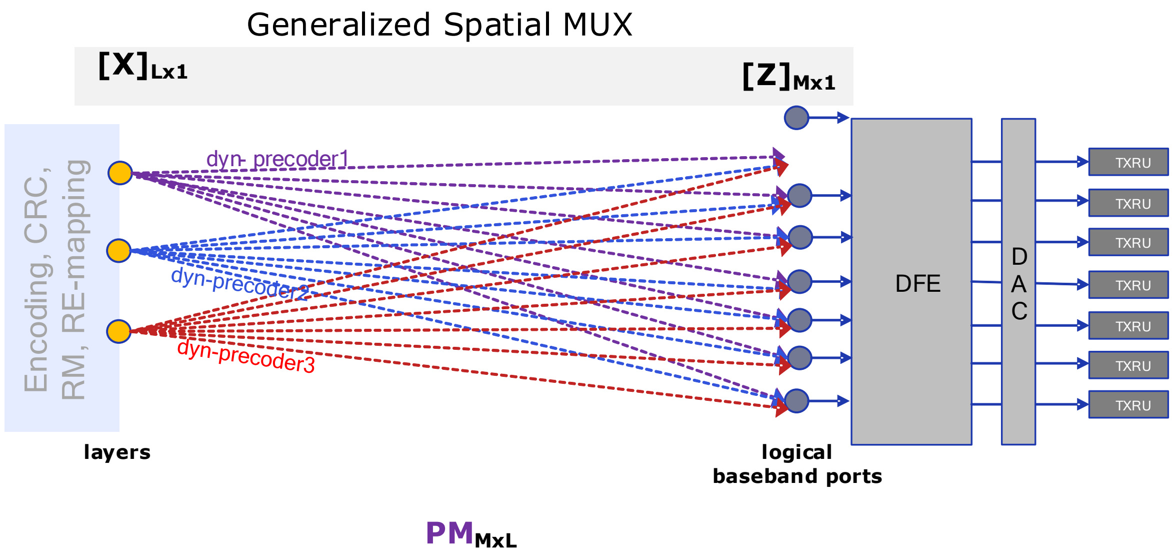


Figure 2-47 Dynamic Precoding flow (Downlink)

X: precoder input vector of size (L×1)
Z: digital beamformer output vector of size (M×1)
PM: Dynamic Precoding matrix of size (P×M)

Z = PM × X

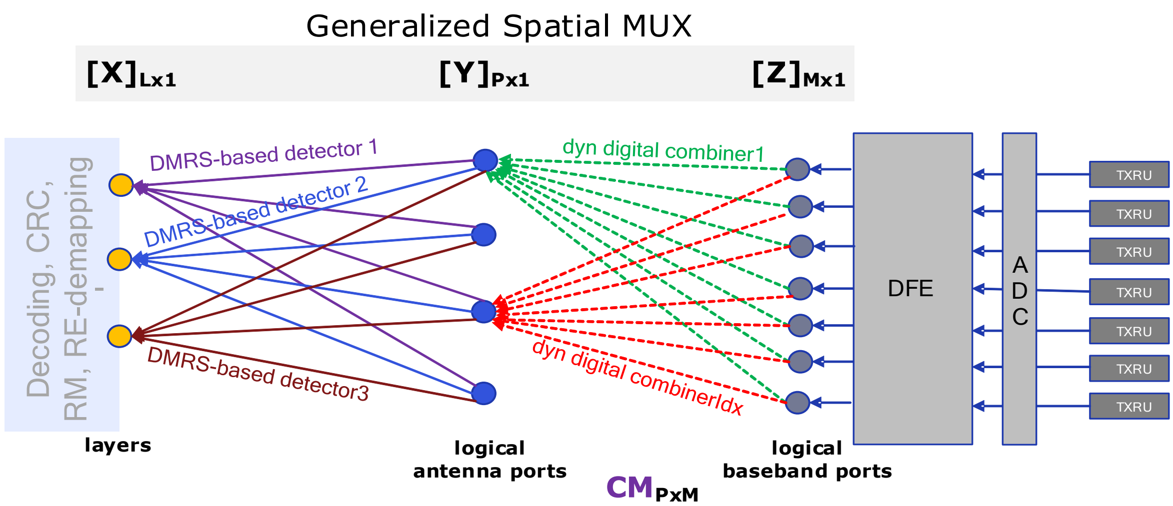


Figure 2-48 Dynamic Combiner flow (Uplink)

X: Layer output vector of size (L×1)
Y: Spatial stream digital combiner output vector of size (Px1)
Z: Digital combiner input vector of size (M×1)
CM: Dynamic Combining matrix of size (P×M)

Y = CM × Z
X = detection(Y) – detection is based on DM-RS detection, not the outcome of combining.

An example call flow for MU-MIMO scheduling of two UEs is illustrated in Figure 2-49:

1. MAC configures PHY to expect Antenna Switch SRS Resources for each UE
   - In this example, the SRS arrive in different slots, but they could have arrived in the same slot, depending on higher-layer configuration and triggering.
2. PHY generates SVD reports (H = U S V) for each of the UEs.
3. MAC schedules the two UEs, with respective layers picked from the per-PRG singular value matrix S from step 2, for each UE).
4. PHY computes (RE-level) joint precoders, corresponding to the MAC UE and layer selection step 3.
   - MU-MIMO Group: assists in joint precoder computation
   - PDSCH PDUs: channel profiles, for each UE

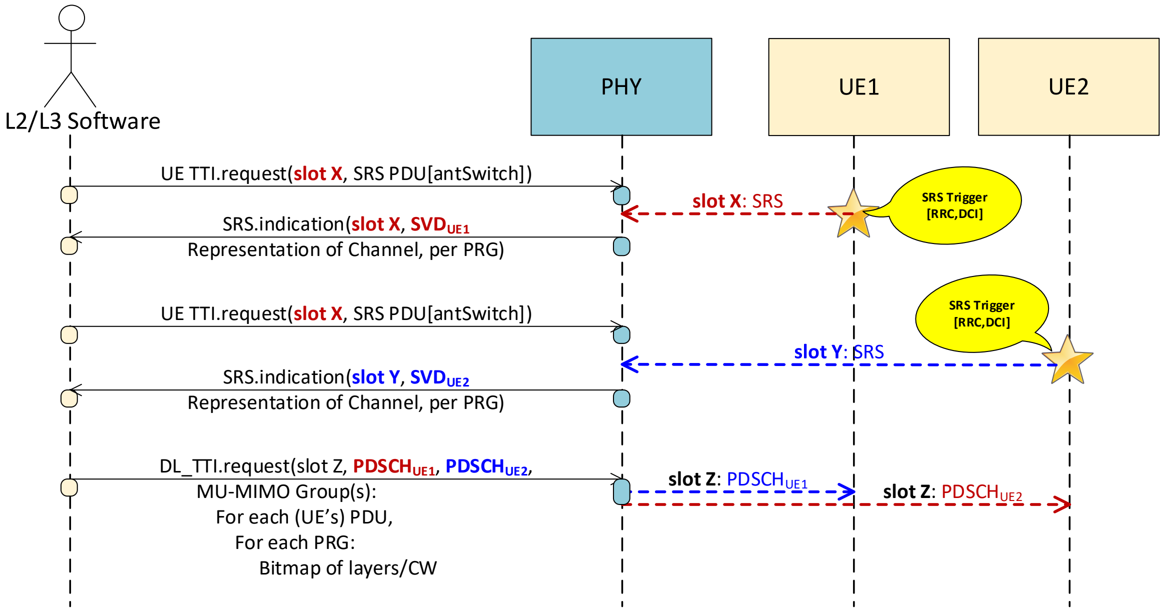


Figure 2-49 Example Call flow for MU-MIMO DL Scheduling

2.2.6.2a MU-MIMO Groups

The set of all the RBs and symbols in a slot over which the same set of UE channels and layers is scheduled is denoted as a "MU-MIMO Group", as illustrated in

Figure 2-50. L2/L3 signals these groups regardless of precoding and combining format (see sections 2.2.7.1, 2.2.7.2, 2.2.6.2b), including for SU-MIMO allocations, but L1 PHY is not required to avail itself of the signalling of MU-MIMO groups.

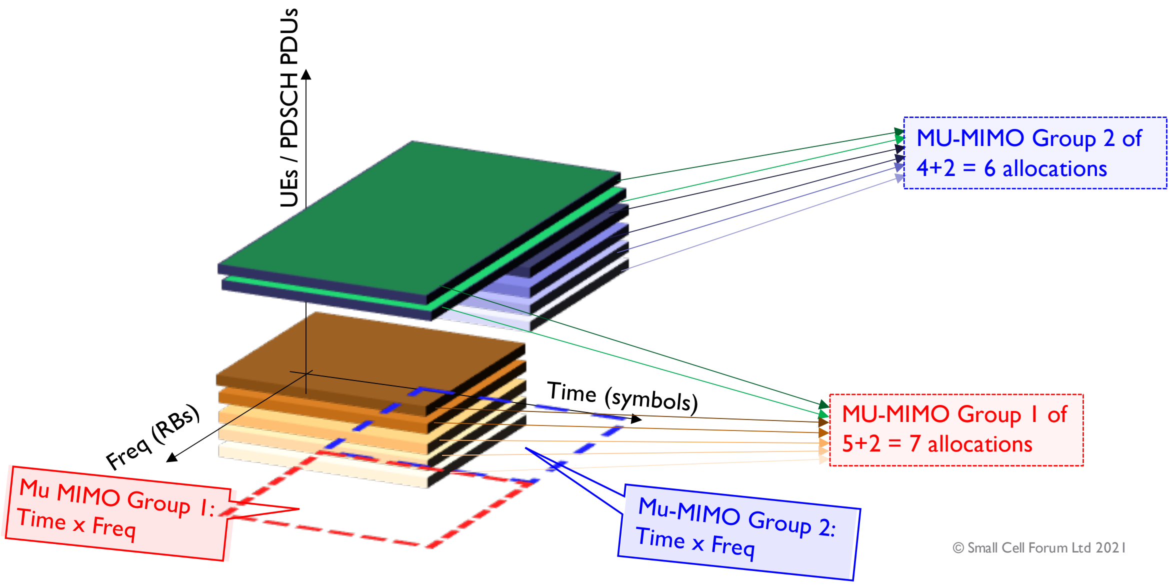


Figure 2-50 MU-MIMO Groups

2.2.6.2b Precoding based on L2/L3-determined weights using channel 2D-DFT representation

Alternatively, downlink precoders/uplink combiners can be determined by L2/L3. As illustrated in the example call flow in Figure 2-51, PHY receives the antenna switched SRS signals from UEs and estimates the channels between each UE antenna and the radio unit antennas. Then, it performs a 2D-DFT on the SRS basis vectors of the estimated channels, where the SRS basis vectors may be orthogonal (e.g. SVD eigenvectors) or not (e.g. SRS ports). The channel projection onto each DFT vector can be considered a *beam*. Each complex value in the 2D-DFT output indicates the amplitude of a beam's projection onto that DFT basis, and PHY finds the strength of the beam by calculating the power of the corresponding complex value. PHY ranks the beams in the 2D-DFT outputs according to their strengths and determines the strongest beams. Then, it conveys the indices of the strongest beams along with their complex values to L2/L3.

PHY also provides frequency subsampled channel estimates to L2/L3 via `SRS.indication`. Channel estimates can be in the form of a channel matrix H for each UE whose SRS are processed in PHY.

L2/L3 may determine the UEs to be spatially multiplexed in the same PRG according to the placement of their strongest beams on the 2D-DFT grid, e.g. it may select the UEs whose strongest beams do not overlap with each other. After user selection, L2/L3 may determine the precoder. For example, it may calculate the precoder for the selected set of UEs using their channel matrices.

Since L2/L3 has the channel estimates as well as the precoder, it may attempt to estimate the SINR at the UE side, for instance by calculating the product of the precoder matrix and the channel matrix. L2/L3 may, for instance, use the SINR estimate in link adaptation.

L2/L3 signals the precoder weights and MCS part of the PDSCH PDU in `DL_TTI.request`.

Uplink combiner call flow is similar to the DL precoder call flow in the sense that the procedure in this section mirrors the `DL_TTI.request` in `UL_TTI.request` and makes similar use of the `SRS.Indication` message.

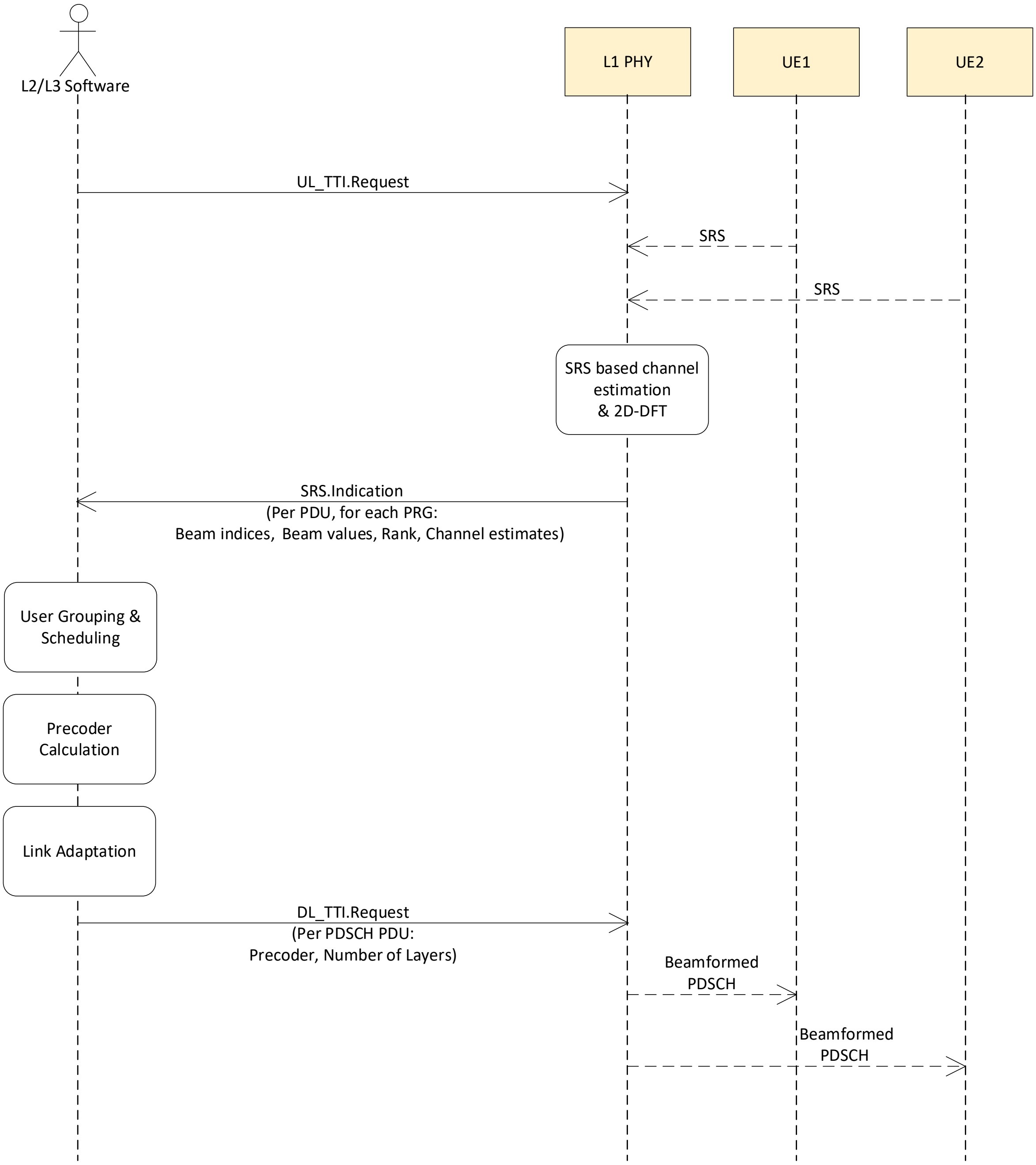


Figure 2-51 Example Call flow for MU-MIMO DL Scheduling using channel 2D-DFT representation

2.2.7.3 Downlink Spatial Stream Limits

In practice, the number of parallel precoders that can be supported by a PHY is limited. Based on PHY capability, MAC assigns DL spatial stream indices between 0 and [(max# downlink streams)-1], so that each RE and symbol output from a precoder is unambiguously associated with a unique DL stream index. The maximum number of DL streams is subject to L1 capability (e.g. maxNumberDlSpatialStreams, maxNumCarriersBWLayersProductDL, maxNumCarriersBWAntennasProductDL).

Downlink spatial streams may – for instance – correspond to downlink data flows, as indexed by "c_eAxC_ID" in section 4.5.4.1 of [26].

To this effect, when spatial streams index layers, per TLV 0x016E, MAC associates DL stream indices per PRG as follows:

- PDCCH PDU: one stream index per DCI.
- PDSCH PDU: one stream index per layer.
- CSI-RS PDU: one stream index per cdm port in a cdm group.
- SSB PDU: one stream index.

More stream indices than may be possible when spatial streams index logical antennas or baseband ports, per TLV 0x016E.

Where L1 capability (e.g. dl-SpatialStreamChannelPriority) allows the same index to be associated with the same RE and symbol in multiple PDUs for a slot, PHY drops all but the RE and symbol for the highest priority channel PDU.

2.2.7.4 Uplink Spatial Stream Limits

In practice, the number of parallel combiners that can be supported by a PHY is limited. Based on PHY capability, MAC assigns UL spatial stream indices between 0 and [(max # uplink streams)-1], so that:

- each RE and symbol output from a combiner is unambiguously associated with a at least one UL channel PDU.
- when ul combiners are signalled as UL logical ports, the number of spatial steams associated with a channel is equal to the number of logical ports for the channel.
- the number of spatial streams associated with an UL channel is at least as large as the number of layers for the UL channel.

The maximum number of UL streams is subject to L1 capability (e.g. maxNumberUlSpatialStreams, maxNumCarriersBWLayersProductUL, maxNumCarriersBWAntennasProductUL).

Uplink spatial streams may – for instance – correspond to uplink data flows, as indexed by "c_eAxC_ID" in section 4.5.4.1 of [26].

To this effect, MAC associates UL stream indices per PDU as follows:

- PUCCH PDU: (at least) one stream index.
- PUSCH PDU: (at least) one stream index per layer.
- PRACH PDU: as many stream indices as logical channels (at least one).
- SRS PDU: as many stream indices as logical channels, except if SRS PDUs are associated with RAW ports. In the latter case, MAC does not assign UL spatial streams to the SRS PDU

2.2.8 Error sequences

The error sequences used for each slot procedure are shown in Figure 2-52 to Figure 2-55. In all slot procedures errors that are detected by the PHY are reported using the

`ERROR.indication` message. In Section 3, the PHY API message definitions include a list of error codes applicable for each message.

The `DL_TTI.request`, `UL_TTI.request`, `UL_DCI.request` and `TX_DATA.request` messages include information destined for multiple UEs. An error in information destined for one UE can affect a transmission destined for a different UE. For each message, the `ERROR.indication` sent by the PHY will return the first error it encountered.

If the L2/L3 software receives an `ERROR.indication` message for `DL_TTI.request`, `UL_TTI.request`, `UL_DCI.request` or `TX_DATA.request`, it should assume that the UE did not receive data and control sent in this slot. This is similar to the UE experiencing interference on the air-interface and 5G mechanisms, such as, HARQ and ARQ, will enable the L2/L3 software to recover.

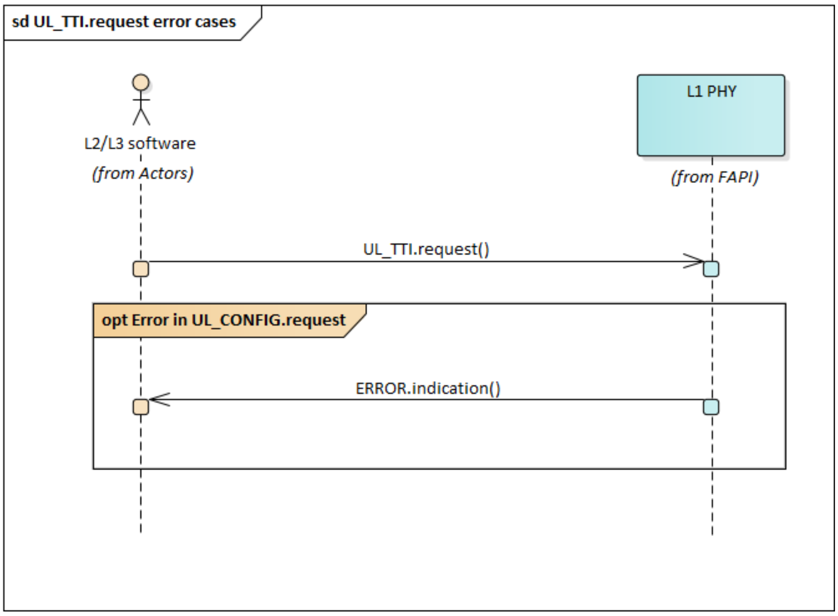


Figure 2-52 **`UL_TTI.request` error sequence**

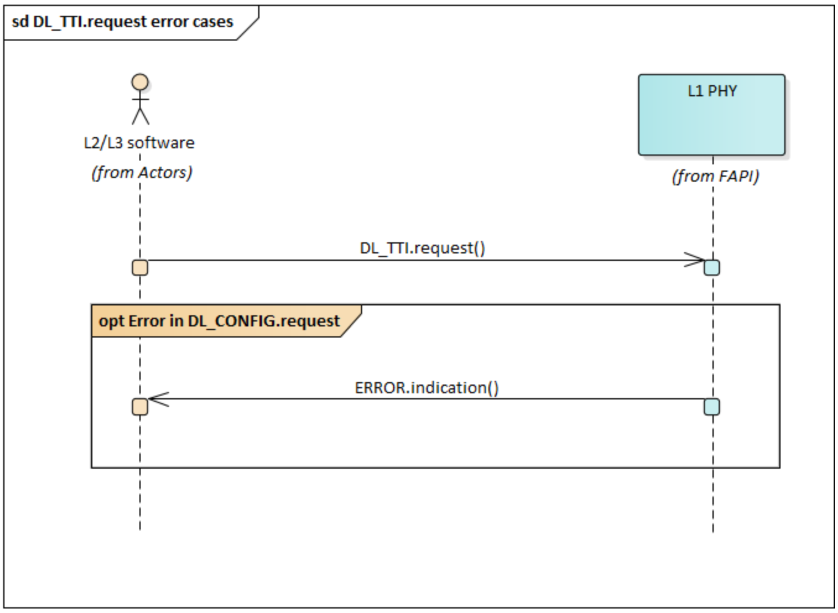


Figure 2-53 `DL_TTI.request` error sequence

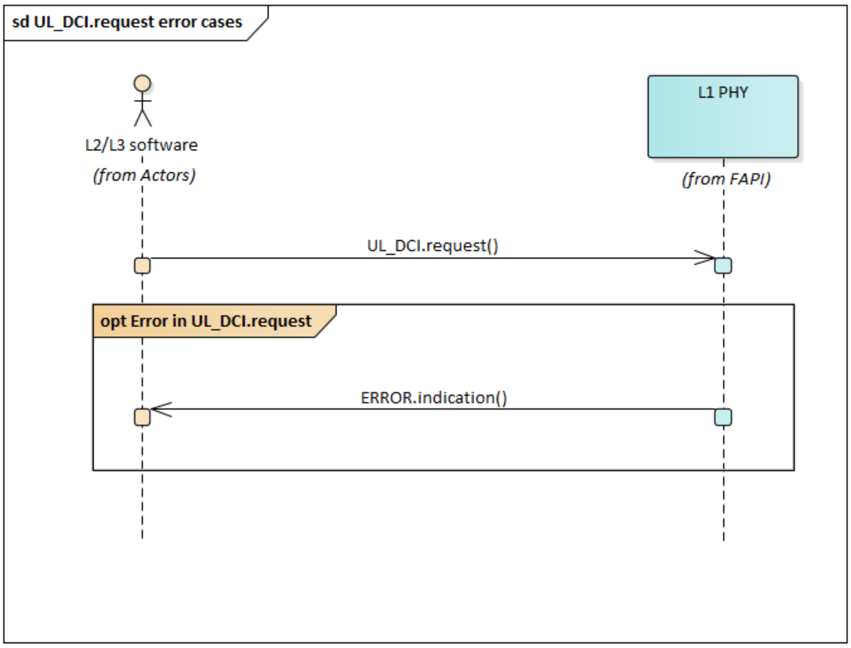


Figure 2-54 `UL_DCI.request` **error sequence**

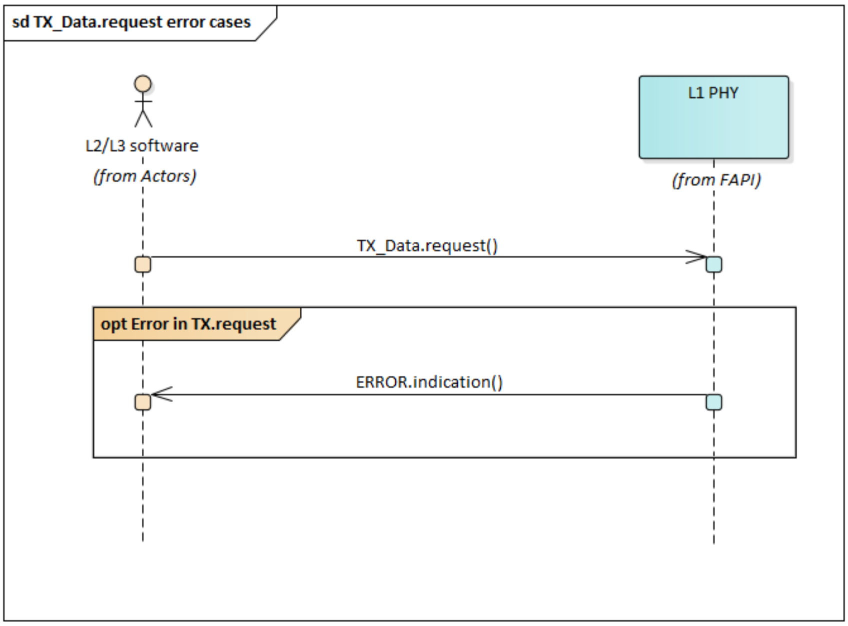


Figure 2-55 `TX_DATA.request` **error sequence**

2.3 Virtual Functions

Multiple distinct L2/L3 functions may be supported by a single PHY Layer implementation, e.g. when L2/L3 functions are virtualized. When that happens:

- Each L2/L3 function is associated with a unique Virtual Function Identifier.
- The Virtual Function Identifier shall be signalled in the headers of all APIs terminated or initiated at the associated L2/L3 function.
- A PHY Group may be orchestrated to at most one L2/L3 function

- APIs shall only be exchanged with the following Contexts:
    - Common Context (P5):
        - signalling shall only be pertinent to the L2/L3 virtual function
    - PHY Group Context (P5)
        - signalling shall only be pertinent to the PHY Groups orchestrated for the L2/L3 virtual function
    - Dedicated PHY (only for the PHYs in the PHY Groups orchestrated for the function) (P5/P7)
        - signalling shall only be pertinent to PHYs in PHY Groups orchestrated for the L2/L3 virtual function.

The orchestration of virtual functions, and of the association of PHY Groups to virtual functions is outside the scope of this release.

3. PHY API Messages

This section provides a description of the PHY API message formats. It defines the PHY API message header, the message bodies and the error codes associated with the PHY API.

3.1 General Message Format

FAPI messages may be define using the Basic (section 3.1.1) of nFAPI (section 3.1.2) message formats.

3.1.1 Simplified Message Format

The general message format of the PHY API is shown in Table 3-1, where it can be seen that there is a top structure of 1 or more PHY API messages. This permits the optional bundling of messages and the inclusion of vendor specific messages.

Description
PHY API Message Header (see Table 3-2)
PHY API Message1 (see Table 3-3)
PHY API MessageLAST
Vendor-specific message

Table 3-1 General PHY API message format

Type	Description
uint8_t	Number of messages included in PHY API message
uint8_t	An opaque handle (purpose not defined, however, usages may include a PHY ID, Carrier ID or designate Common Context or PHY Group Context) . Only P5 PARAM and CONFIG messages may be scoped at Common or PHY Group Context, in the current version of the specification.
uint8_t	Virtual Function Identifier
uint8_t	Reserved

Table 3-2 PHY API message header

Each PHY API message consists of a generic header followed by a message body, as shown in Table 3-3, and consists of a message type ID, a length field and a message body.

Type	Description
uint16_t	Message type ID
uint32_t	Length of message body (bytes)
Message body	

Table 3-3 General PHY API message structure

The current list of message types is given in Table 3-11.

The message body is different for each message type; however, each message body obeys the following rules:

- The first field in each response message is an error code. For each message it is indicated which error codes can be returned. A full list of error codes is given in Section 3.3.6.1.

A full description of each message body is given in the remainder of Section 3.

The API mechanism can use either little-endian byte order, or big-endian byte order. The selection of byte ordering is implementation specific. However, the LSB of the bit stream shall be the LSB of the field defined in the API. For example, the location of a 12-bit bitstream within a 16-bit value would be:

15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
				msb						Bitstream					lsb

This document assumes that the API messages are transferred using a reliable in-order delivery mechanism.

3.1.2 nFAPI Message Format

The general message format of the PHY API is shown in section 2.3.2 of SCF-225 [10], for transport layers other than defined in section 2.3.1 of SCF-225 [10], with the following changes:

- the API mechanism can use either little-endian byte order, or big-endian byte order, as described in section 3.1.1.
- combined messages, as described in section 2.3.2.2 and defined in section 2.3.2.4 of SCF-225 [10] are not supported.
  - **Note**: The absence of combined messages is addressed by the additional param (Table 3-35) and config (Table 3-66) Delay Management TLVs in section 3.3.
- only P7 messages designated as dedicated in SCF-225 [10] are supported, as documented in sections 3.4.12, 3.4.13, and 3.4.14, for PHYs indicating Delay Management capability.
- a 32-byte value shall precede the nFAPI header from section 2.3.2 of SCF-225 [10], with the following value to indicate virtual function scope, as well as the message payloads carried in the nFAPI PDU, as described in Table 3-4:

Description	Type	Subfield
nFAPI adaptation dword (32-bit)	uint8_t	Value: 0: FAPI configuration messages (P5 or P19-C) 1: FAPI slot-based messages (P7 or P19-S)
	uint8_t	Virtual Function ID
	uint16_t	Reserved
nFAPI PDU, as in SCF-225		

Table 3-4 Adaptation of nFAPI structure

3.1.3 Padding

L1 may indicate support for particular padding mechanisms in *MessageEncodingSupport* TLV. If so supported, the following mechanisms are supported:

1. Legacy padding (*Message Encoding Support* = 0): padding of FAPI fields messages is not specified.

2. No padding (*Message Encoding Support* = 1): FAPI fields and messages are concatenated in their order of appearance, without any padding.
3. Aligned padding (*Message Encoding Support* = 2): FAPI messages following the padding rules below:

    a. Each scalar parameter is start-aligned to its size (e.g. 16-bit words start on 16-bit boundaries)
    b. Each parameter that is a scalar array is start-aligned and size-aligned to a 32-bit boundary
    c. Minimal necessary padding to ensure alignment is added before the parameter.
    d. Each structure, PDU, message, and non-scalar loop iteration starts and ends on a 32-bit boundary
    e. Alignment is considered with respect to the start of the FAPI message header; the message headers themselves (or fields thereof) are not subject to padding.

3.1.4 Nomenclature

The PHY API messages follow a standard naming convention where:

- All `.request` messages are sent from the L2/L3 software to the PHY.
- All `.response` messages are sent from the PHY to the L2/L3 software. These are sent in response to a .request.
- All `.indication` messages are sent from the PHY to the L2/L3 software. These are sent asynchronously.
- All message names are capitalized. Underscores may be used.

Starting FAPIv6, FAPI field names shall follow the nomenclature rules for ASN.1 field names in 3GPP TS 38.331 [6], Section A.3.1.2, including the version tag. Version tags are only added for fields introduced or significantly modified since FAPIv3.

3.1.5 Backward Compatibility

FAPI Messages are backward compatible. This means that a parser of FAPI messages implementing SCF-222 for any version 6 and above shall be able to parse FAPI messages generated according to any version 6 and above. If the parser FAPI version is lower than the generator FAPI version, the parser is only expected to interpret parameters for its FAPI version.

3.1.5.1 Extensions

- A consecutive set of parameters preceded by the length of the set.
- The length is a uint16_t, representing the length of the parameter set, in bytes, including any padding. The extension length counts from the beginning of the extension, until the last byte of the extension, including any padding.

    - If the sender chooses to send extension fields beyond the negotiated protocol version (see section 2.1.8), the extension length will account for these fields too.
    - If the negotiated protocol version is:

        - Greater than FAPIv6: A receiver shall be assumed to ignore fields beyond the negotiated FAPI version.

- FAPIv6: receivers are not expected to neither accept nor ignore fields beyond FAPIv6.

- Note: a sender may choose to refrain from send fields beyond the negotiated FAPI version, or may choose to send fields without any expectation that they will influence the receiver behavior.

In this release, extensions can be applied to PDUs or Messages.

3.1.5.2 Extending a PDU or Message

Extending a PDU or Message that does not have a backward compatible extension:

A extension is added to end of the PDU or message. Any newly introduce fields for the PDU or message are appended in this newly-added extension.

The editorial process of extending a legacy Message or PDU is illustrated in Table 3-7, with special colouring to highlight the addition of an extension and a new field.

Field	Type	Description
legacyFieldFirst-vOld	XX	First legacy field of Message or PDU
…	…	…
legacyFieldLast-vOld	XX	Last legacy field of Message or PDU
Extension	structure	See Sample Extension in Table 3-6

Table 3-5 Sample Legacy Message or PDU with newly added extension in a current FAPI Release

Field	**Type**	**Description**
Backward compatible extension to Message or PDU from Sample Table 3-5		
extensionLength-vNew	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.
New Fields in [current FAPI Release]		
newField-vNew	uint8_t	New field in current FAPI Release

Table 3-6 Sample Message or PDU extension in a current FAPI Release

Table 3-7 Example for extending a legacy Message or PDU that did not have an extension

Extending a PDU or Message that already has a backward compatible extension:

Any newly introduce fields for the PDU or message are appended to the existing extension.

The editorial process of extending a legacy Message or PDU is illustrated in Table 3-10, with special colouring to highlight the addition a field to an existing extension.

NOTES:

- this situation occurs when an extension was introduced in a previous FAPI release.

- There is no need to add a new extension, since any receiver of a previous release >= 7 will know to skip from the last parsed field of that release to the end of the extension, based on the extension length.

Field	Type	Description
legacyFieldFirst-vOld	XX	First legacy field of Message or PDU
…	…	…
legacyFieldLast-vOld	XX	Last legacy field of Message or PDU
extension	structure	See Sample Extension in Table 3-9

Table 3-8 Sample Legacy Message or PDU with existing extension from a previous FAPI release

Field	Type	Description
Backward compatible extension to Message or PDU from Sample Table 3-8		
extensionLength-vOld	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.
Existing Fields in [a previous FAPI Release]		
legacyFieldExt-vOld	uint8_t	Legacy field in extension of Message or PDU
…	…	…
New Fields in [current FAPI Release]		
newField-vNew	uint8_t	New field in current FAPI Release

Table 3-9 Sample Existing Message or PDU Extension, with a new field in the current release

Table 3-10 Example for adding a field to an existing extension of a Message or PDU

3.1.6 Upper Layer Payload Transport

Some upper layer payloads (e.g. MAC TBS, DCIs, UCIs) are transferred transparently in message fields or PDUs of this specification. This session describes how these upper loads are represented in this specifications.

Upper layer payloads are represented as bitstreams. Some such payloads may be an octet multiple (e.g. MAC PDUs, as described in in 3GPP TS 38.321[20], and 38.212 [3]), some may not (e.g. DCI payloads, as described in 3GPP TS 38.212 [3]).

Upper Layer Payload Bitstream

Prior to encapsulation into a FAPI field or PDU: the lowest-index bit (here labelled: a_0) of an upper payload bitstream subject to this section is the most significant one ***of the payload***, sometimes also described as "leftmost" (e.g. 3GPP TS 38.331 [6], 3GPP TS 38.212 [3]).

Signalling of an upper-layer payload in FAPI word arrays is based on one of the following bit mapping rules:

- bit *significance* mapping rule, as in section 3.1.6.1, or
- bit *position* mapping rule, as in section 3.1.6.2

The precise rule to follow is described for each field and generally falls in three categories:

- unconditional bit *significance* rule – typically applies to MAC TBS, DCI and UCI payloads.
- conditional on PARAM TLV 0x0177 – typically applies to (p)rb-Bitmap designating PDSCH or PUSCH type0 allocations, or MU-MIMO assignments.
- conditional on PARAM TLV 0x0178 – typically applies to payloads extracted from or emulating from RRC fields (e.g. rate-matching configurations, CORESET frequencyDomainAllocation).

3.1.6.1 Payload mapping by bit significance

This section describes the bit significance mapping rule and illustrates it Figure 3-1. A 12-bit upper layer payload bitstream is illustrated on the top of Figure 3-1.

For signalling of an upper layer payload in a FAPI word array, according to the bit *significance* mapping rule the following steps are applied:

1. padding is appended after the least significant bit, up to the word boundary W (e.g. word = octet, so W = 8 in Figure 3-1)

    - e.g. padding is added after the least significant payload bit a11, in Figure 3-1.
    - Note: this padding rule may be skipped if PHY signals padding capability as 'implementation',

2. The payload of consecutive is split into word sized segments (e.g. octets, in Figure 3-1), so that

    - segment #n contains {a_{n*w}, …, $a_{n*w+n-1}$} – the last segment contains padding bits, if any;
    - e.g. segment #0 contains {a_0, …, a_7}, segment #1 contains {a8, …, a11, *pad*, *pad*, *pad*, *pad*} in Figure 3-1.

3. Segment #n is signalled in the FAPI word array at position #n, with the segment bits mapped according to their *relative significance* in the payload.

    - for the last segment: padding bits, if any, are assumed to be of lower significance than the least significant payload bit;
    - e.g. segment #0 signals a_0 in the FAPI msb(it) position (i.e. #7), per section 3.1.1 => an unsigned integer representation of $\sum_{k=0}^{7} 2^{7-k}\, a_k$, in Figure 3-1;
    - e.g. segment #1 signals the unsigned integer value $a_8*128 + a_9*64 + a_{10}*32 + a_{11}*16$, if zero-padding is used, in Figure 3-1.

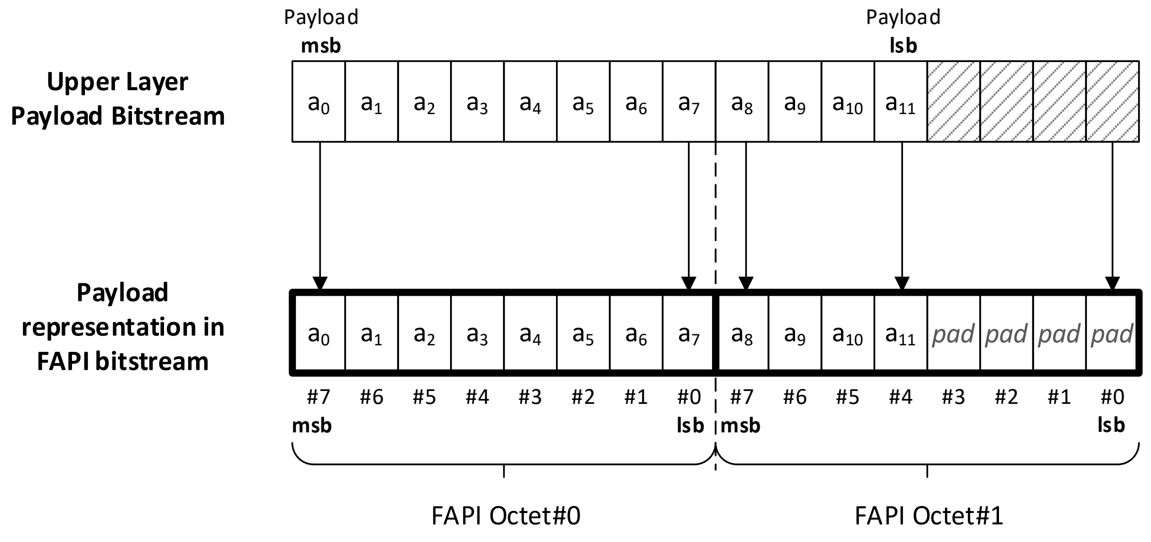


Figure 3-1 Upper layer payload: bit significance mapping rule for unsigned octet string

An example for dword mapping is illustrated in Figure 3-2 for a 16-bit payload, implementing the bit significance mapping rule in this section.

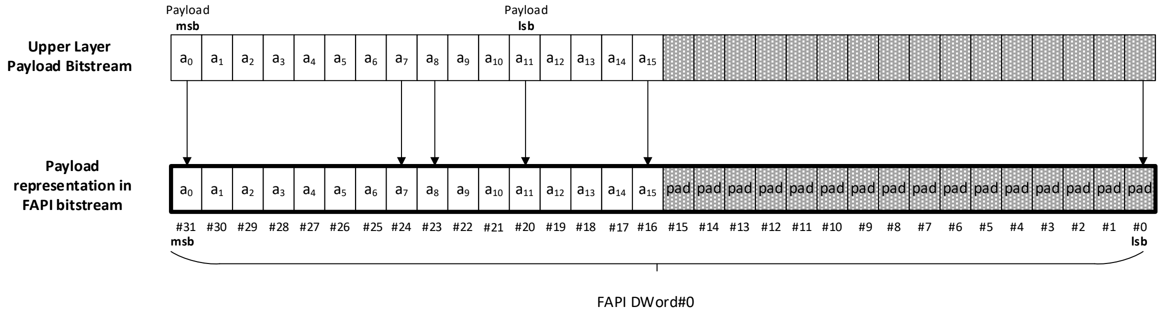


Figure 3-2 Upper layer payload: bit significance mapping rule for dword

3.1.6.2 Payload mapping by bit position

This section describes the bit significance mapping rule and illustrates it Figure 3-3. A 12-bit upper layer payload bitstream is illustrated on the top of Figure 3-3. The only difference from section 3.1.6.1 is in step 3 below:

1. padding to word boundary, as in step 1 of section 3.1.6.1
   - this is illustrated in for an octet string in Figure 3-3;
   - NOTE 1: since the leftmost bit a_0 is in the most payload significant, this step can be described as appending padding after the highest indexed payload position bit (e.g. after a_{11}, in Figure 3-3).
   - NOTE 2: this padding rule may be skipped if PHY signals padding capability as 'implementation',
2. the payload of consecutive is split into word sized segments, as in step 2 of section 3.1.6.1
   - this is illustrated in for an octet string in Figure 3-3.

3. segment #n is signalled in the FAPI word array at position #n, with the segment bits mapped according to their *relative position* in the payload:
   - for the last segment: padding bits, if any, are assumed to be of higher position than the highest position payload bit;
   - e.g. segment #0 signals a_0 in FAPI bit position #0, per section 3.1.1 => an unsigned integer representation of $\sum_{k=0}^{7} 2^k a_k$, in Figure 3-3;
   - e.g. segment #1 signals the unsigned integer value $a_8 + a_9*2 + a_{10}*4 + a_{11}*8$, if zero-padding is used, in Figure 3-3.

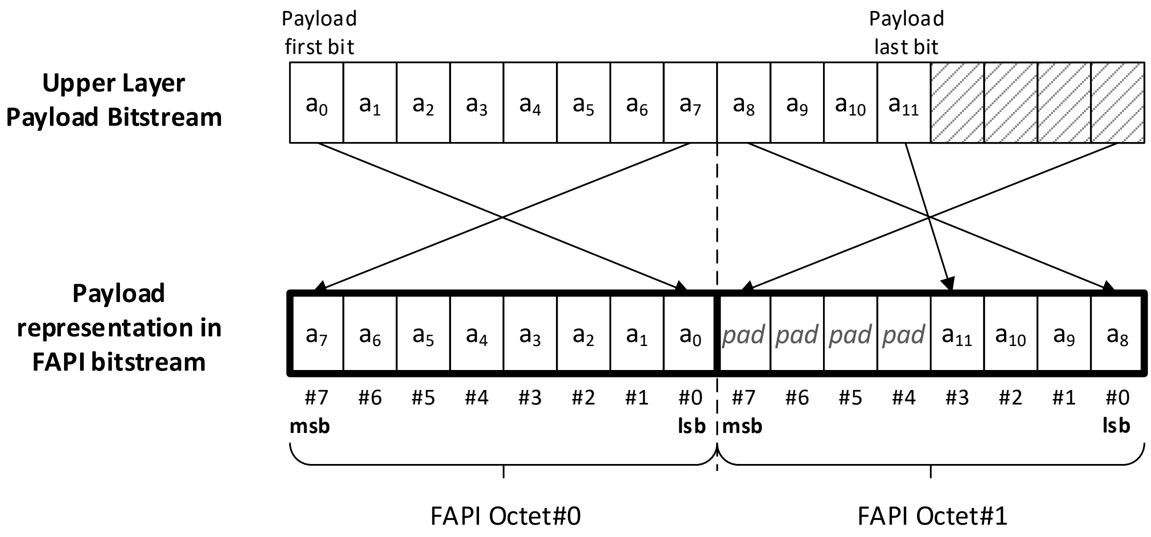


Figure 3-3 Upper layer payload: bit position mapping rule for unsigned octet string

3.2 Message Types

The messages in the 5G PHY API are given in Table 3-11.

Message	Value	Message Body Definition
`PARAM.request`	0x00	See Section 03.3.1.1
`PARAM.response`	0x01	See Section 3.3.1.2
`CONFIG.request`	0x02	See Section 3.3.2.1
`CONFIG.response`	0x03	See Section 3.3.2.2
`START.request`	0x04	See Section 3.3.4.1
`START.response`	See section 2.3.2.7 of SCF-225 [10] Note: 0x108, as of nFAPIv2	See Section 3.3.4.3
`STOP.request`	0x05	See Section 3.3.5.1
`STOP.indication`	0x06	See Section 3.3.5.2
`ERROR.indication`	0x07	See Section 3.3.6.1
`RESET.request`	0x08	See Section 3.3.8.1
`RESET.indication`	0x09	See Section 3.3.8.2
`CCONTEXT_READY.indication`	See section 2.3.2.7 of SCF-225 [10] Note: 0x109 as of nFAPIv3, for PNF_READY.indication	See Section 3.3.8.4
`CONNECTIVITY.indication`	0x0a	See Section 3.3.6.3

Message	Value	Message Body Definition
RESERVED	0x0b-0x7f	
`DL_TTI.request`	0x80	See Section 3.4.2
`DL_TTI.response`	0x8a	See Section 3.4.2b
`UL_TTI.request`	0x81	See Section 3.4.3
`SLOT.indication`	0x82	See Section 3.4.1
`UL_DCI.request`	0x83	See Section 3.4.4
`TX_DATA.request`	0x84	See Section 3.4.6
`RX_Data.indication`	0x85	See Section 3.4.7
`CRC.indication`	0x86	See Section 3.4.8
`UCI.indication`	0x87	See Section 3.4.9
`SRS.indication`	0x88	See Section 3.4.10
`RACH.indication`	0x89	See Section 3.4.11
`TIMING.indication`	0x8b	See Section 3.4.1a Note: 0x8a is taken by DL_TTI.response
RESERVED	0x8c-0xff	
`DL Node Sync`	See section 2.3.2.7 of SCF-225 [10] Note: 0x180, as of nFAPIv2	See Section 3.4.12
`UL Node Sync`	See section 2.3.2.7 of SCF-225 [10] Note: 0x181, as of nFAPIv2	See Section 3.4.13
`Timing Info`	See section 2.3.2.7 of SCF-225 [10] Note: 0x182, as of nFAPIv2	See Section 3.4.14
RESERVED for Vendor Extension Messages	2.3.2.7 of SCF-225 [10] Note: 0x0300-0x03FF, as of nFAPIv2	See Section 2.3.2.10 of SCF-225 [10]

Table 3-11 PHY API message types

3.3 Configuration Messages

The configuration messages are used by the L2/L3 software to control and configure the PHY.

3.3.1 PARAM

The PARAM message exchange was described in section 2.1.1.1

3.3.1.1 PARAM.request

This message can be sent by the L2/L3 when the PHY is in the IDLE state and, optionally, the CONFIGURED state. If it is sent when the PHY is in the RUNNING state, a MSG_INVALID_STATE error is returned in `PARAM.response`. If L2/L3 does not signal any TLVs, the message length in the generic header = 0.

The `PARAM.request` message is given in Table 3-12. From this table it can be seen that `PARAM.request` contains a list of optional TLVs with L2/L3 signalling.

Field	Type	Description
TLVs	Variable	See Table 3-16 PARAM.request TLVs

Table 3-12 `PARAM.request` message body

3.3.1.2 PARAM.response

The `PARAM.response` message is given in Table 3-13. From this table it can be seen that `PARAM.response` contains a list of TLVs providing information about the PHY. When the PHY is in the IDLE state, this information relates to the PHY's overall capability. When the PHY is in the CONFIGURED state this information relates to the current configuration.

The full list of PARAM TLVs is given in Section 3.3.1.4, with the following requirements on the PHY in terms of reporting:

- There is no requirement for every TLV to be reported
- There is no requirement for L1 to provide the TLVs in the order specified in the Tables

Field	Type	Description
errorCode	uint8_t	See Table 3-14
numberOf-TLVs	uint8_t	Number of TLVs contained in the message body.
tlvs	Variable	See Table 3-17.

Table 3-13 **`PARAM.response` message body**

3.3.1.3 PARAM errors

The error codes that can be returned in `PARAM.response` are given in Table 3-14.

Error code	Description
MSG_OK	Message is OK.
MSG_INVALID_STATE	The PARAM.request was received when the PHY was in the RUNNING state.
MSG_INVALID_PHY_ID	The PHY ID is not defined
MSG_UNINSTANTIATED_PHY	The PHY ID is not instantiated
MSG_INVALID_DFE_Profile	No valid DFE Profile exists for the PHY ID
FEU_NOT_INITIALIZED	DFE or RF chains associated with the PHY are not initialized
DFE PROFILE_NOT_SELECTED	No DFE profile was selected for associating DFE and RF chains with the PHY
MSG_PHY_GROUP_UNAVAILABLE	The requested PHY Group is currently unavailable for the virtual function
MSG_PHY_GROUP_CONSTITUENTS_ALTERED	This Error Code is sent to PARAM.request to the L2/L3 from L1 along with PHY PARAM Response containing the updated list of PHY-IDs associated with the requested PHY-Group (when the number of constituents PHY IDs undergoes a change w.r.t the previously shared number for the same PHY-Group).

Table 3-14 **Error codes for `PARAM.response`**

3.3.1.4 PARAM TLVs

PARAM and CONFIG TLVs are used in the PARAM and CONFIG message exchanges, respectively. For both the PARAM and CONFIG TLVs the TLV format is given in Table 3-15. Each TLV consists of:

- Tag parameter of 2 bytes
- Length parameter of 2 bytes
- Value parameter

The length of the Value parameter ensures the complete TLV is a multiple of 4-bytes (32-bits).

The TLVs for the PARAM message exchange are given in this Section 3.3.1.4 and for the CONFIG message exchange in Section 3.3.2.4.

Type	Description
uint16_t	Tag.
uint16_t	Length (in bytes)
variable	Value

Table 3-15 TLV format

To aid additions of future TLVs in later versions of this API, the highest value tag in the Tables below is 0x017F. In addition, the tag value range 0x0100-0x014F only be used for alignment with nFAPI tag value ranges.

PARAM TLVs used in the `PARAM.request` message are given in Table 3-16.

Tag	**Field**	**Type**	**Description**
0x0037	protocolVersion-v3	uint8_t	Indicates if the highest FAPI Protocol version supported by L2/L3 • 0-2: reserved values • 3 = FAPIv3 • 4 = FAPIv4 • 5 = FAPIv5 • 6 = FAPIv6 • 7 = FAPIv7 8-255: reserved values. Future FAPI release will be signalled from this range.
0x0163	phy-GroupIdentifier-v6	uint8_t	Indicates the PHY Group, for PARAM exchanges scoped to PHY Group Context (see sections 2.1.6 and 2.1.10)

Table 3-16 PARAM.request TLVs

PARAM TLVs used in the `PARAM.response` message are given in Table 3-17.

Category	**Description**
Cell Param	See Table 3-19
Carrier Param	See Table 3-20
PDCCH Param	See Table 3-21
PUCCH Param	See Table 3-22
PDSCH Param	See Table 3-23
PRS Param	See Table 3-25
PUSCH Param	See Table 3-26
MsgA-PUSCH Param	See Table 3-27

Category	Description
PRACH Param	See Table 3-28
MsgA-PRACH Param	See Table 3-29
Measurement Param	See Table 3-30
UCI Param	See Table 3-31
Capability Validity	See Table 3-32
PHY Support	See Table 3-33
PHY/DFE Validity Map	See Table 3-34
Delay Management	See Table 3-35
Rel-16 mTRP support	See Table 3-36
Protocol, User Plane and Encapsulation parameters	See Table 3-37
SRS Capabilities	See Table 3-38
Spatial Multiplexing & MIMO Capabilities	See Table 3-39
Semistatic Precoding and Beamforming Capabilities	See Table 3-40
2D-DFT Capabilities	See Table 3-41

Table 3-17 PARAM.response TLV Lists

Tag	Field	Type	Description
This table contains the top-level configuration parameters relating to the cell.			
0x0001	releaseCapability	uint16_t	Bitmap indicating which release the PHY supports For each bit: 0= not supported 1 = supported Bits: 0 = Release15 1 = Release16
0x0002	phy-State	uint16_t	Indicates the current operational state of the PHY. Value: 0 = IDLE 1 = CONFIGURED 2 = RUNNING
0x0003	skipBlank-DL	uint8_t	Indicates if the PHY requires a `DL_TTI.request` message every DL slot or if the message can be skipped if there is nothing to schedule Value: 0 = not supported 1 = supported
0x0004	skipBlank-UL	uint8_t	Indicates if the PHY requires a `UL_TTI.request` message every UL slot or if the message can be skipped if there is nothing to schedule Value: 0 = not supported 1 = supported
0x0005	numConfig-TLVs-ToReport	uint16_t	Indicates the number of CONFIG TLVs which will be reported. For each CONFIG TLV it is indicated if they are optional or mandatory in each PHY state

<table>
<tr><th>Tag</th><th>Field</th><th>Type</th><th>Description</th></tr>
<tr><td></td><td colspan="3">For numConfig-TLVs-ToReport {</td></tr>
<tr><td></td><td>tag</td><td>uint16_t</td><td>Tag value of CONFIG TLV, see Table 3-45 for possible values</td></tr>
<tr><td></td><td>length</td><td>uint8_t</td><td>Length of value field
Value: 1</td></tr>
<tr><td></td><td>value</td><td>uint8_t</td><td>Provides information of optional and mandatory status for each TLV
Value:
0 = Idle state only, optional
1 = Idle state only, mandatory
2 = Idle and Configured states, optional
3 = Idle state mandatory, Configured state optional
4 = Idle, Configured and Running states optional
5 = Idle state mandatory, Configured and Running states optional</td></tr>
<tr><td></td><td colspan="3">}</td></tr>
<tr><td>0x0038</td><td>powerProfilesSupported-v3</td><td>uint8_t</td><td>Supported Power Profiles

Bit Location:
0[LSB] = ProfileNR (same as in FAPIv2) – signals offset are based on standard 3GPP NR parameters.
1 = ProfileSSS (All RS, except possibly PTRS, are offset from SSS)
2-7: reserved, set to 0
Bit values:
0 = Not supported
1 = Supported
Default: 0000 0001b</td></tr>
<tr><td></td><td>numSignals-v3</td><td>uint8_t</td><td>Number of signals
Value: 9 (in FAPIv5)</td></tr>
<tr><td></td><td colspan="2">For signalIndex = 0 to numSignals – 1</td><td>[Signal]’s power range with respect to main RS, when supported, each index corresponds to a signal and its reference RS option(s), per the table below.
Except for the SSS and PRS signals, the ReferenceRS entries are only valid for ProfileSSS, shall be set to all-zero if ProfileSSS is not supported.

<table><tr><th>Idx</th><th>Signal</th><th>RS Index[Notes]: Reference RS</th></tr><tr><td>0</td><td>PDCCH-DMRS</td><td>0: SSS</td></tr><tr><td>1</td><td>PDCCH-Data</td><td>0: SSS
1: PDCCH-DMRS</td></tr><tr><td>2</td><td>PDSCH-DMRS</td><td>0: SSS</td></tr></table></td></tr>
</table>

Tag	Field		Type	Description
				3 | PDSCH-Data | 0: SSS or 1: PDSCH-DMRS 4 | PDSCH-PTRS | 0: SSS or 1: PDSCH-DMRS 5 | CSI-RS | 0: SSS 1: SSS 6 | PSS | 0: SSS 1: SSS 7 | SSS | 0: unscaled SSS 1: SCF-222 v2.0 8 | PRS | 0: SSS **Table 3-18 Signal and Reference RS** Notes: 1. Power offset w.r.t. RS Index 0 is signalled as a value in DB. Power offset w.r.t. RS Index 1 – if allowed and supported – is signalled as a set of look-up table indices 2. e.g. for CSI-RS, the same reference signal (SSS) is indexed twice, the difference being whether the power offset is an explicit value (*csiRsPowerOffsetProfileSSS*) in dB or index (*powerControlOffsetSSProfileNR*) representing one of the values in 3GPP TS 38.214 [5], sec 5.2.2.3.1
		reference-RS-v3	uint16_t	Bit indexes defined per the Reference RS index in the Table 3-18, all other bits shall be set to 0. Valid range: • 0: not supported • 1: supported. If ProfileSSS is supported, at least one bit is set to 1. If RS index 0 is supported, the supported power offset range signalled via [PowerOffsetMin, PowerOffsetMax] with respect to SSS. If RS index 1 is supported, L1 sets the signal[signalIndex] via internal look-up tables, from 3GPP specifications, as described in section 2.2.5.5. As an exception, for index 7 (SSS), RS index 1 refers to the SSS baseband power configuration from SCF-222 v2.0
		powerOffsetMin-v3	int16_t	Value Range for signalIndex ≠ 7: • -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps • -32768: Reference RS 0 not supported. Value Range for signalIndex=7 (SSS):

Tag	Field	Type	Description
			• -11000 -> 12000 representing -110.0 dB to 120.0 dB in 0.01dB, if ssPbchBlockPowerScaling-Unit-Choice = dB • -11000 -> 0 representing -110.0 dBFS to 0.0 dBFS in 0.01dB, if ssPbchBlockPowerScaling-Unit-Choice = dBFS -32768: Reference RS 0 not supported (i.e. fallback to FAPIv2 SSS baseband scaling)
	powerOffsetMax-v3	int16_t	Same as PowerOffsetMin, with PowerOffsetMin ≤ PowerOffsetMax
0x0058	ss-PBCH-BlockPowerScalingUnitChoice-v4	uint8_t	Signals unit PHY assumes for ssPbchBlockPowerScaling: Value: 0 = 'dB' (default from FAPIv3) 1 = 'dBFS' When PHY includes OFH, dBFS values are computed w.r.t. the FS0 reference point given in section 8.1.3.2 of [30]
0x0039	maxNumPDUs-In DL-TTI-v3	uint16_t	Maximum number of PDU that L1 can process in `DL_TTI.request`.
0x003A	maxNumPDUs-In UL-TTI-v3	uint16_t	Maximum number of PDU that L1 can process in `UL_TTI.request`.
0x003B	maxNumPDUs-In UL-DCI	uint16_t	Maximum number of PDU that L1 can process in `UL_DCI.request`.
0x0164	phy-IncludesSplit7-v6	uint8_t	Signals that PHY includes Split 7 support, per [30]

Table 3-19 Cell and PHY parameters

Tag	Field	Type	Description
This table contains the parameters relating to carrier configuration.			
0x0006	cyclicPrefix	uint8_t	Bitmap indicating support. For each bit: 0= not supported 1 = supported Bits: 0 = Normal 1 = Extended
0x0007	subcarrierSpacings-DL	uint8_t	Bitmap indicating support. For each bit: 0= not supported 1 = supported Bits: 0 = 15KHz 1 = 30KHz 2 = 60KHz 3 = 120KHz 4 = 240KHz

Tag	Field	Type	Description
0x0059	subcarrierSpacings-SSB-v4	uint8_t	Bitmap indicating SSB numerology support. If this capability not signalled, SSB numerology capability is signalled by supportedSubcarrierSpacingsDl: For each bit: 0= not supported 1 = supported Bits: 0 = 15KHz 1 = 30KHz 3 = 120KHz 4 = 240KHz All other bit positions (#2, #6, #7) should be set to 0, in this release
0x0008	supportedBandwidth-DL	uint16_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = 5MHz 1 = 10MHz 2 = 15MHz 3 = 20MHz 4 = 25MHz 5 = 40MHz 6 = 50MHz 7 = 60MHz 8 = 70MHz 9 = 80MHz 10 = 90MHz 11 = 100MHz 12 = 200MHz 13 = 400MHz
0x0009	subcarrierSpacings-UL	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = 15KHz 1 = 30KHz 2 = 60KHz 3 = 120KHz
0x000A	supportedBandwidth-UL	uint16_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = 5MHz 1 = 10MHz

Tag	Field	Type	Description
			2 = 15MHz 3 = 20MHz 4 = 25MHz 5 = 40MHz 6 = 50MHz 7 = 60MHz 8 = 70MHz 9 = 80MHz 10 = 90MHz 11 = 100MHz 12 = 200MHz 13 = 400MHz
0x003C	ss-PBCH-MultipleCarriersInABand-v3	uint8_t	0 = disabled 1 = enabled
0x003D	multipleCells-SS-PBCH-InACarrier-v3	uint8_t	Indicates that multiple cells will be supported in a single carrier 0 = disabled 1 = enabled
0x005A	meaningOfCarrierFreq-v4	uint8_t	Indicates the PHY interpretation of dl-Frequency/ul-Frequency parameters: 0 = Point-A [default] 1 = RF centre frequency f0.
0x0175	frequencyShift7p5khz-v7	uint8_t	Indicates support for 7.5KHz frequency shift (in baseband). Value: 0 = not supported 1 = supported
0x0176	dynamicConfigSupport-v7	uint8_t	Signals whether PHY CONFIG.request is possible when another PHY in the same scope is RUNNING. Applies to CONFIG TLVs that cannot be admitted in RUNNING state. Scope: Common or PHY Group Context. Values: • 1: Supported • 0: Not Supported
0x0177	rb-BitMapingMethod-v7	uint8_t	Bit mapping method applied to PDSCH, PUSCH and MU-MIMO RB allocation bitmaps, per section 3.1.6. Value: * 0 = bit-position mapping (i.e. lsb(it) of byte 0 represents VRB0) * 1 = bit-significance mapping (i.e. msb(it) of byte 0 represents VRB0) Bit significance mapping is described in section 3.1.6.1 and bit position mapping is described in section 3.1.6.2.

Tag	Field	Type	Description
			The payload subject to mapping is $a_0 \ldots a_{maxVRBs-1}$, where $a_n = 1$ to indicate VRB#n allocation and $a_n = 0$ otherwise. Bit significance and bit position n FAPI payloads are described in section 3.1.1 NOTE: applies to rb-Bitmap fields in PDSCH and PUSCH PDUs, as well as to prb-Bitmap fields for MU-MIMO Groups in DL_TTI.request and UL_TTI.request.
0x0178	rrc-FieldBitMaping Method-v7	uint8_t	Bit mapping method applied certain RRC-originated payload bitmaps (c.f. 3GPP TS 38.331 [6]), per section 3.1.6. Value: * 0 = bit-position mapping i.e. lsb(it) of byte 0 represents VRB0) * 1 = bit-significance mapping (i.e. msb(it) of byte 0 represents VRB0) Bit significance mapping is described in section 3.1.6.1 and bit position mapping is described in section 3.1.6.2. The payload subject to mapping is $a_0 \ldots a_{max-1}$, where a_0 is the leftmost and most significant bit, as described for relevant payloads in in 3GPP TS 38.331 [6]. Bit significance and bit position n FAPI payloads are described in section 3.1.1 Note: applies to * frequencyDomainAllocation field in CORESET RM or PDCCH PDU * freqDomain-RB and rb-Symbol fields in prb-Symbol-RM in P5 Config or DL_TTI.request or PDSCH PDU.

Table 3-20 Carrier parameters

Tag	Field	Type	Description
This table contains the configuration parameters relating PDCCH configuration.			
0x000B	cce-MappingType	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Interleaved 1 = Non-interleaved

Tag	Field	Type	Description
0x000C	coreset-OutsideFirst3-OFDM-SymsOfSlot	uint8_t	0 = not supported 1 = supported
0x000D	precoderGranularity-CORESET	uint8_t	0 = not supported 1 = supported
0x000E	pdcch-MU-MIMO	uint8_t	0 = not supported 1 = supported
0x000F	pdcch-PrecoderCycling	uint8_t	0 = not supported 1 = supported
0x0010	max-PDCCHs-PerSlot	uint8_t	Value 1->255

Table 3-21 PDCCH parameters

Tag	Field	Type	Description
This table contains the configuration parameters relating PUCCH configuration.			
0x0011	pucch-Formats	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Format 0 1 = Format 1 2 = Format 2 3 = Format 3 4 = Format 4
0x0012	max-PUCCHs-PerSlot	uint8_t	Value 1->255
0x003E	pucch-GroupAndSequenceHopping-v3	uint8_t	Support for signalling mechanism of PUCCH Group & Sequency hopping parameters, per UL BWP ID 0 = dynamic only (explicit parameters in P7) 1 = semi-static only (per linked in P5 to UL BWP ID, with only UL BWP ID signalled in P7 PUCCH PDU) 2 = both dynamic and semi-static
0x003F	maxNum-UL-BWP-Ids-v3	uint8_t	Maximum number of UL BWP IDs supported by PHY, for linking with semi-static PUCCH parameters (see tag 0x003E). 0 = corresponds to value 0 in tag 0x003E > 0: maximum number of semi-static UL BWP IDs configurable to PHY (note that this only relates to semi-static signalling of PUCCH parameters)
0x0040	pucch-Aggregation-v3	uint8_t	Bitmap indicating support for PUCCH aggregation (as signalled by the multiSlotTxIndicator parameter). For each bit: 0= not supported 1 = supported Bits:

Tag	Field	Type	Description
			0 = pucch-Repetition over multiple slots of PUCCH format 1 or 3 or 4

Table 3-22 PUCCH parameters

Tag	Field	Type	Description
This table contains the configuration parameters relating PDSCH configuration.			
0x0013	pdsch-MappingType	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Type A 1 = Type B
0x0014	pdsch-AllocationTypes	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Type 0 1 = Type 1
0x0015	pdsch-VRB-To-PRB-Mapping	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Non-interleaved 1 = Interleaved
0x0016	pdsch-CBG	uint8_t	0 = not supported 1 = supported with segmentation in L2 (if so, then L1 shall be capable of reporting CRC via `DL_TTI.response`) 2 = supported, with segmentation in L1 3 = support with either type of segmentation
0x0017	pdsch-DMRS-ConfigTypes	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Type 1 1 = Type 2
0x0018	pdsch-DMRS-MaxLength	uint8_t	0 = Length 1 (i.e. only single symbol DMRS is supported) 1 = Length 2 (this includes support for double- and single-symbol DMRS)
0x0019	pdsch-DMRS-AdditionalPos	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits:

Tag	Field	Type	Description
			For single-symbol DMRS: 0 = pos0 (1+0) 1 = pos1 (1+1) 2 = pos2 (1+1+1) 3 = pos3 (1+1+1+1) For double-symbol DMRS : 4 = pos0 (1+0) 5 = pos1 (1+1)
0x001A	max-PDSCH-TBs-PerSlot	uint8_t	Value: 1->255 See also TLV 0x0165
0x0165	max-PDSCH-TBs-PerSlot-v6	uint16_t	Value: 1->65535 If L2/L3 uses this TLV, it shall ignore PARAM TLV 0x001A, if present
0x001B	maxNumber-MIMO-Layers-PDSCH-v6	uint8_t	Value: 1->8
0x001C	maxModulationOrder-DL	uint8_t	Value: 0 = QPSK 1 = 16 QAM 2 = 64 QAM 3 = 256 QAM
0x001D	max-MU-MIMO-UsersDL	uint8_t	Value: 1 ->255 Note: 1 indicates no DL MU-MIMO support. This TLV refers to multiple users with layers uniquely identified by DMRS and is fully independent of spatial multiplexing TLVs in Table 3-39.
0x001E	pdsch-DataIn-DMRS-Symbols	uint8_t	0 = not supported 1 = supported
0x001F	preemptionSupport	uint8_t	0 = not supported 1 = supported
0x0020	pdsch-NonSlotSupport	uint8_t	0 = not supported 1 = supported
0x0041	ssb-RateMatch-v3	uint8_t	Rate match via SSB configuration or PDU index reference in PDSCH PDU 0 = not supported 1 = supported via SSB configuration reference 2 = supported via SSB PDU index reference 3 = supported via either configuration or PDU index reference
0x0042	rm-PatternTypeAndMethod-v3	uint16_t	Bitmap indicating support for rate match pattern types and methods: For each bit position, value is 0: not supported 1: supported Bit positions

Tag	Field	Type	Description
			0: scs 15 kHz PRB-symbol bitmap 1: scs 30 kHz PRB-symbol bitmap 2: scs NCP 60 kHz PRB-symb bitmap 3: scs ECP 60 kHz PRB-symb bitmap 4: scs 120 kHz PRB-symbol bitmap 5: by reference in PDSCH PDU, to P5-configured PRB-symbol configuration 6: by value, direct signalling of PRB-symbol patterns in PDSCH PDU. Shall be 0 in this release. 7: CORESET-based @ scs 15 kHz 8: CORESET-based @ scs 30 kHz 9: CORESET-based @ scs NCP 60 kHz 10: CORESET-based @ scs ECP 60 kHz 11: CORESET-based @ scs 120 kHz 12: by reference in PDSCH PDU, to PRB-symbol patterns signalled at the `DL_TTI.request` top level13: by reference in PDSCH PDU, to P5-configured CORESET-based configuration 14: by reference in PDSCH PDU, to CORESET-based patterns signalled at the `DL_TTI.request` top level 15: <reserved> set to 0 in this release. No other method is supported in this release.
0x0043	pdcch-RateMatch-v3	uint8_t	Refers to ability to signal PDCCH rate match resources in PDSCH PDU. 0 = not supported 1 = supported
0x0044	num-RM-Pattern-LTE-CRS-PerSlot-v3	uint8_t	0: no support for LTE-CRS rate matching > 0: max # patterns supported per slot
0x0045	num-RM-Pattern-LTE-CRS-In-PHY-v3	uint8_t	0: no support for LTE-CRS rate matching > 0: max # patterns supported in this PHY
0x0166	num-RM-PatternCORESET-In-PHY-v6	uint8_t	0: no support for Coreset rate matching > 0: max # patterns supported in this PHY
0x0167	num-RM-Pattern-PRB-SymbIn-PHY-v6	uint8_t	0: no support for PrbSymb rate matching > 0: max # patterns supported in this PHY
0x005B	lte-CRS-RM-MBSFN-Derivation-v4	uint8_t	PHY capability to determine MBSFN slots. For each bit position: - 0: method not supported - 1: method supported Bit positions: - [0]: slot classified as MBSFN or non-MBSFN based on MBSFN pattern signalling in MbsfnSubframeConfigList - [1]: slot classified as MBSFN or non-MBSFN based on the P7 lteCrsMbsfnPattern

Tag	Field	Type	Description
0x005C	lte-CRS-RM-Method-v4	uint16_t	Bitmap indicating support for LTE-CRS rate match pattern methods: For each bit position, value is 0: not supported 1: supported Bit positions 0: by reference in PDSCH PDU, to P5-configured LTE-CRS patterns 1: by reference in PDSCH PDU, to LTE-CRS patterns signalled at the `DL_TTI.request` top level 2-15: <reserved> set to 0 in this release. No other method is supported in this release.
0x0046	csi-RS-RM-v3	uint8_t	Refers to ability to signal CSI-RS rate match PDU indices in PDSCH PDU. 0 = not supported 1 = supported
0x0047	pdsch-TransTypeSupport-v3	uint8_t	0 = cannot interpret pdschTransType in PDSCH PDU 1 = requires a valid pdschTransType in PDSCH PDU 2 = can interpret PDSCH PDU resource allocation both with and without pdschTransType
0x0048	pdsch-MAC-PDU-BitAlignment-v3	uint8_t	Alignment of TLV data L1 expects in `TX_DATA.request` MAC PDU. 1 = 8 bit 2 = 16-bit 3 = 32-bit (default) 4 = 64-bit 5 = 128-bit 6 = 256-bit
0x005D	maxNum-PRB-SymBitmapsPerSlot-v4	uint16_t [5]	maximum number of PRB Symbol Bitmaps that can be referenced (in any PDSCH) per slot, for each numerology. [0] = 15KHz [1] = 30KHz [2] = 60KHz [3] = 120KHz [4] = undefined (set to "not applicable", in this release) Value range for each entry: • 0-1000: number of different rate match patterns per slot, for the numerology. • 65535: not applicable (default)
0x005E	maxNum-CSI-RS-RM-PerSlot-v4	uint16_t [5]	maximum number of CSI-RS that can be referenced (in any PDSCH) per slot, for each numerology.

Tag	Field	Type	Description
			[0]: 15KHz [1]: 30KHz [2]: 60KHz [3]: 120KHz [4]: undefined (set to "not applicable", in this release) Value range for each entry: • 0-1000: number of different rate match patterns per slot, for the numerology. 65535: not applicable (default)
0x0155	maxNum-PRS-PuncturingPerSlot-v5	uint16_t [5]	maximum number of PRS that can be referenced (in any PDSCH) per slot, for each numerology. [0]: 15KHz [1]: 30KHz [2]: 60KHz [3]: 120KHz [4]: undefined (set to "not applicable", in this release) Value range for each entry: • 0-1000: number of different rate puncturing patterns per slot, for the numerology. 65535: not applicable (default)
0x005F	maxNum-SSB-PerPDSCH-Slot-v4	uint8_t	Maximum number of SS/PBCH blocks that can be transmitted in a PDSCH slot. Value: • 1 – 16; • 255: not applicable (default)
0x0060	universalRateMatch-v4	uint8_t	Signals whether L1 expects all rate match mechanism to apply universally to all PDSCH allocations, that is: • If time/frequency resources for one PDSCH PDU A overlap with a time/frequency resource X that is unavailable to another PDSCH PDU B, then resource X shall also be unavailable to PDSCH PDU A • MAC ensures that rate match patterns are associated with PDSCH PDUs according to this universality principle. • MAC signals all LTE_CRS, PrbSymRM and CoresetRM patterns applicable to a slot, `DL_TTI.request` for the slot. Value: 0: universal rate match not required. 1: universal rate match required. 2: universal rate match and puncturing required.

Tag	Field	Type	Description
0x0061	require-PDSCH-SignalingAssociations-v4	uint8_t	Signals whether L1 expects L2 to signal in a PDSCH PDU all rate match patterns and signals relevant to a particular PDSCH PDU. Note: this is particularly relevant when some pattern or structure *X* (e.g. SSB) is fully obscured by another pattern or structure Y (e.g. PrbSymb). When required by L1, L2 shall include both structures X and Y, when PDSCH PDU allocation is mutually exclusive with at least one RE and symbol of X. Value: 0: required 1: not required.

Table 3-23 PDSCH parameters

Tag	Field	Type	Description
This table contains the configuration parameters relating PDSCH configuration.			
0x0179	csi-RS-BWP-Support	uint8_t	Bitmap indicating support for CSI-RS BWP signalling in CSI-RS PDU. Value: - 0: not supported - 1: supported

Table 3-24 PDSCH parameters

Tag	Field	Type	Description
This table contains the configuration parameters relating PRS configuration.			
0x0156	puncturingOf-PRS-Symbols-v5	uint8_t	Indicates whether puncturing of PRS symbols is supported (this is needed to allow SSB to puncture PRS) Values: • 0: not supported • 1: supported
0x0157	prs-PDUs-PerSlot-v5	uint16_t	The absolute number of PRS PDUs per slot. Value: 0 … 65534
0x0158	proportion-PRS-RBs-PerSlot-v5	uint8_t	Maximum number of PRS RBs per slot. Value: 0 … 100

Table 3-25 PRS parameters

Tag	Field	Type	Description
This table contains the configuration parameters relating PUSCH configuration.			
0x0021	uci-MUX-UL-SCH-In-PUSCH	uint8_t	0 = not supported 1 = supported
0x0022	uci-Only-PUSCH	uint8_t	0 = not supported 1 = supported
0x0023	pusch-FrequencyHopping	uint8_t	0 = not supported 1 = supported

Tag	Field	Type	Description
0x0024	pusch-DMRS-ConfigTypes	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Type 1 1 = Type 2
0x0025	pusch-DMRS-MaxLen	uint8_t	0 = Length 1 (i.e. only single symbol DMRS is supported) 1 = Length 2 (this includes support for double- and single-symbol DMRS)
0x0026	pusch-DMRS-AdditionalPos	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: For single-symbol DMRS: 0 = pos0 (1+0) 1 = pos1 (1+1) 2 = pos2 (1+1+1) 3 = pos3 (1+1+1+1) For double-symbol DMRS : 4 = pos0 (1+0) 5 = pos1 (1+1)
0x0027	pusch-CBG	uint8_t	0 = not supported 1 = supported
0x0028	pusch-MappingType	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Type A 1 = Type B
0x0029	pusch-AllocationTypes	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Type 0 1 = Type 1
0x002A	pusch-VRB-To-PRB-Mapping	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Non-interleaved 1 = Interleaved In this FAPI release, the puschVrbToPrbMapping[1] entry shall be set to 0,

Tag	Field	Type	Description
			as interleaved VRB-PRB mapping is not supported in 3GPP NR.
0x002B	pusch-Max-PTRS-Ports	uint8_t	Value = 0,1,2
0x002C	maxPUSCH-TBs-PerSlot	uint8_t	Value 1->255
0x0168	maxPUSCH-TBs-PerSlot-v6	uint16_t	Value 1->65535
0x002D	max-MIMO-Layers NonCB-PUSCH	uint8_t	Value: 0 -> 4 L1 should ensure that maxMIMO-LayersNonCB-PUSCH + maxMIMO-LayersCB-PUSCH > 0
0x0049	Max-MIMO-Layers CB-PUSCH-v3	uint8_t	Value: 0 -> 4 L1 should ensure that maxMIMO-LayersNonCB-PUSCH + maxMIMO-LayersCB-PUSCH > 0
0x002E	maxModulationOrder-UL	uint8_t	Value: 0 = QPSK 1 = 16 QAM 2 = 64 QAM 3 = 256 QAM
0x002F	maxMU-MIMO-Users-UL	uint8_t	Value: 1 ->255 Note: 1 indicates no UL MU-MIMO support. This TLV refers to multiple users with layers uniquely identified by DMRS and is fully independent of spatial multiplexing TLVs in Table 3-39.
0x0030	dfts-OFDM	uint8_t	0 = not supported 1 = supported, with pi/2-PBSK support 2 = supported, without pi/2-BPSK support
0x0031	pusch-AggregationFactor	uint8_t	Bitmap indicating support for different aggregation factors. For each bit: 0= not supported 1 = supported Bits: 0 = aggregation factor 1 1 = aggregation factor 2 2 = aggregation factor 4 3 = aggregation factor 8
0x004A	pusch-LBRM-Support-v3	uint8_t	0 = not supported 1 = supported
0x004B	pusch-TransTypeSupport-v3	uint8_t	0 = cannot interpret puschTransType in PUSCH PDU 1 = requires a valid puschTransType in PUSCH PDU 2 = can interpret PUSCH PDU resource allocation both with and without puschTransType
0x004C	pusch-MAC-PDU-BitAlignment-v3	uint8_t	Alignment of TLV data L1 uses for the MAC PDU in `RX_DATA.indication`. 1 = 8 bit

Tag	Field	Type	Description
			2 = 16-bit 3 = 32-bit (default) 4 = 64-bit 5 = 128-bit 6 = 256-bit
0x0169	srs-TLV-BitAlignment-v6	uint8_t	Alignment of TLV data L1 uses for SRS Reports in `SRS.indication`. 1 = 8 bit 2 = 16-bit 3 = 32-bit (default) 4 = 64-bit 5 = 128-bit 6 = 256-bit

Table 3-26 PUSCH parameters

Tag	Field	Type	Description
This table contains the configuration parameters relating to MsgA-PUSCH configuration.			
0x0062	msgA-Num-PUSCH-ResConfig-v4	uint16_t	Maximum number of supported MsgA PUSCH resource configuration index per cell. Value 0-65534 Value 65535 is reserved and shall not be used in this release.
0x0063	msgA-PUSCH-TransPrecoding-v4	uint8_t	Indicating if PHY support transform precoding for MsgA PUSCH transmission Value 0: not supported Value 1: supported
0x0064	msgA-IntraSlot-PUSCH-FreqHop-v4	uint16_t	Indicating if PHY support intra-slot frequency hopping per PUSCH occasion for MsgA PUSCH payload transmission Value 0: not supported Value 1: supported
0x0065	msgA-Max-PUSCH-FD-Occasions-v4	uint8_t	Indicating supported maximum number of FDMed PUSCH occasions for MsgA PUSCH payload transmission Value: 0 = 1 1 = 2 2 = 4 3 = 8
0x0066	msgA-GuardBand-v4	uint8_t	Indicating whether PHY support MsgA PUSCH transmission with configured guard band or not Value 0: support MsgA PUSCH with no zero RB guard band Value 1: support MsgA PUSCH with one RB guard band
0x0067	msgA-GuardPeriod-v4	uint8_t	Bitmap Indicating whether PHY support MsgA PUSCH transmission with configured guard period Bit 0 represents zero symbol guard period Bit 1 represents one symbol guard period Bit 2 represents two symbol guard period Bit 3 represents three symbol guard period

Tag	Field	Type	Description
			Bit set to 0 means not supported, set to 1 means supported
0x0068	msgA-PUSCH-MappingType-v4	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Type A 1 = Type B
0x0069	msgA-PUSCH-DMRS-MaxLen-v4	uint8_t	Value: 0 = Length 1 1 = Length 1 and Length 2
0x006A	msgA-PUSCH-DMRS-CDM-Group-v4	uint8_t	Bitmap indicating support CDM group for MsgA PUSCH Each bit: 0 means not supported, 1 means supported Bit 0 = CDM group 0 Bit 1 = CDM group 1
0x006B	msgA-PUSCH-DMRS-AdditionalPos-v4	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: For single-symbol DMRS: Bit 0 = pos0 (1+0) Bit 1 = pos1 (1+1) Bit 2 = pos2 (1+1+1) Bit 3 = pos3 (1+1+1+1) For double-symbol DMRS: Bit 4 = pos0 (1+0) Bit 5 = pos1 (1+1)Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = pos0 (1+0) 1 = pos1 (1+1) 2 = pos2 (1+1+1) 3 = pos3 (1+1+1+1) Note: MsgA PUSCH only support Type 1 DM-RS
0x006C	msgA-MaxModulationOrderUL-v4	uint8_t	Value: 0 = QPSK 1 = 16 QAM
0x006D	msgA-Max-PUSCH-PerPRU-v4	uint8_t	Maximum number of PUSCH decodable per PUSCH occasion (including associated DMRS). Value: 1 ->255
0x006E	msgA-MaxNum-CDM-Group-v4	uint8_t	Number of MsgA PUSCH DM-RS CDM group supported Value: 0 -> one CDM group Value: 1 -> two CDM group
0x006F	msgA-Max-Num-DMRS-Port-v4	uint8_t	Number of PUSCH DM-RS ports supported per CDM group Value: 0 -> first DM-RS port per CDM group

Tag	Field	Type	Description
			Value: 1 -> first two DM-RS ports per CDM group Value: 2 -> four DM-RS ports per CDM group
0x0070	msgA-PRACH-To-PUSCH-MappingScheme-v4	uint8_t	Indicate which scheme PHY supported to receive MsgA PUSCH configuration information. Value: 1 -> semi-static configuration via P5 Value: 2 -> dynamic configuration via P7 Value: 3 -> both schemes supported

Table 3-27 MsgA-PUSCH parameters

Tag	Field	Type	Description
This table contains the parameters relating PRACH configuration			
0x0032	prach-LongFormats	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Format 0 1 = Format 1 2 = Format 2 3 = Format 3
0x0033	prach-ShortFormats	uint16_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Format A1 1 = Format A2 2 = Format A3 3 = Format B1 4 = Format B2 5 = Format B3 6 = Format B4 7 = Format C0 8 = Format C2
0x0159	prach-ShortFormatNumerologies-v5	uint16_t	Bitmap indicating numerology support. For each bit: • 0= not supported • 1 = supported Bits positions: [0]: 15KHz [1]: 30KHz [2]: 60KHz [3]: 120KHz
0x0034	prach-RestrictedSets	uint8_t	Value: 0 = not supported 1 = Type A and B supported 2 = Type A supported, only 3 = Type B supported, only

Tag	Field	Type	Description
0x0035	maxPRACH-FD-OccasionsInASlot	uint8_t	Value: 0 = 1 1 = 2 2 = 4 3 = 8
0x004D	maxNum-PRACH-Configurations-v3	uint16_t	Value 0 = no Prach configurations supported (can only be signalled if PHY supports both dedicated and shared configurations. PHY must signalled must signal non-zero value for at least one of dedicated and shared capability corresponding to this TLV) 1 = no support for multi-PRACH configuration > 1 = maximum number of PRACH configurations supported by PHY
0x004E	prach-MultipleCarriersInABand-v3	uint8_t	0 = disabled 1 = enabled
0x0071	Prach-FreqOffsetMeaning-v4	uint8_t	1 = Offset between Point-A and a PRB overlapping with the lowest RE of the PRACH signal 2 = k1 definition in [3GPP TS 38.211 [2], sec 5.3.2] in PRB unites 3 = k1 definition in [3GPP TS 38.211 [2], sec 5.3.2] in RE units
0x0072	maxNum-TD-FD-PRACH-OccasionsPerSlot-v4	uint16_t [5]	Maximum number of time-frequency PRACH occasions per PRACH slot. Entries refer to PRACH slot SCS: • [0]: 15 kHz (FR1) • [1]: 30 kHz (FR1) • [2]: N/A (set to 65535) • [3]: 60 kHz (FR2) • [4]: 120 kHz (FR2) Value: • 1-1024; • 65535: not applicable (default)
0x0073	numFormats-v4	uint16_t	Number of formats included for this TLV Value = 14 in this release Note: - if uint16 entries for the processing rate are insufficient, L1 PHY shall use 0x017A instead, which signals uint32 entries.
	prach-RootProcessingRate-v4	uint16_t [n]	n = numberFormats Z-C root processing rate, per RO format type. Word positions: - [0] – undefined (set to 0) - [1] – Format 0 - [2] – Format 1 - [3] – Format 2 - [4] – Format 3 - [5] – Format A1

Tag	Field	Type	Description
			- [6] – Format A2 - [7] – Format A3 - [8] – Format B1 - [9] – Format B2 - [10] – Format B3 - [11] – Format B4 - [12] – Format C0 - [13] – Format C1 - [>13] Undefined in this release, but may be supported in future releases For the specified format, this capability describes the maximum number of Zadoff-Chu roots that can be processed in a frame, after accounting for receive diversity (e.g. 1 root over N logical antenna ports is counted as N roots). There is no default value for this capability.
0x017A	numFormatsExt-v7	uint16_t	Number of formats included for this TLV Value = 14 in this release Note: - the difference from 0x0073 is that the next array has uint32, instead of uint16 entries
	prach-RootProcessingRateExt-v4	uint32_t [n]	n = numberFormats Z-C root processing rate, per RO format type. Word positions: - [0] – undefined (set to 0) - [1] – Format 0 - [2] – Format 1 - [3] – Format 2 - [4] – Format 3 - [5] – Format A1 - [6] – Format A2 - [7] – Format A3 - [8] – Format B1 - [9] – Format B2 - [10] – Format B3 - [11] – Format B4 - [12] – Format C0 - [13] – Format C1 - [>13] Undefined in this release, but may be supported in future releases For the specified format, this capability describes the maximum number of Zadoff-Chu roots that can be processed in a frame, after accounting for receive diversity (e.g. 1 root over N logical antenna ports is counted as N roots). There is no default value for this capability.
0x0074	numFormats-v4	uint16_t	Number of formats included for this TLV Value = 14 in this release

Tag	Field	Type	Description
	prach-OccasionQueueSize-v4	uint32_t [n]	n = numberFormats The sum of weighted RACH Occasions (RO) in the queue size shall not exceed the Cumulative Weight Threshold, with the weights and threshold signalled, at the following word positions: - [0] – Cumulative Weight Threshold - [1] – Weight for Format 0 - [2] – Weight for Format 1 - [3] – Weight for Format 2 - [4] – Weight for Format 3 - [5] – Weight for Format A1 - [6] – Weight for Format A2 - [7] – Weight for Format A3 - [8] – Weight for Format B1 - [9] – Weight for Format B2 - [10] – Weight for Format B3 - [11] – Weight for Format B4 - [12] – Weight for Format C0 - [13] – Weight for Format C1 - [>13] Undefined in this release, but may be supported in future releases For the specified format, this capability describes the weight to assign to each RO in the queue, after accounting for receive diversity (e.g. 1 RO over N logical antenna ports is counted as N ROs). The sum of all weighed ROs is compared against the Cumulative Weight Threshold to determine the queue size. There is no default value for this capability.
0x017B	prach-OccasionsCombining-v7	uint8_t	Indicate support for identical or different spatial combining across PRACH Occasions for a PRACH configuration reflected in a PRACH slot. For identical combining, L1 PHY expects a single PRACH PDU/slot for any one PRACH configuration. Even if different UL combining is supported, L2/L3 should minimize the number of PDUs to match – as much as possible – the number of UL combining options in the slot. Applied to the case of identical combining for a PRACH configuration, this shall result in a single PRACH PDU for the PRACH configuration in a slot. Values: 0: identical spatial combining only 1: different spatial combining supported

Table 3-28 PRACH parameters

Tag	Field	Type	Description
This table contains the parameters relating to MsgA-PRACH configuration			
0x0075	msgA-MaxNum-PRACH-Configs-v4	uint16_t	Value 0 = no support in this PHY > 0 = maximum number of PRACH configurations supported by PHY
0x0076	msgA-PRACH-LongFormats-v4	uint8_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Format 0 1 = Format 1 2 = Format 2 3 = Format 3
0x0077	msgA-PRACH-ShortFormats-v4	uint16_t	Bitmap indicating support format. For each bit: 0= not supported 1 = supported Bits: 0 = Format A1 1 = Format A2 2 = Format A3 3 = Format B1 4 = Format B2 5 = Format B3 6 = Format B4 7 = Format C0 8 = Format C2
0x0078	msgA-PRACH-RestrictedSets-v4	uint8_t	Value: 0 = not supported 1 = Type A and B supported 2 = Type A supported, only 3 = Type B supported, only
0x0079	msgA-Max-PRACH-FD-OccasionsInASlot-v4	uint8_t	Value: 0 = 1 1 = 2 2 = 4 3 = 8
0x007A	msgA-PRACH-MultipleCarriersInABand-v4	uint8_t	0 = disabled 1 = enabled

Table 3-29 MsgA-PRACH parameters

Tag	Field	Type	Description
This table contains the parameters relating Measurement reporting			
0x0036	rssi-Measurement Support	uint8_t	Indicates if RSSI measurement is supported and what units can be used for reporting. The RSSI reported will be total received power summed across all antennas. RSSI includes all noise and interference contributions. Bitmap indicating support. For each bit: 0= not supported 1 = supported Bits: 0 = RSSI reporting in dBm 1 = RSSI reporting in dBFS
0x007B	rsrp-Measurement Support-v4	uint8_t	Indicates if RSRP measurement is supported. The RSRP reported will be *the linear average over the power contributions (in [W]) of the resource elements of the antenna port(s) that carry reference signals configured for RSRP measurements within the considered measurement frequency bandwidth in the configured reference signal occasions*. [3GPP TS 38.215 [18], section 5.1.2] For dBm unit measurements: • *For frequency range 1, the reference point for the RSRP shall be the antenna connector of the FEU.* [3GPP TS 38.215 [19], section 5.1.2] • *For frequency range 2, RSRP shall be measured based on the combined signal from antenna elements corresponding to a given receiver branch.* [3GPP TS 38.215 [19], section 5.1.2] Bitmap indicating support. For each bit: - 0 = not supported - 1 = supported Bits: - 0 = RSRP reporting in dBm - 1 = RSRP reporting in dBFS
0x015A	preamblePowerUnit-v5	uint8_t	Indicates the unit for the preamblePwr in RACH.indication. For dBm unit measurements: - For frequency range 1, the reference point for the measurement shall be the antenna connector of the FEU.

Tag	Field	Type	Description
			- For frequency range 2, the report shall be measured based on the combined signal from antenna elements corresponding to a given receiver branch. Similar to the regular RSRP report, the preamblePwr report will be *the linear average over the power contributions (in [W]) of the resource elements of the antenna port(s) that carry reference signals configured for RSRP measurements within the considered measurement frequency bandwidth in the configured reference signal occasions*. [3GPP TS 38.215 [18], section 5.1.2] Unlike the regular RSRP report, the preamblePwr report represents the preamble power observation averaged over all PRACH Res, at the PRACH numerology Value: - 0 = preamblePwr reporting in dBm - 1 = preamblePwr reporting in dBFS - 2 = preamblePwr reporting in either dBm or dBFS
0x015B	srs-RSRP-PowerUnit-v5	uint8_t	Indicates the unit for the srsRsrp in SRS.indication. For dBm unit measurements: - For frequency range 1, the reference point for the measurement shall be the antenna connector of the FEU. - For frequency range 2, the report shall be measured based on the combined signal from antenna elements corresponding to a given receiver branch. Value: - 0 = srs RSRP reporting in dBm - 1 = srs RSRP reporting in dBFS - 2 = srs RSRP reporting in either dBm or dBFS

Table 3-30 Measurement parameters

Tag	Field	Type	Description
This table contains the parameters relating to UCI reporting			
0x004F	maxNum-UCI-Maps-v3	uint16	Indicates the maximum number of UCI Part1 to Part 2 maps supported in the UCI Configuration Table 3-60. Value: 0 -> 65535 0 (default): no support for FAPIv3 UCI reporting (i.e.no support for CSI part2 reporting)

Tag	Field	Type	Description
0x007C	uci-ReportFormatPUCCH-Format234-v4	uint8	Indicates how UCI for PUCCH Formats 2, 3, 4 is reported For each both position: 0 = PHY supports; 1 = PHY does not support • Bit #0: UCI report includes separate parameters to for SR, HARQ, CSI Part 1 and CSI Part2 (i.e. PHY will only set bits 0-3 in pduBitmap of Table 3-98) • Bit #1: UCI report includes parameters for uciPart 1 and uciPart2 (i.e. PHY will only set bits 2-3 in pduBitmap of Table 3-98) **Note**: In this release exactly one of bit positions 0 or 1 is set.
0x017C	maxTotalSize-P1-P2-UCI-Maps-v7	uint32_t	That maximum cumulative size of all uci part1—part2 maps that can be signalled in TLV 0x1036 (i.e. maximum value for Sum[2^mapBitWidth] over all loops in that TLV) Value: 1 -> 2^32-1

Table 3-31 UCI parameters

Tag	Field		Type	Description
This table describes the validity scopes of capability TLVs in Table 3-17				
0x0050	numCapabilities-v3		uint16	Indicates the number of capabilities for which validity scope is declared. Capabilities are listed by tag. Each tag appears at most once. Value: 0 -> 65535
	For numCapabilites			
		capabilityTag-v3	uint16_t	A tag for any TLV in Table 3-17, except for tag 0x0050. Tag 0x0050 shall be Valid ANYTIME
		validityScope-v3	uint8	0: Valid ANYTIME 1: Valid after PHY Profile Selection 2: Valid after PHY and DFE Profile Selection Default: Tag 0x0050 shall be 0, regardless whether it is listed or not. All other tags default to 2, if not listed.

Table 3-32 Capability validity scope

Tag	Field	Type	Description
This table define PHY and PHY Profile support parameters			
0x0051	This tag describes the supported PHY Profiles. This TLV is only supported at Common or PHY Group Context. Terminology: 3. interface BB port: a baseband (BB) port label known to both PHY and FEU components. BB ports may be either in the UL or DL direction. 4. local BB port: an index used locally in the PHY to refer to BB ports. If a PHY ID is assigned a set of interface BB ports, part of the PHY Profile Selection step of the PHY Instantiation procedure described in section 2.1.9, the local BB port indices are labelled starting from 0, per direction in increasing number of the BB ports in the same direction: 5. the n-th local DL BB port for the PHY corresponds to the n-th largest interface DL BB port assigned to the PHY. 6. the n-th local UL BB port for the PHY corresponds to the n-th largest interface UL BB port assigned to the PHY.		
	num-PHY-Profiles-v3	uint16_t	Indicates the number of supportable PHY Profiles. Value: 0 -> PHY Profiles feature not supported (this value is disallowed when FAPI is carried over nFAPI, as specified in SCF-225 [10]) 1 -> 65535: number of supported PHY Profiles
	num-DL-BB-Ports-v3	uint16_t	Indicates the number of DL interface BB (baseband) ports. These labels are common to all FEU, per SCF-223 [9], and PHYs. DL interface BB ports are indexed 0 … (numDlBbPorts-1) If 0, the number of actual Dl BB ports for a PHY is determined by the numTxAnt configured to the PHY. Value: 0 -> 65535
	num-UL-BB-Ports-v3	uint16_t	Indicates the number of UL interface BB ports. These labels are common to all FEU, per SCF-223 [9], and PHYs. UL interface BB ports are indexed numDlBbPorts … (numDlBbPorts + numUlBbPorts – 1) If 0, the number of actual Ul BB ports for a PHY is determined by the numTxAnt configured to the PHY. Value: 0 -> 65535
	For num-PHY-Profiles *(PHY profiles IDs defined implicitly, as ordered indices of this loop, starting from 1)*		
	maxNum-PHYs-v3	uint8_t	Indicates the maximum number of PHY IDs supported in this PHY profile

Tag	Field	Type	Description
	For maxNumPhys *(PHY IDs for this Profile defined implicitly, as ordered indices of this loop, starting from 1)*		
	DL BB Ports: the set of DL interface ports for this PHY ID is a union of non-overlapping ranges of DL BB interface ports: {dlPortRangeStart[0], …, dlPortRangeStart[0] + dlPortRangeLen[0] - 1} U {dlPortRangeStart[1], …, dlPortRangeStart[1] + dlPortRangeLen[1] - 1} U … The start/length of the above ranges fall in the range of DL BB interface ports.		
	num-DL-Port Ranges-v3	uint8	nD: Number of DL port ranges for this PHYs
	dl-PortRang eStart-v3	uint16_t [nD]	Start of each range of DL ports.
	dl-PortRang eLen-v3	uint16_t [nD]	Length of each range of DL ports.
	num-UL-Port Ranges-v3	uint8_t	nU: Number of UL port ranges for this PHYs
	ul-PortRang eStart-v3	uint16_t [nU]	Start of each range of UL ports.
	ul-PortRang eLen-v3	uint16_t [nU]	Length of each range of UL ports.
0x0052	timeManagement-v3	uint8_t	Type of time management mechanism: Values: 0 – SFN/SL synchronization 1- Delay Management with Timestamps 2- Delay Management without Timestamps
0x015C	dl-TTI-RX-Window ForSSB-v5	uint8_t	Lists the allowed anchoring of the RxWindow for SSB PDUs received in DL_TTI.request (if PHY supports Delay Management) Value: 1 – SSB Numerology 2 – Highest PDSCH numerology, if lower than the SSB numerology 3 – either.
0x015D	ul-TTI-RX-Window For-PRACH-v5	uint8_t	Lists the allowed anchoring of the RxWindow for short PRACH PDUs received in UL_TTI.request (if PHY supports Delay Management) Value: 1 – short PRACH Numerology 2 – Highest PUSCH numerology, if lower than the short PRACH numerology 3 – either.
0x0053	This TLV is mandatory in this specification version, whenever L2 signals its ProtocolVersion in `PARAM.request`.		
	phy-FAPI-Protocol Version-v3	uint8_t	Indicates if the highest FAPI Protocol version supported by PHY • 0-2: reserved values • 3 = FAPIv3

Tag	Field	Type	Description
			• 4 = FAPIv4 • 5 = FAPIv5 • 6 = FAPIv6 • 7 = FAPIv7 • 8-255: reserved values. Future FAPI release will be signalled from this range.
	phy-FAPI-NegotiatedProtocolVersion-v3	uint8_t	Indicates the version FAPI protocol version that PHY uses, for pairing with the MAC protocol version, if any was provisioned. If addressed to supervisory PHY, the assumption applies to all PHYs under the SoC/PNF control. • 0 = No MAC Protocol provisioned PHY will use the same version as ProtocolVersion. • 1 = FAPIv1 • 2 = FAPIv2 • 3 = FAPIv3 • 4 = FAPIv4 • 5 = FAPIv5 • 6 = FAPIv6 • 7 = FAPIv7 • 8-254: reserved values. Future FAPI release will be signalled from this range. • 255 = No support for MAC's FAPI version. PHY will use the same version as ProtocolVersion.

Tag	Field	Type	Description
0x0055	numEntries-v3	uint8_t	Number of entries in this TLV Value: 7 in this release
	moreThanOneIndicationPerSlot-v3	unit8_t [numEntries]	Support flag for transmission of more than one indication per slot. This TLV is supported at Common Context or PHY Group context. Value, for each entry: • 1: at most one message instance per slot & numerology • 2: may generate more than one message instance per slot & numerology • 3: configurable by L2, via the flag indicationInstancesPerSlot – Defaults to at most one message instance per slot & numerology. • other values are reserved Scope of each entry: [0]: `RX_DATA.indication` [1]: `CRC.indication` [2]: `UCI.indication` [3]: `RACH.indication` [4]: `SRS.indication` [5]: `DL_TTI.response` [6]: `TIMING.indication`
0x0056	numEntries-v3	uint8_t	Number of entries in this TLV Value: 4 in this release
	moreThanOneRequestPerSlot-v3	uint8_t [numEntries]	Support flag for reception of more than one request per slot. This TLV is only supported at Common Context or PHY Group Context. Value, for each entry: • 1: at most one message instance per slot • 2: may receive more than one message instance per slot • 3: configurable by L2, via the flag requestInstancesPerSlot – Defaults to at most one message instance per slot & numerology. • other values are reserved Scope of each entry: [0]: `DL_TTI.request` [1]: `UL_TTI.request` [2]: `UL_DCI.request` [3]: `TX_DATA.request`

Tag	Field	Type	Description
0x015E	connectivityObservation-v5	uint32_t	Support bitmap for observations and scope for Connectivity Indication. Value for each bit entry: • 0: not supported • 1: supported Bit positions: [0]: PHY synchronization observable [1]: FEU synchronization observable [2]: PHY/FEU connectivity state observable. [3]: Observations possible for Common Context [4]: Observations possible for dedicated PHYs
0x015F	p5-Timeout-v5	uint16_t	Timeout for P5 procedures, in milliseconds. Value: 1 … 10,000 ms
0x016A	phys-InGroup-v6: This tag indicates the PHY IDs supported by the PHY Group associated with P5 PARAM API. If a single PHY Group exists, the PHY Group ID may be omitted in the P5 API.		
	first-PHY-Id-v6	uint8_t	The first PHY ID in the PHY Group
	num-PHY-Ids-v6	uint8_t	Number of PHY IDs in the PHY Group. The PHY IDs in the Group are designated as: • First PHY ID • First PHY ID + 1 • … • First PHY ID + (Num PHY IDs – 1)
0x016B	phy-GroupIdentifierScope-v6	uint8_t	Indicates the PHY Group Context for the PHY PARAM response. This TLV shall be present for PARAM responses scoped at the PHY Group Context, and reflect the PHY Group ID in the PARAM request (see TLV 0x0163 in Table 3-16) that triggered this PARAM response, otherwise it shall be absent.
0x016C	supportedPHY-Groups-v6: This tag indicates the Supported PHY Groups for the L2/L3 Virtual Function		
	num-PHY-Groups-v6	uint8_t	#pg: Number of supported PHY Groups
	phy-GroupIdentifiers-v6	uint8_t [#pg]	The list of PHY Groups supported for the L2/L3 Virtual Function

Table 3-33 PHY Support

Tag	Field	Type	Description
This table describes the supported pairings of PHY and DFE Profiles. This TLV is only supported at Common or PHY Group Context.			
0x0054	num-PHY-Profiles-v3	uint16_t	Shall be the same as numPhyProfiles in Table 3-33, for the Context.
	num-DFE-Profiles-v3	uint16_t	Number of DFE profiles. The n-th DFE profile corresponds to the n-th largest DFEprofileIdx, as defined in SCF-223 [9], starting from 1.
	profileValidityMap-v3	uint8_t[x]	x = ceil floor(numPhyProfiles * numDfeProfiles/8). (numPhyProfiles*D + P)-th bit of this map determines the validity of the following pair: • [Phy Profile#P, DFE Profile#D] Where bits #n is the (n mod 8)-th LSB bit of profileValidityMap(floor(n/8)).

Table 3-34 PHY / DFE Profile Validity Map

Tag	Field	Type	Description
This table contains parameters pertaining to delay management, or other reserved parameter tags. These parameters shall only be defined by PHYs supporting Delay Management (see the Timing Management PHY capability).			
0x0101 – 0x0105	<reserved>	N/A	See section 3.2.9 of SCF-225 [10]
0x0106	dl-TTI-TimingOffset-v3	uint32_t	See section 3.2.9 of SCF-225 [10]
0x0107	dl-TTI-TimingOffset-v3	uint32_t	See section 3.2.9 of SCF-225 [10]
0x0108	ul-DCI-TimingOffset-v3	uint32_t	See section 3.2.9 of SCF-225 [10]
0x0109	tx-DataTimingOffset-v3	uint32_t	See section 3.2.9 of SCF-225 [10]
0x010A -0x011F	<reserved>	N/A	See section 3.2.9 of SCF-225 [10]
0x011E	timingWindow-v3	uint16_t	See section 3.2.9 of SCF-225 [10]
0x011F	timingInfoOrIndicationMode-v3	uint8_t	See section 3.2.9 of SCF-225 [10] Applies to Timing Info or Timing Indication APIs, depending on whether Delay Management with or without Timestamps is configured
0x0120	timingInfoOrIndicationPeriod-v3	uint8_t	See section 3.2.9 of SCF-225 [10] Applies to Timing Info or Timing Indication APIs, depending on whether Delay Management with or without Timestamps is configured.

Table 3-35 Delay management parameters

Tag	Field	Type	Description
This table contains parameters pertaining to support of the 3GPP Rel-16 mTRP Feature, per 3GPP TS 38.300 [11], section 6.12			

Tag	Field	Type	Description
0x0057	mtrp-Support-v3	uint32_t	Support for the 3GPP mTRP feature. Each bit is set as follows: 0 – not supported 1 – supported Bit positions: 0 (LSB): mTRP SDM, single DCI 1 : mTRP FDM, single DCI scheme 2a 2 : mTRP FDM, single DCI scheme 2b 3 : mTRP TDM, single DCI scheme 3 4 : mTRP TDM, single DCI scheme 4 5 : mTRP multi-DCI 6 : one of two mTRPs (i.e. PHY can represent single TRP in an mTRP scheme) 7 : both mTRPs (i.e. PHY can represent both TRPs, in an mTRP scheme) 8-31: reserved, shall be set to 0 in this release. Note: Support for mTRP requires at least one of bits 6 and 7 to be set to 1. Also, at least one of the bits 0-5 should also be set to 1 if mTRP is supported.

Table 3-36 Rel-16 mTRP parameters

Tag	Field	Type	Description
This table contains parameters pertaining to user plane			
0x007D	up-Encapsulation And-CP-Separation-v4	uint32_t	Support for encapsulation of user plane payloads. Each bit is set as follows: 0 – not supported 1 – supported Bit positions: *[TX_DATA.request]* • 0 (LSB): support for TBS-TLV tag value 0 in TX_DATA.request [payload embedded in message] • 1: support for TBS-TLV tag value 1 in TX_DATA.request [32-bit pointer to payload] • 2: support for TBS-TLV tag value 2 in TX_DATA.request [32-bit offset to payload] • 3: support for TBS-TLV tag value 3 in TX_DATA.request [UP/CP in-message separation & and for non-zero Control Length signalling] • 4: support for TBS-TLV tag value 4 in TX_DATA.request [64-bit pointer to payload]

Tag	Field	Type	Description
			• 5: support for TBS-TLV tag value 5 in TX_DATA.request [64-bit offset to payload] • 6: support for multiple TLVs in TBS-TLVs with tags 1, 2, 4 and 5 • 7: Aggregation of `TX_DATA.request` messages with tag value 3 with other kinds (other than `TX_DATA.request` with tag value 3) messages: ○ If this bit is set to 1, there is NO limit on aggregation, otherwise (if bit is set to 0) `TX_DATA.request` messages with tag value 3 may only be aggregated with other `TX_DATA.request` messages with the same tag value. *[RX_DATA.indication]* • 8: support for TBS-TLV tag value 0 in RX_DATA.indication [payload embedded in message] • 9: support for TBS-TLV tag value 1 in RX_DATA.indication [32-bit pointer to payload] • 10: support for TBS-TLV tag value 2 in RX_DATA.indication [32-bit offset to payload] • 11: support for TBS-TLV tag value 3 in RX_DATA.indication [UP/CP in-message separation & and for non-zero Control Length signalling] • 12: support for TBS-TLV tag value 4 in RX_DATA.indication [64-bit pointer to payload] • 13: support for TBS-TLV tag value 5 in RX_DATA.indication [64-bit offset to payload] • 14: support for multiple TLVs in TBS-TLVs with tags 1, 2, 4 and 5] *[RX_DATA/UCI/SRS.indication]* • 15: Aggregation of uplink messages with tag value 3 with other kinds (other than with tag value 3) messages: ○ If this bit is set to 1, there is NO limit on aggregation, otherwise (if bit is set to 0) uplink messages with tag value 3 may only be aggregated with other uplink messages with the same tag value. *[UCI.indication]* • 16: support for TBS-TLV tag value 0 in UCI.indication [payload embedded in message] • 17: support for TBS-TLV tag value 1 in UCI.indication [32-bit pointer to payload]

Tag	Field	Type	Description
			• 18: support for TBS-TLV tag value 2 in UCI.indication [32-bit offset to payload] • 19: support for TBS-TLV tag value 3 in UCI.indication [UP/CP in-message separation & and for non-zero Control Length signalling] • 20: support for TBS-TLV tag value 4 in UCI.indication [64-bit pointer to payload] • 21: support for TBS-TLV tag value 5 in UCI.indication [64-bit offset to payload] • [22-23]: reserved *[SRS.indication]* • 24: support for TBS-TLV tag value 0 in SRS.indication [payload embedded in message] • 25: support for TBS-TLV tag value 1 in SRS.indication [32-bit pointer to payload] • 26: support for TBS-TLV tag value 2 in SRS.indication [32-bit offset to payload] • 27: support for TBS-TLV tag value 3 in SRS.indication [UP/CP in-message separation & and for non-zero Control Length signalling] • 28: support for TBS-TLV tag value 4 in SRS.indication [64-bit pointer to payload] • 29: support for TBS-TLV tag value 5 in SRS.indication [64-bit offset to payload] • [30-31]: reserved
0x007E	msgEncodingSupport-v4	uint8_t	Describes padding for P7 messages. Values: - 0 (default): Legacy (implementation choice) - 1: P7 message parameters encoded without padding (unless otherwise specified for each individual FAPI parameter) - 2: P7 message parameters encoded with padding for alignment (as described in section 3.1.3) - 3-127: reserved 128-255: vendor range

Table 3-37 User Plane and Encapsulation parameters

Tag	Field	Type	Description
This table contains the parameters relating to SRS PDU and Reports			
0x0080	supported-SRS-Usage-v4	uint32_t	Bitmap indicating support different types of SRS usage; Bit positions relating the allowable values in the Usage field of SRS PDU: - [0] = beamManagement - [1] = codebook - [2] = nonCodebook - [3] = antennaSwitching - [4] = positioning

Tag	Field	Type	Description
			- [5-32] – reserved For each bit position: - Value 0: not supported - Value 1: supported
0x0160	numEntries-v5	uint8_t	Number of entries in this TLV Value: 5 in this release.
	allowed-SRS-ReportsPerUsage-v5	uint16_t [n]	n = numEntries. Bitmap indicating the types of reports supported per SRS usage; Bit positions relating the allowable reports in the Report field of SRS PDU. - [0] = [Per-PRG and Symbol SNR] - [1] = [Normalized Channel I/Q Matrix] - [2] = [Channel SVD Representation] - [3] = [Positioning Report] - [4] = [SU-MIMO Codebook Recommendation] - [5] = [Channel 2D-DFT Representation] - [5-15] – reserved Multiple bits may be set in the bitmap. Array positions correspond to the Usage of the SRS PDU where the report type is signalled: - [0] = beamManagement - [1] = codebook - [2] = nonCodebook - [3] = antennaSwitching - [4] = positioning **Note**: the number of array entries in this TLV may increase in future releases
0x0081	rb-Subsampling-REs-For-SRS-DerivedReport-v4	uint16_t [10]	Minimum number of RB subsampling for any usage: - [0] = beamManagement - [1] = codebook - [2] = nonCodebook - [3] = antennaSwitching - [5-32] – reserved In this release the size is 10, corresponding to the following usages: - [0] = beamManagement - [1] = codebook - [2] = nonCodebook - [3] = antennaSwitching - [4-9] = reserved Value: 1 … 275.
0x0082	max-SRS-ChannelReportsPerSlot-v4	uint16_t [10]	Up to as many PDUs send reports corresponding to those PDUs that have requested the report:

Tag	Field	Type	Description
			In this release the size is 10, corresponding to the following usages: - [0] = beamManagement - [1] = codebook - [2] = nonCodebook - [3] = antennaSwitching - [4] = positioning - [4-9] = reserved Value: 1 … 65535
0x0083	maxNum-SRS-PDUs-PerSlot-v4	uint16_t	Maximum number of SRS PDUs per slot Value: 1 … 65535
0x0084	iq-SampleFormats-v4	uint32_t	Each bit indicates the supported IQ format for SRS reports: Bit positions: - [0]: 16-bit - [1]: 32-bit - [2-31]: reserved Value for each bit position: - 0: not supported - 1: supported
0x0085	singularValueFormats-v4	uint32_t	Each bit indicates the supported singular value format: Bit positions: - [0]: 8-bit linear - [1]: 16-bit linear Value for each bit position: - 0: not supported - 1: supported
0x0086	numEntries-v4	uint8_t	value = 4 in this release
	max-SRS-TX-AntPortsPerUE-v4	uint8_t [n]	n= numEntries Max number of UE ports that can be sounded, per SRS usage and UE. In this release the entries corresponding to the following usages: - [0] = beamManagement - [1] = codebook - [2] = nonCodebook - [3] = antennaSwitching - [4] = positioning **Note**: This TLV may accommodate more entries in the future. Value (per entry): 1 … 16 **Notes**:

Tag	Field	Type	Description
			- This capability limits the number of SRS ports that L1 can be assumed to store for each UE, across all SRS Resources, in all Resource Sets, for a particular usage. - In this release, the usage is reflected by the Usage field in SRS PDU.
0x0087	maxNumConsecutive-SRS-Slots-v4	uint8_t	max number of slots in an SRS bursts Value: 1 .. 255
0x0088	maxDutyCycleInPercentage-v4	uint16_t	#slots in burst/(#slots in SRS burst + #non-SRS slots after burst) Expressed as value/100 (i.e. from 0.01 … 100.00%, in steps of 0.01) Value: 0.001 .. 100.00, represented by 1 … 10,000
0x0089	max-SRS-CombSize-v4	uint8_t	All SRS comb sizes up to this value are supported Value: 2, 4, 8
0x008A	maxNum-SRS-CyclicShifts-v4	uint8_t	Maximum number of SRS cyclic shifts. All numbers up to this value are supported. Value: 1 … 12
0x008B	srs-SymbolsBitmap-v4	uint16_t [2]	14-bit bitmaps, indicating which symbols can be used for SRS. Array element positions are mapped as follows: - [0]: bitmap for normal CP - [1]: bitmap for extended CP Bit position 0 corresponds to symbol position 0. Bit positions 14-15 are reserved, for normal CP. Bit positions 12-15 are reserved, for extended CP. Bit values: - 0: symbol cannot be used for SRS - 1: symbol may be used for SRS
0x008C	maxNum-SRS-SymbPerSlot-v4	uint8_t	maximum number of symbols used by SRS per slot; Value: 1 … 14
0x0161	srs-basedPositioningMeas-v5	uint32_t	Measurements reportable for SRS positioning. Value for each of the bit positions: • 0: not supported • 1: supported Bit positions: [0]: UL-RTOA [1]: Rx-Tx time difference [2]: UL AoA [3]: UL SRS-RSRP

Tag	Field	Type	Description
			[4-31]: reserved
0x0162	puncturingOf-SRS-Symbols-v5	uint8_t	Indicates whether puncturing of SRS symbols is supported (this is needed to allow PUSCH to puncture Rel-16 positioning SRS) Values: • 0: not supported • 1: supported
0x016D	zp-SRS-Resource-PDU-Requirement-v6	uint8_t	Indicates L1 PHY support and requirements for ZP-SRS Resource PDU (in UL_TTI.request). If L1 indicates need or benefit from a ZP-SRS-Resource PDU, L2/L3 is expected to (or may – in case of best effort) signal a ZP-SRS-Resource PDU to indicated unused cyclic shifts, comb offsets. L2/L3 should signal such PDUs to ensure each PRB is subject to a ZP-SRS-Resource PDU signalling at least as often the PRB is subject to sampling by a regular SRS PDU, per usage. Values: * Not Used (0): L1 does not make use of ZP-SRS-Resource PDU * Best Effort (1): L1 can benefit from L2/L3 signalling of ZP-SRS-Resource PDU * Required (2): L1 requires L2/L3 to configure an ZP-SRS-Resource PDU
0x017D	srs-RepetitionSupport-v7	uint8_t	Indicates whether SRS repetition (R>1) is supported. Values: • 0: not supported • 1: supported

Table 3-38 SRS parameters

Tag	Field	Type	Description
This table contains MIMO capabilities introduced since FAPIv4			
0x008E	maxNumCarriers-BW-LayersProduct DL-v4	uint16_t [5]	uint16_t[5] Maximum Num CC * BW in MHz * Layers product for DL across all CC that L1 can process for each of 5 numerologies. In order: 15, 30, 60, 120, 240 kHz. In this release, last entry shall be 0. Value 0xFFFF: not applicable; all other values are supported.
0x008F	maxNumCarriers-BW-LayersProduct UL-v4	uint16_t [5]	uint16_t[5] Maximum Num CC * BW in MHz * Layers product for UL across all CC that L1 can process for each of 5 numerologies. In order: 15, 30, 60, 120, 240 kHz. In this release, last entry shall be 0. Value 0xFFFF: not applicable; all other values are supported.
0x0090	maxNumCarriers-BW-AntennasProductDL-v4	uint16_t [5]	uint16_t[5] Maximum Num CC * BW in MHz * Antennas product for DL across all CC that L1 can process for each of 5 numerologies. In order: 15, 30, 60, 120, 240 kHz. In this release, last entry shall be 0. Value 0xFFFF: not applicable; all other values are supported.
0x0091	maxNumCarriers-BW-AntennasProductUL-v4	uint16_t [5]	uint16_t[5] Maximum Num CC * BW in MHz * Antennas product for UL across all CC that L1 can process for each of 5 numerologies. In order: 15, 30, 60, 120, 240 kHz. In this release, last entry shall be 0. Value 0xFFFF: not applicable; all other values are supported.
0x0092	mu-MIMO-DL-v4	uint32_t	Indicates support for MU-MIMO spatial multiplexing, per DL channel type. Bit positions: - [0]: PDSCH - [1]: PDCCH - [2]: CSI-RS - [3-31] reserved Value for each bit position: - 0: not supported - 1: supported This TLV refers to multiple users separated by spatial multiplexing and is fully independent of TLV 0x001D.

Tag	Field	Type	Description
0x0093	Spatial-MUX-Per-DL-ChannelType-v4	uint32_t [3]	Bitmap indicating support different types of precoding + beamforming. Bit position: - [0]: L2-signaled beams and precoders, using pre-stored PM and DBT - [1]: L1 derived SRS reciprocity based joint Tx precoding Value for each bit position: - 0: not supported - 1: supported Array entries correspond to DL channels types as follows: - [0]: PDSCH - [1]: PDCCH - [2]: CSI-RS
0x0094	maxNumMU-MIMO-LayersPer-DL-ChannelType-v4	uint8_t [3]	Maximum number of MuMIMO layers per DL channel type Value: 1-255; 1 = no Mu-MIMO Array entries correspond to DL channels types as follows: - [0]: PDSCH - [1]: PDCCH - [2]: CSI-RS In this release, non-precoded CSI-RS cannot be spatial multiplexed with other channels
0x0095	crossChannelSpatial-MUX-For-DL-v4	uint16_t	Signals support for spatial multiplexing of different DL channel types Bit positions: - [0]: PDSCH & PDCCH - [1]: PDSCH & CSI-RS - [2]: PDCCH & CSI-RS Value for each bit position: - 0: not supported - 1: supported All other positions are reserved and should be set to 0 in this release

Tag	Field	Type	Description
0x0096	mu-MIMO-UL-v4	uint32_t	Indicates support for MU-MIMO spatial multiplexing, per UL channel type. Bit positions: - [0]: PUSCH - [1]: PUCCH - [2-31] reserved Value for each bit position: - 0: not supported - 1: supported This TLV refers to multiple users separated by spatial multiplexing and is fully independent of TLV 0x002F.
0x0097	spatial-MUX-Per-UL-ChannelType-v4	uint32_t [2]	Bitmap indicating support different types of combining. Bit position: - [0]: L2-signaled beams, using pre-stored DBT - [1]: L1 derived SRS based joint Rx combining Value for each bit position: - 0: not supported - 1: supported Array entries correspond to DL channels types as follows: - [0]: PUSCH - [1]: PUCCH
0x0098	mu-MIMO-LayersPerUL-ChannelType-v4	uint8_t [2]	Maximum number of MuMIMO layers per UL channel type Value: 1-255; 1 = no Mu-MIMO Array entries correspond to UL channels types as follows: - [0]: PUSCH - [1]: PUCCH
0x0099	crossChannelSpatial-MUX-For-UL-v4	uint16_t	Signals support for spatial multiplexing of different UL channel types Bit positions: - [0]: PUSCH & PUCCH Value for each bit position: - 0: not supported - 1: supported All other positions are reserved and should be set to 0 in this release

Tag	Field	Type	Description
0x009A	maxMU-MIMO-GroupsIn-FD-Per-DL-ChannelType-v4	uint16_t [3]	Max number of MuMimo groups over frequency domain RBs for any one time domain symbol, per DL channel type Array entries correspond to DL channels types as follows: - [0]: PDSCH - [1]: PDCCH - [2]: CSI-RS Value:1->273
0x009B	Max-MU-MIMO-GroupsIn-TD-Per-DL-ChannelType-v4	uint8_t [3]	Max number of MuMimo groups over time domain symbols on any frequency domain RB, per DL channel type Array entries correspond to DL channels types as follows: - [0]: PDSCH - [1]: PDCCH - [2]: CSI-RS Value:1->14
0x009C	Max-MU-MIMO-NewPrecodersPerSlot-v4	uint16_t	Max number of precoders that can be refreshed per slot. A refreshed joint precoder corresponds to a PRG over which at least one layer is changed or updated from the previous slot. Value: 1→ 65535
0x009D	all-PRBs-Groups-PRGs-v4	uint8_t	Indicates whether all PRBs are contained in MU-MIMO PRGs (except possibly BWP edge PRBs). If this capability is set to 1, a single PRG can only be referenced once in in prbBitmaps for any one UE's MU-MIMO groups. Applies to both UL and DL MU-MIMO groups. Value: - 0: no need to ensure all PRBs are contained in the PRGs - 1: all PRG PRBs are contained in PRGs.
0x009E	maxMU-MIMO-GroupsInFD-PerUL-ChannelType-v4	uint16_t [2]	Max number of [Channel] MuMimo groups over frequency domain RBs for any one time domain symbol, per UL channel type Array entries correspond to UL channels types as follows: - [0]: PUSCH - [1]: PUCCH Value:1->273

Tag	Field	Type	Description
0x009F	max-MU-MIMO-GroupsIn-TD-PerUL-ChannelType-v4	uint8_t [2]	Max number of MuMimo groups over time domain symbols on any frequency domain RB, per UL channel type Array entries correspond to UL channels types as follows: - [0]: PUSCH - [1]: PUCCH Value:1->14
0x0150	min-RB-ResolutionFor-SRS-DerivedReports-v4	uint16_t	smallest number of RBs for which L1 reports an SVD channel abstraction. Value: 1 .. 275 **Note**: In this release, SVD channel abstractions are reported as indicated in TLV 0x0160
0x0151	maxNum-DL-SpatialStreams-v4	uint16_t	Maximum number of spatial streams per slot, across all channel types. When PHY declares this capability, MAC signals a DL spatial stream ID to each DL layer, logical antenna or baseband port (per TLV 0x016E). For layer assignments, this ensures that each RE in a symbol is associated with a single spatial stream. For non-layer assignments, this ensures that each RE in a symbol is associated with a single channel PDU, unless otherwise allowed by dl-SpatialStreamChannelPriority. If non-unique assignment is allowed, the higher priority spatial stream punctures the lower priority spatial stream. Range: 1-256: the number of spatial streams (MAC labels DL layers with spatial streams 0 … (maxNumberDlSpatialStreams-1)) 65535 (default): no support (MAC need not assign DL spatial streams to DL layers, in this case) Other values: reserved.
0x016E	dl-spatialStreamAssignmentScope-v6	uint8_t	Signals which ports the PHY assigns DL PDU spatial indices to. Range: - 0 – not needed (default) - 1 – spatial streams index baseband ports - 2 – spatial streams index logical ports - 3 – spatial streams index layers
0x016F	ul-spatialStreamAssignmentScope-v6	uint8_t	Signals which ports the PHY assigns UL PDU spatial indices to. Range: - 0 – not needed (default) - 1 – spatial streams index baseband ports - 2 – spatial streams index logical ports

Tag	Field	Type	Description
0x0152	dl-SpatialStreamChannelPriority-v4	uint32_t	For each bit position 0 = not supported / 1 = supported Bit positions: #0: Higher-priority CSI-RS may share a spatial stream ID with a lower priority PDSCH. #1 - #31: reserved, should be set to 0.
0x0153	maxNum-UL-SpatialStreams-v4	uint16_t	Maximum number of spatial streams for receiving UL transmissions per slot, across all channel types. When PHY declares this capability, MAC signals a set of UL spatial stream IDs to each channel, ensuring that each: - #UL spatial streams >= #layers for that UL channel. - an UL spatial stream ID cannot be assigned to channels that overlap in time and frequency domain in a slot. Range: 1-256: the number of spatial streams (MAC labels UL channels with spatial streams 0 … (maxNumberUlSpatialStreams-1)) 65535 (default): no support (MAC need not assign UL spatial streams to UL channels, in this case) Other values: reserved.
0x0154	ul-TpmiCapability-v4	uint8_t	Indicates whether ul schedule must follow the usage of the latest SRS report, or not: - 0 ulTpmiIndex must match on latest SRS report (codebook or non-codebook); - 1: ulTpmiIndex need not match latest SRS report
0x0170	mu-MIMO-PRB-ConfigDifferentFrom-SRS-PDU-v6	uint8_t	Indicates whether the prbBitmap SCS and Offset can be different from the one of the SRS PDUs, or must be configured the same (only relevant for dynamic precoding and combining). Values: • 0 – Exact Match required. All relevant SRS PDUs assumed to have the same offset and SCS. • 1 – full flexibility • [2-255]: reserved
0x017E	mu-MIMO-min-PRG-Size-v7	uint16_t	Minimum MU-MIMO PRG size Value: - 0: Wideband. - 65535: Not Specified - Should be power of 2 otherwise
	mu-MIMO-maxNum-PRG-v7	uint16_t	Maximum number of PRGs per MU-MIMO group Value: - 1…275: max number of PRGs

Tag	Field	Type	Description
0x017F	srs-UsageOverloadSupport-v7	uint8_t	Support for multiple usages in SRS PDU. Value: • 0 – not supported • 1 – supported

Table 3-39 Spatial Multiplexing and MIMO Capabilities

Tag	Field	Type	Description
This table contains parameters pertaining to support of semi-static precoding and beamforming			
0x0171	numBeamWeights-v6	uint32_t	Total number of beam weights that can be stored across all configured beam weight vectors. Value: 0 … 2^32 – 1
0x0172	numPrecodingWeights-v6	uint32_t	Total number of precoding weights that can be stored across all configured beam weight vectors. Value: 0 … 2^32 – 1
0x0173	scopeOfBeamIds-v6	uint16_t	Indicates how L1 supports signalling of semistatic beamIds: Bit positions: - [0]: beamIds in P7 can be looked up in local DBM - [1]: beamIds in P7 can be signalled transparently to LowPHY - [2-15] undefined and shall be set to 0 in this release Bit values: - 0: Not Supported - 1: Supported

Table 3-40 Semistatic Precoding and Beamforming Capabilities

Tag	Field	Type	Description
This table contains parameters pertaining to support of 2D-DFT SRS reporting and beamforming			
0x0174	beamformingDFT-ResolutionAndOversampling-v6: parameters characterizing the DFT beam resolution and oversampling in azimuth and elevation		
	dft-sizeAzimuth-v6	uint16_t	DFT size in azimuth domain Value: 1 ->65355
	dft-sizeElevation-v6	uint16_t	DFT size in elevation domain Value: 1 ->65355
	oversamplingInAzimuth-v6	uint8_t	Oversampling value in azimuth domain Value: 1 ->255 Note: Oversampling in azimuth = DFT size azimuth / num Rx Ant (per polarization) in azimuth
	oversamplingInElevation-v6	uint8_t	Oversampling value in elevation domain Value: 1 ->255

Tag	Field	Type	Description
			Note: Oversampling in elevation = DFT size elevation / num Rx Ant (per polarization) in elevation
	maxNumStrongestBeamsToReport-v6	uint8_t	K_max : The maximum possible number of strongest beams to report at the output of the 2D-DFT operation Value: 1 ->255

Table 3-41 2D-DFT Capabilities

3.3.2 CONFIG

The CONFIG message exchange was described in section 2.1.1.2.

3.3.2.1 CONFIG.request

The `CONFIG.request` message is given in Table 3-42. From this table it can be seen that `CONFIG.request` contains a list of TLVs describing how the PHY should be configured. This message may be sent by the L2/L3 software when the PHY is in any state.

The full list of CONFIG TLVs is given in Section 3.3.2.4 and the PARAM message provided L2/L3 a list of supported CONFIG TLVs and states where these TLVs are valid.

There is no requirement for the L2/L3 software to provide the TLVs in the order specified in the Tables.

Field	Type	Description
Number of TLVs	uint8_t	Number of TLVs contained in the message body.
TLVs	Variable	See Table 3-45

Table 3-42 `CONFIG.request` message body

3.3.2.2 CONFIG.response

The `CONFIG.response` message is given in Table 3-43. If the configuration procedure was successful, then the error code returned will be MSG_OK and no TLV tags will be included. If the configuration procedure was unsuccessful, then MSG_INVALID_CONFIG will be returned, together with a list of TLVs identifying the problem.

Regardless of the error code value, the following fields are always present, even if set to 0:

- Number of Invalid or Unsupported TLVs
- Number of Invalid TLVs that can only be configured in IDLE
- Number of Invalid TLVs that can only be configured in RUNNING
- Number of Missing TLVs

Field	Type	Description
errorCode	uint8_t	See Table 3-44.
numInvalidOrUnsupported-TLVs	uint8_t	N1=Number of invalid or unsupported TLVs contained in the message body. Value: 0->255
numInvalid-TLVs-ThatCanOnlyBeConfiguredIn-IDLE	uint8_t	N2=Number of TLVs contained in the message body, which are valid in IDLE state but not in this PHY state. If PHY is in IDLE state this will always be 0. Value: 0->255
numInvalid-TLVs-ThatCanOnlyBeConfiguredIn-RUNNING	uint8_t	N3=Number of TLVs contained in the message body, which are valid in RUNNING state but not in this PHY state. If PHY is in RUNNING state this will always be 0. Value: 0->255
numMissing-TLVs	uint8_t	N4=Number of missing TLVs contained in the message body. If the PHY is in the CONFIGURED state this will always be 0. Value: 0->255
A list of invalid or unsupported TLVs – each TLV is represented by its TLV tag number.		
tlv	uint16_t [N1]	Array of TLV tags
A list of invalid TLVs that can only be configured in IDLE – each TLV is represented by its TLV tag number		
tlv	uint16_t [N2]	Array of TLV tags
A list of invalid TLVs that can only be configured in RUNNING – each TLV is represented by its TLV tag number		
tlv	uint16_t [N3]	Array of TLV tags
A list of missing TLVs – each TLV is represented by its TLV tag number		
tlv	uint16_t [N4]	Array of TLV tags

Table 3-43 `CONFIG.response` **message body**

3.3.2.3 CONFIG errors

The error codes that can be returned in `CONFIG.response` are given in Table 3-44.

Error code	Description
MSG_OK	Message is OK.
MSG_INVALID_CONFIG	The configuration provided has missing mandatory TLVs, or TLVs that are invalid or unsupported in this state.

Table 3-44 Error codes for `CONFIG.response`

3.3.2.4 CONFIG TLVs

PARAM and CONFIG TLVs are used in the PARAM and CONFIG message exchanges, respectively. For both the PARAM and CONFIG TLVs the TLV format is given in Table 3-15.

The TLVs for the CONFIG message exchange are given in this Section 3.3.2.4 and for the PARAM message exchange in Section 3.3.1.4.

CONFIG TLVs used in the CONFIG message exchange are given in Table 3-45.

Field	Type	Description
PHY Config	struct	See Table 3-46
Carrier Config	struct	See Table 3-47
Cell Config	struct	See Table 3-48
SSB Power and PBCH Config	struct	See Table 3-49
PRACH Config	struct	See Table 3-50
Multi-PRACH Config	struct	See Table 3-51
Multi MsgA-PUSCH Config	struct	See Table 3-54
SSB Resource Config Table	struct	See Table 3-55
Multi-SSB Config	struct	See Table 3-56
TDD Table	struct	See Table 3-57
Measurement Config	struct	See Table 3-58
Beamforming Table	struct	See Table 3-59
Precoding Table	struct	See Table 3-59
Uci Configuration	struct	See Table 3-60
PRB-Symbol Rate Match Patterns	struct	See Table 3-61
LTE-CRS Rate Match Patterns	struct	See Table 3-63
PUCCH Semi-static Config	struct	See Table 3-64
PDSCH Config	struct	See Table 3-65
Delay Management Config	struct	See Table 3-66
Rel-16 mTRP Config	struct	See Table 3-67

Table 3-45 CONFIG TLVs

To aid additions of future TLVs in later versions of this API, the highest value tag in the Tables below is 0x104B. In addition, the tag value range 0x0100-0x014F only to be used for alignment with nFAPI tag value ranges.

Tag	Field	Type	Description
This table contains the configuration parameters relating to PHY ID definition.			
0x102A	phy-Profile-ID-v3	uint16_t	The selected PHY Profile Id (see per Table 3-33) This TLV is only supported for Common or PHY Group Context.
0x102B	numEntries-v3	uint8_t	Number of entries in this TLV Value: 7 in this release
	indicationInstancesPerSlot-v3	unit8_t [numEntries]	Configuration for transmission of more than one indication per slot. This TLV is only supported for Common or PHY Group Context. Value, for each entry: • 0: no (change in) configuration • 1: limit to one message instance per slot & numerology • 2: allow generation of more than one message instance per slot & numerology • other values are reserved Scope of each entry: [0] : `RX_DATA.indication` [1] : `CRC.indication` [2] : `UCI.indication` [3] : `RACH.indication` [4]: `SRS.indication` [5]: `DL_TTI.response` [6]: `TIMING.indication`
0x102C	numEntries-v3	uint8_t	Number of entries in this TLV Value: 4 in this release
	requestInstancesPerSlot-v3	unit8_t [numEntries]	Configuration for reception of more than one request per slot. This TLV is only supported for Common or PHY Group Context. Value, for each entry: • 0: no (change in) configuration • 1: expect at most one message instance per slot & numerology • 2: L2 may send more than one message instance per slot & numerology • other values are reserved Scope of each entry: [0]: `DL_TTI.request` [1]: `UL_TTI.request` [2]: `UL_DCI.request` [3]: `TX_DATA.request`

Tag	Field	Type	Description
0x103D	SFN/SL TLV: when this TLV is included in `CONFIG.request` for a minor reconfiguration procedure, it signals the SFN and slot (for the given numerology and cyclic prefix) where the new configuration should be applied, as described in section 2.1.5.		
	sfn-v4	uint16_t	System frame number where the minor reconfiguration procedure applies. Special value of 0xFFFF indicates earliest SFN available (i.e. ASAP) Range: 0..1023
	slot-v4	uint8_t	Slot index (as per 3GPP TS 38.211 [2], sec 4.3.2) where the minor reconfiguration procedure applies. Special value of 0xFF indicates earliest slot available (i.e. ASAP) Range: 0..159
	subcarrierSpacing-v4	uint8_t	SubcarrierSpacing index as defined in 3GPP TS 38.211 [2], sec 4.2 0: 15 KHz 1: 30 KHz 2: 60 KHz 3: 120 KHz 4: 240 KHz
	cyclicPrefixType-v4	uint8_t	0: Normal CP 1: Extended CP
0x1048	phy-Group-v6	uint8_t	Indicates the PHY Group, for CONFING exchanges scoped to PHY Group Context (see section 2.1.10)
0x1049	timeOffsetToFrame-v6	uint32_t	Offset of Radio Frame with respect to GPS Epoch-relative time, in unit of Tc The Radio Frame shall be determined as: Frame# = floor{[(time in seconds since GPS Epoch – GNSS Offset To Frame*Tc)/(frame period in seconds)] % frame period}

Table 3-46 PHY configuration

Tag	Field	Type	Description
This table contains the configuration parameters relating carrier configuration. The `PARAM.response` message will have indicated which of these parameters is supported by the PHY and in which PHY states these parameters are mandatory or optional.			
0x102D	dl-Bandwidth	uint16_t	Carrier bandwidth for DL in MHz [3GPP TS 38.104 [7], sec 5.3.2] Values: 5, 10, 15, 20, 25, 30, 40,50, 60, 70, 80,90,100,200,400
	dl-Frequency	uint32_t	Absolute frequency of DL in KHz [3GPP TS 38.104 [7], sec5.2 and 3GPP TS 38.211 [2], sec 4.4.4.2] Interpreted as PointA or f0, depending on the PHY capability TLV meaningOfCarrierFrequency Value: 450000 -> 52600000
	dl-CarrierOffset	uint16_t [5]	Carrier offset for the DL carrier, for each of the numerologies [3GPP TS 38.211 [2], sec 5.3.1], as

Tag	Field	Type	Description
			signalled in parameter offsetToCarrier [3GPP TS 38.331 [6], sec 6.3.2] For each numerology, the reference CRB (in e.g. BWPs start RB) for the numerology shall be the CRB at dl-CarrierOffset. Value: 0 ->2199
	dl-GridSize	uint16_t [5]	Grid size $N_{grid}^{size,\mu}$ for each of the numerologies [3GPP TS 38.211 [2], sec 4.4.2] Value: 0->275 0 = this numerology not used
	num-TX-Ant	uint16_t	Number of Tx antennas Value: 0 ->65355 If PHY Profiles are supported, per , numTxAnt cannot exceed the non-zero number of DL BB ports for the PHY, in the selected profile.
	Ul-Bandwidth	uint16_t	Carrier bandwidth for UL in MHz. [3GPP TS 38.104 [7], sec 5.3.2] Values: 5, 10, 15, 20, 25, 30, 40,50, 60, 70, 80,90,100,200,400
	ul-Frequency	uint32_t	Absolute frequency of UL in KHz [3GPP TS 38.104 [7], sec 5.2 and 3GPP TS 38.211 [2], sec 4.4.4.2] Interpreted as PointA or f0, depending on the PHY capability TLV meaningOfCarrierFrequency Value: 450000 -> 52600000
	ul-CarrierOffset	uint16_t [5]	Carrier offset for the UL carrier, for each of the numerologies [3GPP TS 38.211 [2], sec 5.3.1], as signalled in parameter offsetToCarrier [3GPP TS 38.331 [6], sec 6.3.2] For each numerology, the reference CRB (in e.g. BWPs start RB) for the numerology shall be the CRB at ul-CarrierOffset. Value: 0 ->2199
	ul-GridSize	uint16_t [5]	Grid size $N_{grid}^{size,\mu}$ for each of the numerologies [3GPP TS 38.211 [2], sec 4.4.2]. Value: 0->275 0 = this numerology not used
	num-RX-Ant	uint16_t	Number of Rx antennas Value: 0 ->65355 If PHY Profiles are supported, per Table 3-33, numRxAnt cannot exceed the non-zero number of UL BB ports for the PHY, in the selected profile
	frequencyShift7p5 khz	uint8_t	Indicates presence of 7.5KHz frequency shift. Value: 0 = false 1 = true

Tag	Field	Type	Description
	powerProfile-v3	uint8_t	Indicates the selected Power Profile (see section 2.2.5.5 and capability tag 0x0039) 0 = ProfileNR 1 = ProfileSSS
	powerOffset-RS-Index-v3	uint8_t	RS index for configuring power offset, among the RS indices supported, based on capability PowerProfilesSupported (tag 0x0038): Bit index location represents the same channels as in Table 3-18: - Bit#0 [LSB]: PDCCH-DMRS - Bit#1: PDCCH-Data - Bit#2: PDSCH-DMRS - Bit#3: PDSCH-Data - Bit#4: PDSCH-PTRS - Bit#5: CSI-RS - Bit#6: PSS - Bit#7 [MSB]: SSS Value range for each bit: • 0: Power offsets as signalled with respect to RS Index 0 in Table 3-18 • 1: Power offsets as signalled with respect to RS Index 1 in Table 3-18

Table 3-47 Carrier configuration

Tag	Field	Type	Description
This table contains the configuration parameters relating cell configuration. The `PARAM.response` message will have indicated which of these parameters is supported by the PHY and in which PHY states these parameters are mandatory or optional.			
0x100C	phy-Cell-ID	uint16_t	Physical Cell ID, N_{ID}^{cell} [3GPP TS 38.211 [2], sec 7.4.2.1] Value: 0 ->1007
0x100D	frameDuplexType	uint8_t	Frame duplex type Value: 0 = FDD 1 = TDD
0x102E	pdsch-TransTypeValidity-v3	uint8_t	0 = L1 must ignore pdschTransType in PDSCH PDU 1 = L1 must interpret pdschTransType in PDSCH PDU
0x102F	pusch-TransTypeValidity-v3	uint8_t	0 = L1 must ignore puschTransType in PUSCH PDU 1 = L1 must interpret puschTransType in PUSCH PDU
0x104A	n-TimingAdvanceOffset-v6	uint8_t	The UL Timing Advance offset for baseband to be aware of. Corresponds to $N_{TA,Offset}$, as described in 3GPP TS 38.211 [2], section 4.3.1 L1-PHY requiring L2 configuration of this offset in the baseband shall do so by advertising support for this TLV, otherwise L2 may assume that L1-PHY baseband does not require this configuration TLV. Values: • 0: 0 Tc • 1: 25600 Tc • 2: 39936 Tc • 3: 13792 Tc Default value: per the scenarios in 3GPP TS 38.133, section 7.3.1, as determined by PHY from: • FR type: as determined from the centre frequency (P5) or band (P19) • The frame duplex type • The coexistence scenario – if any is configured in P19 band type (if not configured, no coexistence is assumed)

Table 3-48 Cell configuration

Tag	Field	Type	Description
This table contains the configuration parameters relating SSB/PBCH configuration. The `PARAM.response` message will have indicated which of these parameters is supported by the PHY and in which PHY states these parameters are mandatory or optional.			
0x100E	ss-PBCH-Power	uint32_t	SSB Block Power, per the definition of ss-PBCH-Power in 3GPP TS 38.331 [6], section 6. Value: 0 … 110, corresponding to the range (-60..50 dBm), i.e. power value = FAPI value – 60 dBm

Tag	Field	Type	Description
0x1030	ss-PBCH-BlockPowerScaling-v3	int16_t	Baseband power scaling applied to SS-PBCH signal [3GPP TS 38.211 [2], sec 7.4.2.3.1] Values: [ssPbchBlockPowerScaling-Unit-Choice = dB]: -11000 -> 12000 representing -110.0 dB to 120.0 dB in 0.01dB, i.e. scaling is applied to the signal generated in the above sections. [ssPbchBlockPowerScaling-Unit-Choice = dBFS]: -11000 -> 0 representing -110.0 dBFS to 0.0 dBFS in 0.01dB, i.e. scaling is applied with respect to the maximum power for the cell. L1 uses to TLV if configured with PowerOffsetRsIndex bit#7 is set to 0
0x100F	bch-PayloadFlag	uint8_t	Defines option selected for generation of BCH payload, see Table 3-13 (v0.0.011 Value: 0: MAC generates the full PBCH payload 1: PHY generates the timing PBCH bits 2: PHY generates the full PBCH payload

Table 3-49 SSB power and PBCH configuration

Tag	Field	Type	Description
The TLVs from this table signalled directly on `CONFIG.request` apply to PRACH resource configuration index 0. See also the Multi-PRACH Configuration Table 3-51.			
0x1031	prach-ResConfigIndex-v3	uint16_t	A number uniquely identifying the PRACH resource configuration. If signalled directly in `CONFIG.request`, it set to 0. Multi-Prach Configuration Table 3-51 is used if multiple configurations are needed.
	prach-SequenceLength	uint8_t	RACH sequence length. Long or Short sequence length. Only short sequence length is supported for FR2. [3GPP TS 38.211 [2], sec 6.3.3.1] Value: 0 = Long sequence 1 = Short sequence
	prach-SCS	uint8_t	Subcarrier spacing of PRACH. [3GPP TS 38.211 [2], Tables 6.3.3.1-1, 6.3.3.1-2 and 6.3.3.1-7]. Represents Δf_{RA} in [3GPP TS 38.211 [2], section 5.3.2] Value: 0: 15 kHz 1: 30 kHz 2: 60 kHz 3: 120 kHz

Tag	Field	Type	Description
			4: 1.25 kHz 5: 5 kHz
	ul-BWP-PUSCH-SCS-v3	uint8_t	The PUSCH channel subcarrier spacing of UL BWP which associated with this PRACH Configuration. Corresponds to “Δf for PUSCH” in [3GPP TS 38.211 [2], Table 6.3.3.2-1] Value (corresponding to the entries of [3GPP TS 38.211 [2], Table 4.2-1]): 0 -> 3
	restrictedSetConfig	uint8_t	PRACH restricted set config Value: 0: unrestricted 1: restricted set type A 2: restricted set type B
	num-PRACH-FD-Occasions	uint8_t	Number of RACH frequency domain occasions. Corresponds to the parameter M in [3GPP TS 38.211 [2], sec 6.3.3.2] which equals the higher layer parameter msg1FDM Value: 1,2,4,8
	prach-ConfigIndex	uint8_t	PRACH configuration index reflected in RRC parameter for the configuration. Value: 0..255, per 3GPP TS 38.331 [6], section 6.3.2. If indices from range 255..262 are needed, this field shall be signalled as 255, and the actual index value shall be signalled in prach-ConfigurationIndexExtended
	prach-Format-v4	uint8_t	RACH format information for this PRACH configuration. This corresponds to one of the supported formats i.e., 0, 1, 2, 3, A1, A2, A3, B1, B4, C0, C2, A1/B1, A2/B2 or A3/B3. [3GPP 38.211 [2], sec 6.3.3.2] Values: 0 = 0 1 = 1 2 = 2 3 = 3 4 = A1 5 = A2 6 = A3 7 = B1 8 = B4 9 = C0 10 = C2 11 = A1/B1 12 = A2/B2

Tag	Field	Type	Description
			13 = A3/B3
	num-PRACH-TD-Occasions-v4	uint8_t	$N_t^{RA,slot}$: the "*number of Time domain PRACH occasions within a PRACH slot*", per 3GPP TS 38.211 [2] section 6.3.3.2. Set to 1 for long sequences (0-3). Note: this field can be determined directly from prachConfigIndex
	numberOfPreambles-v4	uint8_t	total number of preambles used by the RACH configuration. Corresponds to $N_{preamble}^{total}$ in 3GPP TS 38.213 [4], section 8.1 Value: 1 … 64
	startPreambleIndex-v4	uint8_t	preamble index start for each PRACH occasion in this PRACH configuration, per 3GPP TS 38.213 [4], section 8.1 **Note**: For example, this may be 0 for separate PRACH configurations (type 1 or 2), or 'R' for a type-2 configuration common with a type-1 configuration with 'R' preambles.
	For numPRACH-FD-Occasions {		
	prach-RootSequenceIndex	uint16_t	Starting root sequence index, i, equivalent to higher layer parameter *prach-RootSequenceIndex* [3GPP TS 38.211 [2], sec 6.3.3.1] Value: 0 -> 837
	numRootSequences	uint8_t	Number of root sequences for a particular FD occasion that are required to generate the necessary number of preambles
	prach-FrequencyOffset-v4	int16_t	Frequency offset (from UL bandwidth part) for each FD, with exact meaning based on PARAM prach-FrequencyOffsetMeaning (TLV 0x0071) as one of: • offset between Point-A and a PRB overlapping with the lowest RE of the PRACH signal, measured in PRBs at the minimum numerology between the PRACH & PUSCH numerologies (ulBwpPuschScs) • parameter k_1 [3GPP TS 38.211 [2], sec 5.3.2] in units of PRBs at ul-BWP-PUSCH-SCS • parameter k_1 [3GPP TS 38.211 [2], sec 5.3.2] in units of REs at ul-BWP-PUSCh-SCS
	prach-GuardbandOffset-v4	uint16_t	Offset (in units of PRACH SCS) between the PRB indicated by *prachFrequencyOffset* and the first RE of the PRACH signal.
	prach-ZeroCorrConf	uint8_t	PRACH Zero CorrelationZone Config which is used to derive N_{cs} [3GPP TS 38.211 [2], sec 6.3.3.1] Value: from 0 to 15

Tag	Field			Type	Description
		numUnusedRootSequences		uint16_t	Number of unused sequences available for noise estimation per FD occasion. At least one unused root sequence is required per FD occasion.
		For numUnusedRootSequences {			
			unusedRootSequences	uint16_t	Unused root sequence or sequences per FD occasion. Required for noise estimation.
		}			
	}				
	ssb-Per-RACH			uint8_t	SSB-per-RACH-occasion Value: 0: 1/8 1:1/4, 2:1/2 3:1 4:2 5:4, 6:8 7:16
	prach-ConfigIndexExtended			uint16_t	PRACH configuration index reflected in RRC parameter prach-ConfigurationIndex for the configuration. Value: per 3GPP TS 38.331 [6], section 6.3.2 Note: for values 0..255, this field and prach-ConfigIndex shall signal the same value.

Table 3-50 PRACH configuration

Tag	Field	Type	Description
This table contains multiple PRACH configurations for the carrier.			
0x1032	num-PRACH-Configurations-v3	uint16_t	Number of PRACH configurations supported in the carrier (in addition to PRACH configuration index 0)
	prach-Configuration-TLVs-v3	Set of PRACH configuration TLVs (Table 3-50)	A set of numPrachConfigurations TLVs. Each PRACH configuration TLV shall contain a unique prach-ResConfigIndex between 1 and numPrachConfigurations. **Note**: These numPrachConfigurations TLVs are additional to PRACH configuration index 0 – if any – signalled directly via Table 3-50 TLVs in `CONFIG.request`.

Table 3-51 Multi-PRACH configuration table

Tag	Field	Type	Description
This table contains a MsgA-PUSCH configuration for the carrier. This TLV can be configured to PHYs that indicate support for "semi-static configuration via P5" in msgA-Prach-To-PuschMappingScheme TLV. It shall be signalled part of the Multi-MsgA-PUSCH configuration in Table 3-54.			
0x103E	msgA-PUSCH-ResConfigIndex-v4	uint16_t	A reference index to the MsgA PUSCH resource on which the MsgA PUSCH reception operation is performed. Value: 0 … 65534; value 65535 is reserved.
	msgA-PRACH-ResConfigIndex-v4	uint16_t	A pointer linked to the PRACH resource config which carries the MsgA preamble transmission associated with this MsgA PUSCH configuration
	msgA-groupAorB-v4	uint8_t	The group for this MsgA-PUSCH configuration (c.f. 3GPP TS 38.331 [13], section 6.3.2; 3GPP TS 38.321 [21], sections 5.1.1 and 5.1.2a) Value: 0 = Group A 1 = Group B
	n-ID-MsgA-PUSCH-v4	uint16_t	The scrambling ID used to generate MsgA PUSCH data scramble sequence, corresponding to n_{ID} in 3GPP TS 38.211 [12], section 6.3.1.1 [2]. **Note**: This is used together with RAPID and RA-RNTI to determine the scrambling sequence generator. Value: 0 -> 1023
	dmrs-Ports-v4	uint16_t	Bitmap of all the DMRS ports amongst which ports are picked for each PUSCH occasion of the MsgA PUSCH configuration. [3GPP TS 38.212 [3], 7.3.1.1.2] based on msgA-PRACH → msgA-PUSCH mapping Bit mapping is as follows: • bit 0 : antenna port 1000 • bit 7: antenna port 1007 For each bit • 0: DMRS port not used • 1: DMRS port used numDmrsPorts = sum of all bits in dmrsPorts Note: the actual DMRS port to use is based on the msgA-PRACH → msgA-PUSCH & associated DMRS mapping
	available-DMRS-Sequence-IDs-v4	uint8_t	Bitmap indicating the DMRS sequence IDs available for this MsgA-PUSCH Configuration. For each bit position: • 0: ID not supported • 1: ID supported Bit significance: - [0]: n_{SCID} = 0 available. - [1]: n_{SCID} = 1 available - [2-7]: n/a (bit value shall be set to 0

Tag	Field	Type	Description
			numDmrsSequences = sum of all bits in avaialbleDmrsSequenceIds
	pusch-DMRS-ScramblingIds-v4	uint16_t [2]	The values of parameter N_{ID}^{0} (first entry) and N_{ID}^{1}(second entry), as specified in 3GPP TS 38.211 [12], section 6.4.1.1.1.1 for this MsgA-PUSCH Value: 0 -> 65535
	num-PUSCH-Ocas-FD-v4	uint8_t	The number of frequency domain PUSCH occasions within the UL slot for this Msg-A PUSCH Configuration. Value: 1, 2, 4, 8
	num-PUSCH-Ocas-TD-v4	uint8_t	The number of time domain PUSCH occasions for this Msg-A PUSCH Configuration. Value: 1, 2, 3, 6
	Num-PUSCH-OcasSlots-v4	uint8_t	The number of slots configured for this MsgA-PUSCH configuration; Value: 1…4
	msgA-PUSCH-TimeDomainOffset-v4	uint8_t	The time domain offset, as specified in [3GPP TS 38.213 [22], section 8.1A] in slots between the PRACH Configuration slot where a MsgA-PRACH preamble may be triggered and the corresponding MsgA-PUSCH occasion for this configuration. Value: 1…32
	n-Preambles-v4	uint16_t	$N_{preambles}$, per section 8.1A of 38.213, as computed by L2 for the particular mapping of MsgA-PRACH to Msg-A PUSCH mapping relevant to this MsgA-PUSCH configuration. Values: 1 … 3,584
	associationPatternPeriodInSlots-v4	uint8_t	#ApPSlots = The size of the association pattern period, measured in PRACH slots, for the PRACH configuration pointed to by msgA-prachResConfigIndex Expected to correspond to (10, 20, 40, 80, 160) ms in PRACH slots
	For each {set of PRACH slot} in the association pattern period {		
	num-PRACH-SlotsFor-PRACH-To-PRU-And-DMRS-Map-v4	uint16_t	#sameMapSlots Number of PRACH slots sharing the same PRACH to MsgA-PUSCH&Dmrs map.
	Prach-SlotIndicesModulusAPP-Slots-v4	uint16_t [n]	n = [#sameMapSlots] Set of PRACH slot indices modulus #ApPSlots For this computation: PRACH slot index = SFN * #slots/10ms + (slot indication within 10 ms)

Tag	Field		Type	Description
		msgA-PRACH-To-PRU-And-DMRS-Mapping-v4	See Table 3-53	For this mapping, the following are determined from the PRACH configuration indicated by msgA-prachResConfigIndex: 1. *prachFd1* … *prachFdMax* = 0 … numPrachFdOccasions-1 2. *prachTd1* … *prachTdMax* = 0 … numPrachTdOccasions-1 The preamble group is the group for the MsgA-PUSCH configuration, i.e. 3. *grpStart* = *grpEnd* (= group A or B or a dedicated group; exact index is irrelevant, since only the number of groups matters in Table 3-53) Furthermore, the following parameters are used to interpret the contents of the mapping: • $N_{preambles}$: Npreambles • *numPuschOcasFd*: numPuschOcasFd • *numDmrsPorts*: sum of all bits in dmrsPorts • *numDmrsSequences*: sum of all bits in available-DmrsSequenceIds • *numPuschOcasTd*: numPuschOcasTd • *numPuschOcasSlots*: numPuschOcasSlots • *startPreambleIndex*: startPreambleIndex in the PRACH configuration indicated by msgA-prachResConfigIndex • *numberOfPreambles*: numberOfPreambles in the PRACH configuration indicated by msgA-prachResConfigIndex
	}			

Table 3-52 MsgA-PUSCH configuration table

Field	Type	Description
This table contains MsgA-PRACH-to-(PRU & DMRS) map multiple PRACH configurations for the carrier. This structure is parametrized by: 1. a range of PRACH FD occasions (*prachFd1* … *prachFdMax*), from which preambles are mapped 2. a range of PRACH TD occasions (*prachTd1* … *prachTdMax*), from which preambles are mapped 3. a range of prach preamble groups (*grpStart* … *grpEnd*), to which preambles belog Furthermore, interpretation of the contents requires awareness of the following: • $N_{preambles}$: the number of preambles that can map to a “valid PUSCH occasion and the associated DMRS resource” [3GPP 38.213, section 8.1A]) • *numPuschOcasFd*: number of frequency domain occasions for the MsgA-PUSCH • *numDmrsPorts*: number of DMRS ports for the MsgA-PUSCH • *numDmrsSequences*: number of DMRS sequences for the MsgA-PUSCH • *numPuschOcasTd*: number of time domain occasions for the MsgA-PUSCH • *numPuschOcasSlots*: number of time domain occasions for the MsgA-PUSCH • *startPreambleIndex*: start preamble for each PRACH RO. • *numberOfPreambles*: total number of preambles for each PRACH RO All the above italicized parameters are provided as an input for this structure.		
For fd-Index = *prachFd1* … *prachFdMax*		

Field	Type	Description
For td-Index = *prachTd1* … *prachTd2*		
Valid-RO-v4	uint8_t	Validity of the PRACH RO opportunity corresponding to the fdIndex and tdIndex, as described in [3GPP TS 38.213 [22][4], section 8.1]. Range: 1. 0: not valid (in which case, no preambles from this RO are mapped to MsgA-PRUs) 2. 1: valid (in which case, preambles from this RO may be mapped to valid MsgA-PRUs, and associated DMRS, as described in [3GPP TS 38.213 [22], section 8.1A]
For groupIndex = *grpStart* … *grpEnd*		Group A, B or other dedicated group(s). The PRU preamble map indexing is per MsgA-PUSCH configuration, as described in 3GPP TS 38.213 [22], section 8.1A, hence per PRACH preamble group.
pru-PreambleIndexForFirstPreambleInGroup-v4	uint16_t	Offset for the MsgA-PUSCH mapped preamble in msgA-Pusch-assocDmrs-Occasion[0] that corresponds to a *offsetPreambleFromstartPreambleIndex*. L2 ensures that this offset is consecutive to the MsgA-PUSCH mapped preamble index in the last valid MsgA-PUSCH & Associated DMRS resource. The Subsequent MsgA-PUSCH numPreamblesInGroup indices starting from this offset are mapped consecutively first to the preambles for the same msgA-Pusch-assocDmrs-Occasion[0], then for consecutively ordered msgA-Pusch-assocDmrs-Occasion[1], msgA-Pusch-assocDmrs-Occasion[2], etc. As described in [3GPP TS 38.213 [22], section 8.1A], there are $N_{preambles}$ mappings possible to each PRU and associated DMRS. Note: in alignment with [3GPP TS 38.213 [22], section 8.1A], an L2 implementation may choose to identify pruPreambleIndex = 0 as corresponding to the first preamble mapping to the first valid Msg-A PUSCH time and frequency domain opportunity, associated with the first DMRS resource, for the MsgA-PUSCH configuration.
firstPreambleIndexInGroup-v5	uint8_t	The first preamble index in this group (cannot be less than *startPreambleIndex* for the PRACH occasion) for the MsgA-PRACH PDU that corresponds to pruPreambleIndex-for-firstPreambleInGroup in the MsgA-PUSCH preamble space.
num-PUSCH-Assoc-DMRS-OccasionsForThis-PRACH-RO-v4	uint8_t	**#Pru-AssocDmrs**: number of "*valid PUSCH occasion and the associated DMRS*", as defined in 38.213 [22], section 8.1A, to which this RO's RACH preambles map to.

Field				Type	Description
					Set to 0 for invalid ROs, otherwise L2 ensures that enough MsgA-PUSCH and Associated Dmrs resource occasions signalled to map all of the preambles signalled in catAorB-Bitmap, for this PRACH occasion, except possibly the last one.
			msgA-PUSCH-Assoc-DMRS-Occasion-v4	uint16_t [#Pru-AssocDmrs]	Each entry is coded to correspond to the MsgA-PUSCH and Associated Dmrs resources as follows: Fd_idx + 16*(DmrsPort + 16*(DmrsSeq + 4*(Td_idx + 8*Slot_idx))), • Fd_Idx: LSB 0:3 • DmrsPort: LSB 4:7 • DmrsSeq: LSB 8:9 • Td_idx: LSB 10:12 • Slot_idx: LSB 13:15 The following ranges are assumed: • Fd_idx: ○ 0…7: an individual MsgA-PUSCH index; ○ 15: all *numPuschOcasFd* frequency domain indices for the MsgA-PUSCH • DmrsPort: ○ 0..7: an individual DMRS port associated with the MsgA-PUSCH; ○ 15: all *numDmrsPorts* DMRS ports associated with the MsgA-PUSCH • DmrsSeq: ○ 0…1: an individual DMRS sequence associated with the MsgA-PUSCH; ○ 3: all *numDmrsSequences* DMRS sequences associated with the MsgA-PUSCH. • Td_idx: ○ 0…5: an individual MsgA-PUSCH time index ○ 7: all *numPuschOcasTd* time indices for the MsgA-PUSCH • T_slot: ○ 0..3: an individual MsgA-PUSCH slot index ○ 7: all *numPuschOcasSlots* slots indices for the MsgA-PUSCH Entries are ordered consistently with 3GPP TS 38.213 [22], section 8.1A, to ensure that PRACH preambles map consecutively to the MsgA-PUSCH and associated DMRS occasions signalled in this array. If the wildcard "all" entries is used, the actual number of entries is determined from *numPuschOcasFd*, *numDmrsPorts*, *numDmrsSequences*, *numPuschOcasTd*, *numPuschOcasTd*

Table 3-53 MsgA-PRACH-to-(PRU & DMRS) mapping structure

Tag	Field	Type	Description
This table contains multiple MsgA-PUSCH configurations for the carrier.			
0x103F	num-MsgA-PUSCH-Configurations-v4	uint16_t	Number of MsgA-PUSCH configurations supported in the carrier
	num-PRACH-ConfigurationTLVs-v4	Set of MsgA-PUSCH configuration TLVs (Table 3-52)	A set of nummsgA-Pusch-Configurations TLVs. Each PRACH configuration TLV shall contain a unique msgA-puschResConfigIndex between 0 and numMsgA-Pusch-Configurations-1.

Table 3-54 Multi-MsgA-PUSCH configuration table

Tag	Field	Type	Description
This table contains the configuration parameters used for storing an SSB transmission pattern (up to 64 SSB) in the PHY, which the PHY can then auto-generate. The `PARAM.response` message will have indicated which of these parameters is supported by the PHY and in which PHY states these parameters are mandatory or optional. The TLVs from this table signalled directly on `CONFIG.request` apply to SSB configuration index 0. See also the Multi-SSB Configuration Table 3-56.			
0x1033	ssb-ConfigIndex-v3	uint16_t	A number uniquely identifying the SSB configuration. If not signalled, or signalled directly in `CONFIG.request`, it set to 0. Multi-SSB Configuration Table 3-56 is used if multiple configurations are needed.
	Ssb-OffsetPointA	uint16_t	Offset of lowest subcarrier of lowest resource block used for SS/PBCH block. Given in PRB [3GPP TS 38.211 [2], section 4.4.4.2] Value: 0->2199
	beta-PSS-Profile-NR	uint8_t	PSS EPRE to SSS EPRE in a SS/PBCH block [3GPP TS 38.213 [4], sec 4.1] Values: 0 = 0dB 1 = 3dB L1 uses this TLV if configured with ProfileNR, or if PowerOffsetRsIndex bit#6 is set to 1
	beta-PSS-Profile-SSS-v3	int16_t	Ratio of PSS EPRE to SSS EPRE. EPRE is defined as in section 2.2.5.5. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB L1 uses this TLV if configured with ProfileNR and PowerOffsetRsIndex bit#6 is set to 0 3GPP reference: 3GPP TS 38.213 [4], section 4.1
	ssb-Period	uint8_t	SSB periodicity in msec Value:

Tag	Field	Type	Description
			0: ms5 1: ms10 2: ms20 3: ms40 4: ms80 5: ms160
	ssb-SubcarrierOffset	uint8_t	ssbSubcarrierOffset or k_{SSB} (3GPP TS 38.211 [2], section 7.4.3.1) Value: 0->31
	case-v3	uint8_t	First symbol mapping for candidate SSB locations, per [3GPP TS 38.213 [4], sec 4.1] 0: Case A 1: Case B 2: Case C 3: Case D 4: Case E
	subcarrierSpacing	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] of the SSB Value:0->4
	subCarrierSpacingCommon	uint8_t	subcarrierSpacingCommon [3GPP TS 38.331 [6], sec 6.2.2], as broadcast in MIB. Value:0->3
	l-Max-v3	uint8_t	L_{max}, as defined in 3GPP TS 38.211 [2], section 7.4.1.4.1.
	ssb-Mask	uint32_t [2]	Bitmap for actually transmitted SSB. MSB->LSB of first 32 bit number corresponds to SSB 0 to SSB 31 MSB->LSB of second 32 bit number corresponds to SSB 32 to SSB 63 Value for each bit: • 0: not transmitted • 1: transmitted Note that bits beyond the first L_{max} are expected to be 0.
	beam-ID-Presence-v5	uint8_t	Flag indicating whether the array of beam IDs below is present or not. Value: • 0: absent • 1: present
	Beam-ID-v5	uint16_t [l-Max]	Beam ID for each SSB in ssbMask. For example, if SSB mask bit 26 is set to 1, then beamId[26] will be used to indicate beam ID of SSB 26. Value: from 0 to 65535
	rmsi-Presence-v3	uint8_t	Indicates whether the SSB is cell-defining or not (i.e. is associated with an RMSI). 0: RMSI absent 1: RMSI present (default).

Tag	Field	Type	Description
			When signalled directly in `CONFIG.request`, value 0 is not allowed. RMSI may only be signalled as absent when signalled part of the Multi-SSB configuration (see Table 3-56).

Table 3-55 SSB resource configuration table

Tag	Field	Type	Description
This table contains multiple SSB configurations for the carrier.			
0x1034	num-SSB-Configurations-v3	uint8_t	Number of SSB configurations supported in the carrier (in addition to SSB configuration index 0)
	ssb-Configuration-TLVs-v3	Set of SSB table TLVs (See Table 3-55)	A set of numSsbConfigurations TLVs. Each TLV shall contain a unique SsbConfigIndex. Notes: these numSsbConfigurations TLVs are additional to SSB configuration index 0 – if any – signalled directly via Table 3-55 TLVs in `CONFIG.request`.

Table 3-56 Multi-SSB resource configuration table

Tag	Field	Type	Description
This table contains the configuration parameters used for storing a TDD pattern in the PHY. The `PARAM.response` message will have indicated which of these parameters is supported by the PHY and in which PHY states these parameters are mandatory or optional.			
0x1035	tdd-Period	uint8_t	DL UL Transmission Periodicity Value: 0: ms0p5 1: ms0p625 2: ms1 3: ms1p25 4: ms2 5: ms2p5 6: ms5 7: ms10
	For MAX_TDD_PERIODICITY {		
	For MAX_NUM_OF_SYMBOL_PER_SLOT {		
	slotConfig	uint8_t	For each symbol in each slot a uint8_t value is provided indicating: 0: DL symbol 1: UL symbol 2: Flexible symbol
	}		
	}		

Table 3-57 TDD table

Tag	Field	Type	Description
This table contains the configuration parameters relating Measurement configuration. The `PARAM.response` message will have indicated which of these parameters is supported by the PHY and in which PHY states these parameters are mandatory or optional.			
0x1028	rssi-Measurement	uint8_t	RSSI measurement unit. See Table 3-30 for RSSI definition. Value: 0: Do not report RSSI 1: dBm 2: dBFS
0x1040	rsrp-Measurement-v4	uint8_t	RSRP measurement unit. See Table 3-30 for RSRP definition. Value: 0: Do not report RSRP 1: dBm 2: dBFS
0x1047	preamblePowerUnitSelection-v5	uint8_t	preamblePwr measurement unit. See TLV preamblePowerUnit in Table 3-30 for definition. Value: 0: Do not report preamblePwr 1: dBm 2: dBFS Note: If both units are supported, the default is dBFS
0x1042	srs-PowerUnitSelection-v5	uint8_t	srsRsrp measurement unit. See TLV srsRsrpUnit in Table 3-30 for definition. Value: 0: Do not report srsRsrp 1: dBm 2: dBFS

Table 3-58 Measurement configuration

Tag	Field	Type	Description
This table contains the configuration parameters relating to digital Beamforming and Precoding.			
0x1043	beamformingTable-v5	Table 3-75	The Digital Beamforming table (for DL precoding of logical ports into BB ports and UL combining or BB ports into logical ports)
0x1044	precodingMatrix-v5	Table 3-76	A Digital Precoding matrix (for precoding DL Layers into logical ports)
0x104B	precodingTable-v6	Table 3-77	A table of Digital Precoding matrices (for precoding DL Layers into logical ports)

Table 3-59 Beamforming and Precoding

Tag	Field		Type	Description
This table contains the configuration parameters relating UCI indication.				
0x1036	num-UCI2-Maps-v3		uint16_t	Number of UCI maps indicating how a set of UCI part1 parameters map to a length of a corresponding UCI part2 Value: 0 … maxNumUciMaps
	for mapIndex = 0 … num-UCI2-Maps-1			the map Index
		numPart1Params-v3	unit8_t	1 … 4 in this specification release
		sizesPart1Params-v3	unit8_t [n]	n = numPart1Params Σ(sizesPart1Params[*]) shall not exceed 12. The value of each part1 parameter uses the same MSB0 encoding as in the 3GPP payload.
		mapBitWidth-v5	uint8_t	mapBitWitdh = Σ(sizesPart1Params[*]) Value: 0 … 12
		map-v3	uint16_t [m]	m = 2^mapBitWidth Array indicating the uci Part2 size, as a function the part1 parameters. Interpretation : map[concatenate(param1Vals)] e.g. : - numPart1Params = 2 - sizesPart1Params = [2,6] – determined from the part1 param offsets and sizes. - Σ(sizesPart1Params) = 2+6 = 8 bits. - map has 2^8 = 256 entries of 16 bits each. To determine the size of part2 for the following parameter values in part 1, the example below use makes use of highlighting to illustrate the concatenation: - p1Param[0] = 3 = 11b - p1Param[1] = 26 = 011010b Size of part2 is determined as map[11011010b] = map[218]

Table 3-60 UCI configuration

Tag	Field		Type	Description
This table contains PRB-symbol rate-match patterns, for reference in PDSCH PDU (choice 'bitmap' of RateMatchPattern in 3GPP TS 38.331 [6], section 6.3.2)				
0x1037	numberOf-PRB-SymbRM-Patterns-v3		uint8_t	Number of PRB-Symbol-based rate match patterns configured for this PHY
	For each PRB-SymbRMPattern:			
		prb-Symb-RM-Pattern-ID-v3	uint8_t	0.. (numberOfPrb SymbRateMatch Patterns – 1), in the order in which they appear in this loop; can be signalled via a bitmap in PDSCH PDU.

Tag	Field		Type	Description
		freqDomain-RB-v3	uint8_t [35]	Corresponds to 275 RBs interpreted as resourceBlocks parameter in 3GPP TS 38.331 [6], RateMatchPattern IE definition, with bit mapping as described in section 3.1.6, per TLV 0x0178
		symbolsIn-RB-v3	uint32_t	Corresponds to the oneSlot or twoSlots parameter in 3GPP TS 38.331 [6], RateMatchPattern IE definition, with bit mapping as described in section 3.1.6, per TLV 0x0178.
		subcarrierSpacing-v3	uint8_t	Value 0, 1, 2, 3 for 15/30/60/120kHz respectively or 255 to indicate this is a BWP-level pattern and it uses the same SCS as the PDSCH PDU.

Table 3-61 PRB-symbol rate match patterns bitmap (non-CORESET) configuration

Tag	Field		Type	Description
This table contains CORESET rate-match patterns, for reference in PDSCH PDU (choice 'controlResourceSet' of RateMatchPattern in 3GPP TS 38.331 [6], section 6.3.2).				
0x1041	numberOfCORSET-RM-Patterns-v3		uint8_t	Number of Coreset-based rate match patterns configured for this PHY
	For each CoresetRM-Pattern:			
		coreset-RM-PatternID-v3	uint8_t	0.. (numberOfCoreset RateMatchPatterns – 1), in the order in which they appear in this loop;can be signalled via a bitmap in PDSCH PDU.
		freqDomainResources-v3	uint8_t [6]	45 LSB interpreted same as frequencyDomainResources in ControlResourceSet of 3GPP TS 38.331 [6], with bit mapping as described in section 3.1.6, per TLV 0x0178.
		symbolsPattern-v3	uint16_t	Patterns of symbols to rate match around: 0: no rate match 1: rate match Derived to reflect slot-level gating of the seach space associated with the Coreset, as described in *controlResourceSet* choice of the RateMatchPattern in 3GPP TS 38.331 [6], section 6.3.2
		subcarrierSpacing-v3	uint8_t	Value 0, 1, 2, 3 for 15/30/60/120kHz respectively or 255 to indicate this is a BWP-level pattern and it uses the same SCS as the PDSCH PDU.

Table 3-62 CORESET rate match patterns bitmap configuration

Tag	Field		Type	Description
This table contains LTE CRS rate-match patterns, for reference in PDSCH PDU				
0x1038	numberOf-LTE-CRS-RM-Patterns-v3		uint8_t	Number of LTE CRS rate match patterns configured for this PHY
	For each lte-CRS-RM-Pattern:			
		crs-RM-Pattern-ID-v3	uint8_t	0.. (numberOfLteCRS RateMatchPatterns – 1), in the order in which they appear in this loop; can be signalled via a bitmap in PDSCH PDU.
		carrierFreq-DL-v3	uint16_t	Same interpretation as carrierFreqDL in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]

Tag	Field			Type	Description
		carrierBandwidth-DL-v3		uint8_t	Same interpretation as carrierBandwidthDL in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]
		num-CRS-Ports-v3		uint8_t	Same interpretation as num-CRS-Ports in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]: 1 = n1, 2 = n2, 4 = n4.
		v-Shift-v3		uint8_t	Same interpretation as v-Shift in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]: 0 = n0, 1 = n1, 2 = n2, 3 = n3, 4 = n4, 5 = n5.
		Size-MBSFN-SubframeConfigList-v3		uint8_t	range 0…8
		For sizeMbsfnSubframeConfigList			
			radioframeAllocationPeriod-v3	uint8_t	Same interpretation as radioframeAllocationPeriod in EUTRA-MBSFN-SubframeConfig in 3GPP TS 38.331 [6]: 1 = n1, 2 = n2, 4 = n4, 8 = n8, 16 = n16, 32 = n32.
			radioframeAllocationOffset-v3	uint8_t	Same interpretation as radioframeAllocationOffset in EUTRA-MBSFN-SubframeConfig in 3GPP TS 38.331 [6]
			lte-FrameStructureType-v3	uint8_t	LTE frame structure type for MBSFN subframe allocation, Value 0: FDD Value 1: TDD
			subframeAllocLength-v3	uint8_t	Number of frame and subframe defined as MBSFN subframe in DL Value 0: one_sixe Value 1: one_eight Value 2: four_twenty-four Value 3: four_thirty-two
			subframeAllocationBitmap-v3	uint32_t	MBSFN subframe allocation pattern defined as bitmap, bitmap size is 32, exact valid bits depends on configuration of lteFrameStructureType and subframeAllocLength, If combination of above two parameters are as below, then the interpretation of this field shall follow below description accordingly. **FDD + one_six**: 6 LSB bits valid, each associated with FDD subframe 1,2,3,6,7,8 of one frames, starting from bit 0 map to SF1, bit 1 map to SF2, and so on; **FDD + one_eight**: 8 LSB bits valid, each associated with FDD subframe 1,2,3,4,6,7,8,9 of one frames, starting from bit 0 map to SF1, bit 1 map to SF2, and so on; **FDD + four_twenty-four**: 24 LSB bits valid, each 6 bits associated with FDD subframe 1,2,3,6,7,8 of one frame out of four consecutive frames, starting from bit 0 map to SF1 in first frame, bit 6 map to SF1 in second frame, and so on; **FDD + four_thirty-two**: 32 LSB bits valid, each associated with FDD subframe 1,2,3,4,6,7,8,9 of

Tag	Field			Type	Description
					one frame out of four consecutive frames, starting from bit 0 map to SF1 in first frame, bit 8 map to SF1 in second frame, and so on; **TDD + one_six**: 5 LSB bits valid, each associated with TDD subframe 3,4,7,8,9 of one frames, starting from bit 0 map to SF3, bit 1 map to SF4, and so on; **TDD + one_eight**: does not apply to TDD, L2/L3 should avoid using this combination for DSS with TDD-LTE, and PHY should ignore it otherwise; **TDD + four_twenty-four**: 20 LSB bits valid, each 5 bits associated with TDD subframe 3,4,7,8,9 of one frame out of four consecutive frames, starting from bit 0 map to SF3 in first frame, bit 5 map to SF3 in second frame, and so on; **TDD + four_thirty-two**: does not apply to TDD, L2/L3 should avoid using this combination for DSS with TDD-LTE, and PHY should ignore it otherwise.

Table 3-63 LTE-CRS rate match patterns configuration

Tag	Field		Type	Description
This table contains the configuration parameters relating to semi-static signalling of PUCCH parameters linked to UL BWP ID				
0x1039	num-UL-BWP-IDs -v3		uint8_t	Number of UL BWP IDs to which PUCCH semistatic parameters are linked
	for ul-BWP-ID = 0 … numUL-BWP-Ids – 1			
		pucch-GroupHopping-v3	uint8_t	See the definition and range of pucchGroupHopping in PUCCH PDU (Table 3-136)
		n-ID-PUCCH-Hopping-v3	uint16_t	Cell-specific scrambling ID for PUCCH group hopping and sequence hopping in this UL BWP. Corresponds nID in 3GPP TS 38.211 [2], sec 6.3.2.2.1. For example, when MAC is implemented according to 3GPP TS 38.211 [2] Rel-15 and higher layer(RRC) does not include hoppingId IE for UE, MAC must ensure value of this field aligns with physical cell ID, i.e., L1 always assume the value is aligned with that of used by UE. Valid for formats 0, 1, 3 and 4. Value: 0->1023

Table 3-64 PUCCH semi-static configuration

Tag	Field	Type	Description
This table contains the configuration parameters relating to semi-static PDSCH parameters.			
0x103A	pdsch-CBG-Scheme-v3	uint8_t	Value: 0 = No DL CBG Re-tx 1 = DL CBG Re-tx with CBG segmentation in L2 2 = DL CBF Re-Tx, with segmentation in L1

Table 3-65 PDSCH semi-static configuration

Tag	Field	Type	Description
This table contains configuration delay management parameters defined by reference to [10], or reserved configuration tag ranges			
0x011E	timingWindow-v3	uint16_t	See section 3.2.9 of [10] Applies to Delay Management with or without Timestamps, whichever is configured.
0x011F	timingMode-v3	uint8_t	See section 3.2.9 of [10] Applies to Timing Info or Timing Indication APIs, depending on whether Delay Management with or without Timestamps is configured
0x0120	timingPeriod-v3	uint8_t	See section 3.2.9 of [10] Applies to Timing Info or Timing Indication APIs, depending on whether Delay Management with or without Timestamps is configured
0x1045	dl-TTI-RX-WindowFor-SSB-v5	uint8_t	Configures the anchoring of the RxWindow for SSB PDUs received in DL_TTI.request (if PHY supports Delay Management) Value: 1 – SSB Numerology 2 – Highest PDSCH numerology, if lower than the SSB numerology
0x1046	ul-TTI-RX-WindowFor-PRACH-v5	uint8_t	Configures the anchoring of the RxWindow for short PRACH PDUs received in UL_TTI.request (if PHY supports Delay Management) Value: 1 – short PRACH Numerology 2 – highest PUSCH numerology, if lower than the short PRACH numerology
0x0F00	<reserved>	N/A	See section 3.2.2 of [10]

Table 3-66 Delay management configuration

Tag	Field	Type	Description
This table contains configuration pertinent to Rel-16 mTRP (3GPP TS 38.300 [11], section 6.12) configuration			
0x103B	num-TX-Ports-TRP1-v3	uint8_t	[Rel-16] Number of DL baseband ports corresponding to the first TRP, for a PHY hosting baseband for both TRPs a Rel-16 mTRP scheme: Value 0: only hosting one TRP (and all DL baseband ports are assigned to it). > 0: number of baseband ports assigned to TRP1 (in which case TRP2 is assigned numTxAnt – numTxPortsTRP1 antennas)
0x103C	num-RX-Ports-TRP1-v3	uint8_t	[Rel-16] Number of UL baseband ports corresponding to the first TRP, for a PHY hosting baseband for both TRPs a Rel-16 mTRP scheme: Value 0: only hosting one TRP (and all UL baseband ports are assigned to it). > 0: number of baseband ports assigned to TRP1 (in which case TRP2 is assigned numRxAnt – numRxPortsTRP1 antennas)

Table 3-67 Rel-16 mTRP configuration

3.3.3 Vendor TLVs

The range 0xA000 – 0xAFFF are reserved for vendor-specific TLVs. These TLVs can be either CONFIG or PARAM TLVs.

3.3.4 START

The START message exchange was described in section 2.1.1.3.

3.3.4.1 START.request

This message can be sent by the L2/L3 when the PHY is in the CONFIGURED state. If it is sent when the PHY is in the IDLE, or RUNNING state an `ERROR.indication` message will be sent by the PHY. No message body is defined for `START.request`. The message length in the generic header = 0.

3.3.4.2 START errors

The error codes returned in an `ERROR.indication` generated by the `START.request` message are given in Table 3-68.

Error code	Description
MSG_INVALID_STATE	The `START.request` was received when the PHY was in the IDLE or RUNNING state.
MSG_INVALID_PHY_ID	The PHY ID is not defined
MSG_UNINSTANTIATED_PHY	The PHY ID is not instantiated
MSG_INVALID_DFE_Profile	No valid DFE Profile exists for the PHY ID

Error code	Description
FEU Not Initialized	DFE or RF chains associated with the PHY are not initialized
DFE Profile Not Selected	No DFE profile was selected for associating DFE and RF chains with the PHY

Table 3-68 Error codes for `START.request`

3.3.4.3 START.response

When PHY supports both nFAPI message format and Delay Management, successful initialization requires a `START.response` procedure, as documented in section 3.1.9 of SCF-225 [10].

3.3.5 STOP

The STOP message exchange was described in Figure 2-6.

3.3.5.1 STOP.request

This message can be sent by the L2/L3 when the PHY is in the RUNNING state. If it is sent when the PHY is in the IDLE, or CONFIGURED, state an `ERROR.indication` message will be sent by the PHY. No message body is defined for `STOP.request`. The message length in the generic header = 0.

3.3.5.2 STOP. Indication

This message is sent by the PHY to indicate that it has successfully stopped and returned to the CONFIGURED state. No message body is defined for `STOP.indication`. The message length in the generic header = 0.

3.3.5.3 STOP errors

The error codes returned in an `ERROR.indication` generated by the `STOP.request` message are given in Table 3-69.

Error code	Description
MSG_INVALID_STATE	The `STOP.request` was received when the PHY was in the IDLE or CONFIGURED state.

Table 3-69 Error codes for `STOP.request`

3.3.6 PHY Notifications

The PHY notification messages are used by the PHY to inform the L2/L3 software of an event which occurred.

3.3.6.1 ERROR.indication

This message is used to report an error to the L2/L3 software. These errors all relate to API message exchanges. The format of `ERROR.indication` is given in Table 3-70.

Field	Type	Description
sfn	uint16_t	SFN Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159
message-ID	uint8_t	Indicates which message received by the PHY has an error. Values taken from Table 3-11. Shall be set to 0xFF if the ID of the concerned message is greater than 254, otherwise shall be set to message ID. See also message-ID-Ext.
errorCode	uint8_t	The error code, see Section 3.3.6.2 for values
expected-SFN	uint16_t	If the error code is OUT_OF_SYNC, this field is set to the expected SFN, otherwise it is set to 0xFFFF Value: 0 -> 1023 or 0xFFFF
expectedSlot	uint16_t	If the error code is OUT_OF_SYNC, this field is set to the expected Slot, otherwise it is set to 0xFFFF Value: 0 ->159 or 0xFFFF
extendedStatus-v5	Table 3-71	Field signalling an extended status;
backwardCompatibleExtension-v7	Structure	See Table 3-72 (backward compatible extension, see section 3.1.5)

Table 3-70 `ERROR.indication` **message body**

Field	Type	Description
extendedStatusLength-v5	uint16_t	xlen = Length of the extended status payload Value: 0 … 65535
extendedStatusContents-v5	uint8_t [xlen]	The extended status payload. Content shall depend on the Error Code in ERROR.indication (Table 3-70)

Table 3-71 Extended Status format

Field	Type	Description
Backward compatible extension (first introduced in FAPIv7). See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension (introduced in FAPIv7). See section 3.1.5 for what this length represents.
message-ID-Ext-v7	uint16_t	Indicates which message received by the PHY has an error. Values taken from Table 3-11.

Table 3-72 ERROR.indication Message Extension

3.3.6.2 Error Codes

The list of possible error codes returned in either `.response` messages, or the `ERROR.indication` message is given in Table 3-73

Value	Error code	Description
0x0	MSG_OK	Message is OK.
0x1	MSG_INVALID_STATE	The received message is not valid in the PHY's current state.
0x2	MSG_INVALID_CONFIG	The configuration provided in the request message was invalid
0x3	OUT_OF_SYNC	The `DL_TTI.request` was received with a different SFN or slot than the PHY expected.
0x4	MSG_SLOT_ERR	The `DL_TTI.request` or `UL_TTI.request` had an invalid format.
0x5	MSG_BCH_MISSING	A SSB PDU was expected in the `DL_TTI.request` message for this subframe. However, it was not present.
0x6	MSG_INVALID_SFN	The received `UL_DCI.request` or `TX_DATA.request` message included a SFN/SL value which was not expected. The message has been ignored.
0x7	MSG_UL_DCI_ERR	The `UL_DCI_DCI.request`. had an invalid format
0x8	MSG_TX_ERR	The `TX_DATA.request` had an invalid format.
0x9	MSG_INVALID_PHY_ID	The PHY ID is not defined
0xA	MSG_UNINSTANTIATED_PHY	The PHY ID is not instantiated
0xB	MSG_INVALID_DFE_Profile	No valid DFE Profile exists for the PHY ID
0xC	PHY PROFILE_SELECTION_ INCOMPATIBLE_WITH_ RUNNING_PHY	There is a RUNNING PHY Id whose definition would be changed by the PHY Profile selection
0xD	FEU_NOT_INITIALIZED	DFE or RF chains associated with the PHY are not initialized
0xE	DFE PROFILE_NOT_SELECTED	No DFE profile was selected for associationg DFE and RF chains with the PHY
0xF	MSG_PHY_GROUP_UNAVAILABLE	The requested PHY Group is currently unavailable for the virtual function
0x10	MSG_PHY_GROUP_CONSTITUEN TS_ALTERED	This Error Code is sent in response to PARAM.request to L2/L3 from L1 along with PHY PARAM Response containing the updated list of PHY-IDs associated with the requested PHY-Group (when the number of constituents PHY IDs undergoes a change w.r.t the previously shared number for the same PHY-Group).
0xFF	MSG_EXTENDED_VENDOR_ STATUS	In this release, this value shall be used to signal vendor-specific extended status codes in ERROR.indication

Table 3-73 PHY API error codes

3.3.6.3 CONNECTIVITY.indication

This message is used to report an connectivity event L2/L3 software regarding the PHY. It may be scoped for a particular PHY, in which case the observation applies to that PHY, or may be scoped for the Common or PHY Group Context, in which case the observation applies to any actual or potential PHY in the Context. If an observation applies to the Common, it is also raised – if supported – for all dedicated PHYs (and only applies to PHY-side observations) in the context.

The format of `CONNECTIVITY.indication` is given in Table 3-74.

Field	Type	Description
sfn-v5	uint16_t	SFN: Indicates the time of the connectivity event for which the indication is issued, if the PHY is in RUNNING state, otherwise 65535. Value: 0 -> 1023, 65535
slot-v5	uint16_t	Slot: Indicates the time of the connectivity event for which the indication is issued, if the PHY is in RUNNING state, otherwise 65535. Value: 0 ->159, 65535
observation-v5	uint16_t	An indication of the connectivity status: 0: Connection lost 1: Synchronization lost 2: Synchronization gained

Table 3-74 `CONNECTIVITY.indication` message body

3.3.7 Storing Precoding and Beamforming Tables

The format of the beamforming table is given in Table 3-75 and the format of the precoding table is given in Table 3-77. Table 3-76 can be used part of TLV 0x1044 to configure one entry in the precoding table.

Field	Type	Description
numDigBeams	uint16_t	Number of digital beams Value: 0->32768
numBasebandPorts	uint16_t	Number of ports at the digital beamformer output. Value: 0->65535
for each digBeam {		
beam-ID	uint16_t	Identifying number for the beam index Value: 0->32767
for each Baseband port {		
digBeamWeightRe	int16_t	Real part of the digital beam weight
digBeamWeightIm	int16_t	Imag part of the digital beam weight
}		
}		

Table 3-75 Digital beam table (DBT) PDU

Field	Type	Description
pm-ID	uint16_t	Index which uniquely identifies the precoding matrix (PM). Value: 0->65535
numLayers	uint16_t	Number of ports at the precoder input Value: 0->65535
numAntPorts	uint16_t	Number of logical antenna ports at the precoder output Value: 0->65535
For numLayers {		
For numAntPorts {		
precoderWeightRe	int16_t	Real part of the precoder weight
precoderWeightIm	int16_t	Imag part of the precoder weight
}		
}		

Table 3-76 Precoding matrix (PM) PDU

Field	Type	Description
numPrecodingMatrices-v6	uint16_t	Number of precoding matrices Value: 0->65535

Field				Type	Description
					When signalled part of TLV 0x104B, the range shall be 1 -> 65535
for each PM {				Precoding Matrices are listed in increasing order of numLayers.	
	pm-ID-v6			uint16_t	Index which uniquely identifies the precoding matrix (PM). Configuration ensures no gaps in the table of stored precoding matrices. Value: 0->65535
	numLayers-v6			uint16_t	Number of ports at the precoder input. Value: 1->65535
	numAntPorts-v6			uint16_t	Number of logical antenna ports at the precoder output Value: 1->65535
	For numLayers {				
		For numAntPorts {			
			precoderWeightRe-v6	int16_t	Real part of the precoder weight
			precoderWeightIm-v6	int16_t	Imag part of the precoder weight
		}			
	}				
}					

Table 3-77 Precoding matrix table (PMT) PDU

When a pm-Idx is signalled part of TLVs 0x1044 or 0x104B, the precoding matrix signalled for pm-Idx overwrites the precoding matrix – if any – at location pm-Idx.

3.3.8 RESET

The RESET message exchange was described in section 2.1.4.

3.3.8.1 RESET.request

This message can be sent by the L2/L3 when the PHY is in the RUNNING or CONFIGURED state. If it is sent when the PHY is in the IDLE, state an `ERROR.indication` message will be sent by the PHY. The message length in the generic header may be 0 if the message is not targeted to PHY Group scope. If non-zero, the message body is as below.

Field	Type	Description
scope-v7	uint8_t	Describes the scope of this message: Common, Group or Dedicated Values: 1: PHY Group (in this case the PHY ID in the header is 255) 0: Common or Dedicated
phy-Group-v7	uint8_t	The PHY Group targeted, if the scope is PHY Group

Table 3-78 `RESET.request` message body

Note on backward compatibility: if the agreed protocol version is FAPIv6, then the message length would be 0, per the above rules.

3.3.8.2 RESET.Indication

This message is sent by the PHY to indicate that it has successfully stopped and returned to the IDLE state. No message body is defined for `RESET.indication`. The message length in the generic header may be 0 if the RESET procedure was not targeting PHY Group scope. If non-zero, the message body is as below.

Field	Type	Description
scope-v7	uint8_t	Describes the scope of the RESET procedure reply: Group or Dedicated Values: 1: PHY Group (in this case the PHY ID in the header is 255) 0: Dedicated Note: A successful common scope Reset does not use RESET.Indication
phy-Group-v7	uint8_t	The PHY Group targeted, if the scope is PHY Group

Table 3-79 `RESET.request` message body

Note on backward compatibility: if the agreed protocol version is FAPIv6, then the message length would be 0, per the above rules.

3.3.8.3 RESET errors

The error codes returned in an `ERROR.indication` generated by the RESET`.request` message are given in Table 3-80.

Error code	Description
MSG_INVALID_STATE	The `RESET.request` was received when the PHY was in the IDLE state.

Table 3-80 Error codes for `RESET.request`

3.3.8.4 CCONTEXT_READY.Indication

This message is sent by the Common Context, if implemented, to indicate that Common Context is available at initial boot, or after a Common Context RESET. The message body is as defined in section 3.1.0 in SCF-225 [10] for PNF_READY.indication.

3.4 Slot Messages

The slot messages are used by the L2/L3 software to control the data transmitted, or received, every slot.

3.4.1 Slot.indication

The `SLOT.indication` message is given in Table 3-81, and is sent from the PHY with a periodicity based on the `CONFIG.request` message sent to the PHY. Specifically, the `CONFIG.request` message includes carrier configuration TLVs, which provide values for different numerologies, including options to indicate a numerology is not used. The `SLOT.indication` message is sent from the PHY based on the highest-value numerology configured in `CONFIG.request`. If the highest numerology configured is:

- 0 – then `SLOT.indication` is provided every 1ms
- 1 – then `SLOT.indication` is provided every 500us
- 2 – then `SLOT.indication` is provided every 250us
- 3 – then `SLOT.indication` is provided every 125us
- 4 – then `SLOT.indication` is provided every 62.5us

Field	Type	Description
sfn	uint16_t	SFN Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159

Table 3-81 Slot indication message body

3.4.1a TIMING.indication

The `TIMING.indication` message is given in Table 3-82. It supports Delay Management without Timestamps, as described in section 2.2.1.2, when configured to the PHY.

It is sent from the PHY on an event-triggered basis, or with a periodicity in slots (for the highest configured numerology of this PHY) based on the `CONFIG.request` message sent to the PHY. It is also sent after a successful `START.request`. The `TIMING.indication` message is sent from the PHY based on the highest-value numerology configured in `CONFIG.request`.

Field	Type	Description
sfn-v5	uint16_t	SFN Value: 0 -> 1023
slot-v5	uint16_t	Slot Value: 0 ->159
numObservations-v6	uint8_t	Value 0 … 255
For 0 … numObservations – 1		
message-ID-v5	uint16_t	16-bit Message Type ID to which the observation applies. For PHY messages, the possible messages are: • DL_TTI.request • UL_TTI.request • UL_DCI.request • TX_DATA.request For FEU messages, the possible messages are: • DFE SCHEDULE.request • RF SCHEDULE.request • FEU.CONFIG.RF SCHEDULE.request • FEU.CONFIG. SELECT_BEAM.REQUEST • FEU.CONFIG. SET_BEAM_SLOT_PATTERN.REQUEST
observation-v5	uint8_t	Value: • 0: Too-late (invalid timeFromRxWindowStart) • 1: Too-early (invalid time report) • 2: Too-late (valid time report) • 3: Too-early (valid time report)

Field		Type	Description
			• 8: periodic or initial report • 4-7, 9-255: reserved.
	timeFrom-RX-WindowStart-v5	uint32_t	Absolute value of time difference between message arrival as observed by the PHY (if supported), and the Rx window start, for the message in messageId. Value: • 0 … (10^8): time in nanoseconds measured from the start of the Rx window for the message, if such a metric is supported. (Note that accuracy is not necessarily in nanoseconds). • 10^8 … 2^32-1: reserved

Table 3-82 Timing indication message body

3.4.2 DL_TTI.request

The format of the `DL_TTI.request` message is shown in Table 3-83. A `DL_TTI.request` message indicates the SFN/Slot it contains information for, and this control information is for a downlink slot.

This message can be sent by the L2/L3 when the PHY is in the RUNNING state. If it is sent when the PHY is in the IDLE or CONFIGURED state an `ERROR.indication` message will be sent by the PHY.

- A PDCCH PDU includes 1 or more DCI PDUs

The PDUs included in this structure are signalled in causal order: if PDU A references PDU B, PDU B appears before PDU A in the "Number of PDUs" loop.

Field	Type	Description
sfn	uint16_t	SFN Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159
num-PDUs	uint16_t	Number of PDUs that are included in this message. All PDUs in the message are numbered in order. Value 0 -> 65535 Backward compatibility note: the field was 8 bits in SCF222 v2.0
num-DL-Types-v3	uint8_t	Maximum number of DL PDU types or DCIs supported by `DL_TTI.request`. nDlPduTypes = 5 in this release. Backward compatibility note: SCF222 v2.0 did not have this field

Field	Type	Description
num-PDUs-OfEachType-v3	uint16_t [numDL-Types]	Number of PDUs of each type that are included in this message. Each array entry corresponds to a PDU type as follows: [0]: number of PDCCH PDUs [1]: number of PDSCH PDUs [2]: number of CSI-RS PDUs [3]: number of SSB PDUs [4]: number of DlDCIs across all PDCCH PDUs in this message [5]: number of PRS PDUs Value 0 -> 65535, for each entry Backward compatibility note: SCF222 v2.0 did not have this field
numGroups	uint16_t	Number of PDU Groups included in this message. Value 0 -> 3,822
topLevel-RM-Patterns-v4	struct	See Table 3-84
For Number of PDUs {		
pdu-Type	uint16_t	0: PDCCH PDU, see Section 3.4.2.1 1: PDSCH PDU, see Section 3.4.2.2 2: CSI-RS PDU, see Section 3.4.2.3 3: SSB PDU, see Section 3.4.2.4 5: PRS PDU, see Section 3.4.2.4a Note: value 4 is to be allocated next time a DL_TTI.request PDU is added.
pdu-Size	uint16_t	Size of the PDU control information (in bytes). This length value includes the 4 bytes required for the PDU type and PDU size parameters. Value 0 -> 65535
dl-PDU-Configuration	structure	See Sections 3.4.2.1 to 3.4.2.4
}		
In this message, if numGroups > 0 then for each group { *-- For FAPIv4, groups represent MU MIMO groups, as described in section 2.2.7.2; muMimoGroups are listed in increasing order of their start symbol. muMimogroups starting at the same symbol, are listed channel by channel in increasing order of their first PRB index.*		
num-PDUs-In-Group v4	uint8_t	Number of PDUs (in this release, PDU types in each group are identical) in this group For SU-MIMO, one group includes one PDU only. For MU-MIMO, one group includes up to 32 PDUs. For each PDU type, each PRB in a symbol can be referenced by at most one MuMIMO group. Value 1 -> 32

	Field	Type	Description
	layerMapping-FD-Resolution-v4	uint16_t	This field signals the RB resolution at which layer mapping is conveyed. The resolution shall be consistent with TLV 0x017E. The frequency anchoring of layerMapping-PRG as used in this document follows the definition as PRGs in 3GPP TS 38.214 [5], Section 5.1.2.3, unless otherwise specified, i.e. layerMapping-PRGs start on CRBs that are multiples of layerMapping-FD-Resolution Notes: - there is no defined relationship between layerMapping-PRG and 3GPP PRG, other than using the same frequency anchoring reference (CRB0). PDSCH 3GPP PRG is signalled in PDSCH PDU. - layerMapping-FD-Resolution be should signalled consistent with the Precoding granularity (P'_{BWP}) for the PRB Bundling, for this PDSCH allocation (see 3GPP TS 38.214 [5], section 5.1.2.3); for instance, a boundary between two layerMapping-PRGs with different precoders should not fall inside a PRB Bundle (see 3GPP TS 38.214 [5], section 5.1.2.3) for PDSCH allocation. Values: • 1 → 275 (FAPIv4), • 0 (to mean FAPIv2) • 0xFFFF (wideband only)
	prb-BitmapStart-v6	uint16_u	Offset from reference CRB to PRB0 of prb-Bitmap. Value: 0 … 274. If TLV 0x0170 is set to 1 or numLayerMapping-PRGs = 0, prb-BitmapStart is the offset with respect to reference CRB for the carrier, otherwise it shall be set so that PRB0 of prbBitmap and the UL CRB reference of srs-BandwidthStart of SRS PDUs used for dynamic precoding are the same.
	prb-SCS-v6	uint8_t	subcarrier spacing [3GPP TS 38.211 [2], sec 4.2] common to all PDUs in this muMimoGroup. If TLV0x0170 is set to 1 or numLayerMapping-PRGs = 0, prb-SCS is the numerology corresponding to prbBitmap and prbBitmapStart, otherwise it shall be set the same as for all the SRS PDUs used for dynamic precoding. Value:0->4

Field				Type	Description
	prb-Bitmap-v4			uint8_t [36]	bitmap of PRBs assumed common to all UEs in the group, for joint spatial precoding. 273 rounded up to multiple of 32. This bitmap is in units of PRBs. Bit mapping is described in section 3.1.6, per TLV 0x0177. Note: the lowest numbered PRB in the bitmap corresponds to CRB #prb-BitmapStart

	numLayerMapping-PRGs-v5			uint16_t	Number of layerMapping-PRGs with at least one PRB allocated in prb-Bitmap.
					
					Shall be set to 0 if and only if the PDUs in the MU MIMO group are used with semi-static precoding.
					
					This field shall be consistent with TLV 0x017E, e.g. shall be set to 1 if only Wideband resolution is supported.
					
					If semi-static precoding or wideband resolution do not apply, the following pseudocode illustrates the expected value from L2/L3:
					
					num-layerMapping-PRGs := 0
					`For` *lmPrgIdx* `= 0 … max`
					   `if layerMapping-PRG#`*lmPrgIdx* `has at least one prb bit set to 1 in` prb-Bitmap:
					      num-layerMapping-PRGs `:=`
					         num-layerMapping-PRGs `+ 1`
					`End`

	symbol-Bitmap-v4			uint16_t	bitmap of symbols for the joint spatial precoding group bit location indicates the symbol index, up to 14 symbols. First symbol corresponds to location #0
	For each of the num-PDUs in the group				
		pdu-Index-v4		uint16_t	This value is an index for number of PDU of the ones signalled in this message Value: 0 -> 65535
		dci-Index-v6		uint16_t	This value is the dciIndex parameter associated with a DCI in the PDCCH PDU PduIdx. If the PDU does not correspond to a PDCCH PDU, the dciIndex shall be set to 0xFFFF.
		For each of the num-PRGs layerMapping-PRGs{			Note that only PRGs with allocations in the prbBitmap are included in this loop. PrgIdx *n* refers to the *n*-th layerMapping-PRG with at least one PRB allocated in prbBitmap, in the same order they appear in frequency domain.
			chosenLayerBitmapFor-CW0-v4	uint8_t	All layers scheduled for CW0 of this PDU. Bit indices are in in the same order as the S-matrix singular values for the layers.

Field				Type	Description
			chosenLayerBitmapFor-CW1-v4	uint8_t	All layers scheduled for CW1 this PDU. Bit indices are in in the same order as the S-matrix singular values for the layers.
	}				
}					
backwardCompatibleExtension-v7				structure	Table 3-85 (backward compatible extension, see section 3.1.5)

Table 3-83 `DL_TTI.request` **message body**

Field		Type	Description
Structure for signalling set of all rate match structures. The signalling can be by reference to P5 configuration, or directly embedded in this structure, depending in PHY capability and MAC choice. Each rate match pattern signalled in this structure shall be reference in at least one PDSCH PDU in the same `DL_TTI.request`, otherwise the structure is irrelevant. For PHYs that signal support for universal rate match (per TLV universalRateMatch), MAC ensures that all PDSCH PDUs that overlap with any of the patterns in this structure rate match around that pattern.			
`DL_TTI.request` Top-level: Prb Symbol Rate Match			
num-PRB-Sym-RM-Patterns-v4		uint16_t	Number of PrbSym rate match patterns signalled at the DL_TTI top level
prb-Symb-RM-ConfigurationMethod-v4		uint8_t	Indicates whether the configuration for the PrbSym rate match patterns is in P5, or in P7: Values: • 0: P5-based configuration. Then: ○ numPrbSymbRmP5Ref = numPrbSymRmPatterns ○ numPrbSymbP7Sig = 0 • 1: P7-based configuration. Then: ○ numPrbSymbRmP5Ref = 0 ○ numPrbSymbRmP7Sig = numPrbSymRmPatterns Note: value 0 is only relevant for PHYs that signal support for universal rate match (per TLV universalRateMatch)
prb-Sym-RM-PatternsP5-v6		uint8_t [szPsP5]	szPsP5 = numPrbSymbRmP5Ref Array of PrbSymb -based RM indices. Note: this field should only be signalled for PHYs expecting universal rate match (per TLV universalRateMatch)
prb-Symb-RM-P7-Loop: For 0 … num-PRB-Symb-RM-P7-Sig-1			loop index is the identifier of the PrbSymb-based rate match pattern, when referenced in PDSCH PDU
	subcarrierSpacing-v4	uint8_t	Value 0, 1, 2, 3 for 15/30/60/120kHz respectively or 255 to indicate this is a BWP-level pattern and it uses the same SCS as the PDSCH PDU.

Field	Type	Description
freqDomain-RB-v4	uint8_t [35]	Corresponds to 275 RBsinterpreted as resourceBlocks parameter in 3GPP TS 38.331 [6], RateMatchPattern IE definition, with bit mapping as described in section 3.1.6, per TLV 0x0178.
symbolsIn-RB-v4	uint16_t	Patterns of symbols to rate match around, with bit map formatted the same as in the oneSlot parameter in 3GPP TS 38.331 [6], RateMatchPattern IE definition, each bit indicating: 0: no rate match 1: rate match Bit mapping as described in section 3.1.6, per TLV 0x0178.
DL_TTI.request Top-level: CORESET-based Rate Match		
num-CORESET-RM-Patterns-v4	uint16_t	Number of CORESET-based rate match patterns signalled at the DL_TTI top level
coreset-RM-ConfigurationMethod-v4	uint8_t	Indicates whether the configuration for the CORESET-based rate match patterns is in P5, or in P7: Values: • 0: P5-based configuration. Then: ○ numCoresetRmP5Ref = numCoresetRmPatterns ○ numCoresetRmP7Sig = 0 • 1: P7-based configuration. Then: ○ numCoresetRmP5Ref = 0 ○ numCoresetRmP7Sig = numCoresetRmPatterns Note: value 0 is only relevant for PHYs that signal support for universal rate match (per TLV universalRateMatch)
coreset-RM-Pattern-P5-v6	uint8_t [szCsP5]	szCsP5 = numCoresetRmP5Ref Array of CORESET-based RM pattern indices. Note: this field should only be signalled for PHYs expecting universal rate match (per TLV universalRateMatch)
coreset-RM-P7-Loop: For 0 … num-CORESET-RM-P7-Sig-1		loop index is the identifier of the CORESET-based rate match pattern, when referenced in PDSCH PDU
coreset-RM-SignalingType-v6	uint8_t	Values: * 0-3, 255: 0, 1, 2, 3 for 15/30/60/120kHz respectively or 255 to indicate this is a BWP-level pattern and it uses the same SCS as the PDSCH PDU. * 254: CORESET is per the PDCCH PDU whose pdcchPduIndex is signalled in param16bit. Shall be used, unless a PDCCH PDU cannot be referenced.
freqDomainResources-v6	uint8_t [6]	45 LSB interpreted same as frequencyDomainResources in ControlResourceSet of 3GPP TS 38.331 [6], with bit mapping as described in section 3.1.6, per TLV 0x0178.

Field	Type	Description
		.
param16Bit-v6	uint16_t	Depends on 'CoresetRM Signalling Type: * pdcch-PduIndex Or * Patterns of symbols to rate match around, with bit map formatted the same as in the oneSlot parameter in 3GPP TS 38.331 [6], RateMatchPattern IE definition (but applied to the CORESET), each bit indicating: 0: no rate match 1: rate match Bit mapping as described in section 3.1.6, per TLV 0x0178.
`DL_TTI.request` Top-level: LTE CRS-based Rate Match		
num-LTE-CRS-RM-Patterns-v4	uint16_t	Number of LTE CRS-based rate match patterns signalled at the DL_TTI top level
lte-CRS-RM-ConfigurationMethod-v4	uint8_t	Indicates whether the configuration for the LTE CRS-based rate match patterns is in P5, or in P7: Values: • 0: P5-based configuration. Then: ○ numLteCrsRmP5Ref = numLteCrsRmPatterns ○ numLteCrsRmP7Sig = 0 • 1: P7-based configuration. Then: ○ numLteCrsRmP5Ref = 0 ○ numLteCrsRmP7Sig = numLteCrsRmPatterns Note: value 0 is only relevant for PHYs that signal support for universal rate match (per TLV universalRateMatch)
lte-CRS-RM-Pattern-P5-v6	uint8_t [szLcP5]	szLcP5 = numLteCrsRmP5Ref Array of LTE CRS-based RM pattern indices. Note: this field should only be signalled for PHYs expecting universal rate match (per TLV universalRateMatch)
Lte-CRS-RM-P7-Loop For 0 … num-LTE-CRS-RM-P7-Sig-1		loop index is the identifier of the LTE CRS-based rate match pattern, when referenced in PDSCH PDU
carrierFreq-DL-v4	uint16_t	Same interpretation as carrierFreqDL in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]
carrierBandwidth-DL-v4	uint8_t	Same interpretation as carrierBandwidthDL in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]
num-CRS-Ports-v4	uint8_t	Same interpretation as num-CRS-Ports in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]: 1 = n1, 2 = n2, 4 = n4.
v-Shift-v4	uint8_t	Same interpretation as v-Shift in RateMatchPatternLTE-CRS in 3GPP TS 38.331 [6]: 0 = n0, 1 = n1, 2 = n2, 3 = n3, 4 = n4, 5 = n5.
size-MBSDN-SubframeConfigList-v4	uint8_t	range 0…8 (L2 shall set this to 0, if L1 only supports P7-based MBSFN subframe determination)

Field			Type	Description
	For sizeMbsfnSubframeConfigList			
		radioframeAllocationPeriod-v4	uint8_t	Same description as radioframeAllocationPeriod in Table 3-63
		radioframeAllocationOffset-v4	uint8_t	Same description as radioframeAllocationOffset in Table 3-63
		lte-FrameStructureType-v4	uint8_t	Same description as lte-FrameStructureType in Table 3-63
		subframeAllocLength-v4	uint8_t	Same description as subframeAllocLength in Table 3-63
		subframeAllocationBitmap-v4	uint32_t	Same description as subframeAllocationBitmap in Table 3-63

Table 3-84 `DL_TTI.request` **top-level rate match signalling**

Field			Type	Description
This is a backward compatible extension of DL_TTI.request, introduced in FAPIv7				
extensionLength-v7			uint16_t	Size of this backward compatible extension. See section 3.1.5.
For group = 0 … nGroups-1				MU-MIMO Groups are listed in the same order as in the groups loop in the main DL_TTI.request message body
	extensionLength		uint16_t	Size of this backward compatible extension for the group loop. See section 3.1.5.
	For each of the num-PDUs in the group			PDUs for the group are listed in the same order as in the PDU loop for the group, in the main DL_TTI.request message body
		extensionLength	uint16_t	Size of this backward compatible extension for the PDU loop. See section 3.1.5.
		srs-RNTI-v7	uint16_t	RNTI of the SRS PDU whence channel estimates are extracted for this MU-MIMO Group. If channel estimates are not based on SRS (e.g. are derived from semi-static PMI and beamforming), this field shall be set to 0xFFFF

Table 3-85 `DL_TTI.request` **extension (introduced in FAPIv7)**

3.4.2.1 PDCCH PDU

Each DL DCI PDU includes the information that the L2/L3 software must provide the PHY so it can transmit the DCI described in [3GPP TS 38.212 [3], section 7.3.1], however, the L2/L3 constructs the DCI message.

Each CORESET is related to a specific BWP and there can be more than 1 CORESET per BWP. If there is more than 1 CORESET per BWP, then multiple PDCCH PDUs should be included in the `DL_TTI.request`.

Field	Type	Description
BWP [3GPP TS 38.213 [4], sec 12]		
coreset-BWP-Size	uint16_t	Bandwidth part size [3GPP TS 38.213 [4], sec12]. Number of contiguous PRBs allocated to the BWP

Field	Type	Description
		Note: CoresetBWPStart and CoresetBWPSize are expected to correspond to BWP signalled in RRC Value: 1->275
coreset-BWP-Start	uint16_t	bandwidth part start RB index from reference CRB [3GPP TS 38.213 [4], sec 12] Note: CoresetBWPStart and CoresetBWPSize are expected to correspond to BWP signalled in RRC Value: 0->274
subcarrierSpacing	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
cyclicPrefix	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
Coreset [3GPP TS 38.211 [2], sec 7.3.2.2]		
startSymbolIndex	uint8_t	Starting OFDM symbol for the CORESET Value: 0->13
durationSymbols	uint8_t	Contiguous time duration of the CORESET in number of symbols. Corresponds to L1 parameter $N_{symb}^{CORESET}$ [3GPP TS 38.211 [2], sec 7.3.2.2] Value: 1,2,3
freqDomainResource	uint8_t [6]	Frequency domain resources. This is a bitmap defining non-overlapping groups of 6 PRBs in ascending order. [3GPP TS 38.213 [4], section 10.1]. Also, corresponds to L1 parameter $N_{\mathrm{RB}}^{\mathrm{CORESET}}$ [3GPP TS 38.211 [2], sec 7.3.2.2] Bitmap of uint8 array. 45 bits in the same representation as the RRC parameter frequencyDomainResources, with bit mapping as described in section 3.1.6, per TLV 0x0178 Notes: If this CORESET is defined by *ControlResourceSet* IE for a UE, L2/L3 signals this parameter in a way consistent with *frequencyDomainResources* IE for the *ControlResourceSet* IE for the BWP, per 38.331 [6], section 6.3.2 - If this CORESET is defined by *controlResourceSetZero* IE for a UE, L2/L3 signals this parameter in a way consistent with the RBs indicated by *controlResourceSetZero* IE for BWP#0, by setting CoresetBWPStart and CoresetBWPSize to start point and size of CORESET 0 directly, per 3GPP TS 38.213 [4], section 13. For this case, freqDomainResource always have bit value 1 at start of bitmap.

Field	Type	Description
cceRegMappingType	uint8_t	CORESET-CCE-to-REG-mapping-type [3GPP TS 38.211 [2], sec 7.3.2.2] 0: non-interleaved 1: interleaved
reg-BundleSize	uint8_t	The number of REGs in a bundle. Must be 6 for cceRegMappingType = nonInterleaved. For cceRegMappingType = interleaved, must belong to {2,6} if duration = 1,2 and must belong to {3,6} if duration = 3. Corresponds to parameter L. [3GPP TS 38.211 [2], sec 7.3.2.2] Value: 2,3,6
interleaverSize	uint8_t	The interleaver size. For interleaved mapping belongs to {2,3,6} and for non-interleaved mapping is NA. Corresponds to parameter R. [3GPP TS 38.211 [2], sec 7.3.2.2] Value: 2,3,6
coreset-Type	uint8_t	[3GPP TS 38.211 [2], sec 7.3.2.2 and sec 7.4.1.3.2] 0: CORESET is configured by the PBCH or SIB1 (subcarrier 0 of the CORESET) 1: otherwise (subcarrier 0 of CRB0 for DMRS mapping)
shiftIndex	uint16_t	[3GPP TS 38.211 [2], sec 7.3.2.2] Not applicable for non-interleaved mapping. For interleaved mapping and a PDCCH transmitted in a CORESET configured by the PBCH or SIB1 this should be set to phy cell ID. Value: 10 bits Otherwise, for interleaved mapping this is set to 0-> max num of PRBs. Value 0-> 275
precoderGranularity	uint8_t	Granularity of precoding [3GPP TS 38.211 [2], sec 7.3.2.2] 0: sameAsRegBundle 1: allContiguousRBs
num-DL-DCI	uint16_t	Number of DCIs in this CORESET. Value: 0->MaxDciPerSlot
For number of DL DCIs {		
dl-DCI	structure	See Table 3-87
}		
pdcch Maintenance Parameters added in FAPIv3	structure	See Table 3-88

Field	Type	Description
pdcch PDU parameters FAPIv4	structure	See Table 3-89
backwardCompatibleExtension-v7	structure	See Table 3-90 (see section 3.1.5)

Table 3-86 PDCCH PDU

Field	Type	Description
rnti	uint16_t	The RNTI used for identifying the UE when receiving the PDU Value: 1 -> 65535.
n-ID-PDCCH-Data	uint16_t	Parameter n_{ID} used for PDCCH Data scrambling in [3GPP TS 38.211 [2], sec 7.3.2.3]. Value: 0->65535
n-RNTI-PDCCH-Data	uint16_t	Parameter nRNTI used for PDCCH data scrambling, in [3GPP TS 38.211 [2], sec 7.3.2.3] Value: 0 -> 65535
cce-Index	uint8_t	CCE start Index used to send the DCI Value: 0->135
aggregationLevel	uint8_t	Aggregation level used [3GPP TS 38.211 [2], sec 7.3.2.1] Value: 1,2,4,8,16
Beamforming info		
Precoding and Beamforming	structure	See Table 3-111. If the PHY supports Rel-16 see also Table 3-112. For PHYs supporting FAPIv4 MU-MIMO groups, see also Table 3-113.
Tx Power info		
beta-PDCCH-1-0 [deprecated]	uint8_t	In this FAPI release, for ProfileNR, L2/L3 signals PDCCH power via powerControlOffsetSSProfileNR (next field), regardless of DCI format. PHY may ignore this beta_PDCCH_1_0.
powerControlOffset-SS-Profile-NR	int8_t	PDCCH power value used for all PDCCH Formats. This is ratio of PDCCH and PDCCH DMRS EPRE to SSB/PBCH block EPRE 3GPP TS 38.214 [5] 3GPP Spec reference (informative): - RRC parameter *powerControlOffsetSS*: NZP CSI-RS EPRES to SSB/PBCH block EPRE is given in [3GPP TS 38.214 [5], sec 5.2.2.3.1] and PDCCH EPRE to NZP CSI-RS EPRE is – in cases of link recovery – assumed as 0 dB [3GPP TS 38.214 [5], section 4.1].“UE may assume that the ratio of PDCCH DMRS EPRE to SSS EPRE is within -8 dB and 8 dB when the UE monitors PDCCHs for a DCI format 1_0 with CRC scrambled by SI-RNTI, P-RNTI, or RA-RNTI.” [3GPP TS 38.213 [4], section 4.1] Value range:

Field	Type	Description
		-8 … 8 representing -8 to 8 dB in 1dB steps -127: L1 is configured with ProfileSSS
Payload		
payloadSizeBits	uint16_t	The total DCI length (in bits) including padding bits [3GPP TS 38.212 [3], sec 7.3.1] Range 0-> DCI_PAYLOAD_BTYE_LEN*8
payload	uint8_t[DCI_PAYLOAD_BTYE_LEN]	DCI payload per [3GPP TS 38.212 [3], sec 7.3.1], where the actual size is defined by payloadSizeBits. Bit mapping is per bit significance rules, as described in section 3.1.6.1.

Table 3-87 DL DCI PDU

Field		Type	Description
Set of Coreset and DCI parameters added in FAPIv3			
Coreset parameter			
pdcch-PDU-Index-v3		uint16_t	Unique across all PDCCH PDUs in the slot. The PDCCH PDU indices signalled in a slot (according to the lowest PDSCH numerology) shall be numbered between 0 and [(the number of PDCCH PDUs in the slot)-1], in the order in which they appear in DL_TTI.request. Note: the requirement in the last paragraph means that a FAPIv5 MAC may not index PDCCH PDUs according to the expectations of a FAPIv6 PHY.
n-ID-PDCCH-DMRS-v3		uint16_t	Parameter N_{ID} used for PDCCH DMRS scrambling in [3GPP TS 38.211 [2], sec 7.4.1.3.1]. Note: this field was defined in FAPIv6 and was obsolete before that. Value: 0->65535
DCI-specific parameters			
For number of DL DCIs {			The same number of DL DCIs as
	dci-Index-v3	uint16_t	Unique across all DCIs in this PDU Value: 0→65535
	collocated-AL16-Candidate-v3	unit8_t	true if an AL-16 candidate exists at the same CCE Index, which UE expects to rate match around, per 3GPP TS 38.214 [5], section 5.1.4.1
	pdcch-DMRS-PowerOffsetProfile-SSS-v3	int16_t	Ratio of PDCCH DMRS EPRE to SSS EPRE. EPRE is defined as in section 2.2.5.5. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps -32768: ProfileNR is in use
	pdcch-DataPowerOffsetProfile-SSS-v3	int16_t	Ratio of PDCCH Data EPRE to SSS EPRE. EPRE is defined as in section 2.2.5.4. Values:

Field		Type	Description
			-32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps -32768: L1 determines PDCCH data power from the PDCCH dmrs power.
}			

Table 3-88 PDCCH PDU maintenance FAPIv3

Field		Type	Description
Set of Coreset and DCI parameters added in FAPIv4			
DCI-specific parameters			
For number of DL DCIs {			
	<obsolete>	uint16_t	Obsolete in FAPIv6
}			
MU-MIMO support in FAPIv4			
numSpatialStreamIndices-v4		uint16_t	The same number of DL DCIs, or zero. Must match the number of DL DCIs, if PHY indicates that it requires DL spatial stream index assignment, via capability maxNumberDlSpatialStreams (TLV 0x0151), may be set to zero otherwise.
For numSpatialStreamIndices DL DCIs {			this loop is present if and only if numSpatialStreamIndices > 0
	dci-Index-v4	uint16_t	Unique across all DCIs in this PDU
	spatialStreamIndex-v4	uint16_t	Spatial Stream index for mapping of PDCCH, based on PHY capability indicating need for spatial stream index signalling Value: 0 → (max # spatial streams – 1, per TLV 0x0151)
}			

Table 3-89 PDCCH PDU parameters FAPIv4

Field	Type	Description
Backward compatible extension (first introduced in FAPIv7). See section 3.1.5.		
Extension Length	uint16_t	Size of this backward compatible extension (introduced in FAPIv7). See section 3.1.5 for what this length represents.

Table 3-90 PDCCH PDU Extension

3.4.2.2 PDSCH PDU

The format of the PDSCH PDU is shown in Table 3-91.

The PDSCH PDU includes both mandatory and optional parameters, where the presence of the optional elements is defined in the parameter pduBitmap. The presence of these optional elements is included based on the following:

- pdschPtrs and pdschPtrsV3 are included if and only if PTRS are included in the downlink transmission
- cbgReTxCtrl is included if CBG is supported and retransmit is used.

(It should be noted that some parameters are only applicable for certain modes/types etc. These are still included but their value is ignored by the PHY.)

Field	Type	Description
pdu-Bitmap	uint16_t	Bitmap indicating presence of optional PDUs Bit 0: pdschPtrs and pdschPtrsV3 – Indicates PTRS included Bit 1:cbgRetxCtrl (Present when CBG based retransmit is used.) All other bits reserved
rnti	uint16_t	The RNTI used for identifying the UE when receiving the PDU Value: 1 -> 65535
pdu-Index	uint16_t	PDU index incremented for each PDSCH PDU sent in TX control message. This is used to associate control information to data and is reset every slot. Value: 0 -> 65535
BWP [3GPP TS 38.213 [4], section 12]		
bwp-Size	uint16_t	If L1 uses pdschTransType, this field signals the bandwidth part size [3GPP TS 38.213 [4], sec 12]. Number of contiguous PRBs allocated to the BWP If L1 does not use pdschTransType, this field represents the number of VRBs to which PRBs are mapped for the allocation, according to 3GPP TS 38.211 [2], section 7.3.1.6. Note: The two interpretations align, except when the allocation is scheduled by DCI format 1_0 in CSS. Value: 1->275
bwp-Start	uint16_t	If L1 uses pdschTransType, this field signals the bandwidth part start RB index from reference CRB [3GPP TS 38.213 [4], sec 12] If L1 does not use pdschTransType, this field is represents the CRB corresponding to VRB0, according to 3GGP TS 38.211 [2], section 7.3.1.6. Note: The two interpretations align, except when the allocation is scheduled by DCI format 1_0 in CSS. Value: 0->274
subcarrierSpacing	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4

Field	Type	Description
cyclicPrefix	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
Codeword information		
numCodewords	uint8_t	Number of code words for this RNTI (UE) Value: 1 -> 2
For each codeword {		
targetCodeRate	uint16_t	Target coding rate [3GPP TS 38.212 [3], sec 5.4.2.1 and 3GPP TS 38.214 [5], sec 5.1.3.1]. This is the number of information bits per 1024 coded bits expressed in 0.1 bit units
qam-ModOrder	uint8_t	QAM modulation [3GPP TS 38.212 [3], sec 5.4.2.1 and 3GPP TS 38.214 [5], sec 5.1.3.1] Value: 2,4,6,8
mcs-Index	uint8_t	MCS index [3GPP TS 38.214 [5], sec 5.1.3.1], should match value sent in DCI Value : 0->31
mcs-Table	uint8_t	MCS-Table-PDSCH [3GPP TS 38.214 [5], sec 5.1.3.1] 0: notqam256 1: qam256 2: qam64LowSE
rv-Index	uint8_t	Redundancy version index [3GPP TS 38.212 [3], Table 5.4.2.1-2 and 3GPP TS 38.214 [5], Table 5.1.2.1-2], should match value sent in DCI Value : 0->3
tb-Size	uint32_t	Transport block size (in bytes) [3GPP TS 38.214 [5], sec 5.1.3.2]
}		
nID-PDSCH	uint16_t	Parameter n_{ID} from [3GPP TS 38.211 [2], sec 7.3.1.1] Value: 0->1023
numLayers	uint8_t	Number of layers [3GPP TS 38.211 [2], sec 7.3.1.3] Value : 1->8
transmissionScheme	uint8_t	PDSCH transmission schemes [3GPP TS 38.214 [5], sec 5.1.1]

Field	Type	Description
		0: Up to 8 transmission layers
refPoint	uint8_t	Reference point for PDSCH DMRS "k" – used for tone mapping [3GPP TS 38.211 [2], sec 7.4.1.1.2] Resource block bundles [3GPP TS 38.211 [2], sec 7.3.1.6] Value: 0 -> 1 If 0, the 0 reference point for PDSCH DMRS is at Point A [3GPP TS 38.211 [2], sec 4.4.4.2]. Resource block bundles generated per sub-bullets 2 and 3 in [3GPP TS 38.211 [2], sec 7.3.1.6]. For sub-bullet 2, the start of bandwidth part must be set to the start of actual bandwidth part +*NstartCORESET* and the bandwidth of the bandwidth part must be set to the bandwidth of the initial bandwidth part. If 1, the DMRS reference point is for PDSCH DM-RS is subcarrier 0 of the lowest numbered resource block bandwidth part which given by the field BWPStart. L2/L3 shall only set this value to 1 for PDSCH PDU carrying payload of RMSI. L2/L3 shall ensure that BWP start corresponds to start of a CORESET#0 Resource block bundles generated per sub-bullets 1 [3GPP TS 38.211 [2], sec 7.3.1.6]
DMRS [3GPP TS 38.211 [2], sec 7.4.1.1]		
dl-DMRS-SymbPos	uint16_t	DMRS symbol positions [3GPP TS 38.211 [2], sec 7.4.1.1.2 and Tables 7.4.1.1.2-3 and 7.4.1.1.2-4] Bitmap occupying the 14 LSBs with: bit 0: first symbol and for each bit 0: no DMRS 1: DMRS
dmrs-ConfigType	uint8_t	DL DMRS config type [3GPP TS 38.211 [2], sec 7.4.1.1.2] 0: type 1 1: type 2
pdsch-DMRS-ScramblingId	uint16_t	PDSCH DMRS-Scrambling-ID [3GPP TS 38.211 [2], sec 7.4.1.1.1] as provided by parameter $N_{ID}^{n_{SCID}}$. For example, in reference [12], this field applies to below case: 1. PDSCH transmission with Rel-15 DM-RS configuration 2. PDSCH transmission having DM-RS port(s) within CDM group 0 or 2 with Rel-16 DM-RS configuration

Field	Type	Description
		Value: 0->65535
pdsch-DMRS-ScramblingIdComplement-v4	uint16_t	PDSCH ~~DL-~~DMRS-Scrambling-ID [3GPP TS 38.211 [2], sec 7.4.1.1.1 ~~7.4.1.1.2~~] as provided by parameter $N_{ID}^{\bar{n}_{SCID}^{\lambda}}$ (Rel-16) when $\bar{n}_{SCID}^{\lambda} \neq n_{SCID}$ PHY shall disregard this parameter if lowPaprDmrs = 0 For example, in reference [12], this is the scrambling ID associated with CDM group 1 for PDSCH DM-RS sequence generation and only valid if DM-RS port(s) of the CDM group indicated by antenna ports field in DCI Value: 0->65535
lowPAPR-DMRS-v4	uint8_t	**1:** $\bar{n}_{SCID}^{\lambda}$ depends on DM-RS CDM group **0:** $\bar{n}_{SCID}^{\lambda} = n_{SCID}$ (i.e. independent of DM-RS CDM group)
nSCID	uint8_t	DMRS sequence initialization [3GPP TS 38.211 [2], sec 7.4.1.1.2]], as provided by parameter n_{SCID}. For example, in reference [12], this value is associated with DM-RS scrambling ID given by dlDmrsScramblingId Value : 0->1
numDMRS-CDM-GrpsNoData	uint8_t	Number of DM-RS CDM groups without data [3GPP TS 38.212 [3], sec 7.3.1.2.2] [3GPP TS 38.214 [5], Table 4.1-1] it determines the ratio of PDSCH EPRE to DM-RS EPRE. Value: 1->3
dmrs-Ports	uint16_t	DMRS ports. [3GPP TS 38.212 [3], 7.3.1.2.2] provides description between DCI 1-1 content and DMRS ports. Bitmap occupying the 12 LSBs with: bit 0: antenna port 1000 bit 11: antenna port 1011 and for each bit 0: DMRS port not used 1: DMRS port used
PDSCH allocation in frequency domain [3GPP TS 38.214 [5], sec 5.1.2.2]		
resourceAlloc	uint8_t	Resource Allocation Type [3GPP TS 38.214 [5], sec 5.1.2.2] 0: Type 0 1: Type 1
rb-Bitmap	uint8_t [36]	For resource alloc type 0.

Field	Type	Description
		3GPP TS 38.212 [3], section 7.3.1.2.2 bitmap of RBs, 273 rounded up to multiple of 32. This bitmap is in units of VRBs. Bit mapping as described in section 3.1.6, per TLV 0x0177. VRB to PRB mapping is per section 7.3.1.6 of 3GPP TS 38.211 [2].
rb-Start	uint16_t	For resource allocation type 1. [3GPP TS 38.214 [5], sec 5.1.2.2.2] The starting resource block corresponding to VRB 0 for this PDSCH, per section 7.3.1.6 of 3GPP TS 38.211 [2]. Value: 0->274
rb-Size	uint16_t	For resource allocation type 1. [3GPP TS 38.214 [5], sec 5.1.2.2.2] The number of resource blocks within for this PDSCH. Value: 1->275
vrb-ToPRB-Mapping	uint8_t	VRB-to-PRB-mapping [3GPP TS 38.211 [2], sec 7.3.1.6] 0: non-interleaved 1: interleaved with RB size 2 2: Interleaved with RB size 4
Resource Allocation in time domain [3GPP TS 38.214 [5], sec 5.1.2.1]		
startSymbolIndex	uint8_t	Start symbol index of PDSCH mapping from the start of the slot, S. [3GPP TS 38.214 [5], Table 5.1.2.1-1] Value: 0->13
numSymbols	uint8_t	PDSCH duration in symbols, L [3GPP TS 38.214 [5], Table 5.1.2.1-1] Value: 1->14
PTRS Parameters (existing in FAPIv2)		
pdsch-PTRS	structure	See Table 3-92; this structure is included if and only if pduBitmap[0] = 1. It shall only apply to the first PTRS port.
Spatial Multiplexing		
Precoding and Beamforming	structure	See Table 3-111. If the PHY supports Rel-16 see also Table 3-112. For PHYs supporting FAPIv4 MU-MIMO groups, see also Table 3-113.
Tx Power info		
powerControlOffsetProfile-NR-v3	uint8_t	Ratio of PDSCH EPRE to NZP CSI-RS EPRE [3GPP TS 38.214 [5], sec 5.2.2.3.1] Values: 0->23: representing -8 to 15 dB in 1dB steps

Field	Type	Description
		255: L1 is configured with ProfileSSS
powerControlOffset-SS ProfileNR-v3	uint8_t	Ratio of NZP CSI-RS EPRE to SSB/PBCH block EPRE to [3GPP TS 38.214 [5], sec 5.2.2.3.1] Values: 0: -3dB, 1: 0dB, 2: 3dB, 3: 6dB 255: L1 is configured with ProfileSSS
cbgReTxCtrl: CBG fields for Segmentation in L2. This structure is only included if and only if (pduBitmap[1] = 1 and L1 is configured to support L2 CBG segmentation)		
isLast-CB-Present	uint8_t	Indicates whether the last code block is presented in the CBG re-transmission or not. Bitmap of length 8. LSB 0 is signalled for 1st TB and LSB1 for 2nd TB respectively. LSB 0: 0: Last code block is not presented for 1st TB (e.g. last CB is not present in the CBG-based re-Tx or 1st TB is initial Tx.) 1: Last code block is presented for 1st TB LSB 1: 0: Last code block is not presented for 2nd TB (e.g. last CB is not present in the CBG-based re-Tx or 2nd TB is initial Tx.). 1: Last code block is presented for 2nd TB If CBG Re-Tx includes last CB, L1 will add the TB CRC to the last CB in the payload before it is read into the LDPC HW unit
isInline-TB-CRC	uint8_t	Indicates whether TB CRC is part of data payload or control message 0: TB CRC is part of data payload 1: TB CRC is part of control message This field applies to any and all CWs for which CRC is required, per IsLastCbPresent
dl-TB-CRC-CW-v3	uint32_t [2]	TB CRC: to be used in the last CB, applicable only if last CB is present dlTbCrcCW[0]: CRC for 1st CW dlTbCrcCW[1]: CRC for 2nd CW Crc is defined the same way as in `DL_TTI.response`
PDSCH Maintenance Parameters added in FAPIv3	structure	See Table 3-93
pdschPtrsV3: PDSCH PTRS Maintenance	structure	See Table 3-94 this structure is included if and only if pduBitmap[0] = 1

Field	Type	Description
Parameters added in FAPIv3		
Rel-16 PDSCH Parameters added in FAPIv3	structure	See Table 3-95
PDSCH Parameters added in FAPIv4	structure	See Table 3-96
PDSCH Parameters added in FAPIv5	structure	See Table 3-97
backwardCompatibleExtension-v7	structure	See Table 3-98 (see section 3.1.5)

Table 3-91 PDSCH PDU

Field	**Type**	**Description**
PTRS [3GPP TS 38.214 [5], sec 5.1.6.3]		
ptrs-PortIndex	uint8_t	PT-RS antenna ports [3GPP TS 38.214 [5], sec 5.1.6.3] [3GPP TS 38.211 [2], table 7.4.1.2.2-1] Bitmap occupying the 6 LSBs with: bit 0: antenna port 1000 bit 5: antenna port 1005 and for each bit 0: PTRS port not used 1: PTRS port used
ptrs-TimeDensity	uint8_t	PT-RS time density [3GPP TS 38.214 [5], table 5.1.6.3-1] 0: 1 1: 2 2: 4
ptrs-FreqDensity	uint8_t	PT-RS frequency density [3GPP TS 38.214 [5], table 5.1.6.3-2] 0: 2 1: 4
ptrs-RE-Offset	uint8_t	PT-RS resource element offset [3GPP TS 38.211 [2], table 7.4.1.2.2-1] Value: 0->3
nEPRE-RatioOf-PDSCH-To-PTRS-Profile-NR-v3	uint8_t	PT-RS-to-PDSCH EPRE ratio [3GPP TS 38.214 [5], table 4.1-2] Value : 0->3: index into table 4.1-2 of 3GPP TS 38.214 [5] 255: DL-PTRS uses SSS for power offset

Table 3-92 PDSCH PTRS parameters

Field	**Type**	**Description**
Set of PDSCH PDU parameters added in FAPIv3		

Field	Type	Description
BWP information		
pdsch-TransType-v3	uint8_t	L1 interprets this field as below, if so configured by pdschTransTypeValidity=1, otherwise it ignores it. Indication used to indicate the transmission type of PDSCH PDU. Value 0: non-interleaved PDSCH which is scheduled by DCI format 1_0 in a common search space Value 1: any non-interleaved PDSCH except above case Value 2: interleaved PDSCH which is scheduled by PDCCH DCI format 1_0 in Type0-PDCCH common search space in CORESET 0. If pdschTransType set to 2, then BWP start and size of the PDSCH PDU shall be set to CORESET 0 start and size respectively instead of active downlink BWP start and size. Value 3: interleaved PDSCH which is scheduled by PDCCH DCI format 1_0 in any common search space other than Type0-PDCCH common search space in CORESET 0 when CORESET 0 is configured for the cell Value 4: interleaved PDSCH which is scheduled by PDCCH DCI format 1_0 in any common search space other than Type0-PDCCH common search space in CORESET 0 when CORESET 0 is not configured for the cell Value 5: any interleaved PDSCH which not fall into above 3 interleaved PDSCH categories.
Coreset-StartPoint-v3	uint16_t	L1 interprets this field as below, if so configured by pdschTransTypeValidity=1, otherwise it ignores it. The PRB index of lowest-numbered RB in the CORESET in which PDCCH carrying scheduling info for the PDSCH PDU is received. Refer to 3GPP TS 38.211 [2], section 7.3.1.6 $N_{start}^{CORESET}$. Only valid when pdschTransType field set to 0, 3, 4 Value: 0 to 274
initial-DL-BWP-Size-v3	uint16_t	L1 interprets this field as below, if so configured by pdschTransTypeValidity=1, otherwise it ignores it. The size of initial downlink BWP used for the cell, shall be set to size of CORESET 0 if CORESET 0 is configured for the cell, and initial downlink BWP otherwise. Refer to TS 3GPP 38.211 [2], section 7.3.1.6 $N_{BWP,init}^{size}$. Only valid when pdschTransType field set to 3, 4 value 0 to 274
Codeword information		
ldpc-BaseGraph-v3	uint8_t	LDPC base graph to use for CW generation [3GPP TS 38.212 [3], sec 7.2.2]. Values: 1: LDPC base graph 1

Field	Type	Description
		2: LDPC base graph 2 (other values are reserved)
tb-Size-LBRM-Bytes-v3	uint32_t	Parameter TBS_{LBRM} from 3GPP TS 38.212 [3], section 5.4.2.1, for computing the size of the circular buffer. Value 0 is reserved.
Tb-CRC-Required-v3	uint8_t	Indicates whether L1 should report TB CRC for the PDSCH PDU's TB(s). Bitmap of 8 length and LSB 0 and 1 are valid. LSB 0: 0: TB CRC of 1st TB not required 1: TB CRC of 1st TB required LSB 1: 0: TB CRC of 2nd TB not required 1: TB CRC of 2nd TB required CRC report, if any shall be performed via `DL_TTI.response` (see section 3.4.2b).
Rate Matching references		
ssb-PDUs-For-RM-v3	uint16_t [8]	the SSB PDU indices in the slot, whose RBs are not available for PDSCH allocation. Value < 256 an SSB PDU index 65534: rate matching is indicated only via ssb Configuration Index 65535: no SSB PDU to rate match around, in this position, for this PDSCH allocation; **Notes**: There can be up to eight 240 kHz SSBs to be rate matched around by a PDSCH PDU allocation with SCS 60 kHz.
ssb-ConfigForRM-v3	uint16_t	the SSB PDU indices in the slot, whose RBs are not available for PDSCH allocation. Value < 256 a SSB configuration index. 65534: rate matching is indicated only via ssb PDU Index 65535: no SSB rate matching for this PDSCH allocation
prb-Sym-RM-PatternBitmapSizeByReference-v3	uint8	size of by-reference bit map of Rm Patterns, in bits (set to 0 if L1 is not configured with by-reference rate match patterns.) The size of the bitmap is determined as szRm = ceil(prbSymRmPatternBitmapSizeByReference / 8)
prb-Sym-RM-PatternBitmapByReference-v3	uint8_t [szRm]	each bit maps to one RB-based Rate Match Pattern, bit set to 1 represents rate matching for this RB-based RMP resources, bit set to 0 represents no rate matching for this RB-based RMP resources. If prbSymbRmConfigurationMethod is absent or indicates P5-based references, then this bitmap

Field	Type	Description
		refers to indices in P5 TLV 0x0137 (see Table 3-61), otherwise it refers to the loop indices in prbSymbP7Loop for this `DL_TTI.request`.
num-PRB-Sym-RM-PatternsByValue-v3	uint8	At most 8 patterns may be signalled (set to 0 if L1 is not configured with by-value rate match patterns.) Shall be set to 0 in this release
For num-PRB-Sym-RM-PatternsByValue		
freqDomainRB-v3	uint8_t [35]	Corresponds to 275 RBs interpreted as resourceBlocks parameter in 3GPP TS 38.331 [6], RateMatchPattern IE definition, with bit mapping as described in section 3.1.6, per TLV 0x0178.
symbolsIn-RB-v3	uint16_t	Patterns of symbols to rate match around, with bit map formatted the same as in the oneSlot parameter in 3GPP TS 38.331 [6], RateMatchPattern IE definition, each bit indicating: 0: no rate match 1: rate match Bit mapping as described in section 3.1.6, per TLV 0x0178.
pdcch-PDU-Index-v3	uint16_t	PDCCH PDU index for rate matching Set to 65535 if not applicable.
dci-Index-v3	uint16_t	DCI index in the PDCCH PDU indexm for rate matching Set to 65535 if not applicable.
lte-CRS-RM-PatternBitmapSize-v3	uint8_t	Size of bitmap signalling the LTE CRS pattern(s) to apply. Rel-15: a single pattern per UE Cf. 3GPP TS 38.331 [6], section 6.2.2 Rel-16: multiple patterns supported per UE Cf. 3GPP TS 38.331 [13], section 6.2.2 The of the bitmap array is determined as szCrs = ceil(lteCrsRateMatchPatternBitmapSize / 8)
lte-CRS-RM-Pattern-v3	uint8_t [szCrs]	Each bit map to one CRS Rate Match Pattern, bit set to 1 represents rate matching for this CRS pattern, bit set to 0 means no rate matching for this CRS pattern. If lteCrsRmConfigurationMethod is absent or indicates P5-based references, then this bitmap refers to indices in P5 TLV 0x0138 (Table 3-63), otherwise it refers to the loop indices in lteCrsP7Loop for this `DL_TTI.request`.
num-CSI-RS-ForRM-v3	uint8_t	Number of CSI-RS PDUs that signal REs to rate match around for this PDSCH PDU allocation
csi-RS-For-RM-v3	uint16_t [szCsiRs]	Array of szCsiRs = numCsiRsForRateMatching CSI-RS PDU indices to rate match around
Tx Power info		

Field	Type	Description
pdsch-DM-PowerOffsetProfile-SSS-v3	int16_t	Ratio of PDSCH DMRS EPRE to SSS EPRE. EPRE is defined as in section 2.2.4.5. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps -32768: ProfileNR is in use
pdsch-DataPowerOffsetProfile-SSS-v3	int16_t	Ratio of PDSCH DMRS EPRE to SSS EPRE. EPRE is defined as in section 2.2.5.4. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps -32768: L1 determines PDSCH data power from PDSCH dmrs power 3GPP reference: 3GPP TS 38.214 [5], section 4.1
cbgReTxCtrlL2Seg: CBG fields for Segmentation in L2. This structure is only included if and only if pduBitmap[1] = 1 and L1 is configured to support L2 CBG segmentation)		
cbgReTxCtrlL1Seg: CBG fields for Segmentation in L1 This structure is only included if and only if pduBitmap[1] = 1 and L1 is configured to support L1 CBG segmentation)		
maxNum-CBG-PerTB-v3	uint8_t	Valid bits in the CbgTxInformation field per TB for associated PDSCH payload Value 2,4,6,8 For single TB/CW case, maximum value could be 8; For double TB/CW case, maximum value restricted to 4;
For each codeword {		
cbg-TX-Information-v3	uint8_t	Bitmap with size of 8 bits used to indicate which CBG(s) is/are (re)transmitted. Actual valid bits depend on above parameter maxNumCbgPerTb and numCodewords. LSB Bit 0 corresponding to first CBG in the TB, bit 1 corresponding to second CBG in the TB and so on.

Table 3-93 PDSCH maintenance parameters V3

Field	Type	Description
PDSCH-PTRS parameter added in FAPIv3		
Tx Power info		
pdsch-PTRS-PowerOffsetProfile-SSS-v3	int16_t	Ratio of PDSCH PT-RS EPRE to SSS EPRE. EPRE is defined as in section 2.2.5.4. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps -32768: L1 determines PT-RS data power from PDSCH dmrs power 3GPP reference: 3GPP TS 38.214 [5], section 4.1

Table 3-94 PDSCH PTRS maintenance parameters V3

Field	Type	Description
Rel-16 parameters added in FAPIv3		
repetitionScheme-v3	uint8_t	Signals the Rel-16 single-DCI mTRP repetition scheme. Description: If any fdmSchemeB scheme is signalled: - numCodewords = 2 - Each codeword uses numLayers/2 layers, per the mapping in 3GPP TS 38.214 [5], section 7.3.1 - Per CW RV determination is as described in 3GPP TS 38.214 [5], section 5.1.2.3. - Per CW PRGs allocation is as described in 3GPP TS 38.214 [5], section 5.1.2.3. Values: 1: fdmSchemeB-Prg2 2: fdmSchemeB-Prg4 3: fdmSchemeB-PrgWideband 255: repetition scheme not relevant [other values reserved]
PTRS Parameters this section is included if and only if pduBitmap[0] = 1		
pdsch-PTRS2-v3	structure	See Table 3-92. If the PDSCH is setup to use two PTRS ports, as described in 3GPP TS 38.214 [5], section 5.1.6.4, then PTRSPortIndex in pdschPtrs may be non-zero, otherwise it is all-zero
pdsch-PTRS2-Maintenance-v3	structure	See Table 3-94 this structure is included if and only if pduBitmap[0] = 1 and applies to pdschPtrs2

Table 3-95 Rel-16 PDSCH parameters V3

Field	Type	Description
coreset-RM-PatternBitmapSizeByReference-v4	uint8	size of by-reference bit map of CORESET-based Rm Patterns, in bits (set to 0 if L1 is not configured with by-reference coreset RM patterns.) The size of the bitmap is determined as szRm = ceil(coresetRmPatternBitmapSizeByReference / 8)
coreset-RM-PatternBitmapByReference-v4	uint8_t [szRm]	each bit maps to one CORESET-based Rate Match Pattern, bit set to 1 represents rate matching for this Coreset-based RMP resources, bit set to 0 represents no rate matching for this Coreset-based RMP resource. If coresetRmConfigurationMethod is absent or indicates P5-based references, then this bitmap refers to indices in P5 TLV 0x1041 (see Table 3-62), otherwise it refers to the loop indices in coresetRmP7Loop for this `DL_TTI.request`.

Field	Type	Description
lte-CRS-MBSFN-DerivationMethod-v4	uint8_t	Signals how L1 should derive whether a LTE-CRS pattern overlaps should be MBSFN or non-MBSFN. Value: • 0: Based on P5 signalling (in this case, szCrs-Mbsfn = 0) • 1: Based on lteCrsMbsfnPattern (in this case, szCrs-Mbsfn = szCrs)
lte-CRS-MBSFN-Pattern-v4	uint8_t [szCrs-Mbsfn]	Each bit maps to one CRS Rate Match Pattern, following the same indexing as lteCrsRateMatchPattern. For each bit: * Value = 0: either of the following is true: - thjs slot overlaps with a non-MBSFN subframe *or* - same-index bit in lteCrsRateMatchPattern is 0 (i.e. no rate matching for the corresponding LTE-CRS pattern) * Value = 1: this slot overlaps with an MBSFN subframe for the LTE-CRS pattern corresponding to this index position. **Note**: This field is non-empty if and only if mbsfnDerivationMethod = 1
MU-MIMO support in FAPIv4		
numberCodewords-v4	uint8_t	Set to the number codewords, or zero. Must be set to the number codewords if PHY indicates that it requires DL spatial stream index assignment, via capability maxNumberDlSpatialStreams (TLV 0x0151), otherwise it may be set to 0. Value: 0 … 2
For each CW 0 … numberCodewords - 1)		This for loop is present iff numberCodewords > 0
numSpatialStreamIndicesFor-CW-v4	uint8_t	Number of spatial streams mapping to the CW for the loop. • If TLV 0x016E is set to 3, this is the same as the number of layers; • If TLV 0x016E is set to 2, this is the same as the number of logical antenna ports, if supported by the precoding, otherwise is the same as the number of layers; • If TLV 0x016E is set to 1, this is the same as the number of baseband ports.
spatialStreamIndicesFor-CW-v4	uint16_t [#spatial streams for CW]	Spatial Stream indices for mapping of PDSCH layers, based on PHY capability indicating need for layer index signalling. For reciprocity-based MU-MIMO, these spatial streams correspond to CW layers for each layerMapping-PRG in the MU-MIMO prb-Bitmap (see

Field	Type	Description
		section 3.4.2), in the same order as they appear in the chosenLayersBitmap, for each PRG. E.g. chosenLayerBitmapForCW0 may take value 00000011_b for PRG-A and 00010100_b for PRG-B. In this example: - CW0 spatial stream#0 (index spatialStreamIndicesForCW[0]) uses singular value#0 (=bit position for layer#0) over PRG-A and singular#2 (=bit position for layer#0) for PRG-B - CW0 spatial stream#1 (index spatialStreamIndicesForCW[1]) uses singular value#1 (=bit position for layer#1) over PRG-A and singular value#4 (=bit position for layer#1) for PRG-B If the spatial stream assignment scope is '1' (per TLV 0x016E) the order of indices corresponds to the index of the baseband antenna dimension. If the spatial stream assignment scope is '2' (per TLV 0x016E) the order of indices corresponds to the index of the logical antenna dimension. If the spatial stream assignment scope is '3' (per TLV 0x016E) , the order of the indices corresponds to index of the layer dimension in the Precoding Matrix for the Precoding and Beamforming PDU. Value: 0 → (max # spatial streams-1, per TLV 0x0151)

Table 3-96 PDSCH parameters FAPIv4

Field	Type	Description
Rel-16 parameters added in FAPIv5		
num-PRS-ForPuncturing-v5	uint8_t	Number of PRS PDUs that signal REs to puncture for this PDSCH PDU allocation
prs-ForPuncturing-v5	uint16_t [szPrs]	Array of szPrs = numPrsForPuncturing PRS PDU indices to rate match around

Table 3-97 PDSCH parameters FAPIv5

Field	Type	Description
Backward compatible extension for PDSCH PDU. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.
Spatial Multiplexing		
prg-Size-v7	uint8_t	Precoding granularity (P'_{BWP}) for the PRB Bundling, for this PDSCH allocation (see 3GPP TS 38.214 [5], section 5.1.2.3). Values: • 0: Wideband • 1: 2 RBs • 2: 4 RBs
PDSCH allocation in frequency domain [3GPP TS 38.214 [5], sec 5.1.2.2]		

Field	Type	Description
rbg-Size-v7	uint8_t	Resource Block Group: allocation resolution *P* for this PDSCH allocation (see 3GPP TS 38.214 [5], section 5.1.2.2) • 0: 2 RBs • 1: 4 RBs • 2: 8 RBs • 3: 16 RBs

Table 3-98 PDSCH PDU Extension

3.4.2.3 CSI-RS PDU

The format of the CSI-RS PDU is shown in Table 3-99.

Field	Type	Description
BWP [3GPP TS 38.213 [4], sec 12]		
subcarrierSpacing	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
cyclicPrefix	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
start-RB	uint16_t	PRB where this CSI resource starts from index from reference CRB [3GPP TS 38.213 [4], sec 12]. Only multiples of 4 are allowed. [3GPP TS 38.331 [6], sec 6.3.2 parameter CSI-FrequencyOccupation] Value: 0 -> 2473 = 2199+274
num--RBs	uint16_t	Number of PRBs across which this CSI resource spans. Only multiples of 4 are allowed. [3GPP TS 38.331 [6], sec 6.3.2 parameter CSI-FrequencyOccupation] Value: 24 -> 276
csi-Type	uint8_t	CSI Type [3GPP TS 38.211 [2], sec 7.4.1.5] Value: 0:TRS 1:CSI-RS NZP 2:CSI-RS ZP
row	uint8_t	Row entry into the CSI Resource location table. [3GP TS 38.211 [2], sec 7.4.1.5.3 and table 7.4.1.5.3-1] Value: 1-18
freqDomain	uint16_t	Bitmap defining the frequencyDomainAllocation [3GPP TS 38.211 [2], sec 7.4.1.5.3] [3GPP TS 38.331 [6] CSI-Resource Mapping]

Field	Type	Description
		Value: Up to the 12 LSBs, actual size is determined by the Row parameter
symb-L0	uint8_t	The time domain location l0 and firstOFDMSymbolInTimeDomain [3GPP TS 38.211 [2], sec 7.4.1.5.3] Value: 0->13
symb-L1	uint8_t	The time domain location l1 and firstOFDMSymbolInTimeDomain2 [3GPP TS 38.211 [2], sec 7.4.1.5.3] Value: 2->12, or 255 (i.e. symb-L1 is not signalled)
cdm-Type	uint8_t	The cdm-Type field [3GPP TS 38.211 [2], sec 7.4.1.5.3 and table 7.4.1.5.3-1] Value: 0: noCDM, 1: fd-CDM2, 2: cdm4-FD2-TD2, 3: cdm8-FD2-TD4
freqDensity	uint8_t	The density field, p and comb offset (for dot5). [3GPP TS 38.211 [2], sec 7.4.1.5.3 and table 7.4.1.5.3-1] Value: 0: dot5 (even RB), 1: dot5 (odd RB), 2: one, 3: three
scramb-ID	uint16_t	ScramblingID of the CSI-RS [3GPP TS 38.214 [5], sec 5.2.2.3.1] Value: 0->1023
Tx Power Info		
powerControlOffsetProfile-NR	uint8_t	Ratio of PDSCH EPRE to NZP CSI-RSEPRE [3GPP TS 38.214 [5], sec 5.2.2.3.1] Value :0->23 representing -8 to 15 dB in 1dB steps 255: L1 is configured with ProfileSSS
powerControlOffset-SS-Profile-NR	uint8_t	Ratio of NZP CSI-RS EPRE to SSB/PBCH block EPRE [3GPP TS 38.214 [5], sec 5.2.2.3.1] Values: 0: -3dB, 1: 0dB, 2: 3dB, 3: 6dB 255: L1 is configured with ProfileSSS
Beamforming		

Field	Type	Description
Precoding and Beamforming	structure	See Table 3-111. If the PHY supports Rel-16 see also Table 3-112. For PHYs supporting FAPIv4 MU-MIMO groups, see also Table 3-113.
CSI-RS Maintenance Parameters added in FAPIv3	structure	See Table 3-100
CSI-RS Parameters added in FAPIv4	structure	See Table 3-101
backwardCompatibleExtension-v7	structure	See Table 3-102 (see section 3.1.5)

Table 3-99 CSI-RS PDU

Field	Type	Description
Set of CSI-RS PDU parameters added in FAPIv3		
Basic		
csi-RS-PDU-Index-v3	uint16_t	Unique across all CSI-RS PDUs in the slot
Tx Power info		
csi-RS-PowerOffsetProfile-SSS-v3	int16_t	Ratio of CSI-RS EPRE to SSS EPRE. EPRE is defined as in section 2.2.4.5. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps -32768: ProfileNR is in use

Table 3-100 CSI-RS PDU maintenance parameters from FAPIv3

Field	Type	Description
MU-MIMO support in FAPIv4		
numSpatialStreamIndices-v4	uint8_t	Set to the number of ports in the cdm group, or zero. Must be set to set to the number of ports in the cdm group if PHY indicates that it requires DL spatial stream index assignment, via capability maxNumberDlSpatialStreams (TLV 0x0151), may be set to zero otherwise.
spatialStreamIndices-v4	uint8_t [n]	n = numSpatialStreamIndices Spatial Stream indices for mapping of CSI ports in the cdm group, based on PHY capability indicating need for spatial stream index signalling. In this release, the order of the indices corresponds to index of the layer dimension in the Precoding Matrix for the Precoding and Beamforming PDU. Value: 0 → (max # spatial streams - 1, per TLV 0x0151)

Table 3-101 CSI-RS PDU parameters for FAPIv4

Field	Type	Description
This is a backward compatible extension of CSI RS PDU		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5.
bwp-Size-v7	uint16_t	Bandwidth part size [3GPP TS 38.213 [4], sec 12], i.e. number of contiguous PRBs allocated to the BWP in which CSI-RS corresponding to this PDU is transmitted. This field is only relevant to PHYs supporting TLV 0x0179, otherwise PHY ignores this field. Value: • 1->275 (for PHYs supporting TLV 0x0179) • 0xFFFF (for PHYs not supporting TLV 0x0179)
bwp-Start-v7	uint16_t	Bandwidth part start RB index from reference CRB [3GPP TS 38.213 [4], sec 12]. This field is only relevant to PHYs supporting TLV 0x0179, otherwise PHY may ignore this field. Value: • 0->274 (for PHYs supporting TLV 0x0179) • 0xFFFF (for PHYs not supporting TLV 0x0179)

Table 3-102 CSI-RS PDU Backward Compatible Extension

3.4.2.4 SSB PDU

The format of the SSB/BCH PDU is shown in Table 3-103.

The SSB/BCH PDU has several options for the payload length and contents these are:

- If the MAC generates the full PBCH payload, the payload length is 32 bits and Table 3-105 defines payload
- If the PHY generates the timing PBCH information, the payload length is 24 bits and Table 3-105 defines payload
- If the PHY generates the full PBCH payload, Table 3-106 defines the payload
- The choice of PBCH payload generation will be determined when the PHY is configured.

Field	Type	Description
physCell-ID	uint16_t	Physical cell ID. [3GPP TS 38.211 [2], sec 7.4.2.1] N_{ID}^{cell} Value: 0->1007
beta-PSS-Profile-NR	uint8_t	PSS EPRE to SSS EPRE in a SS/PBCH block [3GPP TS 38.213 [4], sec 4.1] Values: 0 = 0dB 1 = 3dB 255: power offset is conveyed in

Field	Type	Description
		betaPssProfileSSS
ssb-BlockIndex	uint8_t	SS/PBCH block index within a SSB burst set [3GPP TS 38.211 [2], section 7.3.3.1]. Required for PBCH DMRS scrambling. Value: 0->63 (L_{max})
ssb-SubcarrierOffset	uint8_t	ssbSubcarrierOffset or k_{SSB} (3GPP TS 38.211 [2], section 7.4.3.1) Value: 0->31
ssb-OffsetPointA	uint16_t	Offset of lowest subcarrier of lowest resource block used for SS/PBCH block. Given in PRB [3GPP TS 38.211 [2], section 4.4.4.2] Value: 0->2199
bch-PayloadFlag	uint8_t	A value indicating how the BCH payload is generated. This should match the PARAM/CONFIG TLVs. Value: 0: MAC generates the full PBCH payload, see Table 3-105, where bchPayload has 32 bits 1: PHY generates the timing PBCH bits, see Table 3-105, where the bchPayload has 24 bits 2: PHY generates the full PBCH payload, see Table 3-106.
bch-Payload	struct	See Table 3-105 and Table 3-106
Beamforming		
Precoding and Beamforming	structure	See Table 3-111. If the PHY supports Rel-16 see also Table 3-112. For PHYs supporting FAPIv4 MU-MIMO groups, see also Table 3-113.
SSB/PBCH Maintenance Parameters added in FAPIv3	structure	See Table 3-104
SSB/PBCH Parameters added in FAPIv4	structure	See Table 3-107
backwardCompatibleExtension-v7	structure	See Table 3-108 (see section 3.1.5)

Table 3-103 SSB/PBCH PDU

Field	Type	Description
Set of SSB PDU parameters added in FAPIv3		
Basic Parameters		
ssb-PDU-Index-v3	uint8_t	Unique PDU id across all SSB PDUs signalled in this slot. The SSB PDU Indices signalled in a slot (according to the lowest PDSCH numerology) shall be numbered between 0 and [(the number of SSBs PDUs in the slot)-1], in the order in which they appear in `DL_TTI.request`.

Field	Type	Description
		Note: the requirement in the last paragraph means that a FAPIv5 MAC may not index SSB PDUs according to the expectations of a FAPIv6 PHY.
case-v3	uint8_t	First symbol mapping for candidate SSB location, per [3GPP TS 38.213 [4], sec 4.1] 0: Case A 1: Case B 2: Case C 3: Case D 4: Case E
subcarrierSpacing-v3	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
l-Max-v3	uint8_t	L_{max}, as defined in 3GPP TS 38.214 [5], section 7.4.1.4.1.
Tx Power info		
ss-PBCH-BlockPowerScaling-v3	int16_t	Baseband power scaling applied to SS-PBCH signal [3GPP TS 38.211 [2], sec 7.4.2.3.1) Values: -11000 -> 12000 representing -110.0 dB to 120.0 dB in 0.01dB -32768: L1 determines PSS power as in SCF-222 v2.0
beta-PSSProfile-SSS-v3	int16_t	Ratio of PSS EPRE to SSS EPRE. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB -32768: L1 determines PSS power from betaPSSProfileNR 3GPP reference: 3GPP TS 38.213 [4], section 4.1

Table 3-104 SSB/PBCH PDU maintenance FAPIv3

Field	Type	Description
bch-Payload-v4	uint32_t	BCH payload. The valid bits are indicated in the PARAM/CONFIG TLVs. If PARAM/CONFIG TLVs indicate MAC generates full bchPayload: - the payload length is 32 bits with the 8 LSB bits being (SFN, half-radio frame bit, SS/PBCH block idx) , in the same order as specified in [3GPP TS 38.212 [3], sec 7.1.1] - a_0 in 3GPP TS 38.212 [3], section 7.1.1 is mapped to LSB 31 and a_{31} is mapped to LSB 0 of bchPayload field Otherwise:

Field	Type	Description
		- timing PBCH bits are generated by the PHY. - for bchPayload the 24 LSB are used, in the same order as specified in [3GPP TS 38.212 [3], sec 7.1.1]. - a_0 in 3GPP TS 38.212 [3], section 7.1.1 is mapped to LSB 23 and a_{23} is mapped to LSB 0 of bchPayload field

Table 3-105 MAC generated MIB PDU

Field	Type	Description
dmrs-TypeA-Position	uint8_t	The position of the first DM-RS for downlink and uplink. Value: 0 -> 1
pdcch-Config-SIB1	uint8_t	The parameter PDCCH-ConfigSIB1 that determines a common *ControlResourceSet* (CORESET) a common search space and necessary PDCCH parameters. Value: 0 -> 255
cellBarred	uint8_t	The flag to indicate whether the cell is barred Value: 0 -> 1
intraFreqReselection	uint8_t	The flag to controls cell selection/reselection to intra-frequency cells when the highest ranked cell is barred, or treated as barred by the UE. Value: 0 -> 1

Table 3-106 PHY generated MIB PDU

Field	**Type**	**Description**
MU-MIMO support in FAPIv4		
spatialStreamIndexPresent-v4	uint8_t	Indicates whether a spatial stream index is signalled or not. Must be set to true if PHY indicates that it requires DL spatial stream index assignment, via capability maxNumberDlSpatialStreams (TLV 0x0151). 0 = false 1 = true
spatialStreamIndex-v4	uint16_t	This entry is present if and only if spatialStreamIndexPresent is true. Spatial stream index for mapping SSB PDU. In this release there is a single spatial stream per SSB in the Precoding Matrix for the Precoding and Beamforming PDU. Value: 0 → (max # spatial streams-1, per TLV 0x0151)

Field	Type	Description
backwardCompatibleExtension-v7	structure	See Table 3-110 (see section 3.1.5)

Table 3-107 SSB PDU parameters for FAPIv4

Field	Type	Description
This is a backward compatible extension of SSB PDU		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5.

Table 3-108 SSB PDU Backward Compatible Extension

3.4.2.4a PRS PDU

Field	Type	Description
subcarrierSpacing-v5	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
cyclicPrefix-v5	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
nId-PRS-v5	unt16_t	Parameter $n_{ID,seq}^{PRS}$, as provided in [3GPP TS 38.211 [2], sec 7.4.1.7.2] Value: 0 … 4095
pdu-Index-v5	uint16_t	PDU index incremented for each PRS PDU sent in TX control message. This is used to associate control information to data and is reset every slot. Value: 0 -> 65535
combSize-v5	uint8_t	Transmission comb size K_{comb}^{PRS} [3GPP TS 38.211 [2], sec 7.4.1.7.3] Value: 2, 4, 6, 12
combOffset-v5	uint8_t	Transmission comb offset k_{offset}^{PRS} [3GPP TS 38.211 [2], sec 7.4.1.7.3] Value: 0 … (K_{comb}^{PRS}-1)
numSymbols-v5	uint8_t	Number of symbols L_{PRS} [3GPP TS 38.211 [2], sec 7.4.1.7.3] Value: 2, 4, 6, 12
firstSymbol-v5	uint8_t	Symbol offset l_{start}^{PRS} [3GPP TS 38.211 [2], sec 7.4.1.7.3] Value: 0 … 12
numRBs-v5	uint16_t	PRS bandwidth RBs, as provided by parameter dl-PRS-ResourceBandwidth [3GPP TS 37.355 [28], sec 6.4.3] Value: 24 … 272 PRBs, in multiples of 4.

Field	Type	Description
startRB-v5	uint16_t	PRB offset with respect to Point-A, as provided by parameter dl-PRS-StartPRB [3GPP TS 37.355 [28], sec 6.4.3] Value: 0 … 2176
symbolPuncturingBitmap-v5	uint16_t	A bitmap of PRS symbols that are punctured For each bit: • 0: symbol punctured • 1: symbol transmitted Bit positions: • [0]: symbol #0 • [1]: symbol #1 • … • … • [14]: reserved (set to 0, in this release) • [15]: reserved (set to 0, in this release) Note: this bitmaps servers for handling SSB overlaps.
spatialStreamIndexPresent-v5	uint8_t	Indicates whether a spatial stream index is signalled or not. Must be set to true if PHY indicates that it requires DL spatial stream index assignment, via capability maxNumberDlSpatialStreams (TLV 0x0151). 0 = false 1 = true
spatialStreamIndex-v5	uint16_t	This entry is present if and only if spatialStreamIndexPresent is true. Spatial stream index for mapping PRS PDU. In this release there is a single spatial stream per PRS in the Precoding Matrix for the Precoding and Beamforming PDU. Value: 0 → (max # spatial streams-1, per TLV 0x0151)
Beamforming		
Precoding and Beamforming	structure	See Table 3-111. If the PHY supports Rel-16 see also Table 3-112. For PHYs supporting FAPIv4 MU-MIMO groups, see also Table 3-113.
Tx Power info		
prs-PowerOffset-v5	int16_t	Ratio of PRS EPRE to SSS EPRE; see parameter β_{PRS}, [3GPP TS 38.211 [2], sec 7.4.1.7.3]. EPRE is defined as in section 2.2.5.5. Values: -32767 -> 32767 representing -32.767dB to 32.767dB in 0.001dB steps -32768: reserved

Table 3-109 PRS PDU

Field	Type	Description
This is a backward compatible extension of PRS PDU		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5.

Table 3-110 PRS PDU Backward Compatible Extension

3.4.2.5 Tx Precoding and Beamforming PDU

The precoding and beamforming PDU is included in the PDCCH, PDSCH, CSI-RS and SSB PDUs. The format is shown in Table 3-111.

Semistatic beamforming weights (i.e. signalled via beam ids) are applied to a group of consecutive PRBs. These beamforming groups (BFG) are grouped into non-overlapping consecutive BFGSize PRBs, starting from CRB0.

Semistatic precoding weights (i.e. signalled via PM ids) are applied to a group of consecurive PRBs. These Precoding Matrix groups (PMRGs) are grouped into non-opverlapping consecutive PMRGSize PRBs, starting from CRB0.

Note: BFG and PMRG follow the same frequency anchoring as 3GPP PRG (c.f. 3GPP TS 38.214 [5], section 5.1.2.3). Aside from this anchoring, there is no defined relationship between BFG and PRG or between PMRG and PRG. [3GPP] PRG size is conveyed in PDSCH PDU.

The BFGSize is a multiple of the PMRGSize.

Note: there can be variable number of PMRGs per BFG in a channel allocation, due to the non-continuous allocation of PRB's. The BFGs and PMRGs are configured in separate loops.

Field		Type	Description
trp-Scheme-v3		uint8_t	This field shall be set to 0, to identify that this table is used.
dig-BF-Interfaces		uint8_t	Number of logical antenna ports (parallel streams) feeding into the digBF • If digBFInterfaces = 0, the n-th precoding output (logical port) corresponds to the n-th baseband port, i.e. beamforming is straight-wire/identity. Value: 0->255
num-BFGs-v6		uint16_t	number of BFGs spanning this allocation. (i.e. number of BFGs having at least one PRB allocated) value: 0->275
bfg-Size-v6		uint16_t	beamforming group size in RBs, which is a multiple of the prgSize
for number of BFGs {			
	beam-ID	uint16_t [digBF-Interfaces]	Array of digital beam ids (each beam id corresponds to a weight vector pre-stored at cell configuration). The array maps the logical port indexed by digBFInterface to output baseband ports.

Field		Type	Description
			Value: 0->65535 (for each array entry) Only the 15 LSB indicate a beam index 0…32,767 MSB shall be interpreted as follows: • 0: look up in Hi-PHY preconfigured DBM the beam index constructed from 15 LSB • 1: signal to Lo-PHY the beam index constructed from 15 LSB (e.g. over 7.2x fronthaul, if supported by TLV 0x0164)
}			
num-PMRGs		uint16_t	Number of PMRGs spanning this allocation. Value: 1->275
pmrg-Size		uint16_t	Size in RBs of a PM resource block group (PMRG) – to which same precoding and digital beamforming gets applied. Value: 1->275
precodingInUse-v6		uint8_t	0: the n-th layer corresponds to the n-th logical port, i.e. straight wire / identity precoding (and pm-Idx can be ignored in the loop below) 1: precoding per pm-idx
For number of PMRGs {			
	pm-ID	uint16_t	Index to precoding matrix (PM) pre-stored at cell configuration. Note: If precoding is not used, this parameter shall be ignored and is interpreted as identity precoding. Value: 0->65535.
}			

Table 3-111 Tx precoding and beamforming PDU

3.4.2.6 Rel-16 mTRP Tx Precoding and Beamforming PDU

The precoding and beamforming PDU is included in the PDCCH, PDSCH, CSI-RS and SSB PDUs, when PHY hosts both ports of a Rel-16 mTRP configuration, per 3GPP 38.300 [11], section 6.12. The format is shown in Table 3-112.

See section 3.4.2.5 for the PMRG and BFG definitions.

L1 only expects this table if it supports Rel-16 mTRP.

Field	Type	Description
trp-Scheme-v3	uint8_t	0: refer to Table 3-111, instead, for the format of the Tx Precoding and Beamforming PDU 1: sTRP: all baseband ports are used (like in Rel-15) => #TRPs = 1 2: mTRP: single TRP (first) in PHY => #TRPs = 1 3: mTRP: single TRP (second) in PHY => #TRPs = 1 4: mTRP: both TRPs in PHY => #TRPs = 2

Field	Type	Description
dig-BF-Interfaces-v3	uint8_t [#TRPs]	Number of logical antenna ports (parallel streams) feeding into the digBF If digBFInterfaces[x] = 0, the n-th precoding output (logical port) corresponds to the n-th baseband port, i.e. beamforming is straight-wire/identity for TRP x. Value: 0->255
num-BFGs-v6	uint16_t	number of BFGs spanning this allocation. (i.e. number of BFGs having at least one PRB allocated) value: 0->275
bfg-Size-v6	uint16_t	beamforming group size in RBs, which is a multiple of the prgSize
for number of BFGs {		
For each TRP {		
beam-ID-v3	uint16_t [digBF-Interfaces]	Array of digital beam ids (each beam id corresponds to a weight vector pre-stored at cell configuration). The array maps the logical port indexed by digBFInterface to output baseband ports. Value: 0->65535 (for each array entry) Only the 15 LSB indicate a beam index 0…32,767 MSB shall be interpreted as follows: • 0: look up in Hi-PHY preconfigured DBM the beam index constructed from 15 LSB • 1: signal to Lo-PHY the beam index constructed from 15 LSB (e.g. over 7.2x fronthaul, if supported by TLV 0x0164)
}		
}		
num-PMRGs-v3	uint16_t	Number of PMRGs spanning this allocation. Value: 1->275
pmrg-Size-v3	uint16_t	Size in RBs of a PM resource block group (PMRG) – to which same precoding and digital beamforming gets applied. Value: 1->275
precodingInUse-v6	uint8_t [#TRPs]	0: the n-th layer corresponds to the n-th logical port, i.e. straight wire / identity precoding (and pm-Idx can be ignored in the loop below), for TRP x 1: precoding per pm-Idx, for TRP x
For number of PMRGs {		
layers-v3	uint8_t	Bitmap of layers: MSB = layer 1, LSB = layer 8. Bit = 0 => layer precoded via pm-Idx[0] Bit = 1 => layer (if any) precoded via pm-Idx[1] (if #TRPs = 2, else dropped)
pm-Idx-v3	uint16_t [#TRPs]	Index to precoding matrix (PM) pre-stored at cell configuration.

Field		Type	Description
			Note: If precoding is not used, this parameter shall be ignored for the TRP and is interpreted identity precoding for the TRP. Value: 0->65535.
}			

Table 3-112 Tx precoding and beamforming PDU

3.4.2.7 Tx Precoding based on Channel Reciprocity PDU

The Tx Precoding PDU based on Channel Reciprocity sampling; in this release, this PDU may be included, when precoding is performed as described in section 2.2.7.2. The format is shown in Table 3-113.

Field		Type	Description
trp-Scheme-v4		uint8_t	5: Single TRP, based on channel reciprocity
num-MU-MIMO-Groups -v4		uint8_t	Number of DL MuMIMO groups covering the allocation of this PDU. Any PRBs in the allocation for the channel must belong to exactly one MuMIMO group.
For each of the num-MU-MIMO-Groups {			
	mu-MIMO-Group-ID -v4	uint16_t	muMimoGroups are listed in increasing muMimoGroup index order. A layerMapping-PRG is defined in exactly one muMimoGroup (see 3.4.2). Value: 0→3,821 ***Note****: muMimoGroupIdx are defined in increasing order of their start symbol and on a given start symbol they are defined in increasing order of PRB*
}			

Table 3-113 Tx generalized precoding PDU

3.4.2.8 Tx Precoding Based on Dynamic Weight Signalling by L2/L3

The Tx Precoding PDU based on dynamic weight signalling by L2/L3; in this release, this PDU may be included, when precoding is performed as described in section 2.2.6.2b2.2.7.2. The format is shown in Table 3-114 .

Field	Type	Description
trp-Scheme-v6	uint8_t	6: Single TRP, based on dynamic weight signalling by L2/L3
num-MU-MIMO-Group s-v6	uint8_t	Number of DL MuMIMO groups covering the allocation of this PDU. Any PRBs in the allocation for the channel must belong to exactly one MuMIMO group. For SU-MIMO numMuMIMOGroups to be set to 1.
For each of the numMU-MIMO-Groups {		

Field			Type	Description
	mu-MIMO-Group-ID-v6		uint16_t	muMimoGroups are listed in increasing muMimoGroup index order. A layerMapping-PRG is defined in exactly one muMimoGroup. Value: 0→3,821 ***Note**: muMimoGroupIdx are defined in increasing order of their start symbol and on a given start symbol they are defined in increasing order of PRB*
	numPRGs-In-MU-MIMO-Group-v6		uint16_t	Number of layerMapping-PRGs in the MU MIMO group corresponding to mu-Mimo-GroupIdx
	For each of the numPRGs-InMU-MIMO-Group layerMapping-PRGs {			
		numPrecodingWeights-v6	uint16_t	Number of elements in the precoding matrix numPrecodingWeights = numTxAnt x Layers
		precodingWeigths-v6	uint16_t [2 x numPrecodingWeights]	Complex precoding weights for each Tx antenna at the radio unit and for each of the layers in the current muMimoGroup. Weights[layerIdx, antIdx] = array[antIdx + layersIdx x num Tx Ant] • antIdx : 0 … num Tx Ant-1 • layerIdx : 0 … Layers-1
	}			
}				

Table 3-114 Tx Precoding based on dynamic weight signalling by L2/L3

3.4.2b DL_TTI.response

The format of the `DL_TTI.response` message is shown in Table 3-115. It is sent by L1 if L2 requests TB CRC in `DL_TTI.request` PDSCH PDU setting pduBitmap[1] = 1 on a new PDSCH transmission, when L1 is configured for CBG segmentation in L2. The TB CRC returned by L1 can be used for subsequent CBG retransmission

A `DL_TTI.response` message indicates the SFN/Slot of the downlink slot it contains.

This message can be sent by the L1 when the PHY is in the RUNNING state.

Field	Type	Description
sfn-v3	uint16_t	SFN Value: 0 -> 1023
slot-v3	uint16_t	Slot Value: 0 ->159
num-PDUs-v3	uint16_t	Number of PDSCH PDUs for which TBS CRC are supplied in this message. Value 0 -> 65535

Field		Type	Description
For number of PDUs			
	pdu-Index-v3	uint16_t	Used to associate the TB CRC Payload PDU with the DL_TTI PDSCH PDU Value: 0 → 65535
	cw-v3	uint8_t	0 or 1, in this release, to match the number of codewords in the PDSCH PDU with index pduIndex
	tb-CRC-v3	uint32_t	The TB-level CRC calculation results for transport block corresponding to the indicated CW that transmitted in this slot. Only up to 24 LSB of each array element are valid, L2 shall extract the correct TB CRC from the array based on scheduling information for this slot for this UE. See 3GPP TS 38.212 [3], section 7.2.1
backwardCompatibleExtension-v7		structure	See Table 3-116 (see section 3.1.5)

Table 3-115 `DL_TTI.response` **message body**

Field	Type	Description
This is a backward compatible extension of DL_TTI.response. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5.

Table 3-116 `DL_TTI.response` **backward compatible extension**

3.4.3 UL_TTI.request

The format of the `UL_TTI.request` message is shown in Table 3-117. An `UL_TTI.request` message indicates the SFN and slot it contains information for. This control information is for an uplink slot.

This message can be sent by the L2/L3 when the PHY is in the RUNNING state. If it is sent when the PHY is in the IDLE or CONFIGURED state an `ERROR.indication` message will be sent by the PHY.

The following combinations of PDUs are required:

- The PUSCH PDU is present when a UE has been instructed to send uplink data and/or control data on the PUSCH
- The PUCCH PDU is present when a UE has been instructed to send control data on the PUCCH
- The PRACH PDU is present when a UE may transmit RACH.
- The SRS PDU is present when a UE has been instructed to send SRS.

The PDUs included in this structure have no ordering requirements.

Field	Type	Description
sfn	uint16_t	SFN

Field	Type	Description
		Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159
num-PDUs	uint16_t	Number of PDUs that are included in this message. All PDUs in the message are numbered in order. Value: 0 -> 65535 Backward compatibility note: the field was 8 bits in SCF222 v2.0
num-UL-Types-v3	uint8_t	Maximum number of UL PDU types supported by `UL_TTI.request`. nUlTypes = 13 in this release. Backward compatibility note: SCF222 v2.0 did not have this field
num-PDUs-OfEachType-v3	uint16_t [numUL-Types]	Number of PDUs of each type that are included in this message. Each array entry corresponds to a PDU type as follows: [0]: number of PRACH PDUs [1]: number of PUSCH PDUs [2]: number of Format 0/1 PUCCH PDUs [3]: number of Format 2/3/4 PUCCH PDUs [4]: number of SRS PDUs [5]: number of MsgA-PUSCH PDUs [6]: number of RNTI Release PDUs [7]: number of ZP SRS PDUs [8]: number of Format 0 PUCCH PDUs [9]: number of Format 1 PUCCH PDUs [10]: number of Format 2 PUCCH PDUs [11]: number of Format 3 PUCCH PDUs [12]: number of Format 4 PUCCH PDUs Value 0 -> 65535, for each entry Backward compatibility note: SCF222 v2.0 did not have this field
numGroup	uint16_t	Number of PDU Groups included in this message. Value: 0 -> 3,822
For Number of PDUs {		
pdu-Type	uint16_t	0: PRACH PDU, see Section 3.4.3.1 1: PUSCH PDU, see Section 3.4.3.2 2: PUCCH PDU, see Section 3.4.3.3 3: SRS PDU, see Section 3.4.3.4 4: MsgA-PUSCH PDU, see Section 3.4.3.2A 5: RNTI Release PDU, see Section 0

Field	Type	Description
pdu-Size	uint16_t	Size of the PDU control information (in bytes). This length value includes the 4 bytes required for the PDU type and PDU size parameters. Value: 0 -> 65535
ul-PDU-Configuration	structure	See Sections 3.4.3.1 to 0
}		
In this message, if nGroup > 0 then for each group { *-- For FAPIv4, groups represent MU MIMO groups, as described in section 2.2.7.2; muMimoGroups are listed in increasing order of their start symbol. muMimogroups starting at the same symbol, are listed channel by channel in increasing order of their first PRB index.*		
num-PDUs-InGroup	uint8_t	Number of Channel PDU in this group. PDUs in a group must be of the same type. For SU-MIMO, one group includes one UE only. For MU-MIMO, one group includes up to 32 UEs. Value: 1->32
portMapping-FD-Resolution -v4	uint16_t	This field signals the RB resolution for UL beamforming based on dynamic weight signalling by L2/L3 (see section 3.4.3.7), for PHY to use in constructing UL combiners. Only relevant to PHYs supporting TLV 0x0174, other PHYs may ignore this field. For FAPIv2 grouping: • portMapping FD Resolution is set to 0 • num-PDUs is limited to 1…12 • prbBitmap, symbolBitmap fields are omitted. Value: • 1 → 275 (for PHYs supporting TLV 0x0174, when UL beamforming is conveied via dynamic weight signalling by L2/L3 (see section 3.4.3.7)) • 0 (to mean FAPIv2) • 0xFFFF (to mean Not Applicable)
prb-BitmapStart-v6	uint16_u	Offset from reference CRB to PRB0 of prbBitmap. Value: 0 … 274. If TLV 0x0170 is set to 1 or combinerType = 0, prb-BitmapStart is the offset with respect to reference CRB for the carrier, otherwise it shall be set so that PRB0 of prb-Bitmap and the UL CRB0 reference of srs-BandwidthStart of SRS PDUs used for dynamic precoding are the same.
prb-SCS-v6	uint8_t	subcarrier spacing [3GPP TS 38.211 [2], sec 4.2] common to all PDUs in this muMimoGroup. If TLV0x0170 is set to 1 or combinerType = 0, prb-SCS is the numerology corresponding to prb-Bitmap and prb-BitmapStart, otherwise it

Field			Type	Description
				shall be set the same as for all the SRS PDUs used for dynamic combining. Value:0->4
	prb-Bitmap-v4		uint8_t [36]	bitmap of RBs in the group, for joint spatial precoding. 273 rounded up to multiple of 32. This bitmap is in units of VRBs. Bit mapping is described in section 3.1.6, per TLV 0x0177. Note: the lowest numbered PRB in the bitmap corresponds to CRB #prb-BitmapStart
	combinerType-v6		uint16_t	Values: 0: semistatic (the PDUs in the MU MIMO group are used with semi-static combining) > 0: dynamic (the PDUs in the MU MIMO group are used with dynamic combining)
	symbolBitmap-v4		uint16_t	bitmap of symbols for the joint spatial precoding group bit location indicates the symbol index, up to 14 symbols. First symbol corresponds to location #0
	In the group, for each of the num-PDUs-in-Group {			
		pdu-Idx-v4	uint16_t	This value is an index for number of the PDU, out of the PDUs signalled in this message. Value: 0->65535
	}			
}				
backwardCompatibleExtension-v7			structure	See (see section 3.1.5)

Table 3-117 `UL_TTI.request` **message body**

Field	Type	Description
This is a backward compatible extension of UL_TTI.request. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5.

Table 3-118 `UL_TTI.request` **backeard compatible extension**

3.4.3.1 PRACH PDU

The format of the PRACH PDU is shown in Table 3-119.

Field	Type	Description
num-PRACH-Ocas	uint8_t	Number of time-domain PRACH occasions within a PRACH slot, N_t^{RAslot} [3GPP 38.211 [2], sec 6.3.3.2] If only identical combining is supported per TLV 0x017B, this field is set per CONFIG TLV 0x103E field for this slot and PRACH config

Field	Type	Description
		Value: 1->7
prach-Format	uint8_t	RACH format information for the PRACH occasions signalled in this PDU. This corresponds to one of the supported formats i.e., 0, 1, 2, 3, A1, A2, A3, B1, B4, C0, C2, A1/B1, A2/B2 or A3/B3. [3GPP 38.211 [2], sec 6.3.3.2] Values: 0 = 0 1 = 1 2 = 2 3 = 3 4 = A1 5 = A2 6 = A3 7 = B1 8 = B4 9 = C0 10 = C2 11 = A1/B1 12 = A2/B2 13 = A3/B3 If any of the values 11-13 is signalled, the last of the numPrachOcas time domain occasions is of the B type; any occasions preceding it are of the A type. If only identical combining is supported per TLV 0x017B, this field is set per CONFIG TLV 0x103E field for this slot and PRACH config
index-FD-RA	uint8_t	Frequency domain occasion index $n \in \{0,1,.., M-1\}$, where M equals the higher-layer parameter msg1-FDM which can take values {1,2,4,8} [3GPP TS 38.211 [2], sec 6.3.3.2] Values: 0->7
prach-StartSymbol	uint8_t	Starting symbol for the first PRACH TD occasion in the current PRACH FD occasion. Corresponds to the parameter. [3GPP TS 38.211 [2], sec 6.3.3.2 and Tables 6.3.3.2-2 to 6.3.3.2-4] Values: 0->13
num-Cs	uint16_t	Zero-correlation zone configuration number (RRC parameter zeroCorrelationZoneConfig). Corresponds to the L1 parameter N_{CS}. [3GPP TS 38.211 [2], sec 6.3.3.1 and Table 6.3.3.1-5, 6.3.3.1-6 and 6.3.3.1-7] Value: 0->419 This field is set per CONFIG TLV 0x103E field for this slot and PRACH config

Field	Type	Description
is-MsgA-PRACH-v4	uint8_t	Signals whether this msgA PRACH is at least partially used for 2-Step RACH. Value: - 0 = False (no *MsgA-PRACH to MsgA-PUSCH map signalling*) - 1 = True (*MsgA-PRACH to MsgA-PUSCH map signalling* is included in this PDU) - 2 = True (*MsgA-PRACH to MsgA-PUSCH map signalling* is not included in this PDU, since PHY determines it for P5 configuration).
has-MsgA-PUSCH-Beamforming-v4	uint8_t	When IsMsgA-Prach is true, signals whether Beamforming structure is present for the MsgA-PRACH (in which case, the structure signals beams for all PRBs for valid MsgA-PUSCH PDUs. 0: not included 1: included
Beamforming		
Beamforming	structure	See Table 3-151; for PHYs supporting MU-MIMO groups, see also Table 3-152
PRACH Maintenance Parameters added in FAPIv3	structure	See Table 3-120
Uplink spatial stream assignment FAPIv4	structure	See Table 3-121
MsgA-PRACH to MsgA-PUSCH map signalling in FAPIv4	structure	Only included if IsMsgA-Prach = 1 See Table 3-122
MsgA-PUSCH Beamforning	Structure	Only included if isMsgA-PRACH = *True* and hasMsgA-PUSCH-Beamforming = *included* See Table 3-151; for PHYs supporting MU-MIMO groups, see also Table 3-152
backwardCompatibleExtension-v7	structure	See Table 3-123 (see section 3.1.5)

Table 3-119 PRACH PDU

Field	**Type**	**Description**
Set of PRACH PDU parameters added in FAPIv3		
handle-v3	uint32_t	An opaque handle returned in the `RACH.indication`
prach-ConfigScope-v3	uint8	prachResConfigIndex being referred to: 0: the index shall be looked up in the Common or PHY Group Context (Shared by all PHYs in the Context) 1: the index shall be looked up in this PHY's Context
prach-ResConfigIndex-v3	uint16_t	The PRACH configuration for which this PRACH PDU is signalled (c.f. Table 3-51)
numFD-RA-v3	uint8_t	Number of frequency domain occasions, starting with indexFdRa [3GPP TS 38.211 [2], sec 6.3.3.2]

Field	Type	Description
		Values: 1->8
startPreambleIndex-v3	uint8_t	Start of preamble index to monitor in the PRACH occasions signalled in this slot. Value: 0 – 63 255: all preambles from the PRACH configuration linked by prachResConfigIndex Preamble logical indices for a RACH occasion are defined as in 3GPP TS 38.211 [2], section 6.3.3.1, starting from the *prachRootSequenceIndex* of the PRACH configuration for the RO. If only identical combining is supported per TLV 0x017B, this field is set per CONFIG TLV 0x103E field for this slot and PRACH config
numPreambleIndices-v3	uint8_t	Number of preamble logical indices, starting with startPreambleIndex, to monitor in the PRACH occasions signalled in this slot. Value: 1 – 64 If only identical combining is supported per TLV 0x017B, this field is set per CONFIG TLV 0x103E field for this slot and PRACH config

Table 3-120 PRACH maintenance FAPIv3

Field	Type	Description
Set of PRACH PDU parameters added in FAPIv4		
numSpatialStreams-v4	uint8_t	Non-zero if PHY indicates that UL spatial stream assigned is required, in capability maxNumberUlSpatialStreams (TLV 0x0153) The number of spatial streams must be at least as large as the number of layers. When semistatic logical ports are used, the number of spatial streams is the same as the number of logical ports.
spatialStreamIndices-v4	uint16_t [n]	n = [numSpatialStreams] The indices of the spatial streams used by this channel. Value: 0 → (max # spatial streams – 1, per TLV 0x0153)

Table 3-121 Uplink spatial stream assignment FAPIv4

Field	Type	Description
Set of 2-Step RACH PRACH PDU parameters added in FAPIv4		

Field	Type	Description
numberOfPreambleGroups-v4	uint8_t	Indicates the number of preamble groups (#cfg) signalled for this PRACH PDU (c.f. 3GPP TS 38.331 [6], section 6.3.2; 3GPP TS 38.321 [21], sections 5.1.1 and 5.1.2a). These can correspond to common groups (A or B) or dedicated groups, including singular groups (consisting of single preambles)
dmrs-Ports-v4	uint16_t [#cfg]	Bitmap of all the DMRS ports amongst which ports are picked for each PUSCH occasion of the MsgA-PUSCH PDU(s) mapped from this PRACH PDU, [3GPP TS 38.212 [3], 7.3.1.1.2] based on msgA-PRACH → msgA-PUSCH mapping Bit mapping is as follows: • bit 0 : antenna port 1000 • bit 7: antenna port 1007 For each bit • 0: DMRS port not used • 1: DMRS port used numDmrsPorts = sum of all bits in dmrsPorts Note: the actual DMRS port to use is based on the msgA-PRACH → msgA-PUSCH & associated DMRS mapping
available-DMRS-Sequence-IDs-v4	uint8_t [#cfg]	Bitmap indicating the DMRS sequence IDs available for the MsgA-PUSCH PDU(s) corresponding to this PRACH PDU. For each bit position: • 0: ID not supported • 1: ID supported Bit significance: - [0]: n_{SCID} = 0 available. - [1]: n_{SCID} = 1 available - [2-7]: n/a (bit value shall be set to 0 numDmrsSequences = sum of all bits in availableDmrsSequenceIds
num-PUSCH-Ocas-FD-v4	uint8_t [#cfg]	The number of frequency domain PUSCH occasions within the UL slot for the corresponding Msg-A PUSCH PDU(s). Value: 1, 2, 4, 8
num-PUSCH-Ocas-TD-v4	uint8_t [#cfg]	The number of time domain PUSCH occasions for the corresponding MsgA-PUSCH PDU(s). Value: 1, 2, 3, 6
numPUSCH-OcasSlots-v4	uint8_t [#cfg]	The number of slots configured for the corresponding MsgA-PUSCH PDU(s) Value: 1…4
msgA-PUSCH-TimeDomainOffset-v4	uint8_t [#cfg]	The time domain offset, as specified in [3GPP TS 38.213 [22], section 8.1A] in slots between this PRACH PDU slot where a MsgA-PRACH preamble may be triggered and the corresponding MsgA-

Field	Type	Description
		PUSCH PDU(s) signalling the MsgA-PUSCH occasion. Value: 1…32
n-Preambles-v4	uint16_t [#cfg]	$N_{preambles}$, per section 8.1A of 38.213 [22], section 8.1A as associated by L2/L3 for the particular mapping of MsgA-PRACH to Msg-A PUSCH mapping relevant to the preamble group and this PRACH PDU. Values: 1 … 3,072
msgA-PRACH-To-PRU-And-DMRS-Mapping-v4	See Table 3-53	For this mapping, the following are as signalled for this PRACH PDU: • *prachFd1* … *prachFdMax* = indexFdRa … numFdRa-1 • *prachTd1* … *prachTdMax* = 0 … numPrachOcas-1 The preamble group range *grpStart* … *grpEnd* depends on numberOfpreambleGroups: • *grpStart* = 1 • *grpEnd* = numberOfpreambleGroups Furthermore, the following parameters are used to interpret the contents of the mapping, for each group: • $N_{preambles}$: Npreambles • *numPuschOcasFd*: numPuschOcasFd • *numDmrsPorts*: sum of all bits in dmrsPorts • *numDmrsSequences*: sum of all bits in available-DmrsSequenceIds • *numPuschOcasTd*: numPuschOcasTd • *numPuschOcasSlots*: numPuschOcasSlots • *startPreambleIndex*: startPreambleIndex in this PRACH PDU • *numberOfPreambles*: numberOfPreambles in this PRACH PDU

Table 3-122 MsgA-PRACH to MsgA-PUSCH map signalling in FAPIv4

Field	Type	Description
This is a backward compatible extension of PRACH PDU		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5.

Table 3-123 PRACH PDU Backward Compatible Extension

3.4.3.2 PUSCH PDU

The format of the PUSCH PDU is given in Table 3-124.

The PUSCH PDU includes both mandatory and optional parameters, where the presence of the optional elements is defined in the parameter pduBitmap. The presence of these optional elements is included based on the following:

- puschData is set if an ULSCH transmission is expected from the UE
- puschData and puschUCI are both set if ULSCH and UCI transmission on PUSCH is expected from the UE

- puschUCI is set and puschData is not set if UCI information is expected on PUSCH, but ULSCH is not expected
- puschPtrs is included if and only if PTRS are included in the uplink transmission
- dftsOfdm is included for DFT S-OFDM transmission

Only parameters signalled by pduBitmap are optional, all other parameters are mandatory. (It should be noted that some parameters are only applicable for certain modes etc. These are still included but their value is ignored by the PHY.)

Field	Type	Description
pdu-Bitmap	uint16_t	Bitmap indicating presence of optional PDUs Bit 0: puschData (Indicates data is expected on the PUSCH) Bit 1:puschUci (Indicates UCI is expected on the PUSCH) Bit 2: puschPtrs (Indicates PTRS included. L1 behaviour is undefined if this bit is 1 when 3GPP procedures require no PT-RS transmission) Bit 3: dftsOfdm (Indicates DFT S-OFDM transmission) All other bits reserved
rnti	uint16_t	The RNTI used for identifying the UE when receiving the PDU Value: 1 -> 65535
handle	uint32_t	An opaque handling returned in the `RX_DATA.indication` and/or `UCI.indication` message
BWP [3GPP TS 38.213 [4], sec 12]		
bwp-Size	uint16_t	If L1 uses puschTransType, this field signals the bandwidth part size [3GPP TS 38.213 [4], sec12]. Number of contiguous PRBs allocated to the BWP. If L1 does not use puschTransType, this field represents the number of VRBs to which PRBs are mapped for the allocation, according to 3GPP TS 38.211 [2], section 6.3.1.7. Note: for BWPSize, the two interpretations align, except when the allocation is scheduled for msg3 (by RAR UL grant), in which case the first interpretation uses to active BWP,I size, whereas the second one uses the initial BWP,0 size referenced by 3GPP TS 38. 213 [4], section 8.3. Value: 1->275
bwp-Start	uint16_t	If L1 uses puschTransType, this field signals the bandwidth part start RB index from reference CRB [3GPP TS 38.213 [4], sec 12]

Field	Type	Description
		If L1 does not use puschTransType, this field represents the CRB corresponding to VRB0, according to 3GPP TS 38.211 [2], section 6.3.1.7. **Note 1**: For BWPStart, the two interpretations align, except when the allocation is scheduled for msg3 "*by DCI format 0_0 with CRC scrambled by TC-RNTI in active uplink bandwidth part* i *starting at* $N_{BWP,i}^{start}$*, including all resource blocks of the initial uplink bandwidth part starting at* $N_{BWP,0}^{start}$ *, and having the same subcarrier spacing and cyclic prefix as the initial uplink bandwidth part*", in which case the first interpretation uses to active BWP,i start whereas the second one uses the initial BWP,0 start referenced by 3GPP TS 38.211 [2], section 6.3.1.7. **Note 2**: Only msg3 allocations subject to the conditions listed in Note 1 above are referenced to the initial BWP,0 start. Other types of msg3 allocations are referenced to the start of active uplink BWP,I, as specified in 3GPP TS 38.211 [2], section 6.3.1.7 and 3GPP TS 38.213 [4], section 8.3. Value: 0->274
subcarrierSpacing	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
cyclicPrefix	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
PUSCH information always included		
targetCodeRate	uint16_t	Target coding rate [3GPP TS 38.214 [5], sec 6.1.4.1]. This is the number of information bits per 1024 coded bits expressed in 0.1 bit units
qam-ModOrder	uint8_t	QAM modulation [3GPP TS 38.214 [5], sec 6.1.4.1] Value: 2,4,6,8 if transform precoding is disabled Value: 1,2,4,6,8 if transform precoding is enabled
mcs-Index	uint8_t	MCS index [3GPP TS 38.214 [5], sec 6.1.4.1], should match value sent in DCI Value : 0->31
mcs-Table	uint8_t	MCS-Table-PUSCH [3GPP TS 38.214 [5], sec 6.1.4.1] Value: 0: notqam256 [3GPP TS 38.214 [5], table 5.1.3.1-1] 1: qam256 [3GPP TS 38.214 [5], table 5.1.3.1-2] 2: qam64LowSE [3GPP TS 38.214 [5], table 5.1.3.1-3]

Field	Type	Description
		3: notqam256-withTransformPrecoding [3GPP TS 38.214 [5], table 6.1.4.1-1] 4: qam64LowSE-withTransformPrecoding [3GPP TS 38.214 [5], table 6.1.4.1-2]
transformPrecoding	uint8_t	Indicates if transform precoding is enabled or disabled [3GPP TS 38.214 [5], sec 6.1.4.1] [3GPP TS 38.211 [2], 6.3.1.4] Value: 0: enabled 1: disabled
n-ID-PUSCH	uint16_t	parameter n_{ID} [3GPP TS 38.211 [2], sec 6.3.1.1] Value: 0->1023
num-Layers	uint8_t	Number of layers [3GPP TS 38.211 [2], sec 6.3.1.3] Value: 1->4
DMRS [3GPP TS 38.211 [2], sec 6.4.1.1]		
ul-DMRS-SymbPos	uint16_t	DMRS symbol positions [3GPP TS 38.211 [2], sec 6.4.1.1.3 and Tables 6.4.1.1.3-3 and 6.4.1.1.3-4] Bitmap occupying the 14 LSBs with: bit 0: first symbol and for each bit 0: no DMRS 1: DMRS
dmrs-ConfigType	uint8_t	UL DMRS config type [3GPP TS 38.211 [2], sec 6.4.1.1.3] 0: type 1 1: type 2
pusch-DMRS-Scrambling-ID	uint16_t	PUSCH DMRS Scrambling-ID [3GPP TS 38.211 [2] and v16 [12], sec 6.4.1.1.1.1], as provided by parameter $N_{ID}^{n_{SCID}}$ (Rel-15) or $N_{ID}^{\bar{n}_{SCID}^{\bar{\lambda}}}$ (Rel-16). It is only valid when the transform precoding for PUSCH is disabled or when lowPaprDmrs is enabled. For example, in references [2] and [12], this field applies to below case: 1. CP-OFDM PUSCH transmission with Rel-15 DM-RS configuration 2. CP-OFDM PUSCH transmission having DM-RS port(s) within CDM group 0 or 2 with Rel-16 DM-RS configuration 3. Non-MSG3 and non-DCI 0_0 in CSS scheduled DFT-S-OFDM PUSCH transmission with pi/2-BPSK modulation and Rel-16 DM-RS configuration

Field	Type	Description
		4. Otherwise, L2 shall set it to Physical Cell ID Value: 0->65535
pusch-DMRS-ScramblingIdComplement-v4	uint16_t	PUSCH DMRS Scrambling-ID [3GPP TS 38.211 [12], sec 6.4.1.1.1.1], as provided by parameter $N_{ID}^{\bar{n}_{SCID}^{\bar{\lambda}}}$ (Rel-16) when $\bar{n}_{SCID}^{\bar{\lambda}} \neq n_{SCID}$. (i.e. $N_{ID}^{1-n_{SCID}}$) PHY shall disregard this parameter if either of this conditions is satisfied: 1. CP-OFDM PUSCH data and lowPaprDmrs = 0 2. DFTS-OFDM PUSCH data 1. For example, in reference [12], this field applies to below case: CP-OFDM PUSCH transmission having DM-RS port(s) within CDM group 1 with Rel-16 DM-RS configuration Value: 0->65535
low-PAPR-DMRS-v4	uint8_t	**Without Transform Precoding:** **1:** $\bar{n}_{SCID}^{\bar{\lambda}}$ depends on DM-RS CDM group **0:** $\bar{n}_{SCID}^{\bar{\lambda}} = n_{SCID}$ (i.e. independent of DM-RS CDM group) **With Transform Precoding**: When PUSCH Transform precoding is enabled, this parameter indicates whether the DMRS scrambling is computed based on (Gold or CAG) low-PAPR sequence of type 2 (section 5.2.3, as referenced in 3GPP TS 38.211 [12], section 6.4.1.1.1.2), or based on (Z-C) low-PAPR sequence of type 1 (section 5.2.2, as referenced in 3GPP TS 38.211 [12], section 6.4.1.1.1.2). Values: **1**: Time-domain Gold or CAG low-PAPR sequence (type-2), since Rel-16 **0**: based on Frequency-domain Z-C low-PAPR sequence (type-1), since Rel-15
pusch-DMRS-Identity	uint16_t	PUSCH DMRS ID [3GPP TS 38.211 [2] and [12], sec 6.4.1.1.1.2], as provided by parameter n_{ID}^{RS} This field only valid when Transform Precoding is enabled. Value: 0->1007 (Rel-15), 0-> 65535 (Rel-16)

Field	Type	Description
n-SCID	uint8_t	DMRS sequence initialization [3GPP TS 38.211 [2] and [12], sec 6.4.1.1.1.1], as provided by parameter n_{SCID}. When lowPaprDmrs = 1, UE uses $\bar{n}_{SCID}^{\bar{\lambda}} = 1 - n_{SCID}$, as described in the section above. It is only valid when the transform precoding for PUSCH is disabled or when lowPaprDmrs is enabled. For example, in references [2] and [12], this value is associated with DM-RS scrambling ID given by puschDmrsScramblingId Value : 0->1
num-DMRS-CDM-Grps NoData	uint8_t	Number of DM-RS CDM groups without data [3GPP TS 38.212 [3], sec 7.3.1.1] Value: 1->3
dmrs-Ports	uint16_t	DMRS ports. [3GPP TS 38.212 [3], 7.3.1.1.2] provides description between DCI 0-1 content and DMRS ports. Bitmap occupying the 12 LSBs with: bit 0: antenna port 1000 bit 1 1: antenna port 1011 and for each bit 0: DMRS port not used 1: DMRS port used
Pusch Allocation in frequency domain [3GPP TS 38.214 [5], sec 6.1.2.2]		
resourceAlloc	uint8_t	Resource Allocation Type [3GPP TS 38.214 [5], sec 6.1.2.2] 0: Type 0 1: Type 1
rb-Bitmap	uint8_t [36]	For resource allocation type 0. - [3GPP TS 38.214 [5], sec 6.1.2.2.1] [3GPP TS 38.212 [3], 7.3.1.1.2] bitmap of RBs, 273 rounded up to multiple of 32. This bitmap is in units of VRBs. Bit mapping as described in section 3.1.6, per TLV 0x0177. VRB to PRB mapping is per section 6.3.1.7 of 3GPP TS 38.211 [2].
rb-Start	uint16_t	For resource allocation type 1. [3GPP TS 38.214 [5], sec 6.1.2.2.2] The starting resource block corresponds to VRB0 for this PUSCH, per section 6.3.1.7 of 3GPP TS 38.211 [2].

Field	Type	Description
		Value: 0->274
rb-Size	uint16_t	For resource allocation type 1. [3GPP TS 38.214 [5], sec 6.1.2.2.2] The number of resource block within for this PUSCH. Value: 1->275
vrb-To-PRB-Mapping	uint8_t	VRB-to-PRB-mapping [3GPP TS 38.211 [2], sec 6.3.1.7] Value: 0: non-interleaved
intraSlotFrequencyHopping	uint8_t	For resource allocation type 1. [3GPP TS 38.212 [3], sec 7.3.1.1] [3GPP TS 38.214 [5], sec 6.3] Indicates if intra-slot frequency hopping is enabled Value: 0: disabled 1: enabled
tx-DirectCurrentLocation	uint16_t	The uplink Tx Direct Current location for the carrier. Only values in the value range of this field between 0 and 3299, which indicate the subcarrier index within the carrier corresponding to the numerology of the corresponding uplink BWP and value 3300, which indicates“"Outside the carrie”" and value 3301, which indicates“"Undetermined position within the carrie”" are used. [3GPP TS 38.331 [6], UplinkTxDirectCurrentBWP IE] Value: 0->4095
<obsolete field>	uint8_t	This field is ignored by PHY (was uplinkFrequencyShift7p5khz in SCF FAPIv6)
Resource Allocation in time domain [3GPP TS 38.214 [5], sec 5.1.2.1]		
startSymbolIndex	uint8_t	Start symbol index of PUSCH mapping from the start of the slot, S. [3GPP TS 38.214 [5], Table 6.1.2.1-1] Value: 0->13
num-Symbols	uint8_t	PUSCH duration in symbols, L [3GPP TS 38.214 [5], Table 6.1.2.1-1] Value: 1->14
Optional Data only included if indicated in pduBitmap		
pusch-Data	structure	See Table 3-128
pusch-UCI	structure	See Table 3-129
pusch-PTRS	structure	See Table 3-130
dfts-OFDM	structure	See Table 3-131
Beamforming		
Beamforming	structure	See Table 3-151; for PHYs supporting MU-MIMO groups, see also Table 3-152

Field	Type	Description
PUSCH Maintenance Parameters added in FAPIv3	structure	See Table 3-125
PUSCH Parameters added in FAPIv4	structure	See Table 3-132
Obsolete structure; included for backward compatibility	structure	See Table 3-127 Included if and only if pduBitmap[1] = 1 Note: this structure shall be included, for backward compatibility, if pduBitmap[1] = 1
backwardCompatibleExtension-v7	structure	See Table 3-133 (see section 3.1.5)

Table 3-124 PUSCH PDU

Field	Type	Description
BWP [3GPP TS 38.213 [4], sec 12]		
pusch-TransType-v3	uint8_t	L1 interprets this field as below, if so configured by puschTransTypeValidity=1, otherwise it ignores it. Indication used to indicate the transmission type of PUSCH PDU. Value 0: UL BWP start and size of the PUSCH PDU shall be set to initial UL BWP including start and size used for MSG3, instead of active downlink BWP start and size Value 1: UL BWP start and size of the PUSCH PDU shall be set to the active UL BWP start as starting point for RB numbering and initial UL BWP size as RB size for MSG3 Value 2: UL BWP start and size of the PUSCH PDU shall be set to the active UL BWP start and size, for all other UL transmissions
delta-BWP0-StartFromActive-BWP-v3	uint16_t	L1 interprets this field as below, if so configured by puschTransTypeValidity=1, otherwise it ignores it. L1 interprets this field as below, if so configured by puschTransTypeValidity=1, otherwise it ignores it. The value of $N_{BWP,0}^{start} - N_{BWP,i}^{start}$ as described in 3GPP TS 38.211 [2], section 6.3.1.7, representing an offset from active UL BWP start. It is valid if puschTransType = 0 otherwise PHY shall ignore it. Value: 0 to 274
initial-UL-BWP-Size-v3	uint16_t	L1 interprets this field as below, if so configured by puschTransTypeValidity=1, otherwise it ignores it. The initial bandwidth part size [3GPP TS 38.213 [4], sec 12].

Field	Type	Description
		Only valid when puschTransType field set to 0 or 1, otherwise PHY shall ignore it. Value: 0 to 274
DMRS [3GPP TS 38.211 [2], sec 6.4.1.1]		
groupOrSequenceHopping-v3	uint8_t	PUSCH DMRS hopping mode [3GPP TS 38.211 [2], sec 6.4.1.1.1.2]. It is only valid when the transform precoding for PUSCH is enabled. Value: 0, 1 or 2 • 0: neither, neither group or sequence hopping is enabled • 1: enable, enable group hopping and disable sequence hopping • 2: disable, disable group hopping and enable sequence hopping • other values are reserved.
Frequency Domain Allocation [3GPP TS 38.214 [5], sec 6.1.2.2] and Hopping [3GPP TS 38.214 [5], sec 6.3]		
pusch-SecondHop-PRB-v3	uint16_t	Index of the first PRB after intra-slot frequency hopping, as indicated by the value of RB_{start} for i=1, per 3GPP TS 38.214 [5], section 6.3 Valid: when IntraSlotFrequencyHopping Is true Value: 0->274
ldpc-BaseGraph-v3	uint8_t	LDPC base graph to use for UL reception [3GPP TS 38.212 [3], sec 6.2.2]. Value s: 1: LDPC base graph 1 2: LDPC base graph 2 (other values are reserved)
tb-Size-LBRM-Bytes-v3	uint32_t	Parameter TBS_{LBRM} from 3GPP TS 38.212 [3], section 5.4.2.1, for computing the size of the circular buffer. • L2 may set this parameter to 0 to indicate that I_{LBRM} = 0 (i.e. no LBRM)

Table 3-125 PUSCH maintenance FAPIv3

Field	Type	Description
In this specification version, L2 includes this TLV: - to supplement the optional puschUci structure (see Table 3-129), in PUSCH PDU. - for PUCCH formats 3 and 4, in PUCCH PDU; this table is still included for all PUCCH formats, but L2/L3 signals numPart2Types = 0 for PUCCH formats that cannot accommodate UCI part 2. Note on FAPIv7 modifications: • This structure includes was modified for efficiency. • FAPIv7: the changes in this structure are backward compatible, due to the including of this structure in the backward compatible extensions for PUCCH and PUSCH PDU.		

Field	Type	Description
• For the purposes of FAPIv6 backward compatibility, the single-field sub structure in Table 3-127 is used in the location where the FAPIv6 structure was signalled.		
numPart2Types-v3	uint16_t	Max number of UCI part2 types that could be included in the CSI report. Value: 0 -> 100
maxNumPart1Params-v7	uint8_t	Maximum value of numPart1Params in the loop below. Applies to each loop iteration. Should be set to 4 in this specification release. Value: • 4 (in this release) • Other values: reserved If other values used in future FAPI releases, they should be multiples of 4.
For each part2 type		
priority-v3	uint16_t	Priority of the part 2 report; L2 shall signal part2 parameters in increasing order of priorities. For UCI over PUSCH, this corresponds to the CSI priority in 3GPP TS 38.214 [5], section 5.2.3 For UCI over PUCCH, this corresponds to the CSI priority in 3GPP TS 38.214 [5], sections 5.2.4 and 5.2.5
reserved-v7	uint8_t	This field is reserved, shall be set to 0 and may be ignored by PHY in this release. Should be set to 0 in this release
numPart1Params-v3	uint8_t	Number of Part 1 parameters that influence the size of this part 2. Values - 1:maxNumPart1Params in this release (i.e. 1, 2, 3, 4, since maxNumPart1Params = 4 in this release) - Other values: reserved
paramOffsets-v3	uint16_t [maxNumPart1Params]	Ordered list of parameter offsets (offset from 0 = first bit of part1). Note: the size of this vector field is FIXED across all loops. PHY may ignore entries beyond the first *numPart1Params* ones.
paramSizes-v3	uint8_t [maxNumPart1 Params]	Bitsizes of part 1 param in the same order as paramOffsets Note: the size of this vector field is FIXED across all loops. PHY may ignore entries beyond the first *numPart1Params* ones.
part2SizeMapIndex-v3	uint16_t	Index of one of the maps configured Table 3-60, for determining the size of a part2, from the part 1 parameter values. Note, for the indexed map: - The number of parameters must match.

Field		Type	Description
			- The size of each parameter in this table must not exceed the size of the same-index parameter used for the indexed map. If smaller, the part1 lookup value shall be zero-padded for MSB bits.
	part2SizeMapScope-v4	uint8_t	part2SizeMapIndex being referred to: 0: the map index shall be looked up in the Common or PHY Group Context (Shared by all PHYs in the Context) 1: the map index shall be looked up in this PHY's Context
	numPart2s-v6	uint8_t	Number of part2, each of the same size indicated by the map at the part 1 params, at the same priority. Can, for instance, correspond to a number of subbands.

Table 3-126 Uci information for determining UCI Part1 to Part2 correspondence, added in FAPIv3

Field	Type	Description
This structure servers no functional purpose and is only included for backward compatibility with FAPIv6 PHY receivers. See section 3.1.5. In this release, the 'Uci information for determining UCI Part1 to Part2 correspondence' is conveyed in Table 3-126, which is signalled to PHY receivers compatible with this specification release.		
zeroValueObsoleteField-v7	uint16_t	Zero-value field, for backward compatibility (used to be 'numPart2Types' in FAPIv6). Values: • 0 • No other value is allowed in this release Note: this stub replaces a structure for conveying UCI part1 to part2 maps (Table 3-107 in SCF FAPIv6 [32]). The functional role of that structure is served by the optimized Table 3-126 in the current document.

Table 3-127 Backward compatible stub

Field	Type	Description
rv-Index	uint8_t	Redundancy version index [3GPP TS 38.214 [5], sec 6.1.4], it should match value sent in DCI Value: 0->3
harq-Process-ID	uint8_t	HARQ process number [3GPP TS 38.212 [3], sec 7.3.1.1], it should match value sent in DCI

Field	Type	Description
		Value: 0 ->15
newData	uint8_t	Signals whether the PUSCH PDU corresponds to an initial transmission or a retransmission of a MAC PDU for this HARQ process ID for this TB [3GPP TS 38.212 [3], sec 7.3.1.1] [3GPP TS 38.214 [5], sec 5.1.7.2] Value: 0: retransmission 1: new data, i.e. initial transmission **Note**: Unlike NDI, newData does not toggle to indicate new transmission, but is set to 1.
tb-Size	uint32_t	Transport block size (in bytes) [3GPP TS 38.214 [5], sec 6.1.4.2]
num-CB	uint16_t	Number of CBs in the TB (could be more than the number of CBs in this PUSCH transmission). Should be set to zero in any of the following conditions: 1) CBG is not supported or requested 2) newData=1 (new transmission) 3) tbSize=0
cb-PresentAndPosition	uint8_t [ceil(numCb/ 8)]	Each bit represents if the corresponding CB is present in the current retx of the PUSCH. 1=PRESENT, 0=NOT PRESENT.

Table 3-128 Optional puschData information

Field	Type	Description
harq-ACK-BitLength	uint16_t	Number of HARQ-ACK bits [3GPP TS 38.212 [3], section 6.3.2.4] Value: 0 -> 11 (Small block length) 12 ->1706 (Polar) Note: Does not include CRC bits
csi-Part1BitLength	uint16_t	Number of CSI part1 bits [3GPP TS 38.212 [3], section 6.3.2.4 Value: 0 -> 11 (Small block length) 12 ->1706 (Polar) Note: Does not include CRC bits
flag-CSI-Part2-v3	uint16_t	Number of CSI part2 bits [3GPP TS 38.212 [3], section 6.3.2.4] Value: 0 -> No CSI part 2 (in this case, numPart2s = 0, if signalled inTable 3-126 for this PDU) 65535 -> Determine CSI Part2 length based on Table 3-126 (in this case, numPart2s > 0 in that table)

Field	Type	Description
alphaScaling	uint8_t	Alpha parameter, α, used to calculate number of coded modulation symbols per layer. [3GPP TS 38.212 [3], section 6.3.2.4]. Value: 0 = 0.5 1 = 0.65 2 = 0.8 3 = 1
betaOffset-HARQ-ACK	uint8_t	Beta Offset for HARQ-ACK bits. [3GPP TS 38.212 [3], section 6.3.2.4] [3GPP TS 38.213 [4], Table 9.3-1] Value: 0->15
betaOffset-CSI1	uint8_t	Beta Offset for CSI-part1 bits. [3GPP TS 38.212 [3], section 6.3.2.4] [3GPP TS 38.213 [4], Table 9.3-2] Value: 0->18
betaOffset-CSI2	uint8_t	Beta Offset for CSI-part2 bits. [3GPP TS 38.212 [3], section 6.3.2.4] [3GPP TS 38.213 [4], Table 9.3-2] Value: 0->18

Table 3-129 Optional puschUci information

Field	Type	Description
num-PTRS-Ports	uint8_t	Number of UL PTRS ports [3GPP TS 38.212 [3], sec 7.3.1.1.2] Value: 1->2
For numPTRS-Ports {		
ptrs-PortIndex	uint16_t	PT-RS antenna ports [3GPP TS 38.214 [5], sec 6.2.3.1 and 3GPP TS 38.212 [3], section 7.3.1.1.2] Bitmap occupying the 12 LSBs with: bit 0: antenna port 0 bit 1 1: antenna port 11 and for each bit 0: PTRS port not used 1: PTRS port used
ptrs-DMRS-Port	uint8_t	DMRS port corresponding to PTRS. Value: 0->11
ptrs-RR-Offset	uint8_t	PT-RS resource element offset value taken from [3GPP TS 38.211 [2], table 6.4.1.2.2-1] Value: 0->11

Field	Type	Description
}		
ptrs-TimeDensity	uint8_t	PT-RS time density [3GPP TS 38.214 [5], table 6.2.3.1-1] Value: 0: 1 1: 2 2: 4
ptrs-FreqDensity	uint8_t	PT-RS frequency density [3GPP TS 38.214 [5], table 6.2.3.1-2] Value: 0: 2 1: 4 Intel: Follow DLSCH PDU
ul-PTRS-Power	uint8_t	PUSCH to PT-RS power ratio per layer per RE [3GPP TS 38.214 [5], table 6.2.3.1-3] Value: 0: 0dB 1: 3dB 2: 4.77dB 3: 6dB

Table 3-130 Optional puschPtrs information

Field	Type	Description
low-PAPR-GroupNumber	uint8_t	Group number for Low PAPR sequence generation. [3GPP TS 38.211 [2], sec 5.2.2] For DFT-S-OFDM. Value: 0->29
low-PAPR-SequenceNumber	uint16_t	[3GPP TS 38.211 [2], sec 5.2.2] For DFT-S-OFDM.
ul-PTRS-SampleDensity	uint8_t	Number of PTRS groups. [3GPP TS 38.214 [5], sec 6.2.3.2] Value: 1->8
ul-PTRS-TimeDensityTransformPrecoding	uint8_t	Number of samples per PTRS group. [3GPP TS 38.214 [5], sec 6.2.3.2] [3GPP TS 38.211 [2], sec 6.4.1.2.2]. Value: 1->4

Table 3-131 Optional dftsOfdm information

Field	**Type**	**Description**
cb-CRC-StatusRequest-v4	uint8_t	When set to true, PHY – if supported – reports CB CRC status for CBs in cbPresentAndPosition (if numCb >0) or all CBs (if numCb = 0) Shall be ignored in the absence of puschData.

Field	Type	Description
		Range: • 0 = false • 1 = true
UL MIMO support in FAPIv4		
srs-TX-Ports-v4	uint32_t	Bitmap of SRS ports to which the precoding indicated by UlTpmiIndex applies. Same order of antenna ports as in the SRS PDU, depending on the flavour (codebook, non-codebook). For FAPIv3-type precoding: - this bitmap is set to all-zeroes; - ul-TPMI-Index – may be disregarded In this release, for FAPIv4 type precoding, for instance: - Codebook-based precoding: MAC ensures that the bitmap of srsTxPorts is a subset of the SRS TX ports for a Codebook-based SRS Resource. - Non-codebook-based precoding: MAC ensures that the bitmap of srsTxPorts corresponds to single-port non-codebook-based SRS Resources and is equal to the numLayers
ul-TPMI-Index-v4	uint8_t	Tpmi Index, per 3GPP TS 38.211 [2], Section 6.3.1.5. Actual precoder corresponding selected based on TransformPrecoding, numLayers and srsTxPorts. Value: - 0-27: TPMI index - 255: non-codebook
num-UL-SpatialStreamsPorts-v4	uint8_t	Number of ports must be at least as large as the number of layers, if PHY indicates that UL spatial stream assigned is required, in capability maxNumberUlSpatialStreams (TLV 0x0153).
ul-SpatialStreamPorts-v4	uint16_t [n]	n = NumULSpatialStreamsPorts Spatial streams assigned for this PUSCH PDU. Value: 0 → (max # spatial streams–- 1, per TLV 0x0153)

Table 3-132 PUSCH parameters V4

Field	Type	Description
Backward compatible extension (first introduced in FAPIv7). See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.
UCI parameters		
uci-Part1ToPart2Maps-v7	structure	See Table 3-126 Included if and only if pduBitmap[1] = 1
PUSCH allocation in frequency domain [3GPP TS 38.214 [5], sec 6.1.2.2]		

Field	Type	Description
rbg-Size-v7	uint8_t	Resource Block Group: allocation resolution *P* for this PUSCH allocation (see 3GPP TS 38.214 [5], section 6.1.2.2) • 0: 2 RBs • 1: 4 RBs • 2: 8 RBs • 3: 16 RBs
num-DFTS-OFDM-DMRS-Positions-v7	uint8_t	The number of DFTS-OFDM DMRS positions in this PUSCH. Shall be: - 0 if transform precoding is disabled. - consistent with the dl-DMRS-SymbPos bitmap, otherwise
For rsFirstSymbolIdx = 0 … num DFTS OFDM DMRS		
low-PAPR-GroupNumber-v7	uint8_t	Group number for Low PAPR sequence generation. [3GPP TS 38.211 [2], sec 5.2.2], for the rdIdx-th DFT-S-OFDM port, per the dmrs-Ports field. Note: for the first DMRS position, this is the same as in Table 3-131 Value: 0->29
low-PAPR-SequenceNumber-v7	uint16_t	[3GPP TS 38.211 [2], sec 5.2.2], for the rdIdx-th DFT-S-OFDM port, per the dmrs-Ports field. Note: for the first DMRS position, this is the same as in Table 3-131

Table 3-133 PUSCH PDU Backward Compatible Extension

3.4.3.2A MsgA-PUSCH PDU

The format of the Msg-A PUSCH PDU is given in Table 3-134.

The MsgA-PUSCH PDU may includes both mandatory and optional parameters, where the presence of the optional elements is defined in the parameter pduBitmap. No optional elements are specified in this release.

Only parameters signalled by pduBitmap are optional, all other parameters are mandatory. (It should be noted that some parameters are only applicable for certain modes etc. These are still included but their value is ignored by the PHY.)

Field	Type	Description
pdu-Bitmap-v4	uint16_t	Bitmap indicating presence of optional PDUs All bits are reserved, in this release.
prach-To-PRU-MapType-v4	uint8_t	Indicates the mapping that PHY uses to determine the frequency/time/DMRS resources relevant to this MsgA-PUSCH PDU, corresponding to a MsgA-PRACH preamble. - 0 – based on P5 Configuration (mapIndicator signals a MsgA-PUSCHConfigResIndex to the relevant P5

Field	Type	Description
		configuration). Relevant for PHYs that support semi-static configuration via P5 - 1 – based on P7 PRACH PDU mapping (mapIndicator signals the handle). Relevant to PHYs that support dynamic configuration via P7
prach-To-PRU-MapIndicator-v4	uint32_t	Indicates the map for PHY to determine which MsgA-PUSCH & associated DMRS resources are in use. Value: a 16-bit P5 configuration, or a 32-bit PRACH PDU handle, as signalled by prachToPruMapType Regardless of which mapping type is signalled, PHY determines RA-RNTI from the MsgA-PRACH occasion to which the map applies, and determines RAPID from the As a result of the map, PHY can determines the following set of resources and associated parameters signalled in this PDU - MsgA-PUSCH Time Domain index - MsgA-PUSCH Frequency Domain index - DMRS Port - DMRS Sequence (n_{SCID})
msgA-PRACH-SFN-v4	uint16_t	SFN of the MsgA-PRACH PDU whose preambles map to the MsgA-PUSCH occasion signalled by this MsgA-PUSCH PDU. Value: 0 -> 1023
msgA-PRACH-Slot-v4	uint16_t	Slot of the MsgA-PRACH PDU whose preambles map to the MsgA-PUSCH occasion signalled by this MsgA-PUSCH PDU. Value: 0 ->159
handle-v4	uint32_t	An opaque handling returned in the `RX_DATA.indication` and/or `CRC.indication` message
preambleGroupIndex-v5	uint8_t	Index of the preamble group for this MsgA PUSCH PDU, for the case prachToPruMapType = 1 Set to 0 otherwise.
BWP [3GPP TS 38.213 [4], sec 12]		
bwp-Size-v4	uint16_t	If L1 uses puschTransType, this field signals the bandwidth part size [3GPP TS 38.213 [4], sec12]. Number of contiguous PRBs allocated to the BWP. If L1 does not use puschTransType, this field represents the number of VRBs to which PRBs are mapped for the allocation, according to 3GPP TS 38.211 [2], section 6.3.1.7.

Field	Type	Description
		Note: for BWPSize, the two interpretations align, except when the allocation is scheduled for msg3 (by RAR UL grant), in which case the first interpretation uses to active BWP,i size, whereas the second one uses the initial BWP,0 size referenced by 3GPP TS 38. 213 [4], section 8.3. Value: 1->275
bwp-Start-v4	uint16_t	If L1 uses puschTransType, this field signals the bandwidth part start RB index from reference CRB [3GPP TS 38.213 [4], sec 12] If L1 does not use puschTransType, this field represents the CRB corresponding to VRB0, according to 3GPP TS 38.211 [2], section 6.3.1.7. **Note 1**: For BWPStart, the two interpretations align, except when the allocation is scheduled for msg3 "*by DCI format 0_0 with CRC scrambled by TC-RNTI in active uplink bandwidth part i starting at $N_{BWP,i}^{start}$, including all resource blocks of the initial uplink bandwidth part starting at $N_{BWP,0}^{start}$, and having the same subcarrier spacing and cyclic prefix as the initial uplink bandwidth part*", in which case the first interpretation uses to active BWP,i start whereas the second one uses the initial BWP,0 start referenced by 3GPP TS 38.211 [2], section 6.3.1.7. **Note 2**: Only msg3 allocations subject to the conditions listed in Note 1 above are referenced to the initial BWP,0 start. Other types of msg3 allocations are referenced to the start of active uplink BWP,I, as specified in 3GPP TS 38.211 [2], section 6.3.1.7 and 3GPP TS 38.213 [4], section 8.3. Value: 0->274
subcarrierSpacing-v4	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
cyclicPrefix-v4	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
PUSCH information always included		
msgA-MCS-v4	uint8_t	The MCS index used for the MsgA PUSCH configuration and applies to all PUSCH Occasions, as signalled in 3GPP TS 38.331 [13], section 6.3.2 for this MsgA PUSCH. Value: 0 -> 15

Field	Type	Description
transformPrecoding-v4	uint8_t	Indicates if transform precoding is enabled or disabled [3GPP TS 38.214 [5], sec 6.1.4.1] [3GPP TS 38.211 [2], 6.3.1.4] Value: 0: enabled 1: disabled
n-Id-MsgA-PUSCH-v4	uint16_t	The scrambling ID used to generate MsgA PUSCH data scramble sequence, corresponding to n_{ID} in 3GPP TS 38.211 [12], section 6.3.1.1. Note, this is used together with RAPID and RA-RNTI to determine the scrambling sequence generator. Value: 0 -> 1023
DMRS [3GPP TS 38.211 [2], sec 6.4.1.1]		
pusch-DMRS-ScramblingIds-v4	uint16_t [2]	The values of parameter N_{ID}^{0} (first entry) and N_{ID}^{1}(second entry), as specified in 3GPP TS 38.211 [12], section 6.4.1.1.1.1 for this MsgA-PUSCH Value: 0 -> 65535
ul-DMRS-SymbPos-v4	uint16_t	DMRS symbol positions [3GPP TS 38.211 [2], sec 6.4.1.1.3 and Tables 6.4.1.1.3-3 and 6.4.1.1.3-4] Bitmap occupying the 14 LSBs with: bit 0: first symbol and for each bit 0: no DMRS 1: DMRS
dmrs-MappingType-v4	uint8_t	PUSCH mapping Type A or B, as reflected by the mappingTypeMsgA-PUSCH RRC parameter in 38.331 section 6, and further described in 38.213, section 8.1A
Pusch Allocation in frequency domain [3GPP TS 38.214 [5], sec 6.1.2.2, 3GPP TS 38.331 [13], section 6.3.2]		
start-RB-PUSCH-Ocas-v4	uint16_t	The common RB index of first RB configured for first FD msgA-PUSCH occasion within the UL slot for this MsgA-PUSCH PDU. Value: 0 -> 274 Note: the actual start Rb is determined from the parameters in this PDU frequency-domain subsection, and the frequency domain index mapped from MsgA-PRACH.
num-RB-PUSCH-Ocas-v4	uint8_t	The number of PRBs that is consists of the PUSCH occasion on which all these PUSCH Resource Units reside. Value 1 to 32

Field	Type	Description
guardBand-v4	uint8_t	The guard band between two adjacent frequency domain PUSCH occasions for this MsgA PUSCH PDU in unit of PRB. Value: 0 -> 1
intraSlotFrequencyHopping-v4	uint8_t	Indicates if intra-slot frequency hopping is enabled for this MsgA-PUSCH. [3GPP TS 38.212 [3], sec 7.3.1.1] [3GPP TS 38.214 [5], sec 6.3] Value: 0: disabled 1: enabled
intraSlotFrequencyHoppingBits-v4	uint8_t	The value of RRC parameter msgA-HoppingBits used to determine the location of the second hop, as specified in Table 8.3-1 in 3GPP TS 38.213 [22] (per 3GPP TS 38.213 [22], section 8.1A). Values: 0 … 3, as determined by Table 8.3-1 in 3GPP TS 38.213 [22], and the value of BWPSize. The value of this field is only relevant when intraSlotFrequencyHopping is enabled.
tx-DirectCurrentLocation-v4	uint16_t	The uplink Tx Direct Current location for the carrier. Only values in the value range of this field between 0 and 3299, which indicate the subcarrier index within the carrier corresponding to the numerology of the corresponding uplink BWP and value 3300, which indicates“"Outside the carrie”" and value 3301, which indicates“"Undetermined position within the carrie”" are used. [3GPP TS 38.331 [6], UplinkTxDirectCurrentBWP IE] Value: 0->4095
<obsolete field>	uint8_t	This field is ignored by PHY (was uplinkFrequencyShift7p5khz in SCF FAPIv6)
Resource Allocation in time domain [3GPP TS 38.214 [5], sec 5.1.2.1, 3GPP TS 38.331 [13], section 6.3.2]		
startSymbId-PUSCH-Ocas-v4	uint8_t	The starting symbol index of the first-time domain PUSCH occasion within the UL slot for this MSGA PUSCH PDU. Value: 0 -> 13 **Note**: The actual start symbol is determined from the parameters in this PDU time-domain subsection, and the time domain index mapped from MsgA-PRACH.
duration-PUSCH-Ocas-v4	uint8_t	the duration of all time domain PUSCH occasion symbols within the UL slot for this MsgA PUSCH PDU. Value: 1 -> 14 symbols

Field	Type	Description
guardPeriod-v4	uint8_t	The guard period between two adjacent time domain PUSCH occasions in unit of symbols of the MsgA PUSCH SCS Value: 0, 1, 2, 3
MsgA-Pusch Data Information [3GPP TS 38.214 [5], sec 6.1.4.2, 3GPP TS 38.331 [13], section 6.3.2]		
tb-Size-v4	uint32_t	Transport block size (in bytes) [3GPP TS 38.214 [5], sec 6.1.4.2, 3GPP TS 38.331 [13], section 6.3.2]
backwardCompatibleExtension-v7	structure	See Table 3-135 (see section 3.1.5)

Table 3-134 MsgA-PUSCH PDU

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-135 MsgA-PUSCH PDU Backward Compatible Extension

3.4.3.3 PUCCH PDU

The format of the PUCCH PDU is given in Table 3-136.

The PUCH PDU includes information regarding SR, HARQ and CSI and the valid data varies depending on whether it is PUCCH format 0, 1, 2, 3 or 4.

Field	Type	Description
rnti	uint16_t	The RNTI used for identifying the UE when receiving the PDU Value: 1 -> 65535
handle	uint32_t	An opaque handling returned in the `UCI.indication` message
BWP [3GPP TS 38.213 [4], sec 12		
bwp-Size	uint16_t	Bandwidth part size [3GPP TS 38.213 [4], sec12]. Number of contiguous PRBs allocated to the BWP Value: 1->275
bwp-Start	uint16_t	Bandwidth part start RB index from reference CRB [3GPP TS 38.213 [4], sec 12] Value: 0->274
subcarrierSpacing	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
cyclicPrefix	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2]

Field	Type	Description
		0: Normal; 1: Extended
formatType	uint8_t	PUCCH Format Type [3GPP TS 38.211 [2], sec 6.3.2.1] Value: 0 ->4
multiSlot-TX-Indicator	uint8_t	This field is to flush, keep or combine the buffer used for multiple-slot PUCCH transmissions [3GPP TS 38.213 [4], sec 9.2.6] Value: 0: No multi slot transmission 1: Multi slot transmission starts 2: Multi slot transmission continues 3: Multi slot transmission ends This filed being '0' or '1' indicates the previous PUCCH transmission for this UE (RNTI and Handle) is ended in case L2/L3 doesn't send '3' for the previous ongoing multi-slot transmission.
pi2-BPSK	uint8_t	When enabled, indicates that PHY shall use pi/2 BPSK for UCI symbols instead of QPSK for PUCCH formats¾4 for the PUCCH transmission of PUCCH PDU. [3GPP TS 38.211 [2], sec 6.3.2.6.2] Value: - 0: disabled, - 1: enabled for data (3GPP TS 38.211 [2], section 6.3.2.6.2), together with transform precoding for DMRS (3GPP TS 38.211 Rel-16 [12], section 6.4.1.3.3.1), i.e. Rel-16 low-PAPR waveform - 2: enabled for data (3GPP TS 38.211 [2], section 6.3.2.6.2), without DMRS transform precoding (3GPP TS 38.211 [12], section 6.4.1.3.3.1), i.e. Rel-15 low-PAPR waveform
Pucch Allocation in frequency domain [3GPP TS 38.213 [4], sec 9.2.1]		
prb-Start	uint16_t	The starting PRB $RB_{\mathrm{BWP}}^{\mathrm{offset}}$ within the BWP for this PUCCH, or first PRB prior to hopping [3GPP TS 38.213 [4], sec 9.2.1]. Valid for all formats Value: 0->274
prb-Size	uint16_t	The actual number of PRBs used by the UE within this PUCCH. Valid for all formats. Value: 1 -> 16
Pucch Allocation in time domain		

Field	Type	Description
startSymbolIndex	uint8_t	Start symbol index of PUCCH from the start of the slot, S. [3GPP TS 38.213 [4], sec 9.2.2]. Valid for all formats Value: 0->13
numSymbols	uint8_t	PUCCH duration in symbols [3GPP TS 38.213 [4], sec 9.2.2] Values: 1 -2: Valid for formats 0,2 4->14: Valid for formats 1,3,4
Hopping information [3GPP TS 38.211 [2], sec 6.3.2.2.1]		
intraSlotFrequencyHopping	uint8_t	Intra-slot Frequency hopping for a PUCCH resource [3GPP TS 38.211 [2], sec 6.3.2.2.1]. Valid for all formats Value: 0: disabled 1: enabled
secondHop-PRB	uint16_t	Index of the first PRB after frequency hopping. Valid for all formats. Value:0->274
pucch-GroupHopping	uint8_t	Signalling of group and sequence hopping for PUCCH formats 0, 1, 3 and 4 [3GPP TS 38.211 [2], sec 6.3.2.2.1]. Value: 0: neither, neither group nor sequence hopping is enabled 1: enabled, enable group hopping and disable sequence hopping 2: disable, disable group hopping and enable sequence hopping (note: These values are only applicable if both L1 and L2/L3 support FAPIv3, otherwise refer to SCF-222 v2, parameter groupHopFLag)
obsolete8bit-v3	uint8_t	This flag is obsolete in FAPIv3. Refer to parameter sequenceHopFlag in SCF-222 v2 if FAPIv3 is not supported.
n-ID-PUCCH-Hopping	uint16_t	The parameter n_{ID} used for sequence hopping. [3GPP TS 38.211 [2], sec 6.3.2.2.1] Valid for formats 0, 1, 3 and 4. Value: 0->1023
initialCyclicShift	uint16_t	Initial cyclic shift (M0) used as part of frequency hopping. [3GPP TS 38.213 [4], sec 9.2.1 and 3GPP TS 38.211 [2], sec 6.3.2.2.2]. Valid for formats 0, 1, 3 and 4

Field	Type	Description
		Value: 0->11
n-ID-PUCCH-Scrambling	uint16_t	parameter n_{ID} in [3GPP TS 38.211 [2], sec 6.3.2.5.1 and 6.3.2.6.1] Valid for formats 2, 3 and 4. Value: 0->1023
timeDomain-OCC-Index	uint8_t	An index of an orthogonal cover code [3GPP TS 38.211 [2], sec 6.3.2.4.1]. Valid for format 1. Value: 0->6
pre-DFT-OCC-Index	uint8_t	An index of an orthogonal cover code. [3GPP TS 38.211 [2], sec 6.3.2.6.3]. Valid for format 4. Value: 0->3
pre-DFT-OCC-Len	uint8_t	A length of an orthogonal cover code. [3GPP TS 38.211 [2], sec 6.3.2.6.3]. Valid for format 4. Value: 2 or 4
DMRS [3GPP TS 38.211 [2], sec 6.4.1.3]		
add-DMRS-Flag	uint8_t	Flag for additional DMRS. [3GPP TS 38.213 [4], sec 9.2.2]. Valid for formats 3 and 4. Value: 0 = disabled 1 = enabled
n-ID0-PUCCH-DMRS-Scrambling	uint16_t	NID0, as defined for format 2 in 38.211 [2] (Rel-15), section 6.4.1.3.2.1 and referenced in 38.211 [12] (Rel-16) section 6.4.1.3.3.1 for formats 3 and 4 For formats¾4, PHY shall use if and only if pi2Bpsk is enabled. Value: 0->65535
m0-PUCCH-DMRS-CyclicShift	uint8_t	m0, as defined for referenced for formats 3 and 4 in 38.211, section 6.4.1.3.3.1 [3GPP TS 38.211 [2], sec 6.4.1.3.3.1] PHY shall use if and only if pi2Bpsk is disabled. Value: 0 ->9
sr-BitLen-v4	uint8_t	Indicates the number of SR bits expected in the PUCCH transmission. Valid for all PUCCH formats

Field	Type	Description
		For PUCCH Format 0/1:Value: 0 = no SR 1 = SR occasion For PUCCH Format 2/3/4: Value: 0 = no SR; 1 = one SR bit; 2 = two SR bits; 3 = three SR bits; 4 = four SR bits Note: When MAC is implemented according to 3GPP TS 38.213 [4][22], the maximum number of SR bits expected for a single UE is 4.
bitLen-HARQ	uint16_t	Bit length of HARQ payload Valid for all formats. Value: 0 = no HARQ bits 1->2 = Valid for Formats 0 and 1 2 -> 1706 = Valid for Formats 2, 3 and 4 For PUCCH Format 1, this field shall be the same for the HARQ and SR resource, if both are present
csi-Part1BitLength	uint16_t	Bit length of CSI part 1 payload. Valid for formats 2, 3 and 4. Value: 0 = no CSI bits 1 -> 1706
Beamforming		
Beamforming	structure	See Table 3-151; for PHYs supporting MU-MIMO groups, see also Table 3-152
Extension TLVs		
PUCCH Maintenance Parameters added in FAPIv3	structure	See Table 3-137
-Obsolete structure; included for backward compatibility	structure	Obsolete structure; see Table 3-127
Parameters added in FAPIv4	structure	See Table 3-138
backwardCompatibleExtension-v7	structure	See Table 3-139 (see section 3.1.5)

Table 3-136 PUCCH PDU

Field	Type	Description
In this specification version, L2 always includes this TLV to extend PUCCH PDU		
maxCodeRate-v3	uint8_t	Max coding rate to determine how to feedback UCI on PUCCH for format 2, 3 or 4 [3GPP TS 38.213 [4], Table 9.2.5.2-1] Value Range: - 0 -> 7: max code rate per Table 9.2.5.2-1 of 3GPP TS 28.213 [4] - 255: not applicable (e.g., formats 0 or 1) **Note**: In 3GPP TS 38.213 [4], value 7 is reserved
ul-BWP-Id-v3	uint8_t	Index of PUCCH group and sequence hopping parameters to use for this PUCCH PDU. Value: 255: L1 not configured with semi-static signalling of PUCCH group and sequence hopping parameters <255: PHY may assume that the parameters linked to this UL-BWP-ID inn the PUCCH Semi-Static Configuration Table 3-64 apply.

Table 3-137 PUCCH basic extension for FAPIv3

Field	Type	Description
In this specification version, L2 always includes this TLV to extend PUCCH PDU		
uci-ReportFormat-v4	uint8_t	Flag to indicate if joint reporting of different type of UCI payload is required or not. Range: 0: separate reporting (PHY reports HARQ/SR/CSIp1/CSIp2 in separate parameters) 1: combined reporting (PHY reports the uciPayload(s), without attempting to interpret its mapping to HARQ, SR, CSIp1 or CSIp2 bits) PHY support for this flag is signalled in uciReportFormatPucchForamt234 TLV of `PARAM.response`
For MU-MIMO support in FAPIv4		
num-UL-SpatialStreamsPorts-v4	uint8_t	Number of spatial streams used to signal this PUCCH allocation. Non-zero if PHY indicates that UL spatial stream assigned is required, in capability maxNumberUlSpatialStreams (TLV 0x0153).
ul-SpatialStreamPorts-v4	uint16_t [n]	n = numUL-SpatialStreamsPorts Spatial streams assigned for this PUSCH PDU. Value: 0 → (max # spatial streams–- 1, per TLV 0x0153)

Table 3-138 PUCCH basic extension for FAPIv4

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Field	Type	Description
Parameters added in FAPIv7		
uci-Part1ToPart2Maps-v7	structure	See Table 3-126

Table 3-139 PUCCH PDU Backward Compatible Extension

3.4.3.4 SRS PDU

The format of the SRS PDU is shown in Table 3-140.

Field	Type	Description
rnti	uint16_t	UE RNTI Value: 1->65535
handle	uint32_t	An opaque handling returned in the `SRS.indication`
BWP [3GPP TS 38.213 [4], sec 12]		
bwp-Size	uint16_t	Bandwidth part size [3GPP TS 38.213 [4], sec 12]. Number of contiguous PRBs allocated to the BWP Value: 1->275
bwp-Start	uint16_t	Bandwidth part start RB index from reference CRB [3GPP TS 38.213 [4], sec 12] Value: 0->274
subcarrierSpacing	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] Value:0->4
cyclicPrefix	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
numAntPorts	uint8_t	Number of antenna ports $N_{\mathrm{ap}}^{\mathrm{SRS}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.1] Value: 0 = 1 port 1 = 2 ports 2 = 4 ports
numSymbols	uint8_t	Number of symbols $N_{\mathrm{symb}}^{\mathrm{SRS}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.1] Value: 0 = 1 symbol 1 = 2 symbols 2 = 4 symbols 3 = 12 symbols
numRepetitions	uint8_t	Repetition factor R [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 = 1 1 = 2

Field	Type	Description
		2 = 4
timeStartPosition	uint8_t	Starting position in the time domain l_0 [3GPP TS 38.211 [2], Sec 6.4.1.4.1] Note: the MAC undertakes the translation from startPosition to l_0 Value: 0 → 13
configIndex	uint8_t	SRS bandwidth config index C_{SRS} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 63
Sequence-ID	uint16_t	SRS sequence ID $n_{\mathrm{ID}}^{\mathrm{SRS}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.2] Value: 0 → 65535
bandwidthIndex	uint8_t	SRS bandwidth index B_{SRS} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 3
combSize	uint8_t	Transmission comb size K_{TC} [3GPP TS 38.211 [2], Sec 6.4.1.4.2] Value: 0 = comb size 2 1 = comb size 4 2 = comb size 8 (Rel16)
combOffset	uint8_t	Transmission comb offset $\bar{k}_{\mathrm{TC}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 1 (combSize = 0) Value: 0 → 3 (combSize = 1) Value: 0 → 7 (combSize = 2)
cyclicShift	uint8_t	Cyclic shift $n_{\mathrm{SRS}}^{\mathrm{cs}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.2] Value: 0 → 7 (combSize = 0) Value: 0 → 11 (combSize = 1) Value: 0 → 5 (combSize = 2)
frequencyPosition	uint8_t	Frequency domain position n_{RRC} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 67
frequencyShift	uint16_t	Frequency domain shift n_{shift} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 268
frequencyHopping	uint8_t	Frequency hopping b_{hop} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 3
groupOrSequenceHopping	uint8_t	Group or sequence hopping configuration (RRC parameter groupOrSequenceHopping in SRS-Resource IE)

Field	Type	Description
		Value: 0 = No hopping 1 = Group hopping groupOrSequenceHopping 2 = Sequence hopping
resourceType	uint8_t	Type of SRS resource allocation [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0: aperiodic 1: semi-persistent 2: periodic
t-SRS	uint16_t	SRS-Periodicity in slots [3GPP TS 38.211 [2], Sec 6.4.1.4.4] Value: 1,2,3,4,5,8,10,16,20,32,40,64,80,160,320,640,1280,2560
t-Offset	uint16_t	Slot offset value [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value:0->2559
Beamforming		
Beamforming	structure	See Table 3-151; for PHYs supporting MU-MIMO groups, see also Table 3-152
SRS Optimized Signalling Parameters added in FAPIv4	structure	See Table 3-141
SRS Reporting Parameters added in FAPIv4	structure	See Table 3-142
SRS UL Combiner Parameters added in FAPIv4	structure	See Table 3-143
SRS Parameters added in FAPIv5	structure	See Table 3-144
SRS Parameters added in FAPIv6	structure	See Table 3-145
backwardCompatibleExtension-v7	structure	See Table 3-146 (see section 3.1.5)

Table 3-140 SRS PDU

Field	Type	Description
srs-BandwidthSize-v4	uint16_t	$m_{SRS,b}$: Number of PRB’s that are sounded for each SRS symbol, per 3GPP TS 38.211 [2], section 6.4.1.4.3; Value: 4->272

Field		Type	Description
For each symbol in numSymbols			
	srs-BandwidthStart-v4	uint16_t	PRB index for the start of SRS signal transmission. The PRB index is relative to the CRB0 or reference Point A. 3GPP TS 38.211 [2], section 6.4.1.4.3; Value: 0->(2199+268=3467)
	sequenceGroup-v4	uint8_t	Sequence group (u) as defined in 3GPP TS 38.211 [2], section 6.4.1.4.2 Value: 0->29
	sequenceNumber-v4	uint8_t	Sequence number (v) as defined in 3GPP TS 38.211 [2], section 6.4.1.4.2 TS 38.211 Value: 0->1

Table 3-141 FAPIv4 SRS Optimized Signalling Parameters

Field	Type	Description
srs-Usage-v4	uint32_t	Indicates the usage of the SRS Resource signalled by this PDU. Value: - 0 – beamManagement - 1 – codebook - 2 – nonCodebook - 3 – antennaSwitching - 4 – positioning - 5 – 255: reserved. **Note**: - Capability TLV 0x0080 (supportedSrsUsage) signals what usages are supported by PHY.
reportType-v4	uint32_t	Bitmap of the requested Report Type, based on this SRS PDU. Report types are represented as bit positions: • [0] Per-PRG and Symbol SNR • [1] Normalized Channel I/Q Matrix; Note: ○ if requested for *codebook* Usage, represents PRG I and Q channel estimate, per srs Tx port and gNB antenna element ○ If requested for *nonCodebook* Usage, represents PRG I and Q channel estimate, per SRI and gNB antenna element • [2] Channel SVD Representation; Note: ○ If requested for *antennaSwitching* Usage, it is the SVD representation of UE Rx and gNB sets of antenna element • [3] Positioning Report

Field	Type	Description
		• [4] SU-MIMO Codebook Recommendation • [5] Channel 2D-DFT Representation • [6-31] Reserved Value: • 0: This report is not requested • 1: This report is requested **Notes**: - if all bits are set to 0, no report is required. - per-Usage allowed reports are indicated by L1 in Capability TLV 0x0160 (allowedSrsReportsPerUsage)
singularValueRepresentation-v4	uint8_t	0 – 8-bit dB 1 – 16-bit linear 255 – not applicable See Table 3-186 Other values reserved Other values reserved
iq-Representation-v4	uint8_t	0 – 16 bit 1 – 32-bit 255 - not applicable See Table 3-185 and Table 3-186 Other values reserved
srs-Report-FD-Resolution-v4	uint16_t	The resolution (in RB) at which SRS reports shall be made. Applies to those SRS reports that report measurements across one or more PRBs. This resolution corresponds to the number of PRBs constituting an srs-Report-PRG (see section 3.4.10) 1-272 0 – reserved
numTotal-UE-Antennas-v4	uint8_t	1 … 16 in this release. This is the total number of UE antennas being sampled for the SRS Usage. Notes: - a UE's antennas may be sampled via one or multiple SRS PDU - the subset of *numTotalUE-Antennas* antennas sampled by this SRS PDU is indicated in *sampledUE-Antennas*.
sampled-UE-Antennas-v4	uint32_t	Bitmap of UE antenna indices sampled by the SRS waveform corresponding to this PDU's SRS Resource. • Codebook: corresponds to antenna ports in SRS Resource; non-overlapping indices are expected for ports sampled via different SRIs, for a given UE. [3GPP TS 38.214 [5], section 6.1.1.2] • Non-codebook: corresponds to SRIs [3GPP TS 38.214 [5], section 6.1.1.2] • Antennas-switch: indices of UE Rx antennas in the total number of antennas. Depending on the UE capability, multiple SRS PDUs may be

Field	Type	Description
		needed to sample all UE Rx antennas for each PRG. [3GPP TS 38.213 [4], section 6.2.1.2] Notes: - *numAntPorts* bits are set to '1' in *sampledUE-Antennas*. - only bits in positions usageShift + [0 … (*numTotalUE-Antennas*-1)] may be set to '1' to indicate which of the *sampleUE-Antennas* ports for the SRS usage are sampled in this SRS PDU for this UE. ○ usageShift is the index of the first bit for the usage. - across all SRS PDUs for the *SRS Usage* for this UE, it is expected that all of the *numTotalUE-Antennas* will be sampled for all PRBs in the SRS bandwidth. - bits in sampled-UE-Antennas can correspond to SRS ports for multiple usages. If a L2/L3 reuses a bit position, L1 assumes that the previous port samples for that bit position and frequency domain location can be discarded. Example: - Slot 0: One 2-port SRS PDU with *codebook* usage: ○ numTotal-UE-Antennas = 2 ○ sampled-UE-Antennas = 00..000000011b - Slot 4: Another 2-port SRS PDU with *non-codebook* usage: ○ numTotal-UE-Antennas = 2 ○ sampled-UE-Antennas = 00..000001100b - Slot 8: Another 2-port SRS PDU with antennaSwitch usage: ○ numTotal-UE-Antennas = 4 ○ sampled-UE-Antennas = 00..000110000b - Slot 8: Another 2-port SRS PDU with antennaSwitch usage: ○ numTotal-UE-Antennas = 4 ○ sampled-UE-Antennas = 00..011000000b - Result: L1/PHY has collected channel samples for the following SRS ports: <table><tr><td>FAPI ports/bit locations</td><td>Usage</td></tr><tr><td>0</td><td>Codebook #0</td></tr><tr><td>1</td><td>Codebook #1</td></tr><tr><td>2</td><td>NonCodebook #0</td></tr><tr><td>3</td><td>NonCodebook #1</td></tr></table>

Field	Type	Description
		4 — AntennaSwitch#0 5 — AntennaSwitch#1 6 — AntennaSwitch#2 7 — AntennaSwitch#3
reportScope-v4	uint8_t	Which antennas Report (in ReportType) should account for: Value: 0: ports in sampledUE-Antennas (i.e. SRS Resource) 1: all numTotalUE-Antennas ports for the SRS usage. For *SRS Usage = antennaSwith* reports of SVD type for , value 0 is only allowed if the number of bits set in *sampledUE-Antennas = numTotalUE-Antennas* If this PDU conveys one or more hops of the pertinent SRS Resource in this slot, then the L1 report corresponding to this SRS PDU is also expected to only convey report information pertinent to this/these hop/s. If a frequency-aggregate (e.g. wideband) metric is included in the report (e.g. TPMI, or wideband SNR in the Codebook report), L1 may consider all available hops to convey the aggregate report metric.
indexOf-SRS-Port0-v6	uint8_t	The port index inside *sampled-UE-Antennas* corresponding to port 0 for this SRS PDU and usage. Port #n of this SRS PDU shall correspond to port index *indexOfSrsPort0* + #n (out of *numTotalUE-Antennas* for the usage) All *numTotalUE-Antennas* ports for the shall correspond to consecutive bit positions in *sampledUE-Antennas*. Note: this field is expected to be usageShift (i.e. lowest bit index set in *sampled-UE-Antennas*, for the usage) whenever *numTotalUE-Antennas* = number of ports for the SRS PDU and usage.

Table 3-142 FAPIv4 SRS Reporting Parameters

Field	Type	Description
MU-MIMO support for FAPIv4		
num-UL-SpatialStreamsPorts-v4	uint8_t	In this release, L2 may set this number to 0 to leave spatial stream index assignment to L1 (e.g. L1 uses spatial streams reserved for SRS), regardless of the information in capability maxNumberUlSpatialStreams (TLV 0x0153)

Field	Type	Description
ul-SpatialStreamPorts-v4	uint16_t [n]	n = NumULSpatialStreamsPorts Numbers of ports used for signalling this SRS allocation. Value: 0 → (max # spatial streams-- 1, per TLV 0x0153, or 65535 if the TLV is absent)

Table 3-143 FAPIv4 SRS UL Combiner Parameters

Field	Type	Description
symbolPuncturingBitmap-v5	uint16_t	A bitmap of SRS symbols that are punctured For each bit: • 0: symbol punctured • 1: symbol transmitted Bit positions: • [0]: symbol #0 • [1]: symbol #1 • … • … • [14]: reserved (set to 0, in this release) • [15]: reserved (set to 0, in this release) Note: this bitmaps serves for handling PUSCH overlaps with SRS for positioning

Table 3-144 FAPIv5 SRS Parameters

Field	Type	Description
transformPrecoding-v6	uint8_t	Applies to the *SU-MIMO Codebook Recommendation* Report, if configured in ReportType, can be ignored otherwise. Indicates if transform precoding is enabled or disabled for the PUSCH of this UE, per 3GPP TS 38.214 [5], Section 6.1.4.1 and 3GPP TS 38.211 [2], Section 6.3.1.4 Value: • 0: enabled • 1: disabled
codebookSubset-v6	uint8_t	Applies to the *SU-MIMO Codebook Recommendation* Report, if configured in ReportType, can be ignored otherwise. Value: • 0: fullyAndPartialAndNonCoherent • 1: partialAndNonCoherent • 2: noncoherent • 3:N/A

Table 3-145 FAPIv6 SRS Parameters

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.
Parameters added in FAPIv7		
numAdditional-SRS-Usages-v7	uint8_t	Additional usages this SRS PDU is overloaded with. These usages are additional to the usage in Table 3-142 Value: 0 … number of usages Note: in this FAPI release, a usage may be associated at most once to an SRS PDU, to facilitate linkage with the corresponding SRS Report Non-zero values are only allowed if TLV 0x017F indicates support for SRS PDU usage overloading.
For idxAddl-SRS-Usage		
srs-ReportingStructureForTheUsage-v7	structure	See Table 3-142 (the usage is signal in that table, too)

Table 3-146 SRS PDU Backward Compatible Extension

3.4.3.4b ZP-SRS PDU

The format of the ZP-SRS Resource PDU is shown in Table 3-147. There may be multiple such PDUs per slot, describes SRS time, frequency, comb and cyclic shift resources that L2/L3 does not allocate for UE transmission in this slot. L1 PHY may, for instance, use these for noise estimation in the measurement bandwidth.

Field	Type	Description
meas-BW-Size-v6	uint16_t	Number of contiguous PRBs to which this Zero-Power (ZP) SRS Resource applies Value: 1->275
meas-BW-Offset-v6	uint16_t	Start RB Index, for this ZP SRS from reference CRB [3GPP TS 38.213 [4], sec 12] Value: 0->274
measSubcarrierSpacing-v6	uint8_t	subcarrierSpacing [3GPP TS 38.211 [2], sec 4.2] for this ZP SRS resource Value:0->4
cyclicPrefix-v6	uint8_t	Cyclic prefix type [3GPP TS 38.211 [2], sec 4.2] 0: Normal; 1: Extended
numSymbols-v6	uint8_t	Number of symbols $N_{\text{symb}}^{\text{SRS}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.1] Value: 0 = 1 symbol

Field	Type	Description
		1 = 2 symbols 2 = 4 symbols 3 = 12 symbols
numRepetitions-v6	uint8_t	Repetition factor R [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 = 1 1 = 2 2 = 4
timeStartPosition-v6	uint8_t	Starting position in the time domain l_0 [3GPP TS 38.211 [2], Sec 6.4.1.4.1] Note: the MAC undertakes the translation from startPosition to l_0 Value: 0 → 13
configIndex-v6	uint8_t	SRS bandwidth config index C_{SRS} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 63
sequence-ID-v6	uint16_t	SRS sequence ID $n_{\mathrm{ID}}^{\mathrm{SRS}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.2] Value: 0 → 65535
bandwidthIndex-v6	uint8_t	SRS bandwidth index B_{SRS} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 3
combSize-v6	uint8_t	Transmission comb size K_{TC} [3GPP TS 38.211 [2], Sec 6.4.1.4.2] Value: 0 = comb size 2 1 = comb size 4 2 = comb size 8 (Rel16)
combOffset-v6	uint16_t	bit map of ZP Transmission comb offset $\bar{k}_{\mathrm{TC}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.3] For each bit • 0: the corresponding comb is unused. • 1: the corresponding comb is used. The (16-combSize) MSB of the high 16 bits to be set to 1.
cyclicShift-v6	uint16_t	bit map of ZP cyclic shift $n_{\mathrm{SRS}}^{\mathrm{cs}}$ [3GPP TS 38.211 [2], Sec 6.4.1.4.2] For each bit • 0: the corresponding shift is unused. • 1: the corresponding shift is used. The (16-Nshifts) MSB's to be set to 1, here Nshifts is the number of cyclic shifts.
frequencyPosition-v6	uint8_t	Frequency domain position n_{RRC} [3GPP TS 38.211 [2], Sec 6.4.1.4.3]

Field	Type	Description
		Value: 0 → 67
frequencyShift-v6	uint16_t	Frequency domain shift n_{shift} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 268
frequencyHopping-v6	uint8_t	Frequency hopping b_{hop} [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value: 0 → 3
groupOrSequenceHopping-v6	uint8_t	Group or sequence hopping configuration (RRC parameter groupOrSequenceHopping in SRS-Resource IE) Value: 0 = No hopping 1 = Group hopping groupOrSequenceHopping 2 = Sequence hopping
t-SRS-v6	uint16_t	SRS-Periodicity in slots [3GPP TS 38.211 [2], Sec 6.4.1.4.4] Value: 1,2,3,4,5,8,10,16,20,32,40,64,80,160,320,640,1280,2560
t-Offset-v6	uint16_t	Slot offset value [3GPP TS 38.211 [2], Sec 6.4.1.4.3] Value:0->2559
backwardCompatibleExtension-v7	structure	See Table 3-148 (see section 3.1.5)

Table 3-147 Zero-Power SRS Resource PDU

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-148 ZP-SRS PDU Backward Compatible Extension

3.4.3.4c RNTI_RELEASE PDU

The format of the RNTI Release PDU is shown in Table 3-149. L1 may interpret this PDU to apply to all L1 states/memory maintained for the RNTIs in this list.

As soon as the L2 state associated with the RNTI changes so that PHY resources associated with the RNTI are no longer required, L2 shall release the RNTI by including it in this PDU.

Field	Type	Description
num-RNTIs-v6	uint16_t	Number of RNTIs to be released. N
rntis-ToRelease-v6	uint16_t [n]	The list of RNTIs to release
backwardCompatibleExtension-v7	structure	See Table 3-150 (see section 3.1.5)

Table 3-149 RNTI_RELEASE PDU

Field	**Type**	**Description**
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-150 RNTI-RELEASE PDU Backward Compatible Extension

3.4.3.5 Rx Beamforming PDU

The beamforming PDU is included in the PUCCH, PUSCH, PRACH and SRS PDUs. The format is shown in Table 3-151.

Semistatic beamforming weights (i.e. signalled via beam ids) are applied to a group of consecutive PRBs. These beamforming groups (BFG) are grouped into non-overlapping consecutive BFGSize PRBs, starting from CRB0.

Note: BFGs follow the same frequency anchoring as 3GPP PRG (c.f. 3GPP TS 38.214 [5], section 5.1.2.3). Aside from this anchoring, there is no defined relationship between BFG and PRG.

Field	Type	Description
trp-Scheme-v3	uint8_t	This field shall be set to 0, to identify that this table is used.
num-BFGs	uint16_t	Number of BFGs spanning this allocation. Value : 1->275
bfg-Size	uint16_t	Size in RBs of a beamforming resource block group (BFG) – to which the same digital beamforming gets applied. Value: 1->275
dig-BF-Interface	uint8_t	Number of logical antenna ports (parallel streams) resulting from the Rx combining If digBFInterfaces[x] = 0, the n-th precoding input (baseband port) corresponds to the n-th logical port, i.e. beamforming is straight-wire/identity. Value: 0->255
For number of BFGs {		
for each digBFInterface {		
beam-ID	uint16_t	Index of the digital beam weight vector pre-stored at cell configuration. The vector maps baseband ports to the logical port corresponding to dibBFInterface.

Field	Type	Description
		Value: 0->65535 Only the 15 LSB indicate a beam index 0…32,767 MSB shall be interpreted as follows: • 0: look up in Hi-PHY preconfigured DBM the beam index constructed from 15 LSB • 1: signal to Lo-PHY the beam index constructed from 15 LSB (e.g. over 7.2x fronthaul, if supported by TLV 0x0164)
}		
}		

Table 3-151 Rx beamforming PDU

3.4.3.6 Rx Beamforming PDU for SRS-based combining

In this release, this PDU may be included when UL combining is performed as described in section 2.2.7.2. The format is shown in Table 3-113.

Field	Type	Description
trp-Scheme-v4	uint8_t	5: Single TRP, based on SRS
num-MU-MIMI-Groups-v4	uint8_t	Number of UL MuMIMO groups covering the allocation of this PDU. Any PRBs in the allocation for the channel must belong to exactly one MuMIMO group. For SU-MIMO, the number of groups may be 0.
For each of the num-MU-MIMO-Groups {		
mu-MIMO-Group-ID	uint16_t	muMimoGroups are listed in increasing muMimoGroup index order. Value: 0→3,821 ***Note****: mu-MIMO-GroupIdx are defined in increasing order of their start symbol and on a given start symbol they are defined in increasing order of PRB*
}		

Table 3-152 SRS-based Rx beamforming PDU

3.4.3.7 Rx Beamforming based on dynamic weight signalling by L2/L3

The Rx beamforming PDU based on dynamic weight signalling by L2/L3; in this release, this PDU may be included, when beamforming is performed as described in section 2.2.6.2b2.2.7.2. The format is shown in Table 3-153.

Field	Type	Description
trp-Scheme-v6	uint8_t	6: Single TRP, based on dynamic weight signalling by L2/L3
num-MU-MIMO-Groups -v6	uint8_t	Number of UL MuMIMO groups covering the allocation of this PDU. Any PRBs in the allocation for the channel must belong to exactly one MuMIMO group. For SU-MIMO numMuMIMOGroups to be set to 1.

Field			Type	Description
For each of the numMuMimoGroups {				
	mu-MIMO-Group-ID -v6		uint16_t	muMimoGroups are listed in increasing muMimoGroup index order. A portMapping-PRG is defined in exactly one muMimoGroup. Value: 0→3,821 ***Note***: *muMimoGroupIdx are defined in increasing order of their start symbol and on a given start symbol they are defined in increasing order of PRB*
	num-PRGs-In-MU-MIMO-Group-v6		uint16_t	Number of ulMappingResolution-PRGs in the MU MIMO group corresponding to muMimoGroupIdx
	For each of the ulMapping-PRGs {			
		numPrecodingWeights-v6	uint16_t	Number of elements in the precoding matrix numPrecodingWeights = numRxAnt x Layers
		precodingWeigths -v6	uint16_t [2 x numPrecodingWeights]	Complex precoding weights for each Rx antenna at the radio unit and for each of the layers in the current muMimoGroup. Weights[layerIdx, antIdx] = array[antIdx + layersIdx x num Rx Ant] • antIdx : 0 … num Rx Ant-1 • layerIdx : 0 … Layers-1
	}			
}				

Table 3-153 Rx Beamforming based on dynamic weight signalling by L2/L3

3.4.4 UL_DCI.request

The `UL_DCI.request` message includes DCI content used for the scheduling of PUSCH, except whenever the PDCCH PDU is part of a MU-MIMO Group in DL_TTI.request, or is referenced for rate-matching purposes in PDSCH PDUs; in these latter cases, MAC shall transmit UL_DCIs for the grants in the `DL_TTI.request` message, instead. The `UL_DCI.request` message format is shown in Table 3-154.

The separation of DL and UL DCIs introduces greater flexibility into the design and implementation of the MAC scheduler. Specifically, it permits separate DL (generates DL DCI) and UL (generates UL DCI) scheduling and removes the requirement to integrate the UL relevant information with the DL control in `DL_TTI.request`.

Field	Type	Description
sfn	uint16_t	SFN Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159

Field	Type	Description
num-PDUs	uint16_t	Number of PDCCH PDUs that are included in this message. Value 0 -> 65535
num-DL-Types	uint8_t	Maximum number of DL PDU types or DCIs supported by `UL_DCI.request`. nDlPduTypes = 2 in this release. Backward compatibility note: SCF222 v2.0 did not have this field
num-PDUs-OfEachType	uint16_t [nDlTypes]	Number of PDUs of each type that are included in this message. Each array entry corresponds to a PDU type as follows: [0]: number of PDCCH PDUs [1]: number of DCIs across all PDCCH PDUs in this message Value 0 -> 65535, for each entry Backward compatibility note: SCF222 v2.0 did not have this field
For Number of PDUs {		
pdu-Type	uint16_t	0: PDCCH PDU
pdu-Size	uint16_t	Size of the PDU control information (in bytes). This length value includes the 4 bytes required for the PDU type and PDU size parameters. Value 0 -> 65535
pdcch-PDU-Configuration	structure	See Section 3.4.2.1
}		
backwardCompatibleExtension-v7	structure	See Table 3-155 (see section 3.1.5)

Table 3-154 **`UL_DCI.request` message body**

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-155 **UL_DCI.request PDU Backward Compatible Extension**

3.4.5 SLOT errors

The error codes returned in an `ERROR.indication` generated by the `DL_TTI.request` message are given in Table 3-156.

Error code	Description
MSG_INVALID_STATE	The `DL_TTI.request` was received when the PHY was in the IDLE or CONFIGURED state.
OUT_OF_SYNC	The `DL_TTI.request` was received with a different SFN/SL than the PHY expected. The PHY has followed the SFN/SL sync process, see Section 2.2.3.

Error code	Description
MSG_BCH_MISSING	A SSB PDU was expected in the `DL_TTI.request` message for this subframe.
MSG_SLOT_ERR	The `DL_TTI.request` had an invalid format.

Table 3-156 Error codes for `ERROR.indication` generated by `DL_TTI.request`

The error codes returned in an `ERROR.indication` generate by the `UL_TTI.request` message are given in Table 3-157.

Error code	Description
MSG_INVALID_STATE	The `UL_TTI.request` was received when the PHY was in the IDLE or CONFIGURED state.
MSG_SLOT_ERR	The `UL_TTI.request` had an invalid format.

Table 3-157 Error codes for `ERROR.indication` generated by `UL_TTI.request`

The error codes returned in an `ERROR.indication` generate by the `UL_DCI.request` message are given in Table 3-158.

Error code	Description
MSG_INVALID_STATE	The `UL_DCI.request` was received when the PHY was in the IDLE or CONFIGURED state.
MSG_INVALID_SFN	The `UL_DCI.request` message received in subframe N included a SFN/SF value which was not N-1. The message has been ignored.
MSG_UL_DCI_ERR	The `UL_DCI.request` had an invalid format.

Table 3-158 Error codes for `ERROR.indication` generated by `UL_DCI.request`

3.4.6 TX_DATA.request

The format of the `TX_DATA.request` message is described in Table 3-159. This message contains the MAC PDU data for transmission over the air interface. The PDUs described in this message must follow the same order as `DL_TTI.request`.

This message can be sent by the L2/L3 when the PHY is in the RUNNING state. If it is sent when the PHY is in the IDLE, or CONFIGURED, state an `ERROR.indication` message will be sent by the PHY.

Field	Type	Description
sfn	uint16_t	SFN Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159
controlLength-v4	uint16_t	Length (in bytes) of control plane portion, including any padding bytes, if tag value 3 is used for TBS-TLV (i.e. CP/UP in-message separation), may be zero otherwise. See Table 3-37 for the related capability.

Field		Type	Description
			It is measured from the first byte of this message's body until then end of the control portion.
num-PDUs		uint16_t	Number of PDUs included in this message. Value 0-> MaxDlPDUsPerSlot
For each PDU {			
	pdu-Length	uint32_t	The total length (in bytes) of the PDU description and PDU data, without the padding bytes, from the start of this pdu-Length field, until the end of the tlvs field, including any padding. Note: if the tlvs field carries the TBS (e.g. tag value 0), then the pdu-Lengh includes the TBS length.
	pdu-Index	uint16_t	Used to correlate the MAC PDU with the DL_TTI PDSCH PDU Value: 0 → 65535
	cw-Index-v3	uint8_t	The CW in PDSCH PDU with PDU index, to which this MAC PDU corresponds. Value: 0, 1
	num-TLV	uint32_t	The number of TLVs describing the data of the transport block. Value: 0 -> MaxTLVs Shall be set to 1 if tag values 0 or 3 are used in TLVs
	tlvs	TBS-TLV [numTLV]	See Table 3-160
}			
backwardCompatibleExtension-v7		structure	See Table 3-162 (see section 3.1.5)

Table 3-159 **`TX_DATA.request` message**

Field	Type	Description
tag	uint16_t	Value: 0 -> 5 0: Payload is carried directly in the value field 1: 32-bit pointer to payload is in the value field 2: 32-bit offset from first address 3: offset from the end of control portion 4: 64-bit pointer to payload is in the value field 5: 64-bit offset from first address
length	uint32_t	Length of the actual payload in bytes, without the padding bytes
value	Variable or uint32_t or uint64_t	32 bits, or a multiple of the number of bits indicated by the pdschMacPduBitsAlignment capability (default 32-bits). **Tag=0**: Only the most significant bytes of the size indicated by 'length' field are valid. Remaining bytes are zero padded to the nearest bit boundary indicated by the pdschMacPduBitsAlignment capability (default 32-bit) **Tag=1/4**: Pointer to the payload. Occupies 32-bits for tag 1 and 64 bits for tag 4. **Tag=2/5**: Offset from the "first address" to the payload is in the value field. Where the first address is a predefined value. Occupies 32-bits for tag 2 and 64 bits for tag 5. **Tag=3**: Offset from the end of control portion to the payload is in the value field, where the end of control portion is indicated by the controlLength field in `TX_DATA.request`. Occupies 32-bits. Only the most significant bytes of the size indicated by 'length' field are valid. Remaining bytes are zero padded to the nearest bit boundary indicated by the pdschMacPduBitsAlignment capability (default 32-bit). In all cases, the bit mapping of the MAC TBS payload is per bit significance rules, as described in section 3.1.6.1.

Table 3-160 TBS-TLV structure for TX_DATA.request

3.4.6.1 Downlink Data Errors

The error codes returned in an `ERROR.indication` generate by the `TX_DATA.request` message are given in Table 3-161.

Error code	Description
MSG_INVALID_STATE	The `TX_DATA.request` was received when the PHY was in the IDLE or CONFIGURED state.

Error code	Description
MSG_INVALID_SFN	The `TX_DATA.request` message received in subframe N included a SFN/SF value which was not N. The message has been ignored.
MSG_TX_ERR	The `TX_DATA.request` had an invalid format.

Table 3-161 Error codes for `ERROR.indication`

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-162 TX_DATA.request PDU Backward Compatible Extension

3.4.7 RX_DATA.indication

The format of the `RX_DATA.indication` message is shown in Table 3-163.

Field		Type	Description
sfn		uint16_t	SFN Value: 0 -> 1023
slot		uint16_t	Slot Value: 0 ->159
controlLength-v4		uint16_t	Length (in bytes) of control plane portion, including any padding bytes, if Uplink UP/CP in-message separation, may be zero otherwise. See Table 3-37 for the related capability. It is measured from the first byte of this message's body until then end of the control portion.
num-PDUs		uint16_t	Number of PDUs included in this message. Only successfully-decoded PDUs are included. `RX_DATA.indication` shall be sent even if all TBS for the slot have failed to decode, in which case this field is set to 0. Range 0-> MaxULPDUsPerSlot
For each successfully-decoded PDU {			
	handle	uint32_t	The handle passed to the PHY in an `UL_TTI.request` PUSCH PDU or MsgA-PUSCH PDU.
	rnti	uint16_t	The RNTI passed to the PHY in an `UL_TTI.request` PUSCH PDU or MsgA-PUSCH PDU. If Handle corresponds to a MsgA-PUSCH message, this field indicates the RA-RNTI associated with the received PDU.

Field		Type	Description
			Value: 1 → 65535.
	rapid-v4	uint8_t	If Handle corresponds to a MsgA-PUSCH message, this field indicates the RAPID associated with the received PDU, otherwise it is set to 255. Value 0 -> 63, or 255
	harq-ID	uint8_t	HARQ process ID Value: 0->15
	pdu-Length	uint32_t	Length of PDU in bytes, from the start of this pdu-Length field, until the end of the tlvs field, including any padding. Note: if the tlvs field carries the TBS (e.g. tag value 0), then the pdu-Lengh includes the TBS length.
	num-TLV-v5	uint32_t	The number of TLVs describing the data of the transport block. Value: 0 -> MaxTLVs Shall be set to 1 if tag values 0 or 3 are used in TLVs
	tlvs-v5	TBS-TLV [numTLV]	See Table 3-160
}			
backwardCompatibleExtension-v7		structure	See Table 3-165 (see section 3.1.5)

Table 3-163 `RX_DATA.indication` **message body**

Field	Type	Description
tag-v5	uint16_t	Value: 0 -> 5 0: Payload is carried directly in the value field 1: 32-bit pointer to payload is in the value field 2: 32-bit offset from first address 3: offset from the end of control portion 4: 64-bit pointer to payload is in the value field 5: 64-bit offset from first address
length-v5	uint32_t	Length of the actual payload in bytes, without the padding bytes
value-v5	Variable or uint32_t or uint64_t	32 bits, or a multiple of the number of bits indicated by the puschMacPduBitAlignment capability (default 32-bits). **Tag=0**: Only the most significant bytes of the size indicated by 'length' field are valid. Remaining bytes are zero padded to the nearest bit boundary indicated by the puschMacPduBitAlignment capability (default 32-bit) **Tag=1/4**: Pointer to the payload. Occupies 32-bits for tag 1 and 64 bits for tag 2. **Tag=2/5** Offset from the "first address" to the payload is in the value field. Where the first address is a predefined value. Occupies 32-bits for tag 2 and 64 bits for tag 5. **Tag=3**: Offset from the end of control portion to the payload is in the value field, where the end of control portion is indicated by the controlLength field in `RX_DATA.indication`. Occupies 32-bits. Only the most significant bytes of the size indicated by 'length' field are valid. Remaining bytes are zero padded to the nearest bit boundary indicated by the puschMacPduBitAlignment capability (default 32-bit). In all cases, the bit mapping of the MAC TBS payload is per bit significance rules, as described in section 3.1.6.1.

Table 3-164 TBS-TLV structure for RX_DATA.indication

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-165 RX_DATA.indication PDU Backward Compatible Extension

3.4.8 CRC.indication

The format of the `CRC.indication` message is shown in Table 3-166. There can be more than one `CRC.indication` messages per slot.

Field		Type	Description
sfn		uint16_t	SFN Value: 0 -> 1023
slot		uint16_t	Slot Value: 0 ->159
num-CRCs		uint16_t	Number of CRCs (PDUs) with status indication included in this message. Range 0-> MaxULPDUsPerSlot
For each CRC {			
	handle	uint32_t	The handle passed to the PHY in an `UL_TTI.request` PUSCH PDU or MsgA-PUSCH PDU.
	rnti	uint16_t	The RNTI passed to the PHY in an `UL_TTI.request` PUSCH PDU or MsgA-PUSCH PDU. If Handle corresponds to a MsgA-PUSCH message, this field indicates the RA-RNTI associated with the received PDU, otherwise it is set to 255. Value: 1 → 65535
	rapid-v4	uint8_t	If Handle corresponds to a MsgA-PUSCH message, this field indicates the RAPID associated with the received PDU, otherwise it is set to 255. Value 0 -> 63, or 255
	harq-ID	uint8_t	HARQ process ID Value: 0->15
	tb-CRC-Status	uint8_t	Indicates CRC result on TB data. Value: 0 = pass 1 = fail
	num-CB	uint16_t	If CB CRC status is not reported, this parameter is set to zero. Otherwise, the number of CBs requested in PUSCH PDU. Note: in case of initial transmission or for non CBG Re-Tx is not used, this is the total number of CBs, otherwise (CBG Re-Tx) this is the number of CBs actually scheduled. Value: 0->65535
	cb-CRC-Status	uint8_t [ceil(NumCb/8)]	If NumCb=0 this field is not present. Otherwise each bit indicates CRC result on CB data. Value: 0 = pass 1 = fail

Field		Type	Description
			Note: for the special case where the CW is composed of a single CB, PHY may indicate NumCb=1 with CbCrcStatus set to the TB CRC status
	ul-SINR-Metric	int16_t	A metric of channel quality. Up to PHY implementation whether this is Signal-to-Thermal, Signal-to-Interference+Thermal or any other reasonable equivalent interpretation. Value: - 65.534 dB … +65.534 dB in steps of 0.002 dB (0 corresponds to 0 dB). -32768 = invalid.
	timingAdvanceOffset	uint16_t	Timing advance T_A measured for the UE in multiples of 16 * 64 * T_c / 2^μ [3GPP TS 38.213 [4], Section 4.2] Value: 0 → 63 0xffff should be set if this field is invalid
	timingAdvanceOffsetInNanoseconds-v4	int16_t	Timing advance measured for the UE between the reference uplink time and the observed arrival time for the UE Value: - 16800 … +16800 nanoseconds. -32768 should be set if this field is invalid
	rssi	uint16_t	RSSI. See Table 3-30 for RSSI definition. If RSSI is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSSI is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -128dBm to 0dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
	rsrp-v4	uint16_t	RSRP. See Table 3-30 for RSRP definition. If RSRP is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSRP is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -140dBm to -12dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
}			
backwardCompatibleExtension-v7		structure	See Table 3-167 (see section 3.1.5)

Table 3-166 `CRC.indication` **message body**

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-167 CRC.indication PDU Backward Compatible Extension

3.4.9 UCI.indication

The format of the `UCI.indication` message is shown in Table 3-168.

The `UCI.indication` message includes optional parameters and the same message is used for UCI received on both PUSCH and PUCCH. There can be more than one `UCI.indication` message per slot.

The format of the `UCI.indication` message is dependent on whether the UCI is transmitted on:

- PUSCH
- PUCCH Format 0 or 1
- PUCCH Format 2,3 or 4

The optional parameters are included based on information expected on UCI.

- SR parameters are included if a PUCCH PDU included as SR opportunity
- HARQ parameters are included if PUCCH or PUSCH PDU included HARQ
- CSI part 1 parameters are included if PUCCH or PUSCH PDU included CSI
- CSI part 2 parameters are included if PUCCH or PUSCH PDU included CSI part 2

Field	Type	Description
sfn	uint16_t	SFN Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159
controlLength-v4	uint16_t	Length (in bytes) of control plane portion, including any padding bytes, if tag value 3 is used for TBS-TLV (i.e. CP/UP in-message separation), may be zero otherwise. See Table 3-37 for the related capability. It is measured from the first byte of this message's body until then end of the control portion.
num-UCIs	uint16_t	Number of UCIs included in this message. Range 0-> MaxUCIsPerSlot
For each UCI {		
pdu-Type	uint16_t	0: UCI indication PDU carried on PUSCH, see Section 3.4.9.1. 1: UCI indication PDU carried on PUCCH Format 0 or 1, see Section 3.4.9.2.

Field		Type	Description
			2: UCI indication PDU carried on PUCCH Format 2, 3 or 4, see Section 3.4.9.3.
	pdu-Size	uint16_t	Size of the PDU information (in bytes). This length value includes the 4 bytes required for the PDU type and PDU size parameters. Value 0 -> 65535
	uci-PDU-Information	structure	See Sections 3.4.9.1 to 3.4.9.3.
}			
backwardCompatibleExtension-v7		structure	See Table 3-169 (see section 3.1.5)

Table 3-168 `UCI.indication` **message body**

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-169 UCI.indication PDU Backward Compatible Extension

3.4.9.1 PUSCH PDU

The format of a PUSCH PDU is shown in Table 3-170 and used for UCI received on the PUSCH.

Field	Type	Description
pdu-Bitmap	uint8_t	Bitmap indicating presence of optional PDUs. Value: 0 = not present, 1 = present Bit 0: not used Bit 1: HARQ Bit 2: CSI Part 1 Bit 3: CSI Part 2
handle	uint32_t	The handle passed to the PHY in an `UL_TTI.request` PUSCH PDU.
rnti	uint16_t	The RNTI passed to the PHY in an `UL_TTI.request` PUSCH PDU. Value: 1 → 65535.
ul-SINR-Metric	int16_t	A metric of channel quality. Up to PHY implementation whether this is Signal-to-Thermal, Signal-to-Interference+Thermal or any other reasonable equivalent interpretation. Value: - 65.534 dB … +65.534 dB in steps of 0.002 dB (0 corresponds to 0 dB). -32768 = invalid.
timingAdvanceOffset	uint16_t	Timing advance T_A measured for the UE in multiples of 16 * 64 * T_c / 2^μ [[3GPP TS 38.213 [4], Section 4.2] Value: 0 → 63

Field	Type	Description
		0xffff should be set if this field is invalid
timingAdvanceOffsetInNanoseconds-v4	int16_t	Timing advance measured for the UE between the reference uplink time and the observed arrival time for the UE Value: - 16800 … +16800 nanoseconds. -32768 should be set if this field is invalid
rssi	uint16_t	RSSI. See Table 3-30 for RSSI definition. If RSSI is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSSI is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -128dBm to 0dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
rsrp-v4	uint16_t	RSRP. See Table 3-30 for RSRP definition. If RSRP is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSRP is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -140dBm to -12dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
HARQ information	structure	Included if indicated by pdu-Bitmap. See Table 3-179 for details
CSI part1 information	structure	Included if indicated by pdu-Bitmap. See Table 3-180 for details
CSI part2 information	structure	Included if indicated by pdu-Bitmap. See Table 3-181 for details
backwardCompatibleExtension-v7	structure	See Table 3-171 (see section 3.1.5)

Table 3-170 UCI-PUSCH PDU

Field	**Type**	**Description**
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-171 UCI-PUSCH PDU Backward Compatible Extension

3.4.9.2 PUCCH PDU Format 0/1

The format of a PUCCH PDU Format 0/1 is shown in Table 3-172 and used for UCI received on the PUCCH with either Format 0 or 1.

Field	Type	Description
pdu-Bitmap	uint8_t	Bitmap indicating presence of optional PDUs. Value: 0 = not present, 1 = present Bit 0: SR Bit 1: HARQ
handle	uint32_t	The handle passed to the PHY in an `UL_TTI.request` PUCCH PDU.
rnti	uint16_t	The RNTI passed to the PHY in an UL_TTI.request PUCCH PDU. Value: 1 → 65535.
pucch-Format	uint8_t	PUCCH format Value: 0 -> 1 0: PUCCH Format0 1: PUCCH Format1
ul-SINR-Metric	int16_t	A metric of channel quality. Up to PHY implementation whether this is Signal-to-Thermal, Signal-to-Interference+Thermal or any other reasonable equivalent interpretation. Value: - 65.534 dB … +65.534 dB in steps of 0.002 dB (0 corresponds to 0 dB). -32768 = invalid.
timingAdvanceOffset	uint16_t	Timing advance T_A measured for the UE in multiples of 16 * 64 * T_c / 2^μ [3GPP TS 38.213 [4], Section 4.2] Value: 0 → 63 0xffff should be set if this field is invalid
timingAdvanceOffsetInNanoseconds-v4	int16_t	Timing advance measured for the UE between the reference uplink time and the observed arrival time for the UE Value: - 16800 … +16800 nanoseconds. -32768 should be set if this field is invalid
rssi	uint16_t	RSSI. See Table 3-30 for RSSI definition. If RSSI is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSSI is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -128dBm to 0dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
rsrp-v4	uint16_t	RSRP. See Table 3-30 for RSRP definition. If RSRP is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSRP is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value

Field	Type	Description
		represents -140dBm to -12dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
SR information	structure	Included if indicated by pdu-Bitmap. See Table 3-176 for details
HARQ information	structure	Included if indicated by pdu-Bitmap. See Table 3-177 for details
backwardCompatibleExtension-v7	structure	See Table 3-173 (see section 3.1.5)

Table 3-172 UCI PUCCH format 0 or 1 PDU

Field	**Type**	**Description**
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-173 UCI PUCCH format 0 or 1 PDU Backward Compatible Extension

3.4.9.3 PUCCH PDU Format 2/3/4

The format of a PUCCH PDU Format 2/3/4 is shown in Table 3-174 and used for UCI received on the PUCCH with either Format 2, 3 or 4.

Field	Type	Description
pdu-Bitmap	uint8_t	Bitmap indicating presence of optional PDUs. Value: 0 = not present, 1 = present Interpretation depends on the setting of uciReportFormat in the request PUCCH PDU: separate reporting: - Bit 0: SR - Bit 1: HARQ - Bit 2: CSI Part 1 - Bit 3: CSI Part 2 combined reporting (i.e. transparent UCI payloads): - Bit 2: uciPayload (or uciPart1Payload, if uci is composed of two parts) - Bit 3: uciPart2Payload (if uci is composed of two parts)
handle	uint32_t	The handle passed to the PHY in an `UL_TTI.request` PUCCH PDU.
rnti	uint16_t	The RNTI passed to the PHY in an `UL_TTI.request` PUCCH PDU. Value: 1 → 65535.
pucch-Format	uint8_t	PUCCH format

Field	Type	Description
		Value: 0 -> 2 0: PUCCH Format2 1: PUCCH Format3 2: PUCCH Format4
ul-SINR-Metric	int16_t	A metric of channel quality. Up to PHY implementation whether this is Signal-to-Thermal, Signal-to-Interference+Thermal or any other reasonable equivalent interpretation. Value: - 65.534 dB … +65.534 dB in steps of 0.002 dB (0 corresponds to 0 dB). -32768 = invalid.
timingAdvanceOffset	uint16_t	Timing advance T_A measured for the UE in multiples of 16 * 64 * T_c / 2^{μ} [3GPP TS 38.213 [4], Section 4.2] Value: 0 → 63 0xffff should be set if this field is invalid
timingAdvanceOffsetInNanoseconds-v4	int16_t	Timing advance measured for the UE between the reference uplink time and the observed arrival time for the UE Value: - 16800 … +16800 nanoseconds. -32768 should be set if this field is invalid
rssi	uint16_t	RSSI. See Table 3-30 for RSSI definition. If RSSI is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSSI is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -128dBm to 0dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
rsrp-v4	uint16_t	RSRP. See Table 3-30 for RSRP definition. If RSRP is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFS to 0dBFS with a step size of 0.1dB If RSRP is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -140dBm to -12dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
SR information	structure	Included if indicated by pdu-Bitmap. See Table 3-178 for details
HARQ information	structure	Included if indicated by pdu-Bitmap. See Table 3-179 for details

Field	Type	Description
CSI part1 information	structure	Included if indicated by pdu-Bitmap. See Table 3-180 for details
CSI part2 information	structure	Included if indicated by pdu-Bitmap. See Table 3-181 for details
uci-Payload (or Part1 Payload, if uci is composed of two parts)	structure	Included if indicated by pdu-Bitmap. See Table 3-180 for details
uciPart2Payload (if uci Payload, if uci is composed of two parts)	structure	Included if indicated by pdu-Bitmap. See Table 3-181 for details
backwardCompatibleExtension-v7	structure	See Table 3-175 (see section 3.1.5)

Table 3-174 UCI PUCCH format 2, 3 or 4 PDU

Field	**Type**	**Description**
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-175 UCI PUCCH format 2, 3 or 4 PDU Backward Compatible Extension

3.4.9.4 SR, HARQ and CSI Part 1/ 2 PDUs

The UCI tables for the SR, HARQ, CSI Part 1 and CSI Part 2 are provided in this Section.

Field	Type	Description
sr-Indication	uint8_t	Indicates if an SR was detected. Value: 0 = SR not detected 1 = SR detected
sr-ConfidenceLevel	uint8_t	Confidence level of detected SR Indicates the likelihood that the decoded output for this PDU is correct. Value: 0 = Good 1 = Bad 0xff should be set if this field is invalid

Table 3-176 SR PDU for format 0 or 1

Field	Type	Description
numHarq	uint8_t	Number of HARQ bits present in UCI Value: 1->2
harq-ConfidenceLevel	uint8_t	Confidence level of detected HARQ.

Field		Type	Description
			Indicates the likelihood that the decoded output for this PDU is correct. Value: 0 = Good 1 = Bad 0xff should be set if this field is invalid
For numHarq {			
	harq-Value	uint8_t	Indicates result on HARQ data. Value: 0 = NACK 1 = ACK 2 = DTX
}			

Table 3-177 HARQ PDU for format 0 or 1

Field	Type	Description
sr-BitLen	uint16_t	Indicates the number of SR bits which are expected on the PUCCH transmission. Valid for PUCCH Format 2/3/4 Value: 1 = one SR bit; 2 = two SR bits; 3 = three SR bits; 4 = four SR bits **Note**: When MAC is implemented according to 3GPP TS 38.215 [4][22] or [5], then maximum SR bits expected for a single UE is 4.
sr-Payload	uint8_t [ceil(sr-BitLen/8)]	Contents of SR payload, excluding any CRC. Bit mapping for this payload shall be per bit significance rules, as described in section 3.1.6.1, colloquially known as MSB0. NOTE: as described in section 3.1.6.1, if PHY signals padding capability other than 'implementation', padding bits are appended to the payload.

Table 3-178 SR PDU for format 2, 3 or 4

Field	Type	Description
harq-Detection-Status-v4	uint8_t	Indicates CRC result on UCI containing this HARQ data. Value: 1 = CRC Pass (used when CRC was attached)

Field	Type	Description
		2 = CRC Failure (used when UE is expected to attach CRC) 3 = DTX (undetected UCI) 4 = No DTX (indicates UCI detection) 5 = DTX not checked (indicates that PHY did not check for UCI DTX; in this case L2 may make use of other fields – e.g. UL SINR – to determine UCI validity)
harq-BitLen	uint16_t	Length of HARQ payload in bits. Shall be 0 if and only if detection status is 2 (CRC Failure), 3 (undetected UCI). Value: 0 -> 1706
harq-Payload	uint8_t [ceil(harq-BitLen/8)]	Contents of HARQ, excluding any CRC harq-BitLen is non-Zero only if detection status is 1 (CRC Pass), 4 (No DTX) or 5 (DTX not checked), otherwise the size of harq-Payload is zero. Bit mapping for this payload shall be per bit significance rules, as described in section 3.1.6.1, colloquially known as MSB0. - NOTE: as described in section 3.1.6.1, if PHY signals padding capability other than 'implementation', padding bits are appended to the payload.

Table 3-179 HARQ PDU for format 2, 3 or 4 or for PUSCH

Field	Type	Description
This table describes both UCI part 1 and CSI part 1 transport, since they share data structure, yet are mutually exclusive (i.e. L1 either signals CSI part 1 or UCI part 1, not both at the same time)		
part1DetectionStatus-v4	uint8_t	Indicates detection outcome on UCI/CSI containing this UCI/CSI Part1 data. Value: 1 = CRC Pass (used when CRC was attached) 2 = CRC Failure (used when UE is expected to attach CRC) 3 = DTX (undetected UCI/CSI) 4 = No DTX (indicates UCI/CSI detection) 5 = DTX not checked (indicates that PHY did not check for UCI/CSI DTX; in this case L2 may make use of other fields – e.g. UL SINR – to determine UCI/CSI validity)

Field	Type	Description
expectedPart1BitLen	uint16_t	Length of payload in bits Value: 0 ->1706
part1Payload	uint8_t [len]	Contents of UCI/CSI, excluding any CRC. This will be raw data matching UCI/CSI payload built by UE. Len = ceil(part1BitLen/8) or 0 len is derived from Expected Size only if detection status is 1 (CRC Pass), 4 (No DTX) or 5 (DTX not checked),otherwise len is zero (and this part1Payload shall be absent). Bit mapping for this payload shall be per bit significance rules, as described in section 3.1.6.1, colloquially known as MSB0. NOTE: as described in section 3.1.6.1, if PHY signals padding capability other than 'implementation', padding bits are appended to the payload. -

Table 3-180 UCI/CSI Inline Transport PDU

Field	**Type**	**Description**
This table describes both UCI part 2 and CSI part 2 transport, since they share data structure, yet are mutually exclusive (i.e. L1 either signals CSI part 2 or UCI part 2, not both at the same time)		
part2DetectionStatus-v4	uint8_t	Indicates CRC result on UCI/CSI Part2 data. Value: 1 = CRC Pass (used when CRC was attached) 2 = CRC Failure (used when UE is expected to attach CRC) 3 = DTX (undetected UCI) 4 = No DTX (indicates UCI detection) 5 = DTX not checked (indicates that PHY did not check for UCI DTX; in this case L2 may make use of other fields – e.g. UL SINR – to determine UCI validity)
part2BitLen	uint16_t	Length of UCI/CSI payload in bits. Shall be zero if and only if the detection status is 2 (CRC failure) or 3 (undetected UCI). Value: 0 ->1706
part2Payload-v5	*(see below)*	Signalling of part 2 contents, excluding any CRC will be according to the Tag and Value field, per the Tag and Value field descriptions. The actual payload length will be indicated by the *part2BitLen*. This will be raw data matching UCI/CSI payload built by UE. Len = ceil(part2BitLen/8)

Field		Type	Description
			len is non-zero only if detection status is 1 (CRC Pass), 4 (No DTX) or 5 (DTX not checked), otherwise the size of part2Payload is zero, i.e. this part2Payload shall be absent. Bit mapping for this payload shall be per bit significance rules, as described in section 3.1.6.1, colloquially known as MSB0. NOTE: as described in section 3.1.6.1, if PHY signals padding capability other than 'implementation', padding bits are appended to the payload. - This payload and its contents will be signalled as indicated by the Tag and Value fields below.
	tag-v5	uint8	Value: 0 -> 5 • 0: payload is carried directly in the Value: • 1: 32-bit pointer to payload is in the value field • 2: 32-bit offset from first address • 3: offset from the end of control portion • 4: 64-bit pointer to payload is in the value field • 5: 64-bit offset from first address
	value-v5	Variable or uint32_t or uint64_t	32 bits, or a multiple of the number of bits. **Tag=0**: Only the most significant bytes of the size described 'len' are valid. Remaining bytes are Zero padded to the nearest bit 32-bit boundary **Tag=1/4**: Pointer to payload cumulating to length 'len'. Occupies 32-bits for tag 1 and 64 bits for tag 4. **Tag=2/5**: Offset from the "first address" to the payload of size 'len' is in the Value field. Where the first address is a predefined value. Occupies 32-bits for tag 2 and 64 bits for tag 5. **Tag=3**: Offset from the end of control portion to the payload of size 'len' is in the value field, where the end of control portion is indicated by the controlLength field in UCI.indication. The payload occupies 32-bits.

Table 3-181 UCI/CSI TLV transport PDU (for Part2)

3.4.10 SRS.indication

The format of the `SRS.indication` message is given in Table 3-182.

Field	Type	Description
sfn	uint16_t	SFN

Field		Type	Description
			Value: 0 -> 1023
slot		uint16_t	Slot Value: 0 ->159
controlLength-v4		uint16_t	Length (in bytes) of control plane portion, including any padding bytes, if tag value 3 is used for TBS-TLV (i.e. CP/UP in-message separation), may be zero otherwise. See Table 3-37 for the related capability. It is measured from the first byte of the message's body until then end of the control portion.
numPDUs		uint8_t	Number of PDUs included in this message. Value: 0-> 255
For each SRS PDU {			
	handle	uint32_t	The handle passed to the PHY in the `UL_TTI.request` SRS PDU.
	rnti	uint16_t	The RNTI passed to the PHY in the `UL_TTI.request` SRS PDU. Value: 1 → 65535.
	timingAdvanceOffset	uint16_t	Timing advance T_A measured for the UE in multiples of 16 * 64 * T_c / 2^{μ} [3GPP TS 38.213 [4], Section 4.2] Value: 0 → 63 0xffff will be set if this field is invalid
	timingAdvanceOffset InNanoseconds-v4	int16_t	Timing advance measured for the UE between the reference uplink time and the observed arrival time for the UE Value: - 16800 … +16800 nanoseconds. -32768 should be set if this field is invalid
	srs-Usage-v4	uint8_t	0 – beamManagement 1 – codebook 2 – nonCodebook 3 – antennaSwitching 4 – 255: reserved **Note**: This field matches the SRS usage field of the SRS PDU for the Handle, including for the case of usage overloading.
	reportType-v4	uint8_t	The type of report included in or pointed to by Report TLV corresponds to the requested *Report Type* in the SRS PDU for the Handle: Values: • 0: Per-PRG and Symbol SNR (Table 3-184) • 1: Normalized Channel I/Q Matrix (Table 3-185) • 2: Channel SVD Representation (Table 3-186) • 3: Positioning Report (Table 3-187)

Field		Type	Description
			• 4: SU-MIMO Codebook Recommendation (Table 3-153 in SCF-222v6) – may only be used if the agreed protocol is FAPIv6 • 5: Channel 2D-DFT Representation (Table 3–189) • 6: SU-MIMO Codebook Recommendation v2 (Table 3-188) • 255: No report (the length field of Report TLV will be set to 0). Any values not listed above are reserved **Note**: The values above correspond to the bit positions of the Report Type field in the SRS PDU for the Handle.
	reportTLV-v4		See Table 3-183
backwardCompatibleExtension-v7		structure	See Table 3-190 (see section 3.1.5)

Table 3-182 `SRS.indication` **message body**

Field	Type	Description
tag-v4	uint16_t	Value: 0 to 5 0: Report is carried directly in the value field 1: 32-bit Pointer to payload is in the value field 2: 32-bit offset from first address 3: The offset from the end of the control portion of the message to the beginning of the report. 4: 64-bit Pointer to payload is in the value field 5: 64-bit offset from first address Other values are reserved.
length-v4	uint32_t	Length of the actual report in bytes, without the padding bytes
value-v4	Variable or uint32_t or uint64_t	32 bits, or a multiple of the number of bits indicated by the srs-TLV-BitAlignment capability (default 32-bits). **Tag=0**: Only the most significant bytes of the size indicated by 'length' field are valid. Remaining bytes are zero padded to the nearest 32-bit bit boundary **Tag=1/4**: Pointer to the payload. Occupies 32-bits for tag 1 and 64 bits for tag 4. **Tag=2/5**: Offset from the "first address" to the payload is in the value field. Where the first address is a predefined value. Occupies 32-bits for tag 2 and 64 bits for tag 5. **Tag=3** Offset from the end of the control portion of the message to the payload is in the value field, where the end of control portion is indicated by controlLength field in SRS.indication. Occupies 32-bits.

Table 3-183 SrsReport-TLV structure

Nomenclature note: srs-Report-PRG (shortened as PRG, when there is no ambiguity) in the context of SRS Reports is a unit meant to support frequency-domain report subsampling, and is not representative of the (Downlink) PRG definition in 3GPP TS 38.214 [5], Section 5.1.2.3. The only commonality with the latter PRG definition is that both srs-Report-PRG and the definition of PRG in are anchored at CRB0.

The structure below represents the SRS report for Per-PRG and Symbol SNR, as supported in FAPIv3. The only enhancement added in FAPIv4 is the ability to report at PRG resolution, based on the MAC-signalled PRG size in the corresponding SRS PDU.

Field			Type	Description
srs-Report-FD-Resoluition-v4			uint16_t	Size in RBs of an srs-Report-PRG. Value: 1->275
numSymbols			uint8_t	Number of symbols for SRS Value: 1 -> 4 If a PHY does not report for individual symbols, then this parameter should be set to 1.
wideBand-SNR			uint8_t	SNR value in dB measured within configured SRS bandwidth on each symbol. Value: • 0 → 254 representing -64 dB to 63 dB with a step size 0.5 dB • 255 will be set if this field is invalid
numReportedSymbols			uint8_t	Number of symbols reported in this message. This allows PHY to report individual symbols or aggregated symbols where this field will be set to 1. Value: 1->4
For each reported symbol {				
	num-PRGs		uint16_t	Number of PRGs to be reported for this SRS PDU. Value: 0-> 272
	For each PRG {			
		prg-SNR	uint8_t	SNR value in dB. Value: • 0 → 254 representing -64 dB to 63 dB with a step size 0.5 dB • 255 will be set if this field is invalid
	}			
}				

Table 3-184 FAPIv3 Beamforming report, with PRG-level resolution

The structure below is used to represent the complex channel Nu*Ng matrix H estimated between the UE's Nu antenna ports and gNBs Ng antenna ports. **H**[uI, gI] represents the flat fading approximation of the channel between the uI-th antenna at the UE and the gI-th antenna at the gNB.

Field	Type	Description
iq-Representation-v4	uint8_t	0: 16-bit complex number (iqSize = 2) 1: 32-bit complex number (iqSize = 4) See section 3.4.10.1
num-GNB-AntennaElements-v4	uint16_t	Ng: Number of gNB antenna elements Value: 0→511
num-UE-SRS-Ports-v4	uint16_t	Nu: Number of sampled UE SRS ports Value: 0→7
srs-Report-FD-Resolution-v4	uint16_t	Size in RBs of an srs-Report-PRG. Value: 1->272
num-PRGs-v4	uint16_t	Number of PRGs Np to be reported for this SRS PDU. Value: 0-> 272
channelMatrixH-v4	uint8_t [Np * Nu * Ng *iqSize]	Array of (numPRGs*Nu*Ng) entries of the type denoted by iqRepresentation H{PRG pI} [ueAntenna uI, gNB antenna gI] = array[pI*Ng*Np + uI*Np + gI], - pI: 0…Np-1 // PRG index - uI: 0…Nu-1 // UE antenna index - gI: 0…Ng-1 // gNB antenna index

Table 3-185 Channel I/Q Matrix

The structure below is used to represent the SVD decomposition complex channel Nu*Ng matrix H estimated between the gNBs Ng antenna ports and UE's Nu antenna ports; it can correspond to the channel in the uplink direction, or in the downlink direction, under reciprocity assumptions. **H**[uI, gI] represents the flat fading approximation of the channel between the gI-th antenna at the gNB and the uI-th antenna at the UE.

In particular, the table represents $\mathbf{H} = \mathbf{U}\,\Sigma\,V^H$, using the notation in section 2.4.4 of [31] with:

- **U**: *Nu* x *Nu* = matrix of *Nu* orthonormal left eigenvectors.
- **Σ:** *Nu* x *Ng* = rectangular matrix of singular values. Since only at most Nu diagonal entries may be non-zero, the table represents **Σ** as the diagonal entries of the $\mathbf{\Sigma}_F$ matrix of size *Nu* x *Nu*
- V^H: *Ng* x *Ng* = complex conjugate matrix of right eigenvectors. Only those right eigenvectors corresponding to entries in $\mathbf{\Sigma}_F$ are signalled in matrix V_F^H of size *Nu* x *Ng*

Note that, $\mathbf{H} = \mathbf{U}\,\Sigma\,V^H = = \mathbf{U}\,\Sigma_F\,V_F^H$

Field	Type	Description
ip-Representation-v4	uint8_t	0: 16-bit complex number (iqSize = 2) 1: 32-bit complex number (iqSize = 4) See section 3.4.10.1
singularValueRepresentation-v4	uint8_t	0: 8-bit linear representation (sSize = 1) (0 … 1] in steps of size 1/256 1: 16-bit linear representation (sSize = 2). (0 … 1] in steps of size 1/65536 See section 3.4.10.2
singularValueScaling-v4	int8_t	dB-domain representation of singular value scaling. Range: -10 … 50 in 0.25 dB steps. = (val/4) + 22 Values outside this range are undefined. See section 3.4.10.2
num-GNB-AntennaElements-v4	uint16_t	Ng: Number of gNB antenna elements Value: 0→511
num-UE-SRS-Ports-v4	uint8_t	Nu: Number of sampled UE SRS ports Value: 0→7
srs-Report-FD-Resolution-v4	uint16_t	Size in RBs of an srs-Report-PRG. Value: 1->272
num-PRGs-v4	uint16_t	Number of PRGs to be reported for this SRS PDU. Value: 0-> 272
matrixOfLeftEivenvectors**U**-v6	uint8_t [Np*Nu *Nu *iqSize]	Array of (Nu*Nu) entries of the type denoted by 'iq Representation' U{PRG pI}[ueAntenna uI, left eigenvector leI] = array[pI*Nu*Nu + leI*Nu +uI], - pI: 0…Np-1 // PRG index - uI: 0…Nu-1 - leI: 0…Nu-1
diagonalEntriesOfSingularMatrix**Sigma**-v6	uint8_t [Np*Nu *sSize]	Array of (Nu) entries of the type denoted by 'singular Values Representation' **Σ{**PRG pI**}**[left eigenvector leI, right eigenvector reI] = • 0 if leI ≠ reI • 0 if min(leI, reI) ≥ Nu • array[pI*Nu + leI] if leI = reI

complexConjugateOfMatrixOfRightEivenvectors**V**-v4	uint8_t [Np*Nu *Ng *iqSize]	Array of (Nu*Ng) entries of the type denoted by 'iq Representation' V_F^H\{PRG pI\}[right eigenvector reI, gNB antenna gI] = array[pI*Nu*Ng + reI*Ng + gI], - pI: 0…Np-1 // PRG index - reI: 0…Nu-1 - gI: 0…Ng-1

Table 3-186 Channel SVD Representation

The structure below is used to represent the Positioning report.

Field	Type	Description
ul-RelativeTimeOfArrival-v5	uint32_t	$T_{UL-RTOA}$, as recorded in [3GPP TS 38.215 [19], sec 5.1] Value: 0 … 1970049, per the mapping in [3GPP TS 38.133 [29], Table 13.1.1-1] (i.e. -985024 Tc … 985024 Tc) 0xFFFFFFFF: not available
gnb-RX-TX-TimeDifference-v5	uint32_t	gNB Rx – Tx time difference, as recorded in [3GPP TS 38.215 [19], sec 5.1] Value: 0 … 1970049, per the mapping in [3GPP TS 38.133, [29], Table 13.2.1-1] (i.e. -985024 Tc … 985024 Tc) 0xFFFFFFFF: not available
coordinateSystemforUL-AngleOfArrival-v5	uint8_t	0: Local 1: Global
ul-AngleOfArrival-v5	uint16_t	UL AoA, as recorded in [3GPP TS 38.215 [19], sec 5.1] Value: 0 … 3599, , per the mapping in [3GPP TS 38.133 [29], Table 13.4.1-1] (i.e. -180 … 180 degrees in step of 0.1 degrees) 0xFFFF: not available
ul-SRS-RSRP-v5	uint16_t	UL SRS-RSRP, as recorded in [3GPP TS 38.215 [19], sec 5.1] See Table 3-30 for RSRP definition. If RSRP is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -144dBFs to 0dBFS with a step size of 0.1dB If RSRP is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -156dBm to -12dBm, with a step size of 0.1dB Value: 0-1440 0xffff should be set if this field is invalid

Table 3-187 SRS-based Positioning Report

The structure below represents the SRS report for SU-MIMO Codebook Recommendation, as supported in FAPIv7.

Field	Type	Description
srs-Report-FD-Resolutio n -v6	uint16_t	Size in RBs of an srs-Report-PRG. Value: 1->275
wideBand-SNR-v6	uint8_t	SNR value in dB measured within configured SRS bandwidth on each symbol. Value: • 0 -> 254 representing -64 dB to 63 dB with a step size 0.5 dB • 255 will be set if this field is invalid
wideBand-RSRP-v6	uint16_t	UL SRS-RSRP measured for the SRS bandwidth, as recorded in [3GPP TS 38.215 [19], sec 5.1] See Table 3-30 for RSRP definition. If RSRP is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -144dBFs to 0dBFS with a step size of 0.1dB If RSRP is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -156dBm to -12dBm, with a step size of 0.1dB Value: 0-1440 0xffff should be set if this field is invalid
tri-v6	uint8_t	Recommended Transmit Rank Value: 1->4
tpmi-v6	uint8_t	TPMI, per 3GPP TS 38.211 [2], Section 6.3.1.5, for the recommended rank (TRI, above) and signalled TransformPrecoding (signalled in the corresponding UL_TTI.request SRS PDU) and number of SRS ports. Value: 0->27
num-PRGs-v6	uint16_t	Number of PRGs to be reported for this SRS PDU. Value: 0-> 272
For each PRG {		

	prg-SNR-v6	uint8_t	SNR value in dB. Value: • 0 → 254 representing -64 dB to 63 dB with a step size 0.5 dB • 255 will be set if this field is invalid
	prg-RSRP-v7	uint16_t	UL SRS-RSRP, as recorded in [3GPP TS 38.215 [19], sec 5.1], for the prg See Table 3-30 for RSRP definition. If RSRP is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -144dBFs to 0dBFS with a step size of 0.1dB If RSRP is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -156dBm to -12dBm, with a step size of 0.1dB Value: 0-1440 0xffff should be set if this field is invalid
}			

Table 3-188 SU-MIMO Codebook Recommendation

The table below represents the Channel 2D-DFT Representation, as supported in FAPIv6, according to the procedure in section 2.2.6.2b.

Field	Type	Description
numUE-SRS-Ports-v6	uint8_t	Nu: Number of sampled UE SRS ports Value: 1->16
srs-Report-FD-Resolution-v6	uint16_t	Size in RBs of an srs-Report-PRG. Value: 1->273
num-PRGs-v6	uint16_t	Number of PRGs to be reported for this SRS PDU. Value: 1-> 273
k-v6	uint8_t	K : Number of strongest beams to report Value: 1-> K_max **Note:** (informative) K beams may include the beams to serve the UE as well as the beams to avoid interference
dft-SizeAzimuth-v6	uint16_t	DFT size in azimuth domain Value: 1 ->65355
dft-SizeElevation-v6	uint16_t	DFT size in elevation domain Value: 1 ->65355
oversamplingInAzimuth-v6	uint8_t	Oversampling value in azimuth domain Value: 1 ->255

Field		Type	Description
			Note: Oversampling in azimuth = DFT size azimuth / num Rx Ant in azimuth (per polarization)
oversamplingInElevation-v6		uint8_t	Oversampling value in elevation domain Value: 1 ->255 Note: Oversampling in elevation = DFT size elevation / num Rx Ant in elevation (per polarization)
For each PRG {			
	beamIndices-v6	uint8_t [2*K* numUeSrsPorts]	Indices of the strongest beams obtained at the output of the 2D-DFT performed on the SRS based channel estimation Sent for each SRS basis vector; the determination of SRS basis vectors is up the L1 implementation and can correspond, e.g. to SVD eigenvectors or SRS Ports. Value: 1-> # of beams Indices are reported separately for each polarization. 2 polarizations are assumed. **Note:** numUeSrsPorts = # of SRS basis vectors Counting direction of DFT beams is first in the elevation direction. Example scenario: DFT size azimuth = 16, and DFT size elevation = 2, beam index #3 would be the first beam in the second column in the DFT output. Beam indices [ueSrsBasisCounter A, beamCounter B, polarizationCounter P] = array[P + 2*B + 2*K*A] , - A: 0…numUeSrsPorts-1 - B: 0…K-1 - P: 0…1
	beamValues-v6	uint16_t [2*2*K* numUeSrsPorts]	Complex values of the amplitude of the strongest beams obtained at the output of the 2D-DFT performed on the SRS based channel estimation. Sent for each SRS basis vector. **Note:** - numUeSrsPorts = # of SRS basis vectors - in linear scale - L2 might use the Beam values in capacity calculation when selecting users Beam values array is indexed the same way as the Beam indices array.
	rank-v6	uint8_t	SU-MIMO rank suggestion for the UE made by PHY to be used by L2/L3.

Field		Type	Description
			Note: Rank suggestion is sent per PRG, since PHY does not know in which PRGs the UE will be scheduled.
}			

Table 3–189 Channel 2D-DFT Representation

Field	Type	Description
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-190 SRS.indication PDU Backward Compatible Extension

3.4.10.1 I-Q Representation

Based on PHY capability signalled in [TBD Capability TLV], MAC may request `SRS.indication` reports represent I-Q samples in 16-bit or 32-bit representations.

16-bit representation

An array ***a*** of N 16-bit I-Q samples represents the k^{th} normalized complex value entry (k = 0…N-1) in 8-bit array entries 2*k and 2*k+1 as follows:

- Normalized I-sample: (int8_t)a[2*k]/128.0
- Normalized Q-sample: (int8_t)a[2*k+1]/128.0

Note: array ***a*** has size N*2, and the k-th entry is illustrated below:

a[2k+1]	a[2k]
Q-sample	*I-sample*

32-bit representation

An array ***a*** of N 32-bit I-Q samples represents the k^{th} normalized complex value entry (k = 0…N-1) in 8-bit array entries 4*k, 4*k+1, 4*k+2, 4*k+3 as follows:

- Normalized I-sample: (int16_t)(a[4*k + M] << 8 + a[4*k + 1-M]) / 32768.0
- Normalized Q-sample: (int16_t)(a[4*k + 2 + M] << 8 + a[4*k + 3-M]) / 32768.0

Where M = 1 if FAPI uses little-endian notation, otherwise M=0

Note: array ***a*** has size N*4, and the k-th entry is illustrated below:

a[4k+3]	a[4k+2]	a[4k+1]	a[4k]
Q-sample		*I-sample*	

3.4.10.2 Singular Value Representation

Singular values in the diagonal matrix Σ of the channel SVD decomposition are positive values. Post normalization, these values range between 0…1. Each actual singular value is obtained after accounting for normalized representation described in this section, and the singular value scaling (see Table 3-186):

Singular value $v_{linear} = v_{linear,normalized} * 10\text{^}((\text{singular value scaling})/20)$

This section describes two normalized singular value representations supported in this specification:

8-bit linear representation

An array ***a*** of N 8-bit dB values represents the k[th] normalized entry (k = 0…N-1) in 8-bit array entry k as follows:

- $V_{linear,normalized} = (a[k] + 1)/256$

Notes:

- the range covers the following values: 1/256, 2/256, … 255/256, 1

- array ***a*** has size N bytes and the k-th entry is illustrated below:

a[k]
$V_{linear,normalized}$

16-bit linear representation

An array ***a*** of N 16-bit $v_{linear,normalized}$ samples represents the k[th] normalized complex value entry (k = 0…N-1) in 8-bit array entries 2*k, 2*k+1 as follows:

- $v_{linear,normalized} = (uint16\_t)(a[2*k + M] << 8 + a[2*k + 1-M] + 1) / 65536.0$

Where M = 1 if FAPI uses little-endian notation, otherwise M=0

Notes:

- the range covers the following values: 1/65536, 2/65536, … 65535/65536, 1

- the uint16_t 16-bit representation follows the same byte endianness as FAPI

- array ***a*** has size N*2, and the k-th entry is illustrated below:

a[2k+1]	a[2k]
$V_{linear,normalized}$	

3.4.11 RACH.indication

The format of the `RACH.indication` message is given in Table 3-191. There can be more than one `RACH.indication` message per slot.

Field	Type	Description
sfn	uint16_t	SFN Value: 0 -> 1023
slot	uint16_t	Slot Value: 0 ->159
numPDUs	uint8_t	Number of Measurement PDUs included in this message. Value: 0-> 255
For each Measurement:		
handle-v3	uint32_t	Handle for the `UL_TTI.request` RACH PDU to which this report is linked
symbolIndex	uint8_t	The index of first symbol of the PRACH TD occasion. [3GPP TS 38.211 [2], sec 6.3.3.2 and Tables 6.3.3.2-2 to 6.3.3.2-4] Value: 0->13
slotIndex	uint8_t	The index of first slot of the PRACH TD occasion in a system frame Value: 0->79
ra-Index	uint8_t	The index of the received PRACH frequency domain occasion. [3GPP TS 38.211 [2], sec 6.3.3.2] Value: 0->7
avg-RSSI	uint16_t	RSSI. See Table 3-30 for RSSI definition. If RSSI is reported in dBFS (see PARAM and CONFIG measurement TLVs) then this value represents -128dBFs to 0dBFS with a step size of 0.1dB If RSSI is reported in dBm (see PARAM and CONFIG measurement TLVs) then this value represents -128dBm to 0dBm, with a step size of 0.1dB Value: 0-1280 0xffff should be set if this field is invalid
avgSNR	uint8_t	Average value of SNR in dB Value: 0 → 254 representing -64 dB to 63 dB with a step size 0.5 dB. 255 should be set if this field is invalid.
numPreambles	uint8_t	Number of detected preambles in the PRACH occasion. Value: 0->64
For each preamble		
preambleIndex	uint8_t	Preamble Index. Value: 0-> 63

Field			Type	Description
		timingAdvanceOffset	uint16_t	Timing advance for PRACH. [3GPP TS 38.213 [4], sec 4.2] Value:0 -> 3846 in units of in multiples of 16 * 64 * T_c / 2^μ 0xffff should be set if this field is invalid.
		timingAdvanceOffsetInNanoseconds-v4	uint32_t	Timing advance measured for the UE between the observed arrival and the reference uplink time for the UE Value: 0 … 2005000 nanoseconds. 0xffffffff should be set if this field is invalid
		preamblePwr	uint32_t	Preamble Received power in dBm or dBFS, depending on the P5 capability (preamblePowerUnit) and configuration (preamblePowerUnitSelection) Value: 0->170000 representing Reporting range in dBm: -140dBm to 30dBm with a step size of 0.001dB Reporting range in dBFS: -170 dBFS to 0 dBFS with a step size of 0.001dB 0xffffffff should be set if this field is invalid Notes: - step size should not be used as an expected accuracy guideline. - implementations are not expected to report preamblePwr > 15 dBm
		preamble-SNR-v3	uint8_t	Preamble SNR in dB Value: 0 → 254 representing -64 dB to 63 dB with a step size 0.5 dB. 255 should be set if this field is invalid.
		<deprecated>	uint16_t	<this field is deprecated>
backwardCompatibleExtension-v7			structure	See Table 3-192 (see section 3.1.5)

Table 3-191 `RACH.indication` **message body**

Field	**Type**	**Description**
Backward compatible extension. See section 3.1.5.		
extensionLength-v7	uint16_t	Size of this backward compatible extension. See section 3.1.5 for what this length represents.

Table 3-192 RACH.indication PDU Backward Compatible Extension

3.4.12 DL Node Sync

The DL Node Sync message is supported when using the nFAPI message format for a PHY that supports Delay Management. The body of the DL Node Sync message is as documented in section 4.1.1 of SCF-225 [10].

3.4.13 UL Node Sync

The UL Node Sync message is supported when using the nFAPI message format for a PHY that supports Delay Management. The body of the UL Node Sync message is as documented in section 4.1.2 of SCF-225 [10].

3.4.14 Timing Info

The Timing Info message is supported when using the nFAPI message format for a PHY that supports Delay Management. The body of the Timing Info message is as documented in section 4.1.3 of SCF-225 [10].

References

[1] 3GPP TS 38.401 NG-RAN Architecture Description v16.10.0
[2] 3GPP TS 38.211 NR Physical Channel and Modulation v15.10.0
[3] 3GPP TS 38.212 NR Multiplexing and Channel Coding v15.13.0
[4] 3GPP TS 38.213 NR Physical Layer Procedures for Control v15.15.0
[5] 3GPP TS 38.214 NR Physical Layer Procedures for Data v15.16.0
[6] 3GPP TS 38.331 NR Radio Resource Control (RRC) Protocol Specification v15.19.0
[7] 3GPP TS 38.104 NR Base-station (BS) Radio Transmission and Reception v15.18.0
[8] 3GPP TS 38.300 NR Overall Description Stage-2 v15.13.0
[9] SCF 223 5G FAPI P19 RF and Digital Front End Control API v5.0
[10] SCF 225 5G nFAPI Specification v3.1
[11] 3GPP TS 38.300 NR Overall Description Stage-2 v16.12.0
[12] 3GPP TS 38.211 NR Physical Channel and Modulation v16.10.0
[13] 3GPP TS 38.331 NR Radio Resource Control (RRC) Protocol Specification v16.10.0
[14] 3GPP TS 29.274 3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunnelling Protocol for Control plane (GTPv2-C); Stage 3 v15.9.0
[15] 3GPP TS 38.212 NR Multiplexing and Channel Coding v16.11.0
[16] 3GPP TS 38.214 NR Physical Layer Procedures for Data v16.13.0
[17] 3GPP TS 38.104 NR Base-station (BS) Radio Transmission and Reception v16.15.0
[18] 3GPP TS 38.215 NR; Physical layer measurements v15.7.0
[19] 3GPP TS 38.215 NR; Physical layer measurements v16.6.0
[20] 3GPP TS 38.321 NR; Medium Access Control (MAC) protocol specification v15.13.0
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[31] Gene H. Golub et al, Matrix computations, 3rd Edition, Johns Hopkins Studies in Mathematical Sciences
[32] SCF 222 5G FAPI PHY API v6.0